New techniques and insights for burned area monitoring from remote sensing data

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I, James Brennan confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis.

Abstract

Satellite-derived datasets recording the global extent of burned area (BA) have been generated in the past two decades. These products provide vital information to fire-related disciplines. The algorithms used to produce these products generally share a low level of methodological consistency, are highly empirical, and lack some of the sufficient features required to produce consistent and error characterised long-term data records (LTDRs).

This thesis sets out to re-examine the quality of the algorithms used to produce these products with the aim of advancing the quality of future BA products. While BA products have been validated and inter-compared, little is known about the individual qualities and sensitivities of their respective algorithms. A novel sensitivity analysis of six global burned area algorithms provides information on the key sensitivities determining detection of burned area from the observations as well as highlighting the limitations of present algorithms.

This analysis highlights the need for quantitative quality information within BA products through uncertainty characterisation. Best practice frameworks for uncertainty characterisation have been developed for many Essential Climate Variables (ECVs) but not yet burned area. The thesis then goes on to develop suitable methodologies for estimating uncertainties in present BA products and future products. Global uncertainties of 4–6% are calculated for current BA products, and the necessary error propagation frameworks adapted to be amenable to binary ECV records such as burned area.

The information gained from the sensitivity analysis and advancement of uncertainty characterisation provides a framework to prototype multi-sensor algorithms suitable for producing uncertainty characterised LTDR BA datasets. This involves the application of a spectrally invariant model of the burn signal suitable for mapping BA in a probabilistic manner from optical sensors to provide consistent estimates through the satellite record.

Impact statement

The awareness of the effects of biomass burning amongst the public has grown in recent years due to high-profile fire events in Australia (2020), the Amazon basin (2019), the United Kingdom (2018), and many other regions. Burned area (BA) data sets derived from satellite remote sensing provide scientific evidence to contextualise these events over more than twenty years of global fire monitoring.

This thesis provides key advancements to methods for the production and analysis of BA data sets. The development of an algorithm intercomparison exercise provides the fire monitoring science community with a framework to compare algorithms and identify issues for improvement. The replicability of the framework means that the BA science community could adopt it for the continual improvement of BA algorithms. The uncertainty quantification methods developed also provide useful advances for other presence/absence remote sensing variables. The estimated uncertainties for three global BA products have real value for Earth system and vegetation modellers to evaluate the quality of their predictions with the BA observational uncertainties estimated in chapter 5 which have been published in a journal and provided to the community through an online data repository. The multi-sensor BA algorithm also provides advances to the academic community for addressing the aims of programs such as the European Space Agency's Climate Change Initiative which explicitly details requirements for multi-sensor remote sensing data records. Beyond the BA science community, the thesis also provides important contributions. Better uncertainty quantification (UQ) is useful to many stakeholders. Land use management and forestry services can benefit from improved uncertainty information in future BA products arising from advances in UQ made here. This information will be useful for assessing the confidence of individual burned pixel detections and therefore the allocation of resources. The wider UQ for present global products is useful for policymakers to evaluate the strength of evidence and precision of these products for decision making.

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Chapter One

Objectives of thesis

1.1 Introduction

Fire is one of the most significant global disturbances in the terrestrial biosphere (IPCC, 2019; Bowman et al., 2009). It dramatically influences vegetation structure and ecosystem function — altering vegetation succession, accelerating nutrient recycling, inducing serotinous seed germination, and volatilises soil nutrients (Whelan, 1995). As a result, fire plays a significant role in determining global ecosystem geography (Krawchuk et al., 2009). The influence of fire in the Earth system extends to the climate: fire contributes considerably to atmospheric trace gases and aerosol loading, releasing an estimated 1–3 Pg C $year^{-1}$ of carbon emissions (van der Werf et al., 2017). It also changes the surface energy budget by reducing surface albedo (Jin and Roy, 2005; Lyons et al., 2008) and sensible heat fluxes (Liu et al., 2005). At regional scales, biomass burning reduces surface albedo through the deposition of carbon soot on glaciers, sea ice and the Greenland ice sheet (Warren and Wiscombe, 1980; Hadley and Kirchstetter, 2012). Fire has detrimental impacts on human health, most commonly by affecting cardiovascular function from high concentrations of particulate matter and toxins present in smoke (Hansen et al., 2019; Koplitz et al., 2016; Finlay et al., 2012; Mirabelli et al., 2009) (see figure 1.1).

The importance of fire in the terrestrial biosphere necessitates global spatially and temporally explicit datasets of fire occurrence. Since the 1980s, efforts have been made to observe actively burning fires with Earth Observation (EO) data (Dozier, 1981). But these active fire observations provide an incomplete sampling of the true area affected by fire due to the limited temporal sampling of available satellite sensors modulated by cloud cover. The link between actively burning fires and true burned area is hard to determine, being a function of both fire process and the sampling features of the observations (Giglio et al., 2006). An alternative approach has been to determine burned area on the

premise that an identifiable signal remains on the land surface after a fire has occurred. From this principle, multi-decadal EO derived datasets of burned area have now been generated (see figure 1.2) (Chuvieco et al., 2018; Giglio et al., 2018; Alonso-Canas and Chuvieco, 2015; Tansey et al., 2008; Roy et al., 2005a).

Alongside these developments has been a focus on the generation of Climate Data Records (CDR) for a set of so-called Essential Climate Variables (ECVs). The 50 ECVs as defined by the Global Climate Observing System (GCOS) correspond to the set of key variables determined crucial for monitoring the Earth system (Mason et al., 2010; Bojinski et al., 2014). By necessity, these CDRs are required to provide the longest possible record of fire activity and be of the highest quality possible following measurement principles developed in the field of metrology (Bojinski et al., 2014; Merchant et al., 2017; BIPM et al., 1995). Further, it is expected that these CDR products are produced from several contrasting algorithms that have been intercompared to indicate any structural or scientific uncertainties (Bojinski et al., 2014; Bates and Privette, 2012; Nightingale et al., 2019). GCOS have also proposed accuracy targets for each ECV to ensure that the data sets are of the best quality possible. For burned area ECVs these are determined to be maximum commission and omission errors of 15% (GCOS, 2011). At present, burned area products do not reach these standards and it is the primary argument in this thesis that to achieve this task requires methodological advances in the production of these products.

1.2 Aims of thesis

The thesis aims to develop a set of best practices for the design and assessment of burned area algorithms in the "ECV age". The aim of EO based burned area monitoring should be:

A consistent ECV record of global burned area recognising the limitations of observations and algorithms via uncertainty characterisation.

Achieving this task is considered via three objectives:

I Evaluating present practices and developing new methods to assess the qualities of burned area mapping algorithms.

II Development of uncertainty quantification (UQ) methods for burned area products and algorithms following metrological best practice.

III Prototyping an algorithm for the consistent inference of burned area from multiple sensors suitable for generating a long-term CDR of burned area.

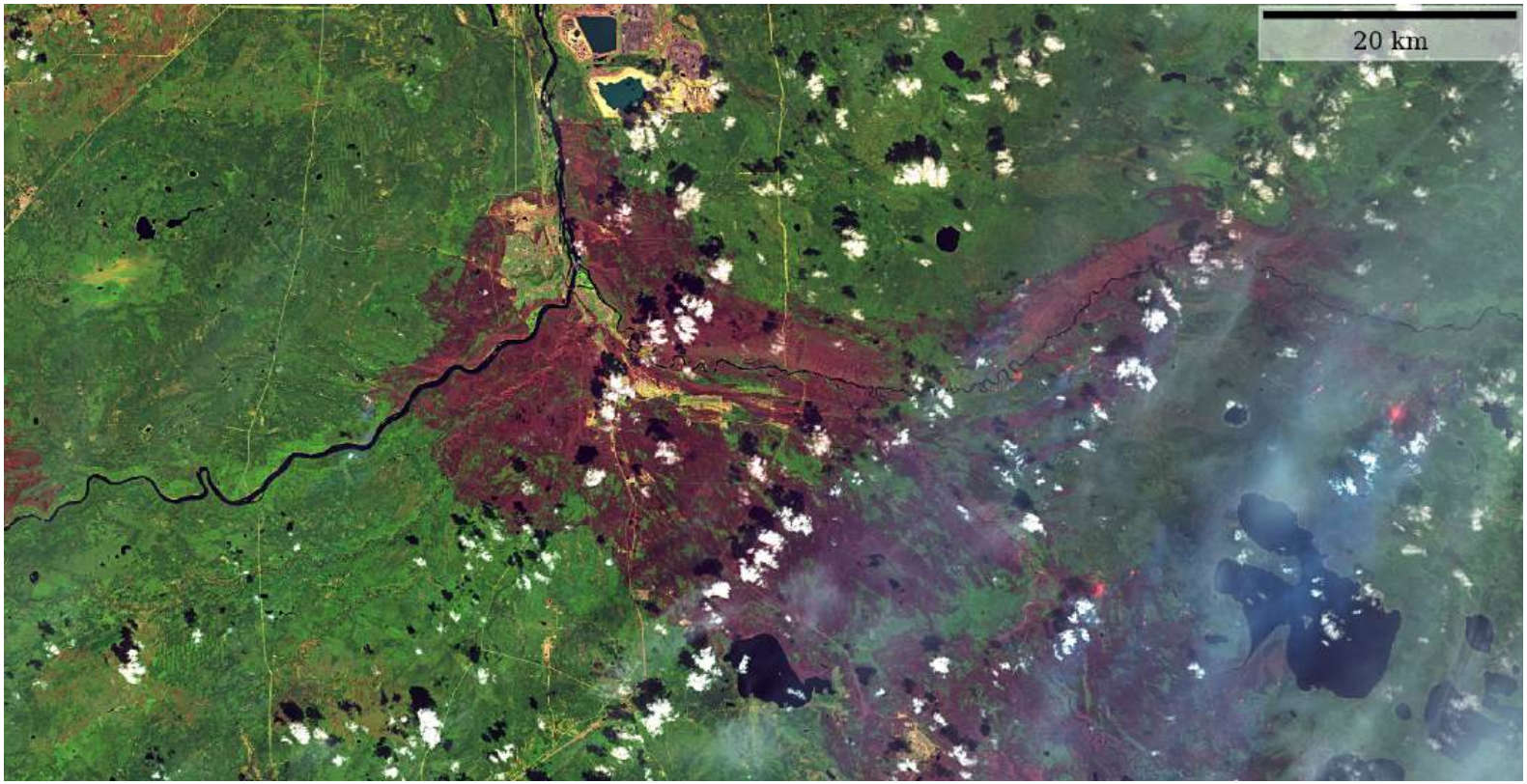


Figure 1.1. Fort McMurray Fire 2016 Acquisition by the Operational Land Imager (OLI) on board Landsat 8 showing the extent of the Fort McMurray Fire burn scar, 2016. Source: NASA/USGS; earthobservatory.nasa.gov/NaturalHazards/view.php?id=88039

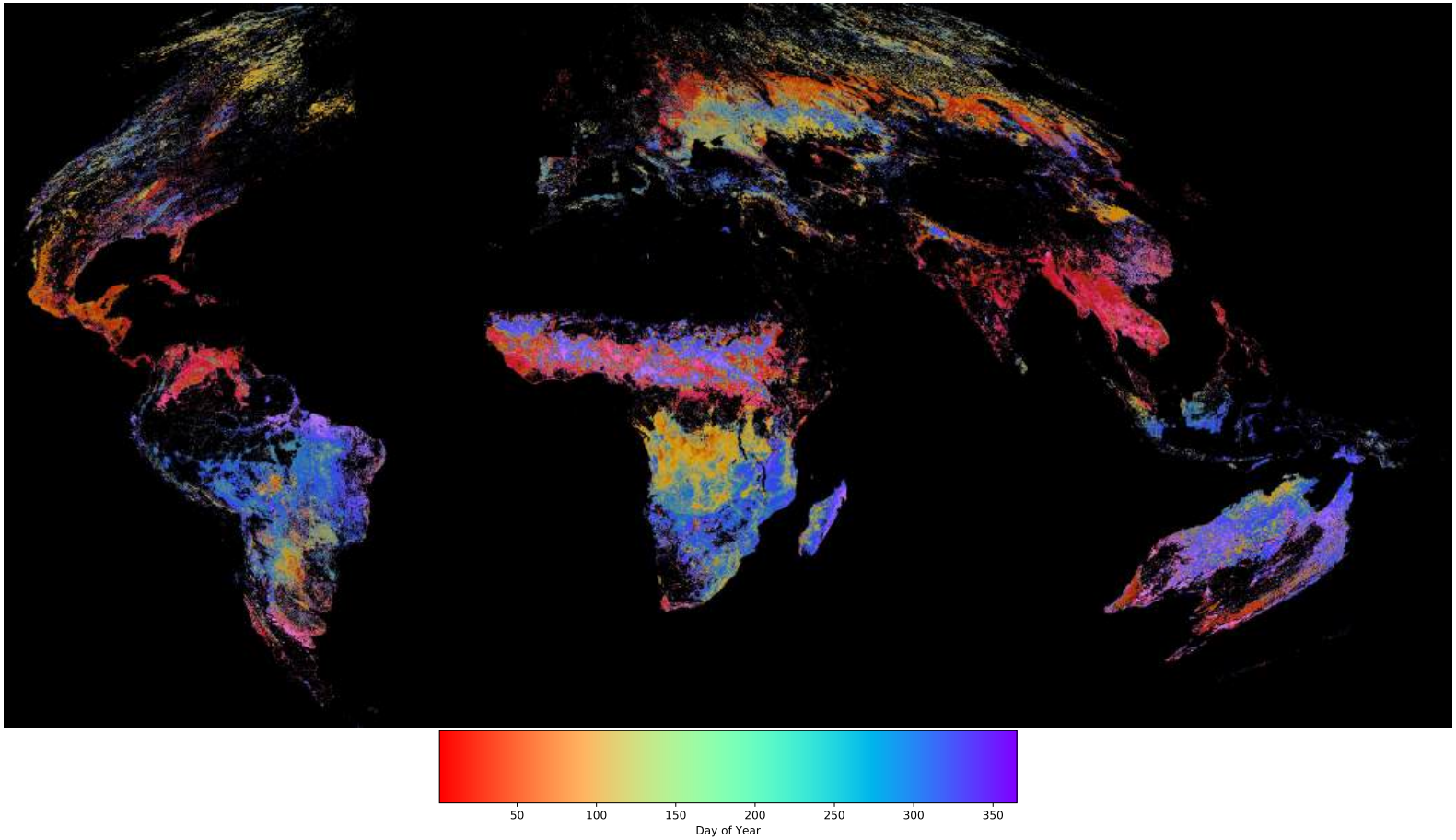


Figure 1.2. 15 years of fire. Visualisation of every 500m burned pixel (by day of burning) from the MODIS Collection 6 burned area algorithm for 2000—2015. Visualisation created with datashader (http://datashader.org) .

Part I New methods for evaluating burned area algorithms

The present generation of global burned area products have been validated against higher resolution data from sensors such as Landsat and Sentinel-2 (Roteta et al., 2019; Chuvieco et al., 2018; Giglio et al., 2018; Padilla et al., 2015). While these initiatives have provided comparative information on the overall accuracy of each product, they have provided limited information about steps to improve the current generation of algorithms. Instead, the focus has generally been on comparing individual algorithm components in isolation (Chuvieco et al., 2005; Miettinen and Liew, 2008; Sousa et al., 2003; Chuvieco et al., 2002; Huang et al., 2016). Algorithm intercomparison initiatives have aided the development of products in other remote sensing fields (Bontemps et al., 2011; Brewin et al., 2015; Ruddick et al., 2010; Nechad et al., 2015; Widlowski et al., 2007). These initiatives attempt to evaluate different algorithms by identifying strengths and limitations and possible refinements. Such initiatives also represent an opportunity to assess the present maturity and optimality of mapping algorithms — the "state of the art" — in comparison to requirements for CDR generation (Nightingale et al., 2019; Bates and Privette, 2012).

Chapter 2 reviews the development of algorithms for global burned area mapping, evaluates the methods used, and the key challenges which determine the performance of these algorithms. Chapter 3 then goes on to develop an intercomparison exercise of six globally-applied burned area algorithms. Based on statistical emulation of the algorithms the key drivers of omission and commission error are quantified and ranked. The examination reveals that only a few features determine the efficacy of each algorithm and provides directions for improving future algorithms.

Part II Uncertainty characterisation for burned area mapping

Developers of Earth system models (including dynamic global vegetation models and fire models) wish to confront their models with observations of global fire extent. This requires model-data comparisons in which the uncertainties of the observations are quantified (Scholze et al., 2017; Rayner et al., 2016; Exbrayat et al., 2019; Raupach et al., 2005). So far such initiatives have been unfeasible because present products do not have sufficient uncertainty information (Mouillot et al., 2014; Yue et al., 2014; Hantson et al., 2016; Knorr et al., 2014). Beyond modellers, other users have also expressed interest in per-pixel uncertainties and probabilistic statements of burned area (Mouillot et al., 2014). Similarly, to generate the highest quality scientific CDRs requires that the principles of metrology are followed (Merchant et al., 2017). And for burned area products, the Global Climate Observing System (GCOS) which

evaluate observing systems for climate monitoring detail that a full error characterisation is necessary (GCOS, 2009).

The generation of uncertainty quantification (UQ) methods for burned area products and algorithms forms the next section of the thesis. Chapter 4 discusses the sources of uncertainty in burned area product generation and outlines the principles of metrology for the generation of CDR satellite products (BIPM et al., 1995; Merchant et al., 2017). Following on from this, two methods of UQ for burned area products are considered. Chapter 5 presents a method for quantifying uncertainties on burned area estimates from three global products. To produce the next generation of uncertainty characterised burned area products requires a more thorough development of UQ methods. Chapter 6 develops an uncertainty framework and demonstrates the methods on an example algorithm. This reveals that a proper treatment of uncertainties reveals additional information about the problem and helps to quantify the available knowledge within mapping algorithms.

Part III Algorithms for consistent multi-sensor burned area mapping

The two advances above open up the path for producing satellite-derived burned area products that fulfil the aims of long-term data records. To produce the longest possible data records of burned area will require the use of multiple satellite sensors. The principal aim of these records is to provide temporally stable and consistent data products of the highest quality possible for diverse users (Merchant et al., 2017; Hollmann et al., 2013). Achieving this aim for burned area requires a transition towards algorithms which can infer burning from multiple sensors in a consistent manner. Practically, this entails algorithms which can deal with the fluctuating spectral, temporal, and spatial resolution of instruments through the coarse-resolution satellite record. But also merge these observational datasets consistently through uncertainty quantification and propagation.

Addressing this objective represents the final part of the thesis. Chapter 7 considers the choices for generating long-term CDR products for burned area, and the necessary mathematical frameworks for merging satellite observational records. The chapter then goes on to prototype an algorithm for the consistent retrieval of burned area with uncertainties from multiple sensors. The algorithm combines a spectrally-invariant model of the optical fire signal with data assimilation methods to merge observations from different instruments. The algorithm is prototyped on observations from the two independent MODIS Aqua and Terra instruments. Chapter 8 then demonstrates the algorithm implemented to run on a multi-sensor system comprised of observations from Suomi-NPP VIIRS, NASA MODIS and Sentinel-3 OLCI. The outputs are validated against independent reference data

collected from Sentinel-2. The algorithm retrieves burned area consistently from different sensors indicating it suitable for the generation of a long-term multi-sensor burned area CDR.

Part I

The science of burned area mapping

Chapter Two

Burned area detection from space

2.1 A need for global monitoring

Biomass burning is a disturbance that affects almost all biomes on Earth, and originates at the dawn of terrestrial plants (~400 million years ago, Scott and Glasspool, 2006). For most of the period since, fire has remained a predominantly natural disturbance; leading to alterations in vegetation structure, driving plant evolution, ecosystem regeneration and biodiversity (Kelly and Brotons, 2017; Bond and Keeley, 2005). In more recent geological history, fire has become a predominantly human-driven phenomenon both through ignition and suppression (Pyne, 1997; Pechony and Shindell, 2010; Andela et al., 2017). Consequences of this shift have been a considerable alteration to vegetation regeneration in fire-prone regions, and the exposure of previously natural ecosystems such as equatorial evergreen forests to fire (Potapov et al., 2017; Cochrane, 2003). Both natural and anthropogenic biomass burning drives considerable fluxes within the global carbon cycle and contributes a significant source of greenhouse gas emissions (van Der Werf et al., 2017; Knorr et al., 2016; Bowman and Johnston, 2005). With a growing awareness of the need to reducing emissions for mitigating climate change, and the role of fire in the Earth system there is a need for global long-term (>30 years) records of fire activity (Bowman and Johnston, 2005; Flannigan et al., 2009; Stocker et al., 2013).

Fire as a disturbance can lead to dynamic shifts in ecosystem function and vegetation type. If the properties of fire frequency and severity vary far from historical conditions, ecosystems may alter to different states. Increases in fire frequency can shift many closed forest ecosystems towards open grasslands (D'Antonio and Vitousek, 1992). Grasses have a higher surface to volume ratio and therefore are more flammable when desiccated. Grasslands also have lower evapotranspiration relative to closed forests, which lead to higher surface temperatures and greater vapour pressure deficits.

These characteristics promote burning, and due to the rapid recovery of grasses, shorter fire cycles which further promote the spread of grassland ecosystems (D'Antonio and Vitousek, 1992). As a result, fire has shaped the global distribution of vegetation. Bond et al. (2005) found that in a world without fire, vaster areas of C_4 grasslands and savannas would develop.

The global variability in fire intensity and frequency entails that the 'fire process' is ecosystem dependent. This complexity has led to classifications of fire "regime". These classifications focus on the diversity in fire impacts on the vegetation (surface, crown, canopy), temporal characteristics (spread rate and return period) and spatial patterns (Bond and Keeley, 2005; Archibald et al., 2013). Even similar biomes show substantially different structures as a result of this variability. Boreal forests in North America tend to have stands of the same age with a small number of species. In contrast, the boreal forests of Eurasia are composed of mixed age stands and a more complex mix of species. The difference arises from the tendency to higher intensity crown fires in North America relative to lower intensity ground fires in the Eurasian forests (Wooster and Zhang, 2004). Biomass burning also accelerates nutrient cycling and alters the local carbon cycle. Nutrients follow three pathways following burning. Most apparent is the volatilisation of nutrients and emission to the atmosphere. The amount and type of nutrient loss is temperature-dependent. Nitrogen and sulphur volatilise at low temperatures (~200°C) relative to phosphorus, potassium and calcium (>500°C) (Boerner, 1982). Nutrients not volatilised may still be removed via convection of ash to neighbouring areas, dependent on wind conditions and the degree of combustion. Rainfall events following a fire also lead to the leaching of nutrients from the ash layer lying on the soil. Any nutrients not lost via these pathways may be reused by vegetation (Richter et al., 1982).

The combustion of biomass is a major source of emissions of greenhouse gasses to the atmosphere. Annual biomass burning provides approximately 38% of atmospheric ozone, 20% of carbon monoxide, and 39% of particulate organic carbon emissions(van Der Werf et al., 2017; Levine, 1991). Globally, fire contributes around 1–3 Pg C $year^{-1}$ of carbon to the atmosphere (van der Werf et al., 2010; van Der Werf et al., 2017). Bowman et al. (2009) calculated that global carbon dioxide emissions from deforestation fires have contributed ~19% to the increased radiative forcing observed since pre-industrial times. Soot carried in wildfire smoke has a significant impact on atmospheric radiative transfer. Soot primarily increases absorption of solar radiation leading to a reduction of light reaching the surface. There is evidence that soot may be the second biggest contributor to present global warming after carbon dioxide (Ramanathan and Carmichael, 2008). Changes in land surface albedo from fire alters local to global surface energy balance. Lyons et al. (2008) observed a 3.2Wm^{-2} increase in shortwave surface radiative forcing over a boreal site following burning. Fire changes the local surface energy balance by

altering sensible heat fluxes. Liu et al. (2005) observed a 33% reduction in annual evapotranspiration over a boreal fire in Alaska. Chambers et al. (2005) observed a 50Wm^{-2} increase in sensible heat exchange over a tundra fire. At the continental scale, Govaerts et al. (2002) indicated that fires reduced local surface albedo by up to 25% relative to unburned areas over the African continent, with potential effects on wind regime and rainfall. Jin and Roy (2005) indicated that a fire-induced reduction in albedo over Northern Australia provided a shortwave surface radiative forcing of 0.52 Wm^{-2}. Emissions of particulates also alter continental-scale albedo through the deposition of soot onto sea-ice and the Greenland ice sheet (Bond et al., 2013; Flanner et al., 2007; Warren and Wiscombe, 1980).

Biomass burning also has detrimental impacts on human health. The most considerable effect is from smoke particulates. Fine $PM_{2.5}$ particulates may be entrained and carried over large distances (>1000km) and have long atmospheric residence times (Kinney, 2008). There is a correlation between long-term exposure to $PM_{2.5}$ and premature deaths related to heart and lung diseases (Pope et al., 2002; Schwartz, 1994). The release of volatile organic compounds from biomass burning is also a key precursor pollutant for the formation of ozone. Ozone reduces lung function and leads to inflammation (Kinney, 2008). Areas blanketed in wildfire smoke see significant rises in hospital admissions for asthma and other respiratory conditions (Bowman and Johnston, 2005). Large fires in southeast Asia in recent years have, for example, affected the health of millions of people (Huijnen et al., 2016; Gaveau et al., 2014; Marlier et al., 2013).

2.2 Monitoring fire from space – active fire detection

The importance of fire within the Earth system and for human health necessitates long-term records of fire activity. The only practical way to achieve a global record of fire disturbance is from satellite-based instruments. From this data, two approaches are taken for monitoring fires. First, actively burning fires are identified based localised increases in temperature from flaming and smouldering of vegetation on the land surface. Second, fire-affected areas are identified primarily through detection of semi-persistent changes in the land surface observable from these instruments. This thesis primarily focuses on improving methods for the second approach, but the separation into the two approaches is somewhat artificial because active fire methods and products are used within burned area detection algorithms. It is now necessary to consider the heritage of these two approaches before delving further into burned area detection.

The earliest attempts at monitoring actively burning fires made use of the Advanced Very High-Resolution Radiometer (AVHRR) instrument in 1970–80s (Matson and Dozier, 1981). Other early studies

followed with a focus on detecting actively burning fires by identifying areas with significantly elevated temperatures from (now) very coarse-resolution instruments such as the Defense Meteorological Satellite Program (DMSP) and the Geostationary Operational Environmental Satellite Imager (GOES) (Elvidge et al., 1996; Prins and Menzel, 1992). Moving into the present, global active fire products have been developed from various sensors, including the Moderate Resolution Imaging Spectroradiometer (MODIS) since 2000, AVHRR since 1981, the Visible Infrared Imaging Radiometer Suite (VIIRS) since 2011, and the Sentinel-3 Sea and Land Surface Temperature Radiometer (SLSTR); as well as algorithms for a much wider range of sensors including geostationary sensors (Giglio et al., 2003; Schroeder et al., 2014; Wooster et al., 2012; Giglio et al., 2008; Calle et al., 2006; Prins and Menzel, 1994; Menzel and Prins, 1996). Active fires have also been used to derive products related to fire severity. Satellite Fire Radiative Power (FRP) products have been developed from active fire detections to estimate the rate of combustion, to the benefit of emissions modelling efforts (Wooster et al., 2013, 2005; Kaiser et al., 2012). The integration of FRP data over the temporal interval of burning provides Fire Radiative Energy (FRE) estimates, which are proportional to the fuel mass combusted (Kaufman et al., 1996). This is important because biomass burning emission estimates form a key dataset for models of the future climate and carbon cycle (Kaiser et al., 2012; Giglio et al., 2006).

Vegetation combustion leads to the release of significant amounts of radiant energy which can be observed by satellite instruments. Active fire products, like the earliest examples, all attempt to identify actively burning fires based on this radiant energy release. This radiant energy release from smouldering and flaming fires is concentrated at infrared (IR, 2–5μm) wavelengths and is well described by Planck's radiation law which describes the emitted radiation density B at wavelength λ as a function of the temperature T:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k_B T)} - 1} \tag{2.1}$$

The functional form of the emitted radiation density is shown in figure 2.1 for three objects at different temperatures T. Smouldering ($T = 600$k) and flaming ($T = 1000$K) vegetation have distinct emission densities when compared with surfaces at lower "normal" temperatures ($T = 300$K corresponds to 27°C). IR-equipped satellite instruments typically have IR thermal channels located in the middle IR atmospheric window (MIR; 3–5 Îijm) and also the longwave IR atmospheric window (LWIR; 8–14 Îijm) where atmospheric absorption of radiation is lowest (shown in figure 2.2).

Active fire algorithms make use of satellite measurements collected in these channels to identify fires based on two physical principles of the signal. First, the greater radiant energy of smouldering ($T = 600$K) and flaming ($T = 1000$K) vegetation in contrast with the ambient background radiation

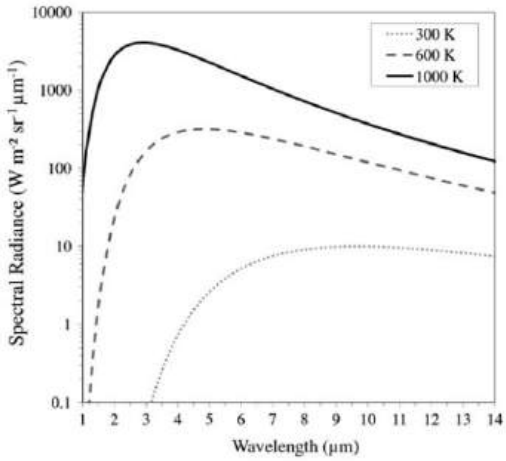


Figure 2.1. Planck's law for thermal emission from 300, 600, and 1000K objects. Planck's Law describes the emitted radiation for a black body at temperature T. 1000K corresponds to a flame, 600k a smouldering fire and 300K an ambient background (Earth temperature). Source: Wooster et al. (2013).

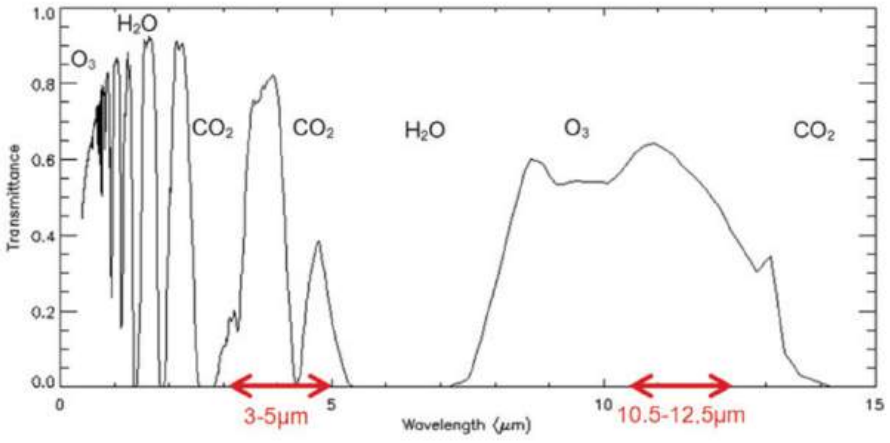


Figure 2.2. Atmospheric windows in the thermal infrared. The two thermal infrared (IR) domains used for infrared sensing of active fires are shown with the red arrows – the middle IR atmospheric window (MIR; 3âĂŞ5 Îijm) and the longwave IR atmospheric window (LWIR; 8âĂŞ14 Îijm). Infrared satellite instruments located channels in these spectral regions because of high atmospheric transmittance (the 'atmospheric windows'). Source: Kuenzer and Dech (2013).

($T = 300$K) as can be seen in figure 2.1. And second, the relative difference between emitted radiation in the two IR windows. This second principle is drawn from the fact that the displacement of the peak of emitted radiation (which is determined by Wien's displacement law) for smoulder and flaming surfaces leads to greater radiance in the MIR than the LWIR region. For ambient temperatures ($T = 300$K), the emitted radiation is greater in the LWIR region than the MIR region. These two principles for the central conditions of all active fire algorithms (Wooster et al., 2013). Most active fire algorithms employ multi-spectral tests by ensuring that radiance in the MIR and LWIR correspond to these physical principles. These tests can be implemented with fixed threshold tests, for example, Baum and Trepte (1999) identified actively burning fires in AVHRR data with four threshold conditions:

$$\mathrm{BT_{MIR}} > 314\ \mathrm{K} \tag{2.2}$$

$$\mathrm{BT_{MIR}} - \mathrm{BT_{LWIR}} > 10\ \mathrm{K} \tag{2.3}$$

$$\mathrm{BT_{MIR}} < 310\ \mathrm{K} \tag{2.4}$$

$$\mathrm{BT_{LWIR}(clear\ sky)} - \mathrm{BT_{LWIR}} < 6\ \mathrm{K} \tag{2.5}$$

where $\mathrm{BT_{MIR}}$ and $\mathrm{BT_{LWIR}}$ are the radiant temperatures (brightness temperatures) in the MIR and LWIR AVHRR channels. The first three conditions enforce the two physical principles listed above. First, that $\mathrm{BT_{MIR}}$ is higher than an'ambient' temperature, and second that emissions in the MIR are greater than in the LWIR (due to the displacement of the emissions peak). The final condition is used to ensure against erroneously flagging cloud as fire.

A similar set of threshold conditions are still used partially within newer active fire algorithms, primarily because of the link back to the physical principles of active fires. For example, the MODIS collection 6 algorithm uses a $\mathrm{BT_{MIR}} > 360$ K threshold which has remained consistent since the first version of the algorithm (Giglio et al., 2016; Kaufman et al., 1998). However, the weakness of fixed thresholds (Giglio et al., 1999) have led to the development of so-called 'contextual' active fire detection algorithms. These algorithms are able to better to adapt to local changes in fire characteristics and background ambient temperatures (Giglio et al., 2003). In contextual detection algorithms, a set of pixels with unambiguous active fires are first detected based on a set of threshold conditions above but weakened to only identify the largest and hottest fires (Wooster et al., 2013). In the contextual stage of the algorithm additional pixels that are close to meeting the threshold, conditions are considered by comparison to a set of statistics for the distribution of ambient background temperatures in the scene in a process termed 'background characterisation' (Roberts and Wooster, 2008). This process refines the detected active fires to include fires which are cooler or smaller that were missed by the restrictive thresholds in the initial stage (Schroeder et al., 2014; Giglio et al., 2016).

Active fire observations provide an incomplete sampling of fire activity. Active fire products, like all remote sensing products, suffer from sources of error. False alarms (commission errors) are most frequently caused by reflective surfaces observed at small solar zenith angles which leads to saturation of the thermal detector of the instrument (Giglio et al., 1999). Other sources of commission errors arise from areas of high surface variance which lead to poor background characterisation, for example around cloud edges or forest clearings (Schroeder et al., 2008; Giglio et al., 2016).

Generally though for active fire products, there is a bias towards omission errors in which many fires are not detected due to properties of the observations or detection algorithm. The primary source of omission errors for active fire products is that the temporal sampling provided by polar-orbiting sensors is often insufficient for detecting actively burning fires. This results in an underestimation of burned area as the instrument passes over after the fire has burned (see figure 2.3). This occurs most frequently when the overpass time of the instrument is not linked with the peak of the diurnal fire cycle (Freeborn et al., 2011). This limitation of active fire products derived from polar-orbiting sensors has lead to the generation of detection algorithms for geostationary instruments. These instruments are able to capture the full diurnal fire cycle which allows for a better estimate of fire activity and FRP (Roberts et al., 2005; Jones et al., 2018; Xu et al., 2017). The limitation of geostationary sensors is, however, a 'coarse resolution bias'' when compared to polar-orbiting sensors which prevents the detection of lower severity/small fires from geostationary instruments (Boschetti et al., 2004). As a result, efforts have been made to merge polar-orbiting and geostationary observational records to provide 'synthesised' active fire records (Freeborn et al., 2009; Roberts et al., 2011).

Due to these issues, researchers have attempted to quantify the 'true' burned area associated with active fire detections. Giglio et al. (2006) estimated that the burned area per active fire detection ranged from 0.29 km^2 in southern-hemisphere South America to 6.6km^2 in Central Asia. They suggested that this variation was primarily driven by difficulties in sampling active fires in closed canopies such as tropical forests. It has been suggested that the relationship between fire counts and burned area is unlikely to be linear due to complexities between cloud cover and average fire size altering the probability of detection (Kasischke et al., 2003). Variations in fire spread rate and burning-to-smouldering time will also alter the likelihood of detection. Eva and Lambin (1998a) indicated that fire counts should be adjusted due to large variations in fire activity diurnally in tropical regions. Nevertheless, there has been some effort in using active fires products to estimate burned area. A common approach has been to relate fire counts to burned area at broad spatial resolutions (typically 0.25–1°) via a proportionality constant α. Giglio et al. (2006) generated multi-year global estimates of burned area with a regression tree approach between active fire counts and burned area. They found

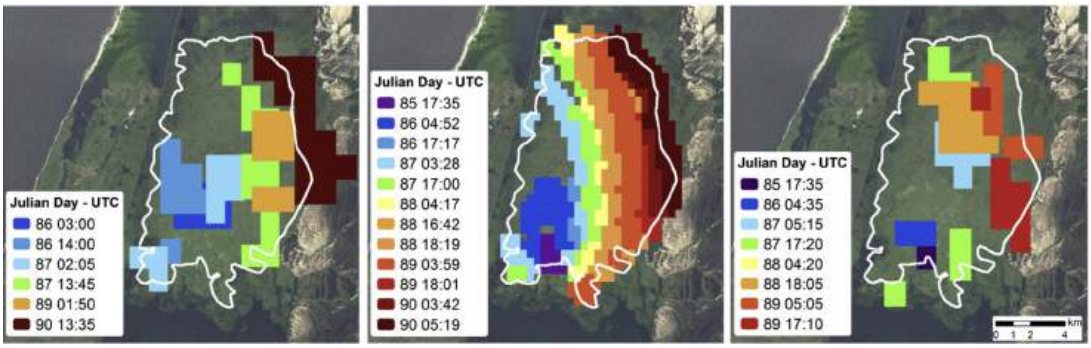


Figure 2.3. Active fire sampling of burned area. Active fires often provide an incomplete sampling of burned area depending on the characteristics of the fire and satellite instrument. Shown are active fire acquisitions for a burned area in Southern Brazil (-32.7° latitude, -52.55° longitude) for the period 26–31 March 2013. The perimeter of the burned area identified from 30 m Landsat-7 data is shown in white. Left) active fire acquisitions from 1 km Terra/MODIS, middle) 375 m VIIRS, and right) 1km Aqua/MODIS. Source: Schroeder et al. (2014).

that varying α as a function of the tree and herbaceous vegetation cover improved estimates. van Der Werf et al. (2003) also found that decreasing α as a function of tree cover improved estimates. Scholes et al. (1996) used an inverse relationship between α and NDVI from a similar viewpoint. But due to the sampling limitations of active fire products, Giglio et al. (2006) suggested that such approaches be seen as an interim method until a long-term data record of burned area became available.

2.3 From active fires to burned area products

Considering the sampling issues inherent to active fire products, algorithms which attempt to infer fire-affected area from optical satellite instruments have also been developed. These so-called burned area algorithms share the common aim of detecting evidence that a fire has affected a pixel based on the assumption that fire leads to a detectable and semi-persistent change in the land surface. From the application of these algorithms, several global satellite-derived datasets of burned area have been generated (summarised in table 2.1).

Roy et al. (2008a) developed an algorithm which became the MODIS Collection 5 burned area product (MCD45). Giglio et al. (2009) later proposed an algorithm for the MODIS sensor which became the MODIS MCD64 Collection 6 burned area product. The European Space Agency (ESA) Climate Change Initiative (CCI) has led to the production of several products from a range of European and American satellites. Alonso-Canas and Chuvieco (2015) developed a BA product (FireCCI41) at 300m using observations from the MEdium Resolution Imaging Spectrometer (MERIS) combined with the

MODIS active fire product. (Chuvieco et al., 2018) later refined this algorithm to produce a 15-year global product from observations from the MODIS sensor (FireCCI51). The CCI project has also produced a long-term product from the AVHRR instruments (FireCCILT10) for 1982-2017 (Oton and Chuvieco, 2018). The Copernicus Global Land Service (CGLS) has also led to the production of products from Proba-V and SPOT-VGT instruments (Tansey and Wolfs, 2017). Several earlier products were also generated as initial prototypes for developing global products. Carmona-Moreno et al. (2005) produced a 17 year burned area product from the NOAA Advanced Very High-Resolution Radiometers (AVHRR) missions. Tansey et al. (2008) developed L3JRC, a global burned area dataset using SPOT-VEGETATION data covering 2000-2007 at 1km. Simon et al. (2004) produced a global product using observations from the Along Track Scanning Radiometer (ATSR-2). Tansey et al. (2004) used SPOT VEGETATION data to produce a global BA dataset for the year 2000.

Beyond global products, others have also developed many algorithms for regional-scale automated detection of burned area. For example, algorithms using AVHRR data have been developed for Alaska, Canada, Siberia, Spain and North America (Kasischke and French, 1995; Fraser et al., 2000; Pu et al., 2007; Sukhinin et al., 2004; Fernández et al., 1997). Several algorithms have also been developed using AVHRR data for burned area estimates for Africa (Barbosa et al., 1999; Roy, 1999). Loboda et al. (2007) presented a regionally adaptive algorithm using MODIS data and applied it to North America and Siberia. Brivio et al. (2003) used SPOT-VEGETATION data for an algorithm developed over Africa. SPOT-VEGETATION has also been used to produce burned area estimates for Canada and Northern Australia (Fraser et al., 2000; Stroppiana et al., 2003).

2.4 Approaches to detecting burned area from space

The focus of this section is on automated algorithms for the detection of fire-affected areas. There are several general features that can be drawn. Primarily, algorithm developers use expectations about the spectral properties of burned surfaces relative to unburned surfaces. There is also a general view that fire as a disturbance leads to a sudden and semi-persistent change in the land surface. This feature has led to the use of approaches which search for discontinuities in surface reflectance observations either in time or space. As a result, many algorithms involve elements of time-series change detection or image processing. Algorithms also frequently use active fire detections to locally train algorithm parameters and provide an initial inference on burned area.

Table 2.1. Global burned area products.

Product	Time span	Spatial Resolution	Sensor	Algorithm Description	Reference
GFED	1997-present	0.5° x 0.5°	MODIS 500m, TRIM/VIRS, ATSR	2000-present: MCD64 algorithm. Pre-2000: prediction from active fire observations using regression between MCD64 and active fires	Giglio et al. (2013a)
MCD64	2000-present	500m	MODIS 500m, MODIS 1km	Multi-temporal change detection with a spectral index. Generation of burned and unburned classes from active fires and change index. Probabilistic Bayesian algorithm	Giglio et al. (2009)
FireCCI50 MODIS	2001-2016	250m	MODIS 250m, MODIS 1km	Hybrid, non-change based algorith. Monthly composites of NIR reflectance,generated observations following active fire observations.Active fires used to generate distributions of burned and unburned pixels.,Region growing algorithm following burned area 'seeding' with active fires.	Chuvieco et al. (2018)
FireCCI40 MERIS	2001-2008	300m	MERIS 300m, MODIS 1km	Similar to above	Alonso-Canas and Chuvieco (2015)
FireCCI10 AVHRR-LTDR	1982-2017	5km	NOAA-AVHRR	Generation of BA index from monthly multi-temporal composites. Classification of burned areas with random forest classifier trained from reference to MCD64 burned areas.	Oton and Chuvieco (2018)
MCD45	2000-2016	500m	MODIS 500m	Change detection modelling scheme in which the post fire reflectance is predicted and compared with the true reflectance.	Roy et al. (2008a)
L3JRC	2000-2007	1km	SPOT VEGETATION	NIR reflectance based multi-temporal change index.	Tansey et al. (2008)
GIOGL1-PROBA-V	2014-present	300m	PROBA-V	Similar to above.	Tansey and Wolfs (2017)
GLOBSCAR	2000	1km	ERS2-ATSR2	Combination of two algorithms. K1: contextual algorithm using bi-spectral properties of burned pixels in the NIR and thermal infrared. E2: Combination of several spectral threshold tests.	Eva and Lambin (1998a); Piccolini (1998)
GBA2000	2000	1km	SPOT VEGETATION	Multi-temporal algorithm involving thresholds for changes in surface reflectance spectral indicies	Grégoire et al. (2003)
GBS	1982-1999	8km	NOAA-AVHRR	Multi-temporal change detection algorithm. Generation of weekly maximum NDVI composites. Thresholding tests to identify burned areas.	Carmona-Moreno et al. (2005)

2.4.1 Spectral discrimination of burned surfaces

All algorithms make use of the spectral characteristics of burned surfaces. Burned surfaces are primarily constituted of white/black ash, char and exposed soil (see figure 2.4 which depicts images of two post-fire sites) (Eva and Lambin, 1998a; Trigg and Flasse, 2000). Most algorithms have made use of changes in near-infrared (700–1400nm; NIR) and shortwave infrared reflectance (1400nm–3000; SWIR). These optical regions have been shown to provide the best discrimination between burned and unburned surfaces (Pereira et al., 1999; Roy et al., 2002). The combustion of vegetation and deposition of char leads to a reduction in visible (300–700nm; VIS) to NIR reflectance. The presence of NIR spectral channels on older sensors (i.e. AVHRR) also meant that the initial burned area remote sensing studies focused on characterising NIR variations (Kasischke and French, 1995). NIR reflectance variations have since been used in many mapping algorithms. Alonso-Canas and Chuvieco (2015) used NIR reflectance as they found it provided the best discrimination for the MERIS sensor which observed only the VIS–NIR region. Tansey et al. (2008) also used daily variations in NIR reflectance for spectral discrimination in their algorithm. The SWIR region has also been used because it shows good discrimination between burned and unburned surfaces — especially for savanna ecosystems (Roy et al., 2002). Eva and Lambin (1998a) developed an algorithm around the temporal evolution of SWIR reflectance. The SWIR region is also used in the global algorithm developed by Giglio et al. (2009). If possible, many algorithms typically use both NIR and SWIR reflectance. Roy and Landmann (2005) used both NIR reflectance and SWIR reflectance. They found that this reduced commission errors from changes in canopy water content or other disturbances (such as flooding). The VIS optical domain has not been exclusively used within algorithms as burned surfaces display poor discrimination here with certain land covers (e.g. water bodies, dark exposed soils, wetlands) (Pereira, 1999).

2.4.1.1 Spectral Indices

Beyond using spectral channels explicitly, a widely used approach has been the generation of mathematical transformations of reflectance that are reportedly more sensitive to burned surfaces. These spectral or vegetation indices (VIs) have been developed over many years — many derived from the correlation of ground studies with satellite data (Verstraete, 1994). VIs attempt to take advantage of the preferential scattering of light across wavelengths driven by the properties of the land surface. The earliest studies into observing burned area from space made use of the Normalised Difference Index, which is sensitive to the amount of green vegetation (Tucker, 1979):



Figure 2.4. Spectral complexity of the burn signal. Examples of two different burn signals. Left) prescribed burning of savanna woodland in Kruger National Park, South Africa. Deposition of ash and black char due to combustion of grasses. Larger vegetation remains leading to a mixture of burned and unburned surfaces. Right) Pine forest in Arizona, USA following a severe fire. Deposition of white ash, char and exposed soils. Source left: Disney et al. (2011), right: Keeley (2009).

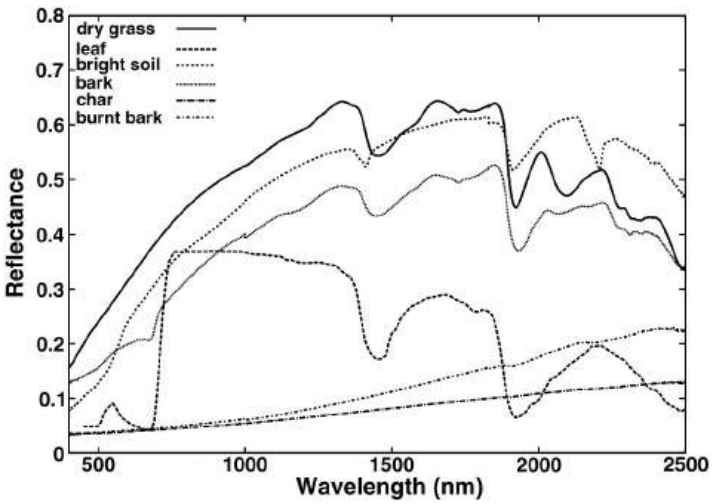


Figure 2.5. Spectra of burned and unburned materials in a savanna ecosystem (Kruger National Park, South Africa). Source: Disney et al. (2011).

$$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}} - \rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}} + \rho_{\mathrm{Red}}} \tag{2.6}$$

where ρ_{NIR} and ρ_{Red} are reflectance in near-infrared and visible red channels respectively. NDVI is sensitive to the contrast between the predominant scattering of photons by leaves in the near-infrared to the absorption of light in the visible by chlorophyll. The magnitude of this differential red-NIR scattering is related to the amount of green vegetation present. As a result, relative changes in NDVI over time give an indication of vegetation development and senescence. Sudden changes in NDVI indicate changes in vegetation structure — in some cases due to fire (Kasischke and French, 1995; Isaev et al., 2002).

More frequently used for fire studies is the normalised burn ratio (NBR) which uses short-wave infrared (SWIR) reflectance:

$$\text{NBR} = \frac{\rho_{\text{SWIR}} - \rho_{\text{NIR}}}{\rho_{\text{SWIR}} + \rho_{\text{NIR}}} \tag{2.7}$$

NBR is sensitive to two features. Following a fire, decreases in vegetation cover and greenness lead to a decline in the near-infrared (reduced scattering from vegetation) and a rise in the shortwave infrared due to a reduction in water content. The change pre-to-post fire is considered with the difference normalised burn ratio (dNBR):

$$\text{dNRB} = \text{NBR}_{\text{burned}} - \text{NBR}_{\text{unburned}} \tag{2.8}$$

While originally developed for Mediterranean ecosystems (García and Caselles, 1991), NBR has been used across a broad range of ecosystems. Harris et al. (2011) found that NBR performed best out of sixteen indices in the Chaparral shrublands of California. It has also been especially used for boreal forests (Murphy et al., 2008; Epting et al., 2005; Allen and Sorbel, 2008).

Pinty and Verstraete (1992) proposed the Global Environmental Monitoring Index (GEMI). GEMI is a non-linear index designed to be less sensitive to variations in soil properties and atmospheric effects:

$$\text{GEMI} = \eta(1 - 0.25\eta) - \frac{\rho_{\text{Red}} - 0.125}{1 - \rho_{\text{Red}}} \tag{2.9}$$

where

$$\eta = \frac{2(\rho_{\text{NIR}}^2 - \rho_{\text{Red}}^2) + 1.5\rho_{\text{NIR}} + 0.5\rho_{\text{Red}}}{\rho_{\text{Red}} + \rho_{\text{NIR}} + 0.5} \tag{2.10}$$

Alonso-Canas and Chuvieco (2015) and Chuvieco et al. (2018) used GEMI alongside NIR reflectance to characterise burned surfaces in the FireCCI algorithms, and Barbosa et al. (1999) used GEMI to generate a burned area product for continental Africa. Stroppiana et al. (2002) found that GEMI was the most suitable index for detecting the impact of fire on vegetation cover in Australia. GEMI has also been seen as a potential index to extend the burned area ECV back to older sensors which only have VIS and NIR spectral channels. Indeed, Pereira (1999) suggested it was the most suitable index in the red-NIR space for burned area detection.

The Mid-Infrared Bispectral Index (MIRBI) was defined by Trigg and Flasse (2001) as:

$$\text{MIRBI} = 10\rho_{\text{SWIR}} - 9.8\rho_{\text{LNIR}} + 2 \tag{2.11}$$

where ρ_{LNIR} corresponds to long near-infrared reflectance. MIRBI was designed especially for shrubland and savanna areas where the sensitivity of NIR reflectance to fire is reduced as a result of vegetation senescence.

The Char Soil Index CSI (Smith et al., 2005) is defined as:

$$\text{CSI} = \frac{\rho_{\text{NIR}}}{\rho_{\text{SWIR}}} \tag{2.12}$$

The index was designed for African savannas and has generally since been applied to savannas and shrubland ecosystems. Boschetti et al. (2010) used CSI in combination with a set of other indices for burned area mapping in Mediterranean environments. Harris et al. (2011) also found that CSI performed well in the Chaparral shrublands of California.

Chuvieco et al. (2002) developed the Burned Area Index (BAI). The index is computed as the distance to a reference point in NIR-red spectral space:

$$\text{BAI} = \frac{1}{(c_{\text{Red}} - \rho_{\text{Red}})^2 + (c_{\text{NIR}} - \rho_{\text{NIR}})^2} \tag{2.13}$$

The reference point ($c_{\text{Red}} = 0.1, c_{\text{NIR}} = 0.06$) represents a char spectral end-member. Chuvieco et al. (2002) highlight that there is a possible non-uniqueness of the reference point against similar dark reflectance targets such as water bodies. Similarly, the magnitude of the signal is dependent on the temporal persistence of the char signal on the land surface as Dempewolf et al. (2007) found that post-fire value of BAI was persistent for only up to 30 days in savannas.

The MODIS Collection 6 Burned Area product uses an index which like MIRBI takes advantage of differential scattering in the shortwave infrared and long near-infrared:

$$\frac{\rho_{\text{LNIR}} - \rho_{\text{SWIR}}}{\rho_{\text{LNIR}} + \rho_{\text{SWIR}}} \tag{2.14}$$

where ρ_{LNIR} and ρ_{SWIR} correspond to MODIS bands 5 (1230–1250 nm) and 7 (2105–2155nm). Giglio et al. (2009) found that the index displayed a temporally persistent drop following a fire. It is also similar to the index defined in Roy et al. (2002) who analysed burned-unburned separability over South African savannas.

2.4.2 Tests for change and persistence in time and space

Alongside these spectral assumptions about fire, numerous time series and image processing techniques have been developed for identifying burned areas. A general principle of these methods in both the

temporal and spatial domains are that fires are localised events which lead to a sudden and semi-persistent change in the land surface.

A very common approach has been the use of differencing presumed pre-fire and post-fire signals to identify changes in the land surface. For temporal approaches, either subsequent observations or image composites are in some form differenced from each other. The concept is shown in figure 2.6 for a time series of NIR reflectance collected by MODIS over a fire. The first burned area algorithms followed this approach. Kasischke and French (1995) developed an algorithm around differencing of NDVI composites. They generated maximum value composites from daily NDVI values for two weeks, as they found this to be sufficient time to obtain cloud-free images. Martin and Chuvieco (1995) also used differencing of maximum value composites of NDVI. Fernández et al. (1997) similarly generated maximum value composites of NDVI collected from ten days of observations, while Fraser et al. (2000) used a normalised difference of two NDVI composites as part of their algorithm. The ESA FireCCI products used a similar procedure in which observations are collected over a month to produce pre- and post-fire composites of NIR reflectance (Alonso-Canas and Chuvieco, 2015; Chuvieco et al., 2018). The MODIS Collection 6 burned area product used pre-fire and post-fire compositing windows to form a temporal vegetation index (Giglio et al., 2009). Two sliding windows with a width of ten days were convolved with the vegetation index. A temporal stability metric generated from these two windows was used to identify sudden and persistent changes in the reflectance time series. The earlier MODIS Collection 5 burned area product (MCD45) used a temporal change detection technique based on modelling the expected evolution of the land surface over a short window (Roy et al., 2005a). Within the algorithm, the prediction of the surface was compared to observations in the forecast window. Windows where the difference between the prediction and the observations exceeded a threshold determined potential burned areas. Tansey et al. (2008) applied a temporal change index in which the daily reflectance was compared to a long-term running average composite. Roy (1999) also developed a multi-temporal algorithm based on the maximum change in a vegetation index between subsequent acquisitions. Often the length of the compositing window is allowed to vary as a function of the available observation sampling. Roy et al. (2005a) extended the pre-fire window up to 32 days to ensure there were sufficient observations. Alonso-Canas and Chuvieco (2015) also considered the preceding month if no observations were available within the processing month of the algorithm.

Many of these algorithms also implemented a persistency test on any change. Roy et al. (2005a) enforced a persistency test by ensuring that several successive prediction windows indicated burning. If an individual window indicated that a fire had not occurred, the pixel was subsequently classified as unburned. This process is shown in figure 2.7. The temporal index used by Giglio et al. (2009) is most

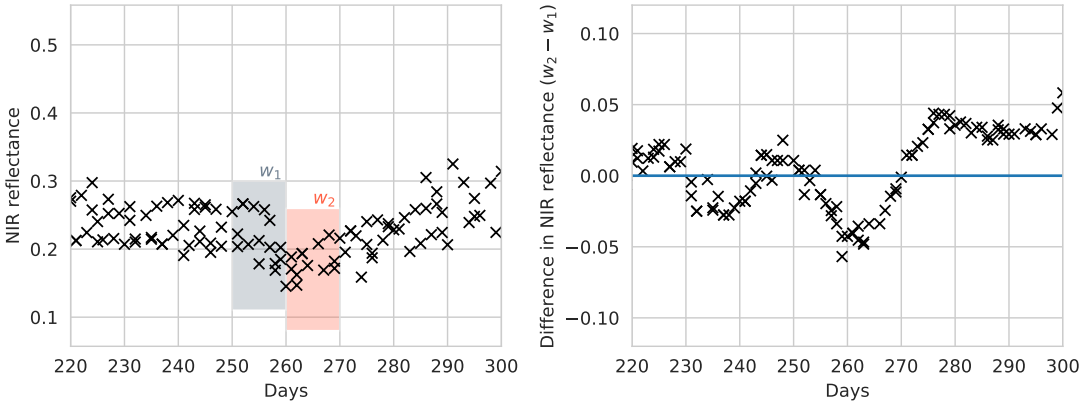


Figure 2.6. Detecting semi-persistent changes for burned area detection. A pre-fire window w_1 and post-fire window w_1 are moved through the observations to produce a composited change signal from the reflectance measurements (right panel).

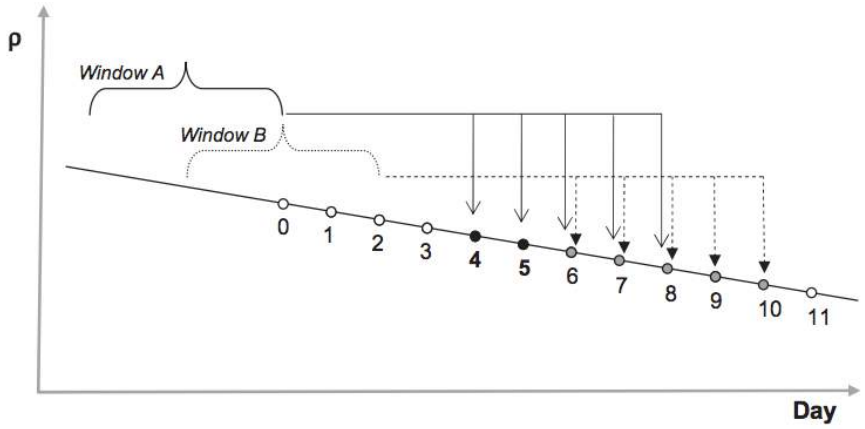


Figure 2.7. Example of persistency tests within burned area algorithms from Roy et al. (2005a). Each consecutive change between window A and B is considered, if the persistency criterion is met in several consecutive windows (filled circles) the algorithm classifies the pixel as burned.

sensitive to large differences between the pre-fire and post-fire windows but weighted according to the persistence of the change. The use of compositing also adds an inherent persistence constraint to algorithms with the length of persistency at least equal the number of days within the window.

While most algorithms are focused on the identification and confirmation of changes in reflectance time series, image processing routines are also used to identify burning from the expectation of fire across space. There are several general properties about fire spread which algorithm developers have attempted to exploit. Primarily, there is an expectation that fires are rare and clustered in the landscape. Iterative contextual algorithms have been developed which grow a burned classification from some

initial burned pixels. These methods follow on from Adams and Bischof (1994) who first proposed seeded region growing approaches within remote sensing. These methods attempt to grow a burn scar based on the similarity of neighbouring pixels while not increasing the variance of the burn scar signal. Bastarrika et al. (2011) considered the use of 3x3 neighbourhood region growing algorithm. In their algorithm, pixels neighbouring already burned pixels were evaluated as to whether they matched a threshold in NIR reflectance. Pu et al. (2007) implemented a region growing procedure which connected nearby pixels already classified as burned. Alonso-Canas and Chuvieco (2015) developed a region growing procedure in which burn scars were grown from active fire detections.

An alternative to region growing routines are global spatial filters which operate over the whole image to refine the classification. Giglio et al. (2009) used a modified majority filter as part of their final classification. Their filter assessed whether pixels within a 3x3 neighbourhood showed temporal consistency in the proposed burn date. If there was a large variance in the proposed day of burn (DOB) in the neighbourhood, the pixel was reclassified as unburned. Conversely, if the surrounding pixels had a small variance in proposed DOB a pixel was reclassified as burned. Bastarrika et al. (2011) evaluated a global method based on the application of a low-pass Gaussian filter combined with a Sobol edge filter to delineate the edges of burn scars. Fraser et al. (2000) used a modal 3x3 filter to remove detections which were not surrounded by at least 4 neighbouring burned pixels. Tansey et al. (2008) approximated a spatial dependence by ascribing pixels which fall two standard deviations below the mean of a 200x200 km window as potentially burned pixels.

In several algorithms, drawing a separation between time series and image processing methods is artificial. Instead, the discrete and clustered nature of fire is exploited in time and space concurrently. Generally, these approaches exploit the spatial-temporal consistency of burned surfaces relative to the greater diversity in unburned surfaces. Fernández et al. (1997) used both the difference between two subsequent NDVI images and a constraint that burned pixels fell two standard deviations below a local 100km window. Giglio et al. (2009) implemented a temporal texture metric which they defined as the day of maximum change in a time series composite. They found that regions of burned pixels showed a smaller variation in this date relative to unburned pixels. Brivio et al. (2003) developed a spatial-temporal classification algorithm with the use of a multi-layer perceptron neural network. Their approach used a two-stage neural network scheme. In the initial stage, each image was classified separately to generate daily classifications. The neural network is then used to learn the local spatio-temporal characteristic of burned pixels relative to unburned pixels. Alonso-Canas and Chuvieco (2015) generated composites of potential post-fire reflectance based on an algorithm which selected the date following the occurrence of the nearest active fire observation. They argued that as active

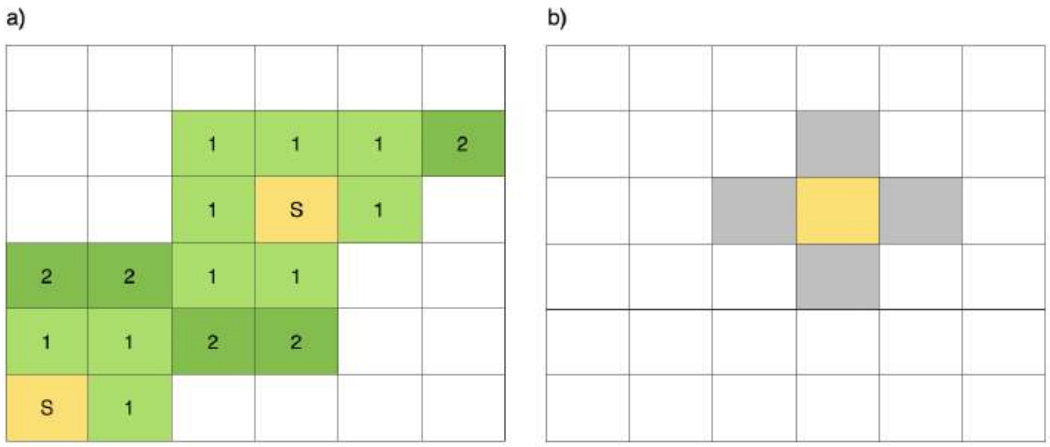


Figure 2.8. Examples of spatial image processing routines. a) iterative region growing. Initial seed pixels (S) are grown iteratively to adjacent pixels which meet conditions (numbers represent each iteration of the routine). b) spatial filters operate over the entire extent of the image and refine the classification (of the yellow pixel) based on local statistics of the stencil (shown in grey).

fire observations provide very reliable timing information on DOB, this timing information could be spread to generate a realistic post-fire reflectance composite.

2.4.3 Characterising burned area with active fire observations

Active fire products have also been used within burned area mapping algorithms. These products are now routinely used in so-called 'hybrid' algorithms in several ways. Active fire products provide a set of temporally and spatially distributed detections which allow for supervised classification methods to be used within algorithms. Pixels with coincident active fire detections are used to refine classification thresholds and to learn the spectral characteristics of burned surfaces.

Early examples of 'hybrid' algorithms were proposed by Roy (1999) and Fraser et al. (2000). Roy (1999) employed active fire detections to refine a statistical threshold on the difference between two composites from AVHRR data. Fraser et al. (2000) detailed an algorithm which used active fire detections to refine a threshold on changes in NDVI. Giglio et al. (2009) used active fire observations to learn the distribution of VI values belonging to burned surfaces for a classifier. This classifier was then applied to predict the occurrence of fire in other pixels. The algorithm by Fraser et al. (2000) learned the properties of a discrimination threshold based on active fires. Alonso-Canas and Chuvieco (2015) also used active fire observations to learn the cumulative distribution function of NIR reflectance of burned surfaces.

Active fire observations are also routinely used to provide an initial spatial estimate of burned area. Alonso-Canas and Chuvieco (2015) used pixels attributed with active fire detections as initial seed pixels from which to grow burn scars. This involved iteratively evaluating the similarity of adjacent pixels to the edge of the burn scar. Such region-growing procedures are common to hybrid burned area algorithms. Fraser et al. (2000) implemented a similar model in which burned areas were initially determined by active fire detections and then grown via a region-growing algorithm. In their algorithm, they applied a relaxation of the detection threshold in surrounding pixels and then evaluated the connectedness of burned pixels via a modal filter. Giglio et al. (2009) implemented a multi-rule region growing procedure. Primarily, the neighbouring pixels must have a similar proposed burn date and be within 10km of the initial seed pixel. Giglio et al. (2009) also used active fire detections as part of a prior probability model of a pixel being fire affected. This was based on a spatial distance decay model of the form:

$$\exp\left(-\frac{d_B^2}{2\sigma^2}\right) \tag{2.15}$$

where d_B is the distance to the nearest burned pixel and σ determines the decay rate of the model. This represents a simpler spatial model of fire spread than region-growing procedures which tend to modulate any spatial decay based on inter-pixel similarities. Alonso-Canas and Chuvieco (2015) also used active fires to discriminate unburned pixels. They used a threshold in the distance to the nearest active fire observation beyond which pixels were considered to be unburned. Active fires are rarely used to explicitly characterise burning temporally, however. One exception is Alonso-Canas and Chuvieco (2015) who used the date of the active fire detection to discriminate between pre- and post-burn compositing windows. They generated a post-burning reflectance composite from observations following the date of the most spatially proximal active fire detection. Recently Campagnolo et al. (2019) proposed a patch-based algorithm which used the spatiotemporal distribution of active fires and a vegetation index to infer connected fire events (i.e. burn scars) using graph theory algorithms (see figure 2.9). This involved the clustering of pixels based on the similarity of a set of spatio-temporal features including the temporal evolution of a vegetation index.

2.4.4 Burned area algorithms as classifiers

A unifying view of these components (e.g. spectral tests, time series and spatial/image processing techniques) is that they represent routines which make up algorithm decision structures that determine the mapped burned area. For most of the algorithms presented the task of mapping burned pixels is not

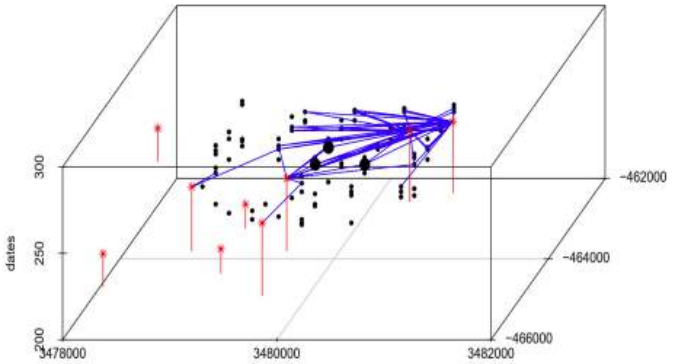


Figure 2.9. Patch based methods (from (Campagnolo et al., 2019)). The temporal and spatial clustering of active fires are used to identify individual fire patches which helps to spatially refine burn scars. The x and y axes display spatial coordinates and the z axis day of active fire detection.

achieved with traditional classification methods such as logistic regression or support vector machines (Suykens and Vandewalle, 1999; Cheng et al., 2006). Instead algorithms are typically composed of a set of consecutive decision rules based on scientific understanding of fire remote sensing which classify each pixel.

Within most algorithms, these decisions are typically phrased as threshold conditions. Temporal constraints imposed on the signal influence whether a pixel is classified as burned in several algorithms. Giglio et al. (2009) thresholded values in their temporal VI to provide an initial classification of burned pixels. This meant classifying pixels burned if the change in the VI within a defined temporal window was judged sufficient to indicate disturbance. Similarly, the persistency test in Roy et al. (2005a) enforces a threshold in which pixels are mapped burned when consecutive observations show a persistent change in the land surface. Functionally these algorithm structures represent the 'classifier' element of the burned area detection problem as opposed to statistical classifiers such as logistic regression or support vector machines (Mountrakis et al., 2011; Suykens and Vandewalle, 1999; Cheng et al., 2006).

Spatial components also contain an implicit constraining of the signal to burned and unburned classes. Region growing procedures include thresholds which determine the expected variance between neighbouring pixels for a pixel to be burned. Fernández et al. (1997) used a threshold derived from a local window of temporal VI difference to classify burned pixels. If a pixel was two standard deviations from the local variability it was classified as burned. These decision boundaries are also evident in the stopping conditions of iterative procedures. Alonso-Canas and Chuvieco (2015) imposed several stopping conditions which are determined based on constraining the variance of the burned class. Similarly, global filtering operations also include expectations of the spatial relation between burned pixels defined by their neighbourhood size and structure. Giglio et al. (2009) generated a prior

classification of burned pixels based on whether a pixel showed good temporal consistency in potential DOB with its neighbours. The definition of the burned class was then determined by this temporal consistency parameter, with pixels classified as burned if they met the constraint. Similarly, if a pixel was further away from an active fire, Giglio et al. (2009) reduced the likelihood that it was burned. The definition of the burned and unburned classes is then partly determined by the parameterisation of this decay.

Frequently, these decision structures are formulated as *ad hoc* thresholds/parameters which are chosen based on the assumptions of fire process by the algorithm developer. An alternative is for these thresholds/parameters to be locally trained from the data. Issues with fixed thresholds have been long known for burned area mapping (Barbosa et al., 1998). The most advanced methods tend to be the local training of burned and unburned classes from active fire observations. In these approaches information about each class is derived locally and should, therefore, be more representative than a globally defined threshold. Fraser et al. (2000) proposed an early example of this with the Hotspot and NDVI Differencing Synergy (HANDS) algorithm. The algorithm computed a set of local and regional thresholds trained on pixels with active fire observations. Similarly, Roy (1999) used active fires to train a statistical threshold in a change based vegetation index. Giglio et al. (2009) utilised active fire observations as part of training a land cover based classification of burned pixels. Alonso-Canas and Chuvieco (2015) proposed an algorithm which also made use of active fires to locally train a set of thresholds. This involved generating cumulative distribution functions (CDFs) of burned and unburned NIR reflectance partially from active fire observations before estimating the threshold from these two distributions.

2.5 Challenges to detecting burned area from space

Despite evident progress in the detection of fire-affected areas from space, important issues remain that are likely to impede further progress. These can be thematically classified as 1) limitations of the remote sensing data available and 2) issues with algorithm assumptions.

2.5.1 Limitations of the observations

The detection of burned area is reliant on the timely observation of the land surface following a fire. What constitutes timeliness is determined by:

1. The longevity of the observable impact of the fire on the land surface.

2. The sensitivity of the sensor to this signal.

3. The observational requirement of the algorithm.

Depending on the severity of the fire as well as the type of vegetation present, an observable signal may last days, weeks or years. The detection of burning in temperate or boreal forests is possible over a longer period following a fire than in savannas and grasslands where fast vegetation regrowth occurs (Sukhinin et al., 2004; Trigg and Flasse, 2000). Other issues arise in croplands as subsequent ploughing quickly removes evidence for burning (Hall et al., 2016). Figure 2.10 displays the median temporal persistence in days of an observable burn signal as estimated by Melchiorre and Boschetti (2018). Large differences in persistence can be seen between different vegetation types and biomes. Melchiorre and Boschetti (2018) indicate that global median persistence to be only 29 days and that persistence is predominantly determined by land cover.

The availability of observations is therefore of considerable importance. Many algorithms are designed with fixed length compositing windows over which observations are collected. These windows enforce a requirement on the minimum number of observations required to map the surface. For example, if there is no observation within ten days, the algorithm of Giglio et al. (2009) cannot map burned areas. Similarly, seven observations within 32 days are required in the algorithm of Roy et al. (2005a). These requirements on the availability of observations ultimately determine the global success of the algorithms. Figure 2.11 shows the impact of these constraints on the unmapped areas of these two algorithms for June 2008. The difference in observational requirements leads to large unmapped areas of South East Asia for the Roy et al. (2005a) algorithm.

Temporal sampling also influences active fire detections which affect the efficacy of hybrid algorithms. The likelihood of observing an active fire increases in higher latitudes due to the overlapping swaths of polar-orbiting sensors. As an example,latitudes above >40° receive up to four observations per day from the MODIS Aqua and Terra satellites, while equatorial regions receive fewer than two (Giglio et al., 2006). This sampling will determine the performance of hybrid algorithms. It would be expected, for example, that the representativeness of thresholds trained with active fires will be better at higher latitudes, where the sampling is better. And in the spatial context of these algorithms, seeding and region growing procedures based on active fire locations will behave inconsistently as a function of sampling.

Considering the importance of sampling, it is perhaps surprising that few previous algorithms have used observations from multiple satellite instruments. Some notable exceptions include Boschetti et al. (2015) who proposed a MODIS-Landsat detection algorithm, Alonso-Canas and Chuvieco (2015) who

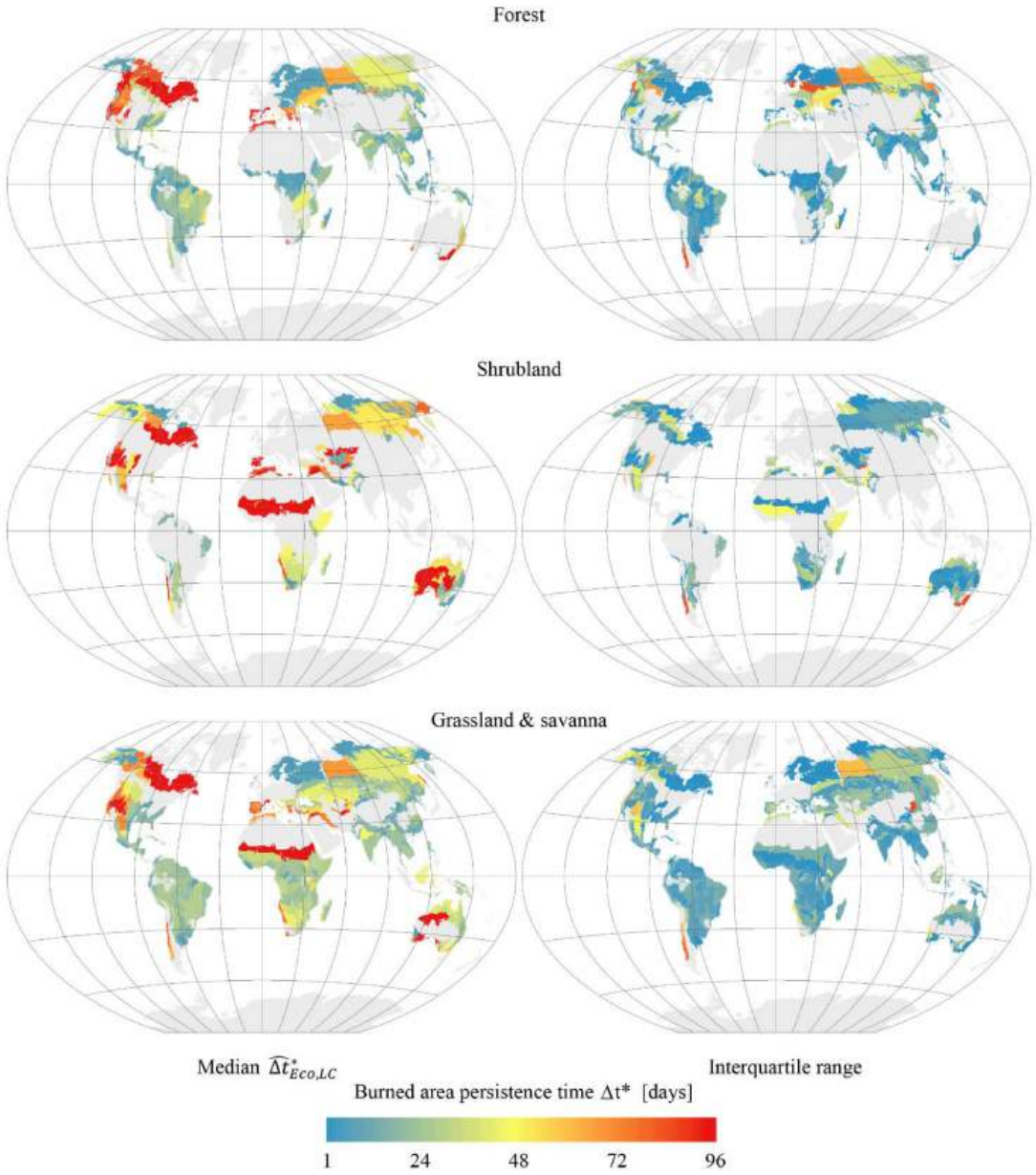
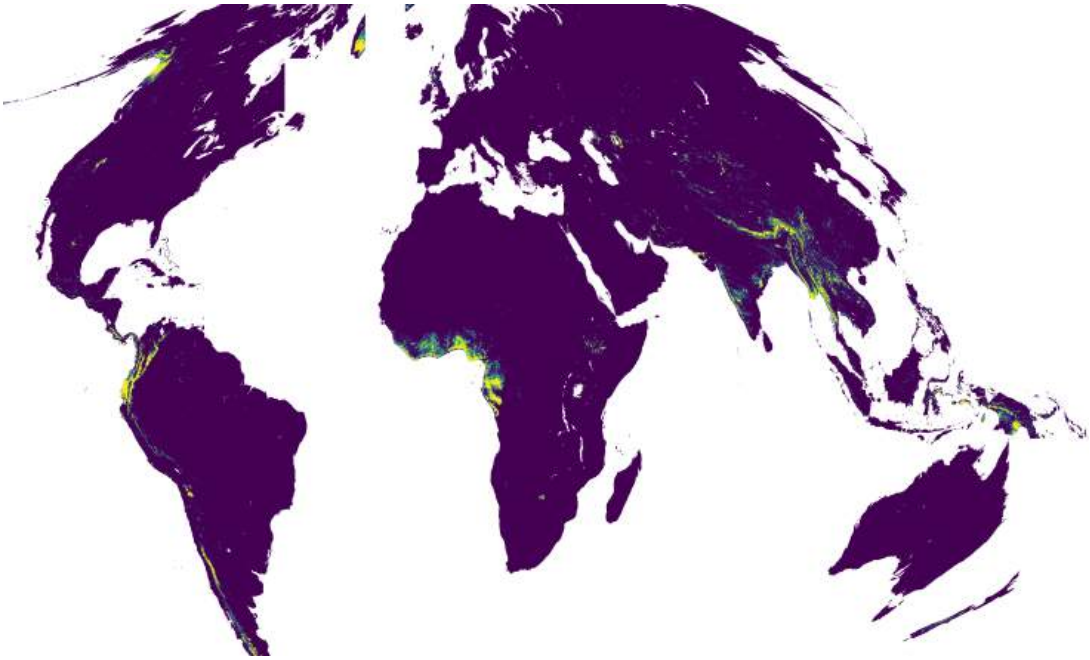
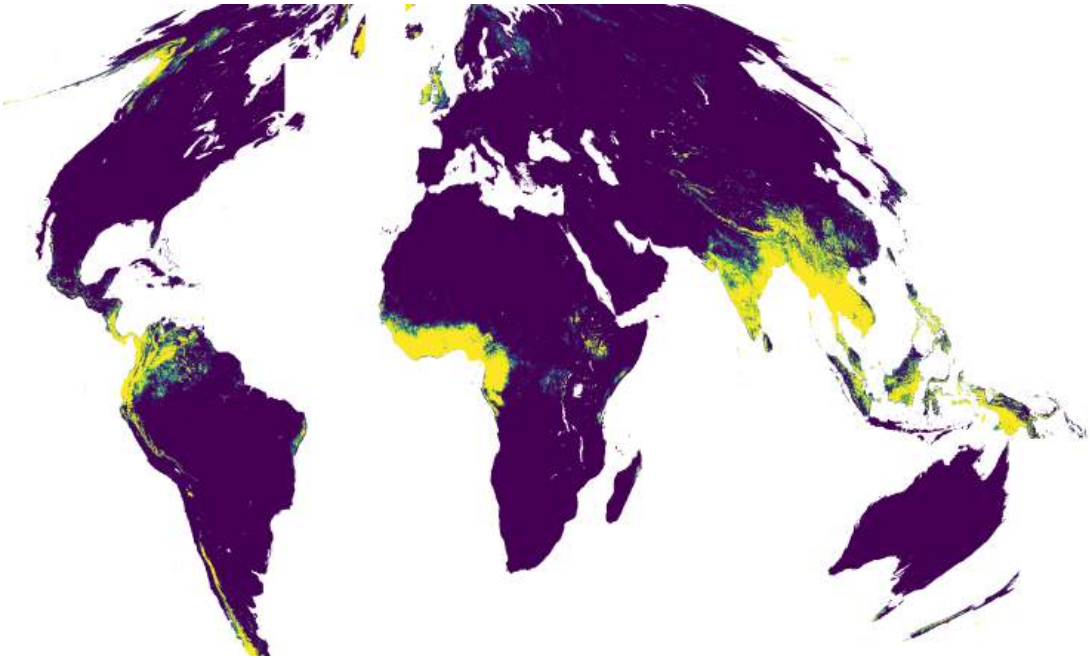


Figure 2.10. Median persistence of an observable burn signal [days] derived from MODIS surface reflectance. (Source: Melchiorre and Boschetti (2018)).



(a) MCD64



(b) MCD45

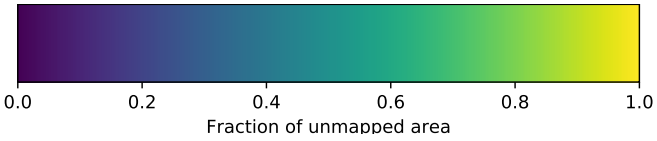


Figure 2.11. Importance of sampling. Fraction of unmapped area in June 2008 due to insufficient observations for a) the MCD64 algorithm (Giglio et al., 2009) and b) the MCD45 algorithm (Roy et al., 2008a). MCD64 requires at least one observation within 10 days. While MCD45 has a larger unmapped extent due to the requirement for seven observations within 32 days to perform the BRDF inversion.

used MERIS reflectance observations with active fire observations from the MODIS sensors, and Roy et al. (2019) recently presented a multi-sensor algorithm for Landsat and Sentinel-2. There remains a considerable opportunity to improve future BA products by the merging of remote sensing datasets.

The trade-off for frequent observations of the land surface from coarse-resolution instruments such as MODIS is observations with highly varying acquisition geometries of the Sun and sensor. For the wide-swath polar-orbiting sensors used for global burned area mapping this leads to variations in measured reflectance only due to the acquisition geometry. This scattering anisotropy is an inherent property of the land surface and is described by the Bidirectional Reflectance Distribution Function (BRDF) (Schaepman-Strub et al., 2006). The BRDF is defined as:

$$\mathrm{BRDF}(\Omega;\Omega') = \frac{dL_r(\Omega;\Omega')}{dE_i(\Omega)} \tag{2.16}$$

where Ω is the angular geometry of the Sun and Ω' the sampling geometry of the sensor. dL_r is the radiance of the surface per unit steradian [sr^{-1}] and dE_i the irradiance from the Sun per unit steradian at the surface. Multi-angular sampling of the BRDF is provided by the wide-swath polar-orbiting instruments frequently used for BA detection. Without proper treatment, variations arising from differences in angular sampling can complicate the detection of burned surfaces (Roy et al., 2002). There is evidence, for example, that these variations are larger than the typical change signal for savanna fires (Stroppiana et al., 2003; Roy et al., 2002). Figure 2.12 shows the evolution of NIR reflectance collected by the MODIS Terra instrument for a savanna fire in South Africa. The apparent day-to-day fluctuations in reflectance are a function of variations in the view zenith angle of the measurement. These fluctuations can be seen to be of a similar magnitude to the change in reflectance due to the fire.

A particular reason that vegetation indices have been used for burned area detection is that they help to reduce BRDF effects. This results from their formulation as band ratios, as BRDF effects are fairly similar across wavelengths. Nevertheless, spectral anisotropy is not spectrally invariant and therefore BRDF effects remain to some extent after the transformation (Huete et al., 1992; Qi et al., 1995).

Algorithms which use multi-temporal compositing of observations may also be affected by BRDF effects due to the mixing of acquisition geometries across the composite. An example of this is shown in figure 2.13, which displays a true colour composite of southern Africa produced via a maximum monthly NDVI compositing criterion. The visible striping between parts of the composite arises from the selection of observations from different sampling geometries. These artefacts may impact many algorithm routines, especially those based on the persistence of the signal in time and space. For

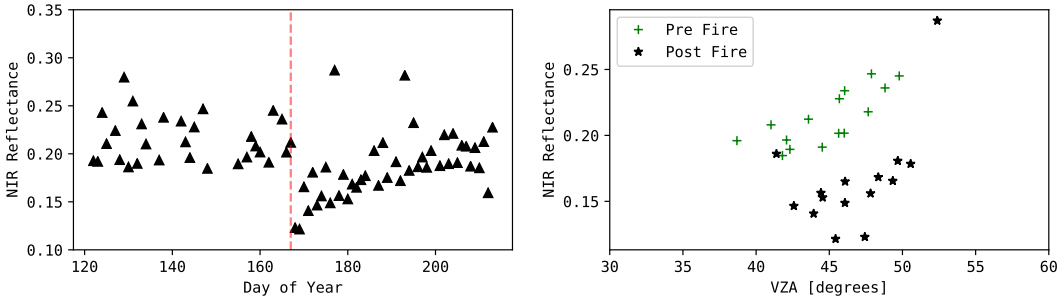


Figure 2.12. Implications of BRDF for detection. Left) NIR reflectance collected by MODIS Terra over a savanna fire in South Africa occurring on day 165 (red line). Right) NIR reflectance as a function of view zenith angle (VZA) pre- and post-fire. The apparent changes in reflectance day-to-day are due to variations in the acquisition angle of the sensor that complicate the detection of the fire event. When considered as a function of VZA, the separation between pre-fire and post-fire is more easily discriminated.

example, region-growing routines which grow burn scars may behave differently across the composite only as an effect of varying sampling geometries.

Given these considerations, it is clear that algorithms should move towards implementing BRDF correction routines. Very few present algorithms, with the notable exception of Roy et al. (2005a), make use of BRDF correction methods. A major concern will be the necessity to develop techniques that preserve the true properties of the pre-to-post fire change, as present BRDF correction algorithms typically smooth over sudden changes (Quaife and Lewis, 2010; Gomez-Dans et al., 2013).

2.5.2 Algorithms

The characteristic spectral features of burned surfaces vary somewhat across biomes. While many algorithms make use of an expected reduction in NIR reflectance, Landmann (2003) highlight that at higher spatial resolutions (<10m), the deposition of white ash can lead to a rise in NIR reflectance. While they suggest that at coarser resolutions this effect will be reduced, this may not be the case for the most severe fires in high biomass regions. The SWIR region also has various modalities associated with the pre-fire state of vegetation. Increases in post-fire SWIR reflectance occur due to the removal of moisture at the surface present in vegetation. However, if the vegetation has senesced prior to burning — which is common in savannas and grasslands during the dry season — the change in SWIR reflectance pre- to post-fire may be small (Trigg and Flasse, 2000; Eva and Lambin, 1998a).

This variability of the expected spectral signature of burning presents a considerable challenge to burned area detection algorithms. Primarily, VIs may not perform as expected. Deposition of white

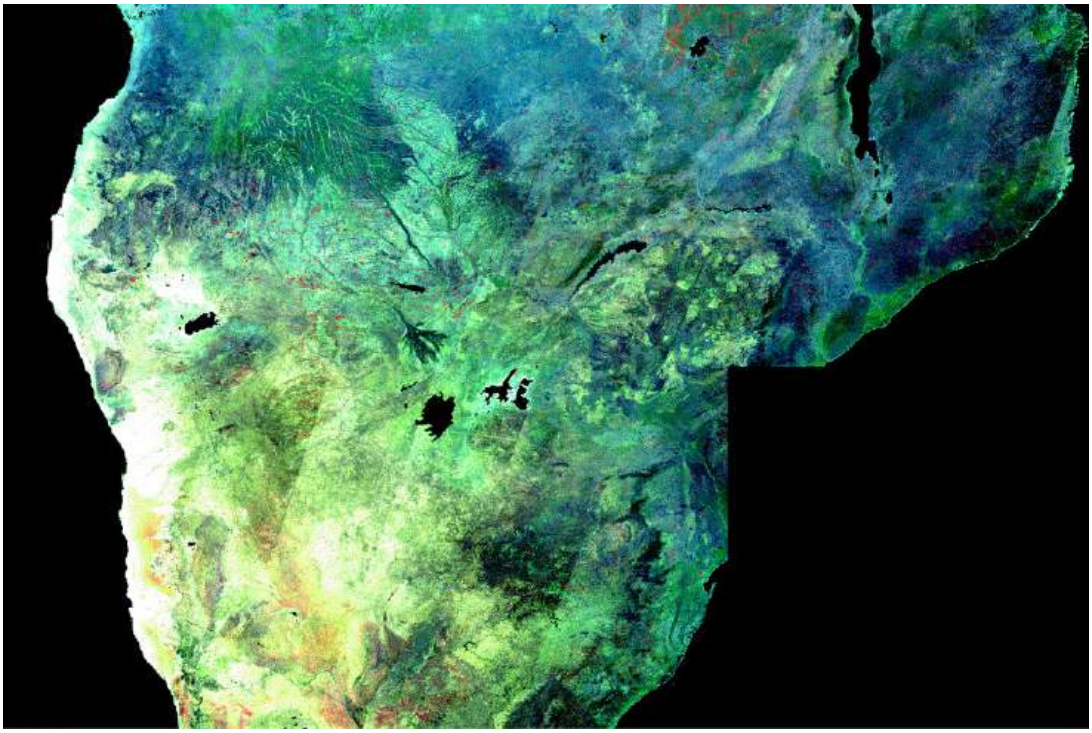


Figure 2.13. Implications of BRDF for detection. True colour image (RGB: 645, 555, and 469 nm) of monthly maximum NDVI composite for September 2000 from the MODIS sensor. The visible striping is caused by BRDF sampling effects. Source: Roy et al. (2002).

ash for very intense fires may lead to increases in NDVI post-fire. Similarly, the variability in the direction of change in the SWIR will reduce the magnitude of SWIR VIs. Therefore any algorithm that makes use of hard thresholds on the magnitude and direction of changes to VIs may inadvertently increase omission errors. Giglio et al. (2008) found that their SWIR-based VI showed inconsistent behaviour in around 20% of cases in African and Australian grasslands. In these areas the VI increased post-fire against the primary assumption within the algorithm of decline in the VI.

Similarly, there are general issues with the appropriateness of present VIs for burned area detection. The effectiveness of a VI for burned area monitoring is ultimately dependent on how sensitive it is to the properties of the land surface which indicate the occurrence of fire (Verstraete and Pinty, 1996). For mapping burned area this can be understood as the level to which the VI discriminates between burned and unburned surfaces (for an example of this see figure 2.14). It should be expected that VIs designed to be most sensitive to the quantity of live-biomass, the presence of ash and char, or exposed soil may be most useful for observing change related to the occurrence of fire. But the complexity of pre-fire and post-fire surface conditions may make the interpretation of VIs difficult. As an example,

the assumption of a post-fire decline in greenness inherent to NDVI may not always be met. Eva and Lambin (1998a) indicated that for regions where vegetation has senesced before burning there will be only a marginal change in greenness. Further, the sensitivity of the VI will vary as a function of fire severity. Chuvieco et al. (2008) evaluated BAI over boreal forests and found that it performed well over the largest fires but performed poorly over smaller and less severe fires. Similarly, the design of present VIs in terms of their sensitivity to burned surfaces is questionable. Roy et al. (2006) highlighted that NBR was not designed following the spectral index theory of Verstraete and Pinty (1996). Using both field data and satellite observations, the optimality of NBR over savannas was found to be low ($< 20\%$) as compared to an 'optimal' index. This was because the displacement in NIR-SWIR space following the fire occurred parallel to the isolines of NBR. Murphy et al. (2008) also found that NBR performed poorly against ground measurements of fire severity in Alaskan Boreal forests, especially over low severity fires. This agrees with Cocke et al. (2005) who found that NBR was poor at discerning low severity fires. These properties make the interpretation of VIs as a signal to classify pixels as burned or unburned more challenging, potentially increasing commission and omission errors.

2.5.2.1 Thresholds and parameter values

Another broad category of issues includes algorithm parameterisations. All of the algorithms explored generally contain sets of semi-arbitrary thresholds. Thresholds are most obviously applied in the classification of burned pixels. However, there are more subtle parameter choices that are made, often implicitly. For example, the use of the distance to the nearest active fire in Giglio et al. (2009) involved a probabilistic computation that used a decay constant of 5km. Similarly, they set a maximum distance constant of 10km from an initial seed as part of a region-growing algorithm. Instead of prescribed numeric thresholds, some algorithms instead apply *ad hoc* statistical thresholds. A clear example is that several algorithms apply a threshold of two standard deviations as part of their classifications. Tansey et al. (2008) applied such a threshold as part of the classification: in their algorithm, a pixel was classified as burned if it fell two standard deviations below the mean of the local spatial region. Fernández et al. (1997) also used such a threshold over a 100km window as part of their classification. While adaptive, such thresholds will add constraints on burned area that need rigorous validation, as depending on the associated statistical assumptions this will ascribe particular burned areas. For example, a normal assumption indicates that $\sim 2.5\%$ of pixels will, therefore, be assumed fire-affected. Alternatively, there are 'data-learnt' parameters. Most commonly the values of a VI or another signal for burned pixels are learnt from active fire detections. Alonso-Canas and Chuvieco (2015) produced empirical distributions for the burned and unburned classes from active fire observations. Giglio et al.

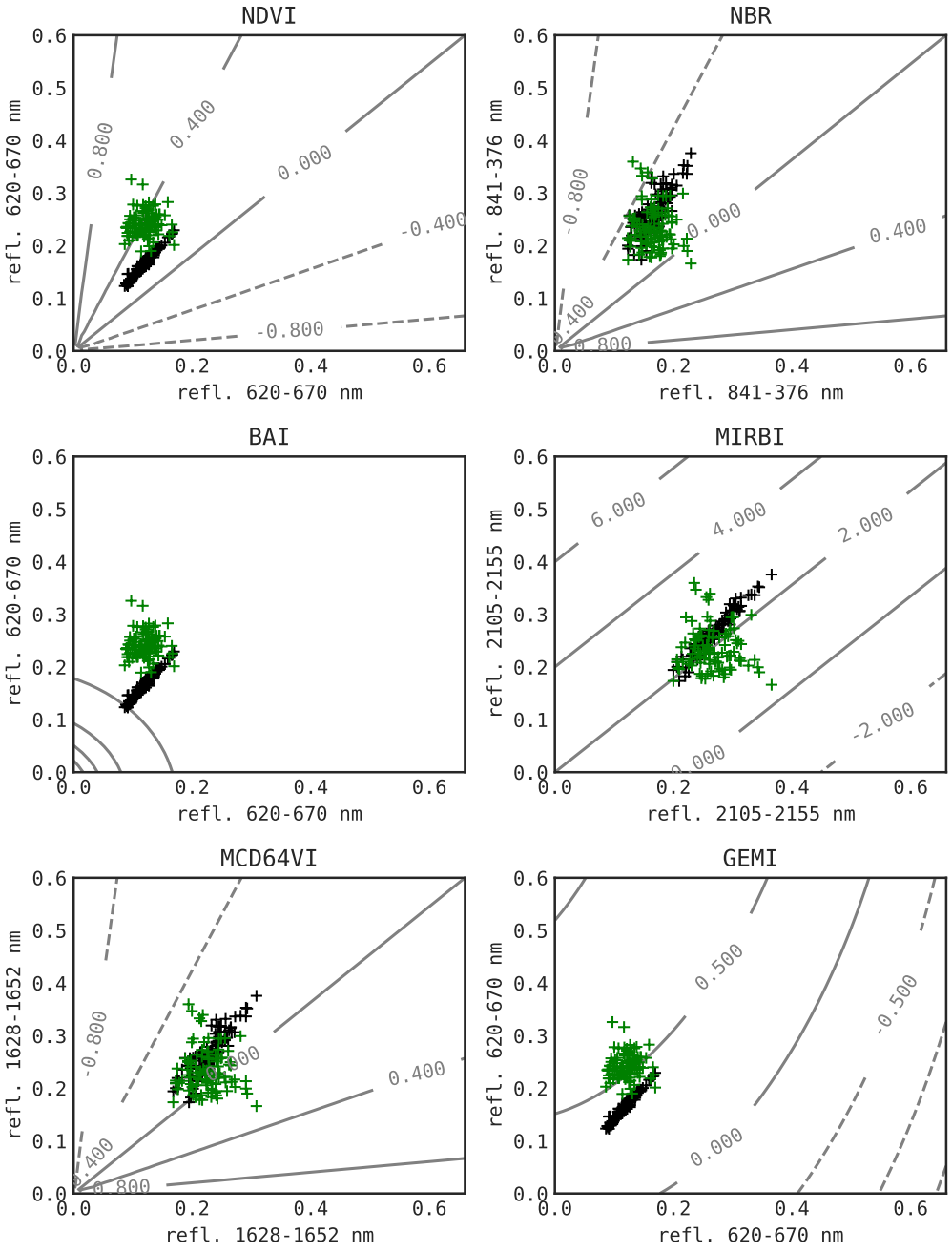


Figure 2.14. Spectral transformations of reflectance Separation between burned (black) and unburned (green) reflectance and mapping to vegetation indicies (VIs). The contours are the isolines of the VI which show the value of the VI at that location in spectral space. Reflectance measurements over a burn scar in Northern Australia, collected by the MODIS sensor.

(2009) also derived empirical distributions with a kernel density estimator based at least partially on active fire sampling. One particular challenge of using active fires to characterise these distributions is again the impact of sampling. With poorer active fire sampling the distribution of the burned class will be more poorly defined because there are functionally less samples to define the distribution. Giglio et al. (2009) attempted to control for this by assessing how separable the burned-unburned distributions are with a statistical distance metric. However, this density estimate was not conducted if the number of samples was less than 100. Comparatively, Alonso-Canas and Chuvieco (2015) noted this limitation and indicated that they will consider also using additional criteria to derive these distributions in future algorithm versions.

The *ad hoc* nature of parameterisations raises concerns about the performance of algorithms beyond the data and geographic regions they were developed on. The suitability of an algorithm for global burned area mapping is dependent on its adaptability to local fire properties and observation characteristics. Present algorithms show varying performance for different biomes. Giglio et al. (2009) acknowledged that their algorithm has a detectability threshold of around 120ha, and indicated it would therefore fail to detect small agricultural and deforestation fires. Similarly, they highlighted that tropical deforestation shows similar properties to burned surfaces in their algorithm and may produce commission errors. Simon et al. (2004) indicated that the value of a parameter defined within their algorithm may lead to varying omission and commission errors across biomes. Roy et al. (2005a) suggested their algorithm misses fires with relatively low severity due to a lack of sufficiently large spectral change, especially in savanna fires.

An example of good practice can be found in the preliminary sensitivity analysis carried out for the algorithm by Campagnolo et al. (2019). They considered the sensitivity of the mapped burned area to the value of the central threshold in the algorithm Q across different geographic regions. Figure 2.15 shows the relative change in burned area with the value of this Q threshold. Campagnolo et al. (2019) indicated that the total mapped burned area varied by only 3% with the value of Q.

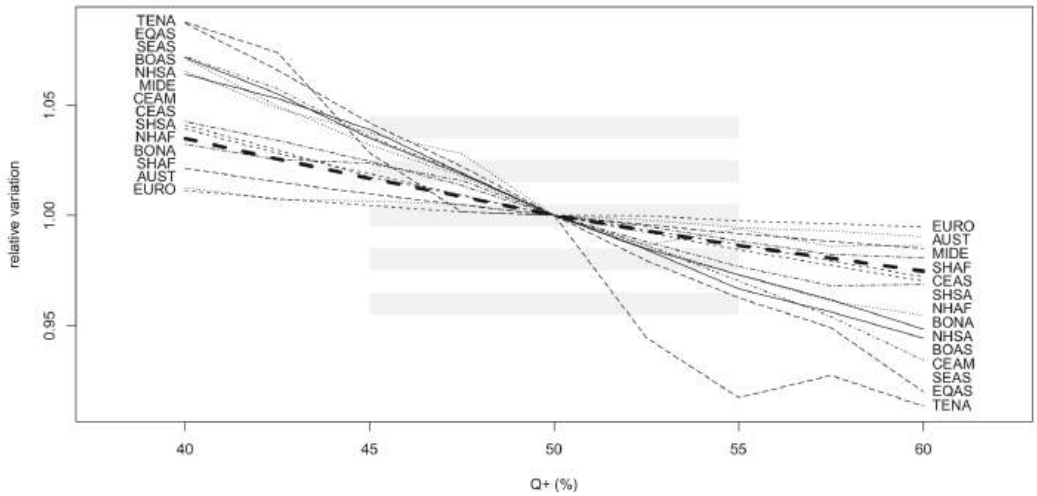


Figure 2.15. Sensitivity analysis carried out in Campagnolo et al. (2019). The dependence of the map burned area on the value of the main parameter within the algorithm Q was explored via a sensitivity analysis.

2.6 Discussion: knowledge gaps and new directions for EO fire monitoring

Considerable progress in mapping burned area from optical and thermal remote sensing instruments has been made over the past 40 years. Satellite-derived estimates of global burned area have been produced from coarse resolution instruments for 2000-present. But, some key questions remain about the methodologies used within mapping algorithms, the uncertainties of these products and whether present products and algorithms are sufficient for the generation of the highest quality long-term satellite records of burned area.

Present global burned area products have been compared against higher resolution images from instruments such as Landsat and Sentinel-2. Following community-agreed protocols, these exercises have validated products against burned area mapped at higher resolutions (Boschetti et al., 2009). Recent validation efforts have focused on retrieving globally representative validation data via stratified random sampling designs (Padilla et al., 2014; Boschetti et al., 2016; Padilla et al., 2015; Chuvieco et al., 2016; Giglio et al., 2018). These studies have indicated the current generation of products do not meet the GCOS target of a 15% commission and omission errors (GCOS, 2009). Padilla et al. (2015) showed that present products display commission errors for burned area greater than 40% and omission errors in excess of 65%. Chuvieco et al. (2018) also reported commission errors greater than 35% and omission errors greater than 62% for MCD64C6 and FireCCI50. While these validation studies are beneficial to users for comparing the accuracy of different products they do not necessarily provide information to

product producers. There has been limited activity to compare the individual qualities and limitations of the algorithms used to produce these products. Instead, previous research has generally focused on contrasting the efficacy of individual algorithm components such as compositing decisions or the sensitivity of different spectral channels and VIs (Chuvieco et al., 2005; Miettinen and Liew, 2008; Sousa et al., 2003; Chuvieco et al., 2002; Huang et al., 2016). Algorithm intercomparison initiatives have aided the development of products in other remote sensing fields (Bontemps et al., 2011; Brewin et al., 2015; Ruddick et al., 2010; Nechad et al., 2015). Further, there is an expectation that the algorithms used to produce EO derived Essential Climate Variables (ECVs) should be generated by at least several contrasting methodologies to indicate any structural or scientific uncertainties (Bojinski et al., 2014). Likewise, the public availability and peer scrutiny of algorithm code is regarded as best practice for ECV development (Bates and Privette, 2012). This indicates that for burned area algorithms to reach the required maturity for ECVs, a serious effort to compare methodologies is required. Chapter 3 therefore presents a novel intercomparison of six global burned area algorithms towards this task.

Given the limited accuracy of the products demonstrated from validation studies, there is a need to better understand and quantify the uncertainties in present products. A key requirement for ECVs is error traceability which has also been surveyed as an interest of BA product users (Bojinski et al., 2014; Mouillot et al., 2014). There is also evidence that the users of BA products want pixel-level uncertainty information. Users would also prefer a probabilistic interpretation of whether a pixel has burned as opposed to a binary classification (Mouillot et al., 2014). Achieving this also presents the opportunity of addressing the need for uncertainty quantification (UQ) of burned area algorithms so that they meet recognised standards for scientific measurements and ECV data products (BIPM et al., 1995; Bojinski et al., 2014; Nightingale et al., 2019; Merchant et al., 2017). Considerable progress has been made within other remote sensing fields by re-phrasing retrieval algorithms away from empirical relations and towards solving semi-empirical or physical models (with quantified uncertainties) via inverse retrieval theory (Combal et al., 2003; Tarantola, 2005; Wikle and Berliner, 2007). More mature remote sensing products — such as atmospheric composition or Leaf Area Index — have been developed on top of these mathematical frameworks which combine observations with a priori scientific knowledge, most commonly referred to as Data Assimilation methods (Rodgers, 2000; Kaminski et al., 2017; Lewis et al., 2012; Gómez-Dans et al., 2016; Disney et al., 2016; Pinty et al., 2007; Gobron et al., 2008). Rephrasing burned area mapping within such frameworks may help to address and quantify the limitations of mapping algorithms and observations discussed in section 2.5. As an example, little is presently known about the sensitivity of burned area estimates to the parameters and values of thresholds chosen

within these algorithms. Developing UQ methods for burned area mapping can reveal the additional uncertainties induced into products and therefore help refine algorithms (Povey and Grainger, 2015).

An ultimate aim of the producers of global burned area products is to provide the longest and highest quality possible record of fire activity. This drive comes from the importance of burned area products as Climate Data Records (CDRs), crucial for monitoring the state of the climate over decades (Merchant et al., 2017; Bojinski et al., 2014). To produce the longest possible burned area CDRs will require a transition towards algorithms which can optimally merge observations from multiple sensors. In practical terms, this will require new algorithms which can treat observations from sensors with different spectral channels and different error characteristics. Ensuring stability and consistency of CDRs of burned area over decades will require a full propagation of uncertainties from the initial satellite measurements through the mapping algorithm. The present generation of global burned area algorithms have only been designed to operate on one instrument, most typically the two MODIS instruments (Chuvieco et al., 2018; Giglio et al., 2018; Alonso-Canas and Chuvieco, 2015; Roy et al., 2005a). Work is underway within the NASA Land science team and the ESA FireCCI science team to extend these algorithms for newer missions such as Suomi-NPP VIIRS and Sentinel-3 OLCI (Giglio et al., 2013; Campagnolo et al., 2019). But ensuring a consistent burned area record between the present MODIS-derived products and future products will require careful consideration of the differences between these sensors and the degree to which algorithms have to be adjusted for new instruments. Instead, a fruitful avenue may be to adopt algorithms which are adaptive to differences in the available spectral, spatial and temporal sampling. This will likely mean using multi-spectral models of the burn signal which can be solved with different spectral sensors combined with data assimilation frameworks to optimally combine observations with sensor-dependent uncertainties.

Chapter Three

Mapping sensitivities of six global burned area algorithms

In this chapter, six global burned area algorithms are compared within a testing and benchmarking framework. The analysis goes beyond previous intercomparisons of mapping algorithms and reveals how differences in algorithm performance are driven by fire properties and characteristics of the remote sensing observations. This addresses aim I outlined previously in section 1.2.

3.1 Introduction

Burned area has been designated an Essential Climate Variable (ECV) by the Global Climate Observing System (GCOS). The European Space Agency (ESA) Climate Change Initiative (CCI) Fire project (FireCCI) aims to produce a global ECV record of burned area derived from optical and thermal remote sensing measurements spanning several decades (1981-2016). GCOS has provided accuracy targets for each ECV determined by the community to ensure that ECV data records are of sufficient quality for climate studies. For burned area ECVs these are determined to be maximum commission and omission errors of 15% (GCOS, 2011). To assess whether present burned area products achieve this target they have been validated against reference data from higher resolution satellite data (e.g Landsat ETM+ imagery) according to community agreed international standards (Boschetti et al., 2009). These validation activities have indicated that the present generation of products are not comparable to these targets, with omission and commission errors in excess of the 15% target (Chuvieco et al., 2018; Giglio et al., 2018; Padilla et al., 2015). Wider intercomparisons of burned area products have also indicated considerable differences in inter- and intra-annual estimates of burned area (Giglio et al.,

2010; Humber et al., 2018). While these validation exercises provide a suitable overview of product accuracy, they provide limited information about the direct drivers of commission/omission errors within the algorithms used to produce these products. Because of the focus on evaluating the accuracy of products without reference to the input observations or the algorithms themselves, little has been learned about the causes for differences in performance between algorithms (i.e. due to their treatment of the observations and 'modelling' of fire properties). The GCOS targets are set to be an ambitious target for the community and are considered unlikely to be met for present data. However, the extent to which these targets can be met in certain circumstances with present data has not been considered. Here the results from a round-robin exercise (RRE), an intercomparison and evaluation of six of global burned area mapping algorithms are presented. The RRE aims to go beyond previous validation exercises and attempt to synthesise a benchmarking procedure for the current generation of mapping algorithms.

3.2 The FireCCI Round-Robin Exercise (RRE)

The purpose of the RRE is to provide a framework to identify any limitations in present burned area mapping algorithms and quantify any potential inter-algorithm differences. An emergent problem of comparing different mapping algorithms is the dependence on the sampling and quality characteristics of the sensor for which the algorithm was designed. Identifying reasons for inter-algorithm differences in performance, therefore, has to be carried out in a 'sensor-agnostic' manner. A majority of global burned area mapping algorithms have been developed for the Moderate Resolution Imaging Spectroradiometer (MODIS) instrument due to the multi-decadal availability of optical and thermal observations MODIS has provided (Kaufman et al., 1998; Justice et al., 2002). To achieve a 'sensor-agnostic' RRE, algorithms reliant on different sensors are reimplemented to ingest observations from the MODIS instruments. Enforcing this reliance on the MODIS sensor consequently provides each algorithm with the same observations on which to map burned areas, and therefore allows a fairer investigation of the algorithms without sensor-dependent biases.

3.2.1 Algorithms considered

The RRE considered six burned area algorithms, each of which has been used to generate global burned area products. These algorithms have been selected for several reasons. Primarily, they represent the most commonly used products by users (Mouillot et al., 2014). They also provide a broad spread of methods which should help to indicate any variations in performance arising from differences in

algorithm assumptions. The inclusion of multiple versions of the same algorithm also helps to assess whether an improvement in performance can be observed. The key properties of each algorithm are now briefly outlined.

3.2.1.1 MCD64 Collection 6 & Collection 5.1 algorithms

Giglio et al. (2009) developed an algorithm based on observations from the MODIS sensors. This algorithm was used to produce the MODIS burned area Collection 5.1 product. The algorithm involves the generation of a temporal change spectral index and the application of a classifier which is locally trained via active fire observations from the MODIS sensors. The spectral index, which uses variations in the two MODIS SWIR channels, evaluates the stability of the surface over a period of 20 days; while moderating the index based on the variability of the compositing window due to, for example, failed cloud masking or BRDF effects. Active fire observations are used to refine the spectral characteristics expected of burned surfaces, as well as to provide spatial information about the extent of burning. Giglio et al. (2018) presented an updated MCD64 (Collection 6) algorithm with some notable changes. Refinements to the compositing of observations were made, particularly reducing the compositing window to 16 days, and relaxations of cloud masking to increase the number of observations considered cloud-free. There were also changes to several parameter values and algorithm thresholds. They reported that the collection 6 algorithm detects 26% more burned area globally than the collection 5.1 algorithm.

A global validation of the collection 6 product by Boschetti et al. (2019) indicated that a 72.6% omission error and a 40.2% commission error. Padilla et al. (2015) indicated comparable omission (68%) an commission (42%) errors for the collection 5.1 product. An earlier validation by (Giglio et al., 2018) on a smaller validation data set indicated that omission errors of 40% and 45% for the collection 6 and 5.1 products respectively. Commission errors were comparable (22% and 23%) between collection 6 and 5.1.

3.2.1.2 FireCCI Collection 5 (FireCCI50) & Collection 4 (FireCCI40) algorithms

Alonso-Canas and Chuvieco (2015) developed an algorithm for the MEdium Resolution Imaging Spectrometer (MERIS) sensor while also utilising active fire observations from the MODIS sensor. The algorithm includes several steps. Firstly, active fire observations are used to produce post-fire composites of NIR reflectance. Burned areas are grown from active fire observations after comparison to a set of conditional tests on the reflectance composite. This algorithm was used to produce the ESA FireCCI collection 4 product (FireCCI40). An updated algorithm was used to produce the FireCCI

collection 5 product (FireCCI50). The FireCCI50 algorithm was designed for use with MODIS 250m surface reflectance observations given the loss of MERIS in 2012 (Chuvieco et al., 2018). Refinements to the algorithm included the use of land cover information to remove non-burnable areas, improvements to the compositing procedure to reduce the noise characteristics of the NIR composites, and the introduction of spatial morphological filters in post-processing to refine the final burned area map. Padilla et al. (2015) estimated a global commission error of 64% and omission error of 76% for the FireCCI40 product. Lizundia-Loiola et al. (2020) indicated lower commission 51% and omission (70%) errors for the FireCCI50 product.

3.2.1.3 MODIS MCD45 Collection 5.1 algorithm

Roy et al. (2008b) prototyped an algorithm for the MODIS sensor. The algorithm is based around the application of a temporal model for the persistence of a drop in NIR reflectance. The algorithm is unique here in that it involves the application of a physically-based model to correct for BRDF effects in the observations and to aid the detection of burned areas. Unlike, the MCD64 and FireCCI algorithms, the MCD45 algorithm does not use active fire observations. Instead, the algorithm exclusively uses changes in BRDF-normalised surface reflectance observations from the MODIS NIR (841–876nm) and SWIR (1230–1250nm) channels. A statistical test is evaluated for each new observation, which accounts for the noise in the observations and the quality of the BRDF retrieval. Pixels which meet the test in both spectral channels are then considered burned subject to a few spectral threshold tests. This algorithm was used to produce the heritage MODS burned area (MCD45) collection 5.1 product — which is superseded by the MCD64 Collection 6 algorithm described in 3.2.1.1. Giglio et al. (2018) estimated a global omission error of 45% and a 23% commission error for the MCD45 collection 5.1 product. Padilla et al. (2015) indicated considerably higher omission (72%) an commission (46%) errors.

3.2.1.4 L3JRC

Tansey et al. (2008) produced an algorithm using surface reflectance measurements from the SPOT Vegetation (VGT) sensor. The algorithm utilises variations in a temporal change spectral index of NIR reflectance. The index responds to immediate changes in daily reflectance as compared to a composite generated as the average of the preceding observations. To reduce the impact of BRDF on the composite, each new observation is assessed via a phase angle test between the illumination and view geometries of the new observation and the previous composite. Spatial assumptions are included by calculating the mean and standard deviation of the index over a 200km spatial window. Pixels within this window are then classified as burned when the index is at least two standard deviations

below the mean of the spatial window, conditional on two threshold tests on NIR and SWIR reflectance. The algorithm was used to produce global burned area estimates for 2000—2007. Padilla et al. (2015) estimated an omission error of 91% for the L3JRC product and a commission error of 74% based on a global validation data set of Landsat derived fire perimeters.

3.2.2 Re-implementation of L3JRC & FireCCI40

Given the stated aim of the RRE to consider algorithms in a 'sensor-agnostic' manner, it was necessary to reimplement the algorithms for L3JRC and FireCCI40. This involved adapting these algorithms to use the MODIS surface reflectance products and associated quality assurance (QA) information. Given that the MCD45 and MCD64 products have already been produced with the same MODIS data, it was not necessary to reimplement these algorithms. Instead, the MCD45 Collection 5.1 and MCD64 Collection 6 products were acquired over the reference sites. Given the presence of sensor-specific thresholds in the L3JRC algorithm, and the use of non-similar bands from the MERIS sensor within FireCCI40, a spectral mapping procedure was implemented to convert MODIS spectral reflectance to an apparent reflectance of the MERIS and SPOT-VGT sensors (Chernetskiy et al., 2018). The surface reflectance for a sensor B at wavelength λ_b can be modelled as a mixture of the spectral reflectance recorded by sensor A:

$$\rho^B(\lambda_b) = c + \sum_{i}^{K} a_i \rho^A(\lambda_i) + \epsilon \tag{3.1}$$

Where a_i is a spectral mapping coefficient and c is a sensor mapping offset. The coefficients of the mapping between the MODIS Aqua and TERRA instruments and MERIS and SPOT-VGT were derived from the ASTER spectral library (Baldridge et al., 2009).

3.2.3 Data

The RRE aims to consider algorithm performance determined by an augmented validation approach. This involved deriving a large collection of validation burned and unburned areas from medium-resolution (30m) Landsat images. A collection of 93 reference sites were considered (shown in figure 3.1). Burned area in each site was delineated from the analysis of two multi-temporal Landsat 7 Enhanced Thematic Mapper Plus (ETM+) acquisitions. The selection of Landsat tiles followed a stratified random sampling to ensure that the selection of validation was globally representative. A two-stage process was conducted by Padilla et al. (2015) to generate the validation data. First, sites were distributed according to the major Olson biomes (Olson et al., 2001). In each biome, strata were split into two regions corresponding to areas with high global fire occurrence as provided by active

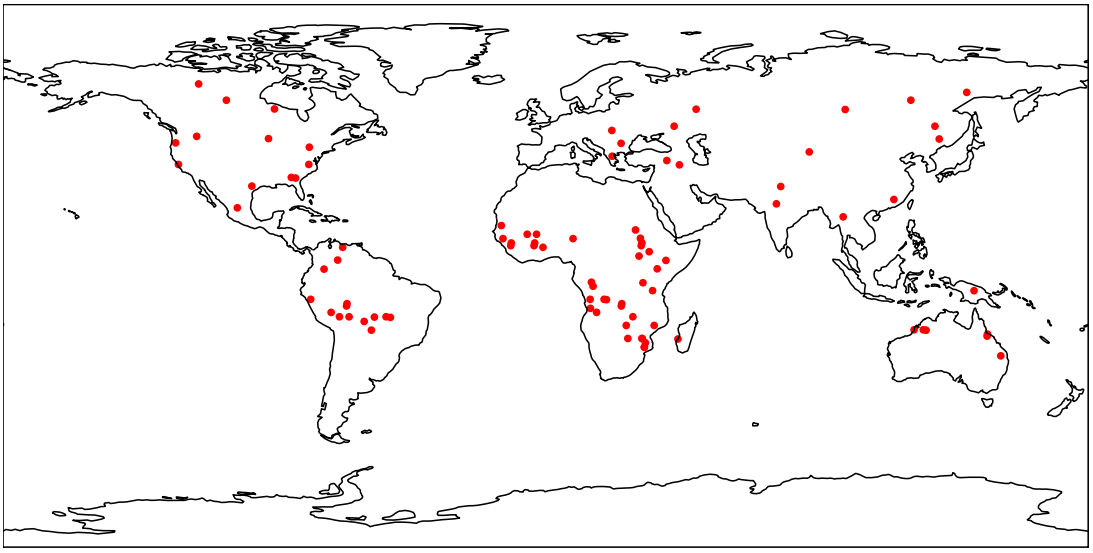


Figure 3.1. Locations of the 93 validation sites derived from Landsat TM/ETM+ acquisitions used within the RRE.

fire observations and areas with low fire occurrence. The combination of biome and fire occurrence stratification should, therefore, provide a globally representative sampling (Padilla et al., 2015).

3.2.3.1 Determination of conditioning features

In the previous chapter, section 2.5 considered challenges for burned area mapping algorithms. Building on this, the RRE focuses on deriving a set of conditioning variables which determine algorithm performance. A broad set of factors were identified and quantified, ranging from the magnitude of the reflectance burn signal, the available observational sampling and local morphological features of burn scars. These features are described below and summarised in table 3.1.

3.2.3.2 Physical fire features

Several physical features of the mapped burned areas were quantified within each reference scene. A primary driver of algorithm detection may be the fractional burned area of the coarse-resolution MODIS pixel. The per-pixel burned area fraction f was estimated from the aggregation of the 30m Landsat burned area to the MODIS 500m sinusoidal grid. Two further conditioning features were derived which were expected to influence algorithm performance. Generally, it might be expected that large clusters of burned or unburned pixels would be detected more effectively. The connectedness of the burn scar was quantified on a per-pixel basis by counting the number of adjacent burned pixels

(sp-counts) from the reference burn maps. To encode the broader site-level density of fire activity the fraction of pixels burned within 10km of the central pixel (BA10) was also considered.

In addition to these factors, the structure and vegetation cover of burned and unburned pixels in the Landsat reference was also considered. The MODIS Collection 6 vegetation continuous fields (MOD44B) product was used to quantify the structural mix of tree and shrub cover for each pixel (DiMiceli and Townshend, 2011). From MOD44B, the percentage tree (perc-Tree) and shrub (perc-Shrub) cover were used as two conditioning variables.

3.2.3.3 Spectral signal & sampling

A key determinant on algorithm performance will be the treatment of the optical reflectance signal. To explore this, the sensitivity of the algorithms to properties of the optical signal and availability of observations was considered. Roy and Landmann (2005) proposed that the post-fire reflectance of a surface can be modelled as a linear combination of two spectral end-members; one a "burn signal" reflectance ρ_{burned} and the other an unburned surface reflectance ρ_{unburned}:

$$\rho_{+} = fcc \cdot \rho_{\text{burned}} + (1 - fcc) \cdot \rho_{\text{unburned}} \tag{3.2}$$

fcc is the multiplication of the fractional burned area f and the combustion completeness cc of the fire. Gomez-Dans et al. (2013) developed a simple model for ρ_{burned} by analysing a spectral database of burned surfaces. A principal component analysis indicated that 80% of the variation in the database can be described with two spectra: a char/ash spectrum, and a dry soil spectrum. Gomez-Dans et al. (2013) parameterised the ash spectra with a flat spectrum a_0 and the dry soil spectrum with a quadratic function a_1:

$$\rho_{\text{burned}}(\lambda) = a_0 + a_1 \cdot \left[2 - (\lambda - \lambda_0) - \frac{(\lambda - \lambda_0)^2}{\lambda_{\text{max}}}\right] \tag{3.3}$$

Given the pre -and post-fire reflectance, fcc, a_0 and a_1 can be estimated from:

$$\rho_{+} - \rho_{-} = fcc \cdot \left(a_0 + a_1 f(\lambda) - \rho_{-}\right) \tag{3.4}$$

where $f(\lambda)$ is the quadratic function in equation 3.3. Given that fcc is multi-spectral, it provides a suitable generic radiometric term to compare algorithms which use different spectral regions and expectations. This allows for a spectrally agnostic comparison of the sensitivity of each algorithm to the integrated magnitude of the reflectance change signal. fcc was also computed for unburned pixels as a general indicator of an apparent fire-like reflectance signal.

The computation of fcc requires the use of BRDF-adjusted measurements of reflectance pre- and post-fire (Gomez-Dans et al., 2013). Two daily surface reflectance products from the MODIS Aqua and

Terra sensors (MYD09A1.006 and MOD09A1.006 respectively) were used to estimate BRDF-adjusted reflectance and estimate fcc (Schaaf et al., 2002). Observations of surface reflectance were collected over the period covering the two Lansat ETM+ acquisitions and 14 days before and after. Observations from before the fire and after the fire were then BRDF corrected in separate observation 'windows' to provide BRDF-adjusted pre- and post reflectance. To split observations to each of the two windows, fcc was optimised over all possible day of burns (DOB) lying between the dates of the two Lansat ETM+ acquisitions. This involved maximising the fcc recorded for the BRDF-adjusted pre- and post-fire windows to provide the most probable DOB.

Given the potential importance of the availability of observations temporally close to the DOB, the influence of observational sampling on algorithm performance was also considered. The number of clear-sky reflectance observations collected in the pre-fire and post-fire BRDF window was recorded as pre-14-days and post-14-days. An artificial DOB was associated with unburned pixels based on the middle date between the two Landsat acquisition dates.

The acquisition and illumination geometries of the observations can dramatically alter the surface reflectance due to varying sampling of the BRDF. Only one of the four algorithms explicitly accounts for the BRDF properties of the observations i.e. Roy et al. (2008b). To consider how angular sampling affects performance, the Anisotropic Flat Index (AFX) was used to quantify the relative magnitude of BRDF in each pixel. The AFX is an index designed to capture the anisotropic scattering information provided by the linear kernel BRDF models (Jiao et al., 2014). The AFX is defined as:

$$AFX = 1 + \frac{f_{vol}(\lambda)}{f_{iso}(\lambda)} \times H_{vol} + \frac{f_{geo}(\lambda)}{f_{iso}(\lambda)} \times H_{geo} \tag{3.5}$$

where f_{iso}, f_{vol} , f_{geo} are the isotropic, volumetric, and geometric scattering coefficients respectively. H_{vol} and H_{geo} are the bi-hemispherical integrals for the volumetric and geometric kernels. For each pixel in the Landsat ETM+ scene, the AFX value was computed from the post-fire BRDF composite. For the RossThick and LiSparse kernels used in the BRDF correction, the values of the integrals are $H_{vol} = 0.18$ and $H_{geo} = -1.37$. The AFX index was computed for both the MODIS NIR (860nm) and SWIR (1640nm) channels.

3.2.3.4 Active fire information

Several of the algorithms compared use active fire observations within a hybrid framework to localise algorithm parameters with the aim of improving mapping accuracy. To evaluate the benefit of active fire observations to the mapping process, two active fire sampling traits were considered. First, for each MODIS pixel in the reference ETM+ scene, the distance to the nearest active fire observation

(AF-dist) was computed. The nearest active fire is only specified spatially, in that the timing of the active fire is not considered in the selection. AF-dist indicates the benefit of active fire observations for mapping nearby burned areas. The effectiveness of hybrid algorithms was also expected to vary with the coverage of burned areas by active fires. To examine this, the local density of active fire sampling was calculated relative to the local burned area (AF-frac). Values of AF-frac closer to 1 indicate that a similar number of active fires were detected to the number of burned pixels in the Landsat ETM+ scene. Values closer to 0 instead indicate that burned pixels are more poorly sampled by active fire detections. Only the occurrence of an active fire between the two Landsat ETM+ acquisition dates is regarded as a suitable detection.

Table 3.1. Predictor variables (conditional features) used within the RRE.

Variable	Description
f	Fractional burned area of the pixel
fcc	Multi-spectral fire impact signal
AF-dist	Distance [km] to the nearest MOD/MYD14 active fire observation from the pixel.
AF-frac	Local density of active fire observations. This is defined as the number of active fires within 10km of the pixel divided by the number of burned pixels within 10km of the pixel (Burnt10).
pre-14-days	Number of optical observations available from Aqua and Terra in the 14 days before the day of burn (DOB).
post-14-days	Number of optical observations available from Aqua and Terra in the 14 days after the day of burn (DOB).
pre-NDVI	Pre-fire normalised difference vegetation index (NDVI)
Burnt10	Percentage of burned pixels within 10 km of the pixel.
sp-counts	Number of adjacent burned pixels to the pixel [0-8 pixels]
AFX-NIR	Anisotropic flat index (AFX) of the MODIS NIR channel post-fire.
AFX-SWIR	Anisotropic flat index (AFX) of the MODIS SWIR channel post-fire.
perc-Tree	Percentage tree vegetation cover of the pixel from the MOD44B product (DiMiceli and Townshend, 2011).
perc-Shrub	Percentage non-tree vegetation cover of the pixel from the MOD44B product (DiMiceli and Townshend, 2011).

3.2.4 Evaluation of algorithms

3.2.4.1 Surrogate error models for algorithm sensitivity analysis

Of primary interest within the RRE is the analysis of how conditioning variables influence the mapping characteristics of the algorithms. To quantify this, statistical emulation models of each algorithm were developed. Random forests (RF) provide non-parametric classification models based on an ensemble of decision trees (Breiman, 2001). The ability of RF classifiers to train on high-dimensional and large datasets in a computationally efficient manner has seen their popularity grow within remote sensing (Belgiu and Drăguţ, 2016). RF classifiers have also recently been demonstrated as a direct method for global burned area mapping, as well as to explore interactions between climate, vegetation and burned area (Ramo et al., 2018; Forkel et al., 2019).

The RF classifiers were used to generate surrogate statistical models of the mapping decisions of each burned area mapping algorithm. For each algorithm, two unique RF classifiers were trained. The first for pixels identified as burned in the reference data and the second for unburned pixels in the reference data. The RF classifier for burned pixels, therefore, provides an emulation of the mapping process within the algorithm and should provide the same mapping decision for a pixel: e.g. the accurate detection of burning or an omission error.

The level to which the surrogate RF models responded to each conditioning variable was quantified via an analysis of the properties of the trained RF classifiers. As a general feature, it would not be expected for algorithms to be equally sensitive to all the conditioning variables. Instead, each algorithm is likely to be sensitive to only a small subset of the variables in its mapping process. In regards to this, RF classifiers have a useful property in that they perform implicit feature selection. The structure of each decision tree within the RF classifier provides information on the relative ranking of each predictor for improving the prediction of the target, e.g. detection or omission of burned pixels (Louppe, 2014). By leveraging the averaging properties of the ensemble of decision trees within an RF, the variance of the estimated ranking is improved. The mean decrease in impurity (MDI) was chosen as the estimator of the feature importance for each variable. MDI provides an estimate of the fraction of samples which are split at a node in each tree by a selected feature averaged across all trees in the RF (Louppe, 2014).

While RF classifiers have typically been considered ‘black-box’ algorithms, the decision paths through the classifier can be decomposed into a series of individual contributions to the predicted probability of class membership. An individual decision tree f within a RF is formed by a decision

function $f(x)$ which across M leaves partitions the feature space X into M regions:

$$f(x) = \sum^{M} c_m I(x, R_m) \tag{3.6}$$

where R_m is a region within the feature space defined for leaf m. c_m is a constant corresponding to the leaf and I is an indicator function for whether the observation $x \in R_m$. As the nodes of the decision tree f are traversed, each region contributes to the prediction of $f(x)$. As only one feature is used to split the decision path, $f(x)$ can be formed as a summation of the contributions of each predictor k:

$$f(x) = c_0 + \sum^{K} c(x, k) \tag{3.7}$$

where c_0 is the contribution at the root node of the tree. The prediction from a random forest $F(x)$ is simply the average prediction of each of its N trees:

$$F(x) = \frac{1}{N} \sum^{N} f_n(x) \tag{3.8}$$

This means that the predicted probability for a class (e.g. omission error) can be decomposed into the contribution provided by each conditioning variable k. This property of the RF classifiers was used to explore the contribution of each conditioning variable to the mapping accuracies of the algorithms. The method is illustrated in figure 3.2.

Finally, the performance of each RF classifier for correctly replicating the mapping decision of the algorithm was evaluated based on splitting the dataset between a training and testing set. 75% of the valid pixels were chosen from across all Landsat ETM+ scenes for training each classifier and the remaining 25% were used to evaluate the performance of each classifier. To produce more efficient RF classifiers, the samples were stratified to achieve a similar balance of correct and incorrect mapping decisions (e.g. detection and omission/commission). The predictive accuracy of each classifier (the fraction of predictions which match the mapping algorithm decision) was evaluated on the test dataset. The area under the receiver operating characteristic (ROC) curve was also calculated for the test set based on the predicted probabilities of each RF classifier.

3.2.4.2 Comparison to GCOS requirements

As specified in section 3.1, GCOS has provided accuracy targets developed by the user community for burned area products. These are determined to be maximum commission and omission errors of 15% (GCOS, 2011). To relate the RRE to the GCOS requirements, the commission error probability C_e and omission error probability O_e of each algorithm as a function of the conditioning variables were

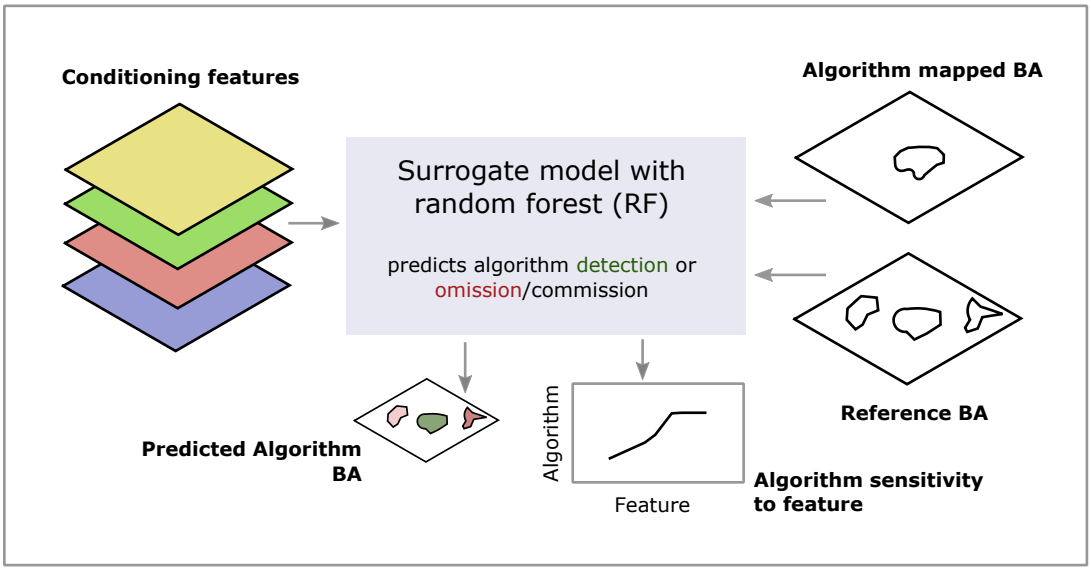


Figure 3.2. Illustration of the round-robin exercise. RF classifiers are trained on the mapping accuracies of the algorithms compared to the reference burned areas. These RF classifiers behave as surrogate models for the algorithms in the way that the conditioning features determine the mapped burned area and the algorithm omission or commission errors.

estimated. These were then used to assess the circumstances in which the GCOS targets were met. To do this, the conditioning variable x was binned and estimates of $\hat{C}_e$ and $\hat{O}_e$ for each bin i were calculated:

$$\hat{O_e}^i = 1 - n_B^i / N_B^i \tag{3.9}$$

$$\hat{C_e}^i = 1 - n_U^i / N_U^i \tag{3.10}$$

where n_B^i is the number of correct burned detections and N_B^i the true number of burned pixels for bin i, and n_U^i and N_U^i the number of correct unburned pixel detections and the true number of unburned pixels respectively. Given the sampling provided by the reference scenes is limited, and each bin will have a different number of samples, confidence intervals on these probabilities were also estimated. A normal approximation for the confidence interval of a binomial proportion is provided by:

$$\hat{p}^i \pm = z\sqrt{\frac{(1-\hat{p})\hat{p}}{N^i}} \tag{3.11}$$

where N^i is the number of samples and $\hat{p^i}$ is the estimate of the probability (e.g. $\hat{O_e}^i$) for bin i. z is the $1 - \frac{\alpha}{2}$ quantile of the standard normal distribution (Wallis, 2013).

3.3 Results

3.3.1 Drivers of omission errors

Table 3.2 reports the feature importance for the RF omission error models for each algorithm. All algorithms follow a similar pattern with the strongest correspondence with the reflectance change signal fcc followed by the fractional burned area of the pixel f. This correspondence is strongest for the MCD45C5.1 algorithm (54%) and the FireCCI50 algorithm (42%). The L3JRC algorithm shows the smallest sensitivity to fcc (13%). The relationship between fcc and the detection skill of the algorithms is shown in figure 3.3, which shows the marginal contribution of fcc to the probability of detection of burned pixels. Low to moderate values of fcc are penalised within algorithms. Beyond some threshold value of fcc, most algorithms then become influenced by the magnitude of the fcc signal. This relationship plateaus for most algorithms when $fcc > 0.6$, beyond which higher values of fcc do not contribute to an increased probability of detection. The threshold value of fcc beyond which an algorithm is positively influenced by fcc to classify pixels as burned is similar for most algorithms at around 0.35. A reduction in this threshold value between FireCCI41 (0.4) and FireCC50 (0.36) can be observed.

Table 3.2. Feature importances [%] for the omission error RF models for each algorithm.

	FireCCI41	**FireCCI50**	**L3JRC**	**MCD45C51**	**MCD64C51**	**MCD64C6**
AF-DIST	8.44	5.53	5.45	2.16	3.10	3.49
AFX-NIR	4.77	1.58	10.19	2.01	1.42	1.63
AFX-SWIR	4.35	1.34	7.64	2.39	1.07	1.14
BA10	4.47	2.72	5.89	1.24	2.73	2.55
fcc	16.82	41.62	12.64	53.78	40.69	38.76
frac-AF	12.60	5.14	5.44	1.57	2.54	3.31
fraction f	4.99	12.73	5.60	15.92	18.41	17.07
perc-Shrub	3.79	2.35	6.79	0.88	3.69	3.53
perc-Tree	5.18	1.86	9.53	0.98	1.92	2.28
post-14-days	15.15	5.18	8.17	3.26	3.23	4.40
pre-14-days	7.98	8.05	7.27	5.36	5.46	6.73
pre-NDVI	9.69	4.85	13.19	2.12	3.45	3.62
sp-counts	1.78	7.04	2.21	8.31	12.29	11.48

In contrast to fcc, the fractional area burned f has a lower feature importance for all algorithms. Both versions of the MCD64 algorithm show the highest correspondence with f, with feature importances of 18% and 17% for the collection 5.1 and 6 algorithms respectively. The FireCCI50 algorithm also shows a larger importance of f compared to the older algorithm version (13% and 5% respectively). Figure 3.4 displays the contribution of f to algorithm mapping decisions. As with fcc, there appears

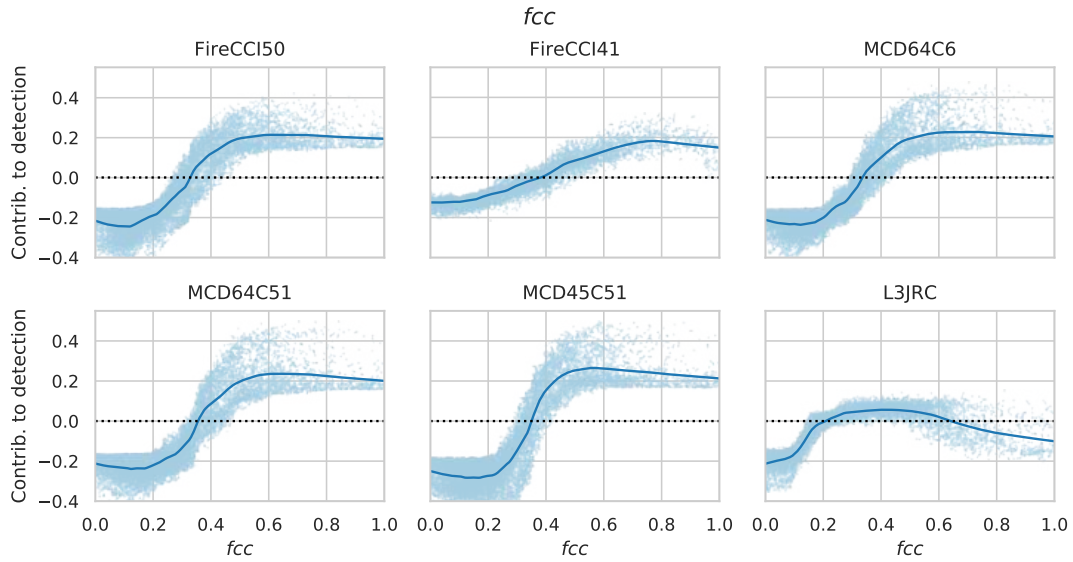


Figure 3.3. Marginal contribution to the probability of algorithm detection of burned pixel with fcc.

to be a nominal burned fraction f above which algorithms classify pixels as burned. Below this value algorithms are insensitive to the value of f, and tend to actually reduce the chance of mapping a pixel as burned. Both MCD64 algorithms respond most strongly to f; displaying a monotonically increasing sensitivity to higher burn coverage. This value of the detection threshold is lowest for MCD64C6 ($f > 0.51$) which is an improvement on the collection 5.1 algorithm ($f > 0.55$). FireCCI41 and L3JRC appear insensitive to f, and L3JRC shows an unexpected monotonically decreasing trend with f. FireCCI50 shows improvement over its predecessor in the sensitivity of the algorithm to f and shows a similar dependence to the MCD64 algorithms. It is apparent that both the physical burn coverage f and reflectance change signal (fcc) drive the majority of the variations in algorithm burned mapping accuracies. The combined f and fcc feature importances for all algorithms (except L3JRC and FireCCI41) exceed 50%.

Beyond fcc and f, the relevance of each conditioning feature varies between algorithms. The local spatial clustering of burned pixels (sp-counts) seems to play a role in some algorithms, and especially the MCD64 algorithm where sp-counts is the third most important feature. Figure 3.5 shows the marginal relationship between sp-counts and detection of burned pixels. Both versions of the MCD64 algorithm respond most strongly to this feature with contributions of 15% to the algorithm prediction when all of the surrounding pixels are burned. The FireCCI50 algorithm also displays some sensitivity to this parameter with a 5% contribution, which is greater than the preceding FireCCI41

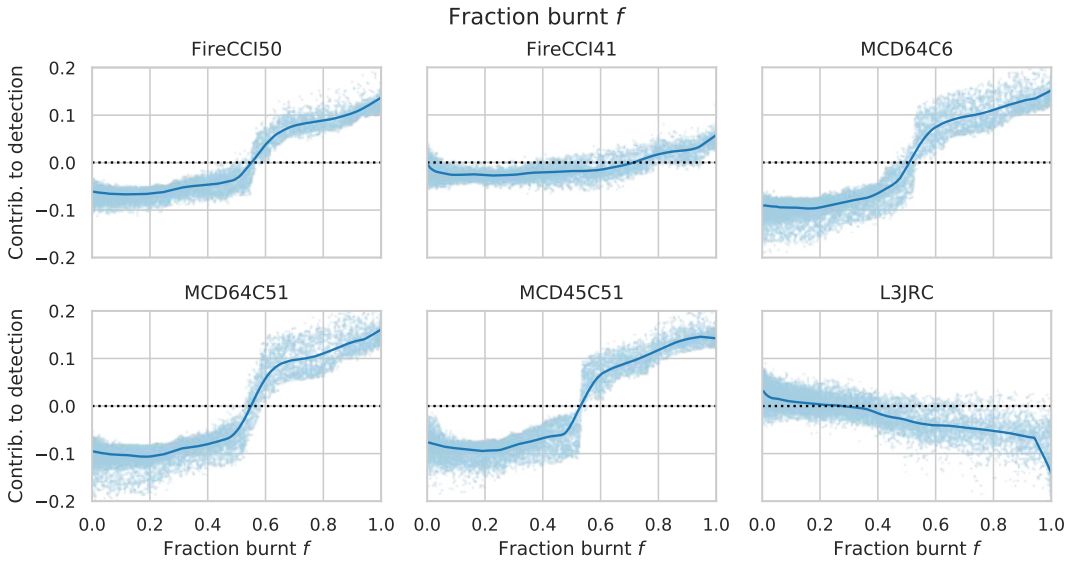


Figure 3.4. Marginal contribution to the probability of algorithm detection of burned pixels with fraction of the pixel burned f.

(<1%). MCD45C5.1 has a lower feature importance attributed to sp-counts, but a comparable probability contribution as the FireCCI50 algorithm. The predictions of L3JRC and FireCCI41 show very little response to sp-counts, perhaps indicating that these algorithms are not exploiting spatial adjacency information of burned pixels.

The sampling density of active fires (AF-frac) also plays a role in determining detection (see figure 3.6). This relationship appears greatest with the hybrid algorithms which actually use active fire observations to aid burned area detection (the FireCCI and MCD64 algorithms). For the FireCCI50 algorithm, a low sampling density of active fires reduces algorithm detection by around 5%. As the fractional coverage of active fires improves, the algorithm is more likely to map burned pixels by around 5%. This relationship is more pronounced in FireCCI41 with a 10% reduction in the detection probability for low fractional coverage. The MCD64 algorithms show a similar sigmoidal relationship with an overall smaller effect than the FireCCI algorithms. For low AF-frac values, both versions of MCD64 display a 4–5% reduction in the probability of identifying burned pixels. MCD45 also shows a residual sigmoidal relationship but has the smallest effect, and L3JRC shows an unexpected decrease with AF-frac indicating that the algorithm is more likely to detect burned areas with poor active fire sampling.

In terms of variations between algorithm versions, the FireCCI50 algorithm shows better sensitivity to both burned fraction f and fcc over the older FireCCI41 algorithm. The newer algorithm has higher

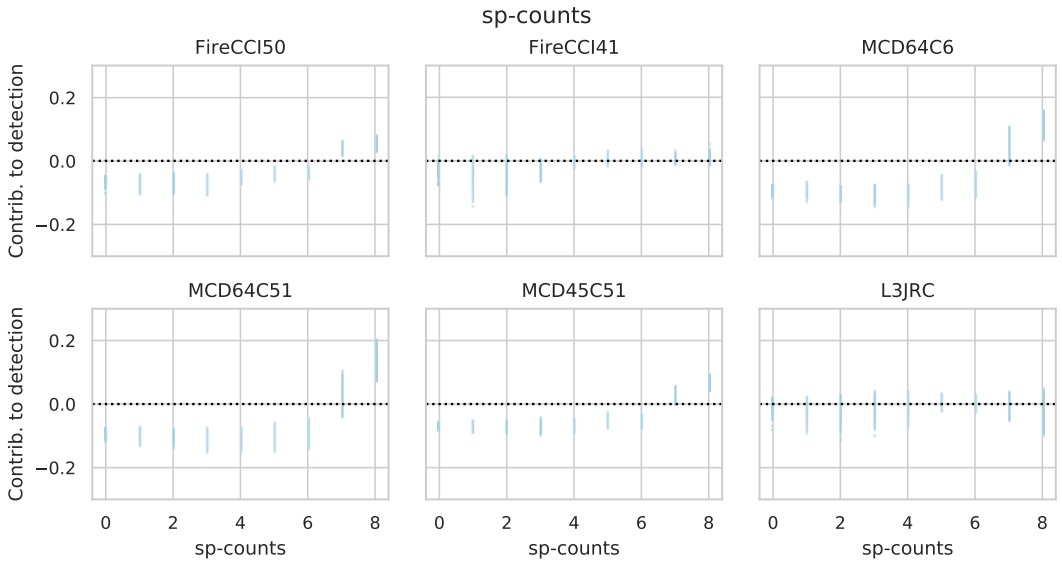


Figure 3.5. Marginal contribution to the probability of algorithm detection of burned pixels with the number of neighbouring burned pixels (sp_counts).

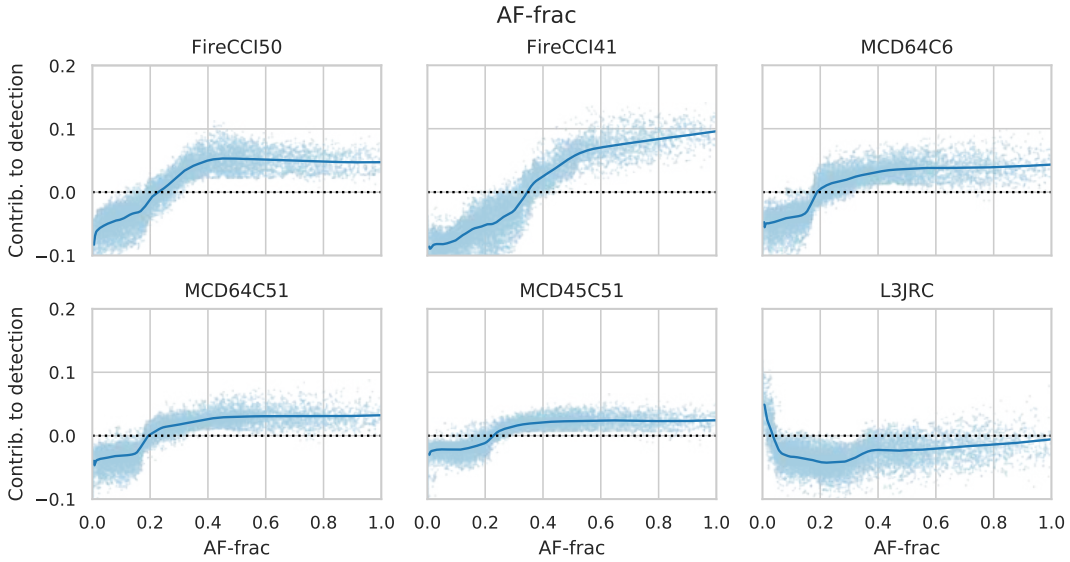


Figure 3.6. Marginal contribution to the probability of algorithm detection of burned pixels with active fire density (frac-AF)

feature importances associated with fcc and f. The newer algorithm also displays less sensitivity to the number of observations available in the pre-fire and post-fire periods. It can also be noted that the MCD64C6 algorithm has a slightly lower sensitivity to both f and fcc than the collection 5.1 algorithm, which indicates a better robustness to the available sampling.

3.3.2 Drivers of Commission errors

Table 3.3 shows the feature importance for the RF commission error models for each algorithm. For most algorithms, the most important feature is again the reflectance change signal fcc. MCD64C5.1 has the largest importance attributed to fcc (52%), followed by MCD64C6 (50%), and MCD45C51 (49%). The FireCCI50 algorithm is more sensitive to spurious fire-like reflectance changes (as indicated by fcc) than its predecessor algorithm. L3JRC has the lowest importance attributed to the fcc signal of all the algorithms (19%), but this is not surprising considering the poor sensitivity to fcc for burned pixels as demonstrated in figure 3.3. Figure 3.7 shows the marginal contribution that fcc makes to the probability of commission errors. The general strength of the contribution for larger values of fcc indicates that an apparent 'fire-like' reflectance change influences algorithm commission errors. As with the omission models, the relationship for most algorithms is generally sigmoidal, with low values of fcc decreasing the probability of commission errors. For larger values of fcc most algorithms become increasingly likely to produce commission errors. Typically above moderate values of fcc (0.3–0.4), there is no additional contribution to the probability of algorithm commission errors.

Table 3.3. Feature importances [%] for the commission error RF models for each algorithm.

	FireCCI41	**FireCCI50**	**L3JRC**	**MCD45C51**	**MCD64C51**	**MCD64C6**
AF-DIST	12.23	7.37	3.80	5.77	6.26	6.71
AFX-NIR	5.37	3.03	6.44	2.66	2.92	2.56
AFX-SWIR	4.42	2.06	3.64	2.01	2.09	1.91
BA10	6.81	3.23	2.55	1.90	2.65	2.94
fcc	16.96	45.29	18.98	49.05	52.08	49.73
frac-AF	8.79	3.11	1.23	1.20	2.92	3.12
perc-Shrub	5.57	4.37	6.38	4.87	3.42	3.33
perc-Tree	6.62	3.97	6.87	4.06	3.14	3.57
post-14-days	10.11	12.41	20.29	13.57	9.13	9.63
pre-14-days	11.59	6.92	9.77	7.61	6.74	7.25
pre-NDVI	10.50	7.85	19.46	6.53	7.94	8.61
sp-counts	1.04	0.39	0.62	0.76	0.71	0.64

In terms of the other conditioning variables, the post-fire observation sampling (post-14-days) appears to be an important feature for several algorithms. This feature contributes 20% to L3JRC

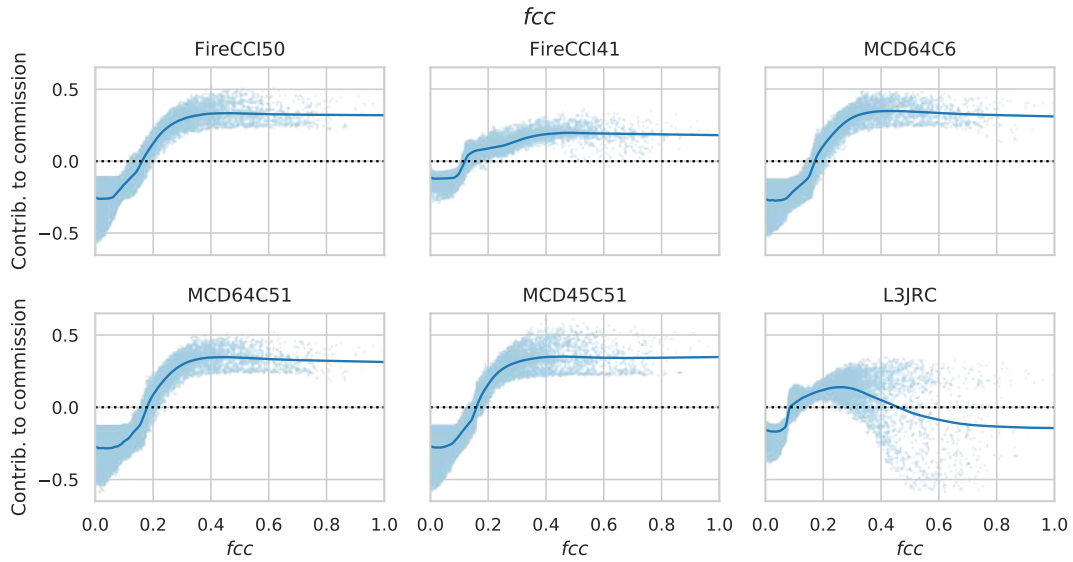


Figure 3.7. Marginal contribution to the probability of algorithm commission of burned pixels with fcc.

algorithm and 14% to the MCD45C5.1 algorithm. Figure 3.8 shows the contribution that post-14-days makes to the commission probability of each algorithm. Most algorithms show a reduction in the probability of commission errors with a reduced number of observations. For pixels with the highest number of post-fire observations, there is a positive contribution to commission errors, ranging from 20% for FireCCI50 and the MCD64 algorithms and 10% for L3JRC. Increased commission error with better sampling will be driven by algorithms incorrectly classifying pixels in which a change signal is detected but incorrectly labelled as burned. The change signal instead may originate from of other sources of change (for example harvesting, removal of vegetation) or spurious sources of signal change (for example BRDF effects, cloud shadow).

The magnitude and nature of BRDF effects in the NIR channel of MODIS (860nm) also plays a role in commission and omission errors for the L3JRC algorithm. Feature importance attributed to this (AFXNIR) is 6% and 10% for commission and omission errors respectively, compared to generally smaller contributions with other algorithms. Figure 3.9 shows the contribution that AFXNIR makes to the commission error probability of each algorithm. All algorithms show a general property of an increased chance of commission errors with AFX values exceeding 1, and a reduction for AFX<1. This relationship is most pronounced with the L3JRC algorithm. An interesting feature is the low importance attributed to the presence of neighbouring burned pixels (sp-counts). All algorithms have

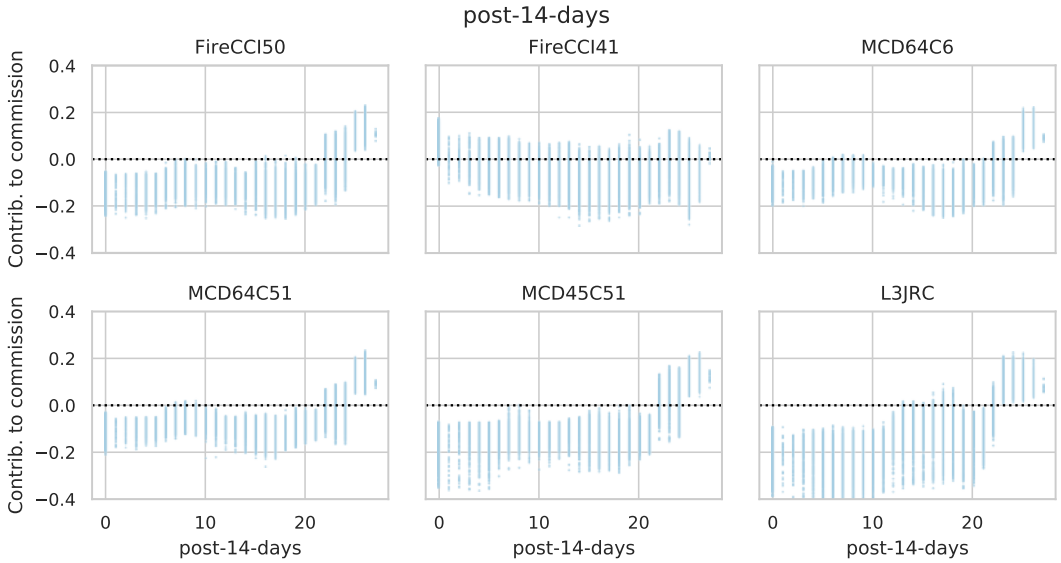


Figure 3.8. Marginal contribution to the probability of algorithm commission of burned pixels with post-fire sampling (post-14-days).

feature importances ≤1% for sp-counts. This would indicate that algorithms are not making additional commission errors at the edges of burn scars (for example where sp-counts >3).

3.3.3 Performance of the RF classifiers

Table 3.4 reports the validation statistics of the RF classifier omission and commission models for each algorithm. For the omission models, the accuracy scores all exceed 0.84 indicating that the RF classifiers can accurately predict over 84% of algorithm omission or detection decisions. The highest accuracy omission model is for MCD64C5.1 (0.88) and the lowest accuracy model belongs to the FireCCI50 algorithm (0.84). Commission error models have higher accuracies, with all RF classifiers predicting in excess of 88% of the algorithm labels. AUC values for omission models exceed 0.88 and are highest for the MCD45C5.1 algorithm. For commission models, AUC values range from 0.89 for FireCCI41 and up to 0.97 for the MCD64C5.1 algorithm.

3.3.4 Algorithm performance in relation to GCOS accuracy requirements

The GCOS requirement of upper limits of 15% omission and commission errors are only met in certain limited circumstances, and not for all algorithms. No product meets the GCOS 15% omission error target for the fractional burned area of the mapped products (see figure 3.10a). All products do

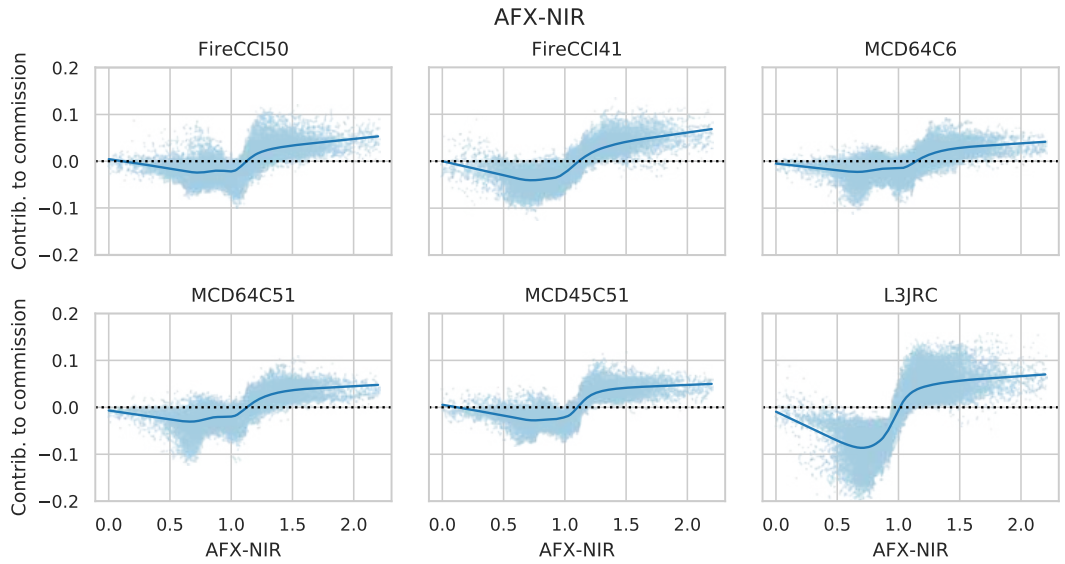


Figure 3.9. Marginal contribution to the probability of algorithm commission of burned pixels with Anisotropic Flat Index (AFX) in the MODIS NIR channel (841–876 nm) (AFXNIR).

Table 3.4. Validation statistics for the omission and commission error models for each algorithm.

	Omission		**Commission**	
	Accuracy	**AUC**	**Accuracy**	**AUC**
FireCCI50	0.84	0.92	0.88	0.95
FireCCI41	0.87	0.88	0.90	0.89
MCD64C6	0.86	0.94	0.90	0.96
MCD64C5.1	0.87	0.95	0.91	0.97
MCD45C5.1	0.88	0.95	0.89	0.96
L3JRC	0.86	0.89	0.88	0.94

display increasing accuracies with increasing fraction burned. FireCCI50 performs best for the lowest fractional areas and MCD64C6 has the lowest omission error beyond around 50% of the pixel being burned. For high fractional burned pixels ($f > 0.75$) omission errors for the four best-performing algorithms are within 50–60%. At best, omission errors are still in excess of the GCOS target for all products, although MCD45C5.1 is within 10% of the target. The L3JRC and FireCCI41 algorithms perform worse than the other algorithms in this regard, showing a more limited sensitivity to f.

The magnitude of the reflectance change signal (fcc) also influences whether an algorithm attains the GCOS requirement. At the global scale, four products meet and exceed the GCOS target (see figure 3.10b). MCD45C5.1 shows the best performance for values of fcc (0.4–0.7). As a considerable proportion of the burned pixels fall within this range, this represents an important difference in

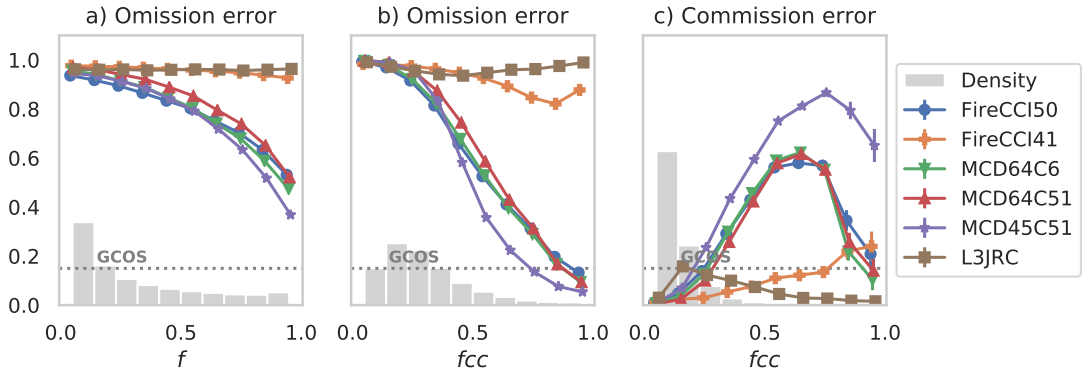


Figure 3.10. Estimated algorithm omission and commission errors in relation to the GCOS targets. a) Algorithm omission error rate with the fractional burned area f. b) omission error rate with the fire spectral signal fcc and c) commission error with fcc. The shaded grey region corresponds to the frequency distribution across burned or unburned pixels in the reference dataset.

detection ability. Improvements within the MCD64C6 algorithm have led to a reduction in omission errors for lower values of fcc. For these burns, there is typically a 10% reduction in omission errors over the older MCD64C5.1 algorithm. The FireCCI50 algorithm performs considerably better than the FireCCI41 algorithm for all fcc impact levels. The FireCCI50 algorithm meets the GCOS requirement for burned areas with $fcc > 0.85$. For the highest impact fires ($fcc > 0.95$), omission errors for the three best-performing products are well within the 15% GCOS requirement. However, these fires represent less than 5% of reported fires in the reference data. fcc also shows a definitive relationship with algorithm commission errors for four of the six algorithms (figure 3.10c). These four algorithms exceed the GCOS requirement for even low to moderate values of $fcc > 0.3$. Fortunately, the frequency of such spurious fcc signals is relatively uncommon, with about 80% of unburned pixels having fcc values less than 0.25. For these pixels, all algorithms apart from MCD45C5.1 are within the GCOS target (<15%).

3.4 Discussion

This exercise represents a new intercomparison of burned area algorithms with the GCOS requirements in mind. Previous efforts have been made to compare individual algorithm components in isolation, such as compositing methods, spectral choices and classification thresholds (Chuvieco et al., 2005; Barbosa et al., 1998; Chuvieco et al., 2002; Huang et al., 2016). From the other side, products have been compared but only based on aggregated validation statistics (Padilla et al., 2015; Roy and Boschetti, 2009). The FireCCI round-robin exercise (RRE) presented here sought to conduct an evaluation of the

sensitivity of individual burned area mapping algorithms and assess performance in relation to the GCOS benchmarks.

What emerges is the notion of detection thresholds in mapping algorithms, most notably in relation to both pixel burned fraction f and the magnitude of the reflectance change signal (fcc). The RF surrogate models indicated that algorithms have inherent thresholds which determine whether a sufficient signal exists to map burned pixels. For example, algorithms have effective thresholds in burned fraction f of approximately 50% — above which algorithms declare sufficient evidence to classify pixels as burned. This 50% threshold has been previously identified as an achievable lower bound on coarse-resolution burned area detection across multiple biomes and for some of the algorithms discussed here (Tsela et al., 2014; Giglio et al., 2009; Eva and Lambin, 1998b). Effective detection thresholds for fcc are slightly lower at around 0.35 for all algorithms. This, and the higher feature importances attributed to fcc, generally indicates that algorithms are most sensitive to the combined effect of the physical fraction burned and the radiometric changes associated with burning. Given that both f and fcc can be calculated easily for coarse-resolution sensors, they present useful heuristics for defining the detection capabilities of current and future burned area mapping algorithms (Roy and Landmann, 2005).

Areas for improvement were identified for all algorithms. For example, a reported sensitivity to pre-fire NDVI indicates that the L3JRC and FireCCI41 suffer in situations involving senescent vegetation. Sensitivity to BRDF effects should also be addressed within the L3JRC algorithm to aid a reduction in commission errors. Most notably, a reduction in sensitivity to spurious burn-like reflectance changes will help to reduce commission errors for all algorithms. The most effective way to reduce omission errors may be improving sensitivity to burns with small reflectance changes (i.e. small values of fcc). For example, improvements in pre-processing and refinements to the spectral decisions used to classify burned areas may address this. Noise in coarse-resolution reflectance measurements (e.g. arising from atmospheric effects and geolocation errors) will ultimately determine the lower boundary for fcc values which can be associated with confident burned area identification (Roy and Landmann, 2005). There is naturally the caveat that a balance between omission and commission errors must be struck and changes which reduce omission errors may lead to greater commission errors.

A novel application of RF classifiers for the production of surrogate statistical decision models for remote sensing retrieval algorithms was demonstrated. The surrogate models allowed for predictions which matched well to the mapping decisions of the algorithms themselves, as demonstrated by the validation of the RF models (see table 3.4). It would be reasonable however to place more trust in the interpretability of the omission models. Primarily, this is based on the difficulty of sampling

from commission errors due to their relative infrequency. Further, the ability to define and quantify sufficiently informative conditioning predictors for commission errors was a difficulty. There may also be a few occasions where burned pixels are missed in the validation Landsat ETM+ data which are correctly identified by the algorithms.

The use of harmonised input datasets derived from the MODIS sensors makes the RRE unique amongst intercomparisons of burned area products. This, however, adds an additional source of error into the generation of MODIS-derived products from the L3JRC and FireCCI41 algorithms dependent on the quality of the spectral mapping procedure. Similarly, MODIS based algorithms were not reimplemented for the RRE. There are some differences between the MODIS input products used within these algorithms. For example, the FireCCI50 algorithm uses active fire observations from the MCD14ML Collection 6 product while the MCD64C6 algorithm utilises the MOD14A1/MYD14A1 active fire products (Giglio et al., 2016). The MCD64C6 algorithm also makes some relaxations of the MODIS quality assurance fields provided with the MODIS surface reflectance product (Giglio et al., 2018). These differences were generally considered of a smaller magnitude to the reliance on different sensors in the L3JRC and FireCCI41 products.

The GCOS omission and commission error targets present a considerable challenge to the present generation of mapping algorithms. Four of the six considered algorithms meet the GCOS 15% omission error target at the global scale. This achievement is only reached in the case of the most spectrally impactful fires where $fcc > 0.8$, which corresponds to only 3% of the observed burning in the reference dataset. In terms of fractional burned area f, no current algorithm is within the 15% target at the global level, even for complete burning ($f = 1$) of the pixel. Improving the sensitivity of algorithms to f and related quantities such as fcc should be an aim for algorithm developers in attaining the GCOS target. Spurious burn-like reflectance changes can cause commission errors for the algorithms considered which exceed the GCOS target, but these were found to be generally infrequent. Algorithms do well in over 80% of cases to constrain commission errors to be less than the 15% GCOS requirement. A success of the RRE has been the demonstration of improvement between algorithm versions, as requested by GCOS (GCOS, 2011). This can be observed in the greater sensitivity of FireCCI50 to both f and fcc compared to FireCCI41. A comparative ranking of different algorithms was something not considered within the RRE, as comprehensive accuracy statistics for each algorithm are available from several previous validation studies (Chuvieco et al., 2018; Padilla et al., 2015, 2014; Humber et al., 2018; Roy and Boschetti, 2009).

3.5 Conclusion

This chapter has presented results from the ESA FireCCI round-robin exercise of six globally-applied burned area algorithms. The exercise represents the first comparison specifically designed to test the sensitivities of the respective mapping algorithms across a globally representative sampling of fire activity. A comparison framework was developed based on the surrogate modelling of the decision characteristics of each mapping algorithm. Random forest (RF) classifiers were trained on mapped omission and commission errors for the algorithms across a globally representative set of Landsat ETM+ burned area validation maps. These RF classifiers, therefore, provided a surrogate model of the mapping decisions of each algorithm determined by their sensitivity to a set of thirteen quantifiable conditioning variables related to the expected performance of each algorithm. The RF classifiers showed predictive ability for all algorithms, with accuracies exceeding 84%. Feature importances from the RF classifiers indicated, that for all algorithms, the magnitude of the post-fire reflectance change was the primary driver of omission errors. The sub-pixel burned area of the pixel also provided a clear relationship to algorithm detection of burned areas. In terms of the decision structures of the mapping algorithms, the presence of apparent classification thresholds between these variables and mapped burned area was demonstrated for all algorithms. The value of these thresholds, therefore, represent quantifiable heuristics to measure the improvement of future generations of mapping algorithms. A similar relationship between the magnitude of reflectance changes and commission errors was also identified for all algorithms. Issues with the exercise were discussed, most primarily the ability to draw sufficient samples of algorithm commission errors. The RRE has the potential to be routinely implemented and help improve the current and next generation of burned area mapping algorithms.

This chapter has presented a new method to examine the sensitivity of burned area mapping algorithms. This was achieved by using a statistical emulation of the mapping decisions of each algorithm in comparison to validation data. The purpose of this is to address the aim specified in chapter 1 to “evaluate present practices and develop new methods to assess the qualities of burned area mapping algorithms”. The work reveals that the error characteristics of algorithms are determined by the quality of the observations available and characteristics of the fire signal. Quantifying this mapping is considered within the field of uncertainty characterisation and quantification, which forms the next section of this thesis.

Part II

Uncertainty characterisation for burned area mapping

Chapter Four

Uncertainty characterisation for burned area mapping

4.1 Introduction

The Global Climate Observing System (GCOS) — which evaluates the state of observing systems for climate monitoring — has specified the need to systematically produce and validate global satellite burned area products. These Climate Data Records (CDRs) prove vital in calibrating and validating the fire models within dynamic global vegetation models (DGVMs) (Hantson et al., 2016; Thonicke et al., 2001; Poulter et al., 2015); as well as to drive 'bottom-up' estimates of fire emissions (van Der Werf et al., 2017; Seiler and Crutzen, 1980); both with the ultimate aim of better quantifying the land carbon cycle (Bowman et al., 2009; Stocker et al., 2013). Because of this, satellite burned area products have vital research, policy and management implications. The detection of (often subtle) signals in Earth's climate requires CDRs with suitable information on the precision of the measurements they contain (Merchant et al., 2017; Krawchuk and Moritz, 2014). The precision of a measurement — or its uncertainty — is vitally important for making decisions about the measurement's worth. But at present, quantitative descriptions of the uncertainties of burned area products are sparse. Practically, this has been driven by difficulties in both the definition of uncertainty for burned area products and appropriate methods to deal with particular routines within burned area mapping algorithms (e.g. use of thresholds, arbitrarily defined parameters — see section 2.5.2 chapter 2). As a result, the implementation needs for burned area products as specified by GCOS explicitly dictate greater uncertainty characterisation. Action tasks requested of the Earth observation (EO) fire community specify a need to *"provide better than*

Table 4.1. The Committee on Earth Observation Satellites (CEOS) Land Product Validation (LPV) group validation hierarchy (from CEOS-LPV (2019)).

Stage	Validation definition	Variable
0	No validation. Product accuracy has not been assessed. Product considered beta.	
1	Product accuracy is assessed from a small (typically <30) set of locations and time periods by comparison with in-situ or other suitable reference data.	Snow, Fire Radiative Power
2	Product accuracy is estimated over a significant set of locations and time periods by comparison with reference in situ or other suitable reference data. Spatial and temporal consistency of the product and consistency with similar products has been evaluated over globally representative locations and time periods. Results are published in the peer-reviewed literature.	fAPAR, Phenology, Land Cover, LAI
3	Uncertainties in the product and its associated structure are well quantified from comparison with reference in situ or other suitable reference data. Uncertainties are characterized in a statistically rigorous way over multiple locations and time periods representing global conditions. Spatial and temporal consistency of the product and with similar products has been evaluated over globally representative locations and periods. Results are published in the peer-reviewed literature.	Vegetation Indices, Albedo, Soil Moisture, LST & Emissisvity, Phenology, **Burned Area**
4	Validation results for stage 3 are systematically updated when new product versions are released and as the time-series expands.	Active Fire

existing uncertainty characterisation" within the design and analysis of burned area products (GCOS, 2016, pg.229).

The Committee on Earth Observation Satellites (CEOS) provides an organisation for the international coordination of satellite products (including to serve the needs specified by GCOS). The CEOS Land Product Validation (LPV) group coordinate the validation of terrestrial satellite products including burned area (CEOS-LPV, 2019). The CEOS LPV have identified four validation levels which correspond to a hierarchy of improving specification and quality assurance for terrestrial satellite products (reproduced in table 4.1). Presently, burned area products achieve level 3 validation. This has been achieved by focusing on the systematic validation of burned area products against sets of global Landsat image pairs (Padilla et al., 2014, 2015).

These analyses have provided statistics on commission and omission error for several burned area products. Such validation initiatives are indeed useful for helping to understand the quality of burned area products. However, they may not accurately inform on the uncertainties of these products. Further, even if the sampling of reference data provides a sufficiently representative value of the global error in burned area products; there remains a need for per-pixel uncertainties to attain

Table 4.2. Maturity of Climate Data Records (CDR) in relation to uncertainty quantification (Schulz et al. (2015), table adapted from Merchant et al. (2017).)

	Standards	**Uncertainty quantification**
1	None	None
2	Standard uncertainty nomenclature is identified or defined	Limited information on uncertainty arising from systematic and random effects in the measurement
3	Score 2 +; standard uncertainty nomenclature is applied	Comprehensive information on uncertainty arising from systematic and random effects in the measurement
4	Score 3 +; procedures to establish SI traceability are defined	Quantitative estimates of uncertainty provided within the product characterising more or less uncertain data points
5	Score 4 +; SI traceability partly established	Temporal and spatial error covariance quantified
6	Score 5 +; SI traceability established	Comprehensive validation of the quantitative uncertainty estimates and error covariance

a higher maturity of burned area CDRs. This is from the view that the requirements of uncertainty information in CDRs is much broader than the validation definition (Merchant et al., 2017). Schulz et al. (2015) proposed a set of standards for assessing the maturity of CDRs. This includes a six-point scale for uncertainty quantification, which is reproduced in table 4.2. Each level corresponds to an increasing maturity of the uncertainty quantification in the CDR. Merchant et al. (2017) provides further interpretation of the scale. Lower levels of maturity (2 & 3) correspond to generic statements of the uncertainty in the CDR — for example for the whole dataset. This is the level at which present burned area products can be placed — due to work by Padilla et al. (2015) & Chuvieco et al. (2018) in quantifying global errors. To achieve higher maturities requires that burned area products contain unique uncertainties for each measurement, allowing users to discriminate between more and less certain data. The purpose of this section is to work towards this aim for burned area CDRs.

4.2 User need for uncertainty quantification in burned area mapping

Mouillot et al. (2014) evaluated the state of present products and indicated several challenges raised by the user community of burned area products. A key focus was the growing requirement for uncertainty quantified products. Current products show considerably different inter- and intra-annual estimates (Giglio et al., 2010; Padilla et al., 2015). Figure 4.1 shows the spread of present estimates from

three widely-used burned area products. Humber et al. (2018) highlighted the problem by showing the range of present products. The total global burned area reported for 2005–2011 varied by 90% (15 million km^2) between the highest and lowest estimates. Such large differences underline the limited confidence users can presently place in burned area estimates. Similarly, with the growing interest in producing a longer ECV record of burned area, uncertainty quantification becomes an important concern. Such CDRs will require the fusion of estimates from multiple satellite missions, and potentially differing mapping algorithms. It can be expected that such products will be of variable quality, for example as a function of the qualities of the satellite instrument used. Polar-orbiting instruments suitable for the generation of a multidecadal burned area ECV begin with the Advanced Very High Resolution Radiometer (AVHRR) instruments (James and Kalluri, 1994), then advance through the Moderate Resolution Imaging Spectroradiometer (MODIS) era and now include the new Sentinel 2 and 3 missions (Drusch et al., 2012; Donlon et al., 2012). The differential characteristics of these sensors must be considered in the generation of a consistent multidecadal burned area CDR. Differences in spatial resolution, temporal revisit and spectral characteristics will ultimately determine the quality of any burned area retrieval. It would be expected that the uncertainty in estimates from older sensors may be greater than from newer sensors; with Giglio et al. (2010) suggesting these to be an order of magnitude larger in the pre-MODIS era due to the reduced mapping opportunity and spatial resolution. To generate ECV records of lengths approaching climate temporal scales (>30 years), the focus will be on the detection and analysis of subtle signals in the burned area time series. But the detection of these signals necessitates a consistent data record through time (Merchant et al., 2017). With the lack of uncertainty methods within present mapping algorithms, there is no true ability to generate a consistent long-term CDR of fire disturbance from multiple sensors.

Many researchers have previously highlighted the need for uncertainty estimates for burned area. The fire modelling community has indicated issues arising from the lack of uncertainty information. Rabin et al. (2017) concluded that substantial but unknown uncertainties in burned area estimates determine the effectiveness of fire model intercomparison and evaluation. They specified that presently the best estimate of uncertainty comes from the range of burned area estimates. Hantson et al. (2016) also highlighted the importance of uncertainty in burned area for model benchmarking activities. They indicated that the largest differences between products tend to occur in regions with little burning, increasing the uncertainty in model evaluation and concluded that the explicit inclusions of observational uncertainties are necessary for benchmarking scores. Yue et al. (2014) raised concerns about the calibration of parameters in empirical fire models against burned area products and whether the agreement between simulations and observations is down to error compensation between the two.

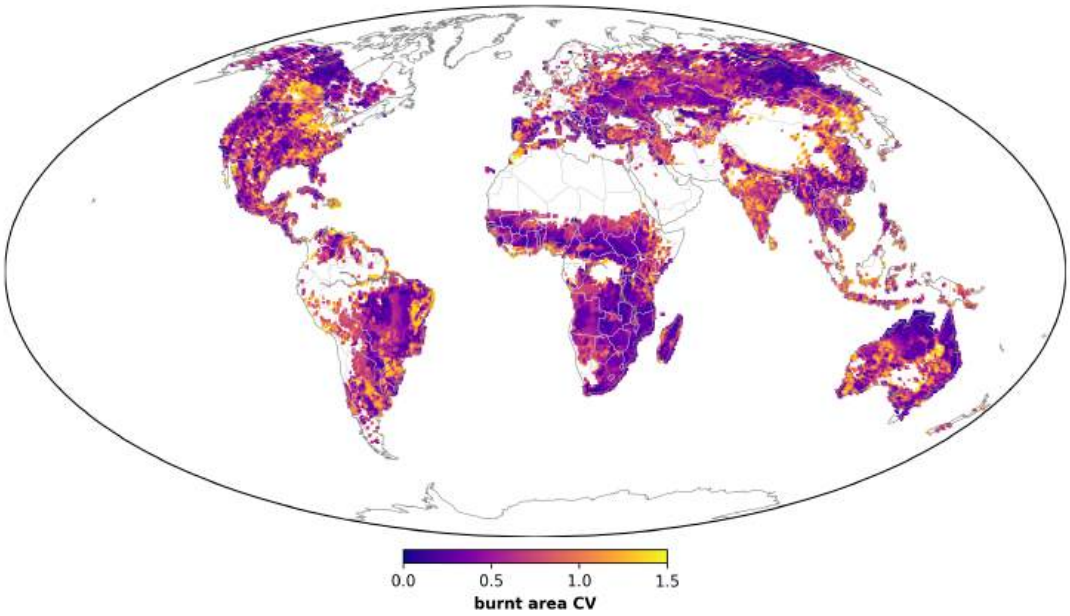


Figure 4.1. Coefficient of variation (CV) between mean annual burned area km^2 (2005–2011) for three global BA products, MODIS Collection 6 (MCD64), MODIS Collection 5 (MCD45) and ESA FireCCI (FCCI41).

Poulter et al. (2015) considered a sensitivity analysis of a dynamic global vegetation model to the driving satellite burned area product used. They indicated that the model displayed considerable sensitivities to divergences between the satellite products and greater uncertainty characterisation would also help to drive improvements in model development and benchmarking. Knorr et al. (2014) demonstrated several of these concerns when calibrating an empirical fire model relating human population density with fire occurrence. They found that the uncertainty in optimised model parameters was highest over regions with the lowest burned area. In such regions, the impact of the uncertainty in burned area on models can be substantial (Hantson et al., 2016). Similarly, Knorr et al. (2014) found that the main driver of differences in model predictions was the burned area observational dataset used.

A common thread across many satellite-model studies is that users frequently make use of advanced techniques to optimally combine uncertain observations with models within model-data fusion or data assimilation frameworks (Kaminski and Pierre-Philippe, 2017; Scholze et al., 2017; Rayner et al., 2016; Evensen, 2009; Tarantola and Valette, 1982). The use of such approaches allows for a full analysis of the sources of uncertainties in model estimates as well as the specification of requirements for future satellite missions to reduce uncertainties (Reichle, 2008). Presently however, it is difficult to fully utilise burned area estimates within these frameworks due to the lack of quantitative uncertainty

information provided in the products. This can be understood by considering the nature of these mathematical frameworks. For example, often an uncertainty weighted cost function between the model $\mathbf{H}$ and the observations of burned area $\vec{y}$ is solved as:

$$\mathrm{J}(\vec{x}) = \frac{1}{2}\left[\mathbf{H}(\vec{x}) - \vec{y}\right]^{\top} \mathbf{C}_{\mathrm{obs}}^{-1}\left[\mathbf{H}(\vec{x}) - \vec{y}\right] \tag{4.1}$$

where $\vec{x}$ is the parameters of interest. $\mathbf{C}_{\mathrm{obs}}$ is the covariance matrix for the observations of burned area which specifies the uncertainty in the observations. The definition of $\mathbf{C}_{\mathrm{obs}}$ is of paramount importance within equation 4.1 because it determines the influence each individual observation of burned area y has on the value of $\vec{x}$ which minimises J. Crucially, it also determines the uncertainty in $\vec{x}$ at the minimum of J, which provides information on the precision of the retrieval and therefore the strength of the evidence in any scientific inference based on particular values of $\vec{x}$.

4.3 Present uncertainty estimates

At present, burned area products provide no suitable uncertainty information for such a scheme to be truly consistent and users of these products have had to infer suitable uncertainties for $\mathbf{C}_{\mathrm{obs}}$. Le Page et al. (2015) for example, used relative uncertainties of 25–50% in burned area from GFED4. Previous estimates of product uncertainties have been largely driven by validation initiatives. In these analyses, product commission and omission errors have been computed in comparison to reference datasets, which are typically generated by the manual or semi-automated mapping of area burned from higher resolution images. The extents of these validation exercises range from regional comparisons against a few selected sites to larger global validation designs (Roy and Boschetti, 2009; Boschetti et al., 2016; Padilla et al., 2017, 2015). The derived validation statistics are then interpreted as providing estimates of the errors of the product in light of these commission/omission statistics. The clearest example of this method is the estimate of burned area standard errors σ_A provided in the Global Fire Emissions Database (GFED) 4 product (Giglio et al., 2010):

$$\sigma_A^2 = c_B A \tag{4.2}$$

where A is the total burned area. c_B serves as an uncertainty coefficient which scales the standard error based on an analysis of residuals against Landsat validated burned area. A natural concern that arises out of these approaches is the quality of the sampling provided by such validation datasets. Even larger and more systematic validation efforts may still provide only a limited sampling of the true uncertainties. For example, the large systematic sampling validations by Padilla et al. (2015) are derived from active fire observations, which display their own issues and uncertainties (Giglio

et al., 2006). Similarly, the challenge of generating sufficient validation data to enumerate the range of probable c_B values across the globe is considerable. The estimated uncertainties provided by GFED4 are derived from only three unique values for c_B (covering Siberia, Southern Africa and the western United States), and regions not sharing sufficient similarities with these are given a median value of c_B (Giglio et al., 2010).

An exception to this approach is provided by the FireCCI version 5 product (FireCCI50) which provides per-pixel estimates of uncertainty in the detection of burned areas (Chuvieco et al., 2018; Brennan et al., 2018). These uncertainties are computed by considering several features of the detection problem such as the number of observations available, and the magnitude of the reflectance change signal. These pixel-level uncertainties are then aggregated into the lower resolution FireCCI50 product to provide per 0.25° grid cell standard errors. The validity of these standard errors will be dependent upon the quality of the per-pixel uncertainty estimates (in terms of modelling the true uncertainty) and the aggregation process from pixel to coarser grid cell scales (Bellprat et al., 2017; Brennan et al., 2018).

In the absence of product provided uncertainty estimates, others have also derived estimates of uncertainties. Le Page et al. (2015) proposed relative uncertainties of 25–50% in burned area as provided by GFED4 based on an inspection of the GFED data. Most frequently the range in burned area reported by different products has been used to provide upper and lower bounds on global burned area (Rabin et al., 2017; Forkel et al., 2019; Poulter et al., 2015; Knorr et al., 2012). The large uncertainty in global burned area implied by this figure contributes considerably to emissions uncertainties (Knorr et al., 2012). It also introduces additional problems into the evaluation of the performance of fire models against satellite-derived observations (Rabin et al., 2017).

4.4 Sources of uncertainty in burned area mapping

To address this task first requires an understanding of the mapping of burned area from Earth observation data as the collection of a ‘measurement’. The production of global records of burned area involves the processing of considerable volumes of coarse-resolution satellite observations. Burned area products lie at the top of a measurement process involving the transformation of initial satellite measurements to the higher-level burned area inferences (Merchant et al., 2017). Figure 4.2 depicts a schematic of a burned area retrieval system. As can be seen there are many individual processes along the chain to produce the final inference. Uncertainties enter this measurement process at all levels. The initial satellite measurements are not error-free and these uncertainties are thus propagated

through the estimation of the land surface state X (e.g. surface reflectance). In addition, a subtlety of remote sensing classification problems such as burned area detection is the they naturally involve an uncertain inference on the state of the land surface X. The delineation of the surface as fire affected B is only observable *through* the properties of the land surface state: the conditional inference $B \mid X$. This conditional inference is the primary challenge within burned area mapping algorithms.

4.4.1 Measurement errors

Merchant et al. (2017) highlight that the generation of level 2/3 EO products involves a set of transformations from the initial level 0 (L0) measurement. Each of these transformations contributes additional uncertainties into the measurement. At the L0 level, errors enter the measurement through random fluctuations in the properties of the sensor. The deterioration of the sensor over time may alter the properties of this noise. As a result, sensors are calibrated through time to remove systematic errors. This most frequently involves vicarious calibration in which the sensor is calibrated based on observing target reference sites which have a stable surface reflectance through time e.g. deserts (Smith et al., 2002). Some sensors also carry calibration sources which allow for a continual on-board calibration and measurement of sensor degradation (Guenther et al., 1996). Burned area mapping algorithms typically make use of derived level 2 (L2) surface reflectance products. These products themselves are derived geophysical variables which involve the application of a retrieval algorithm, which contributes additional uncertainties. For example, the gridding process and atmospheric retrieval both contribute uncertainties to reflectance measurement (Vermote et al., 2002). As a result of these transformations, measurements of surface reflectance are corrupted by random and systematic errors.

4.4.2 Observation sampling

The collection of measurements by satellite-borne instruments naturally involves a sampling process. The sampling provided by the sensor has several features which introduce additional uncertainty into the problem. Each individual detector of the instrument samples an area of the land surface defined by its point spread function (PSF). The recorded value is the result of integration across the PSF. These measurements are then mapped into a pixel grid which is consistent between different acquisitions to produce level-2 products. This process typically introduces additional uncertainties due to decisions in the gridding process as well as geolocation errors in the level-1 products (Wolfe et al., 1998; Tan et al., 2006). As the sensor PSF is typically larger than the resolution of the gridding pixel, a significant proportion of the recorded signal originates from outside of the pixel — which effectively 'blurs' the

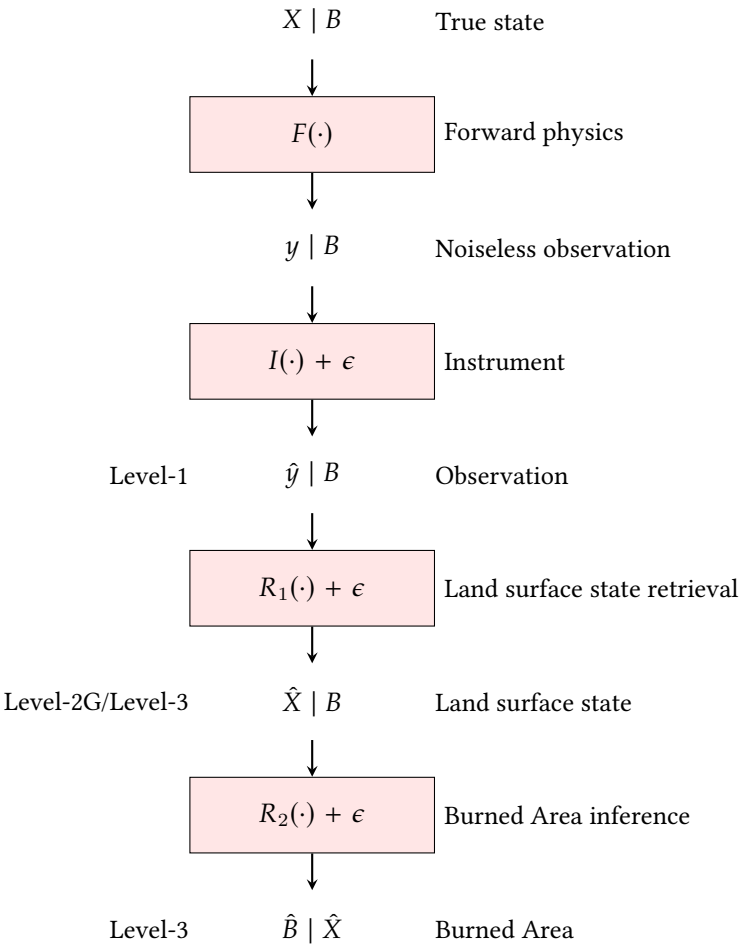


Figure 4.2. Abstraction of the burned area retrieval problem — schematic adapted from Turmon and Braverman (2019). The presence or absence of fire determines the properties of the true land surface $X \mid B$ (e.g. surface reflectance, burned vegetation, presence of ash and char, exposed soil, reduced leaf area). The true value of a measurand $y \mid B$ (e.g. radiance) is measured at the top of the atmosphere by a satellite instrument $I(\cdot)$ to produce a noisy measurement $\hat{y} \mid B$. These initial measurements are transformed into estimates of the land surface state or geophysical product (e.g. surface reflectance) $\hat{X} \mid B$ by a retrieval algorithm R_1 (e.g. an atmospheric correction algorithm). Burned area algorithms R_2 then typically use these products $\hat{X}$ to determine whether a fire occurred via a conditional inference $\hat{B} \mid \hat{X}$.

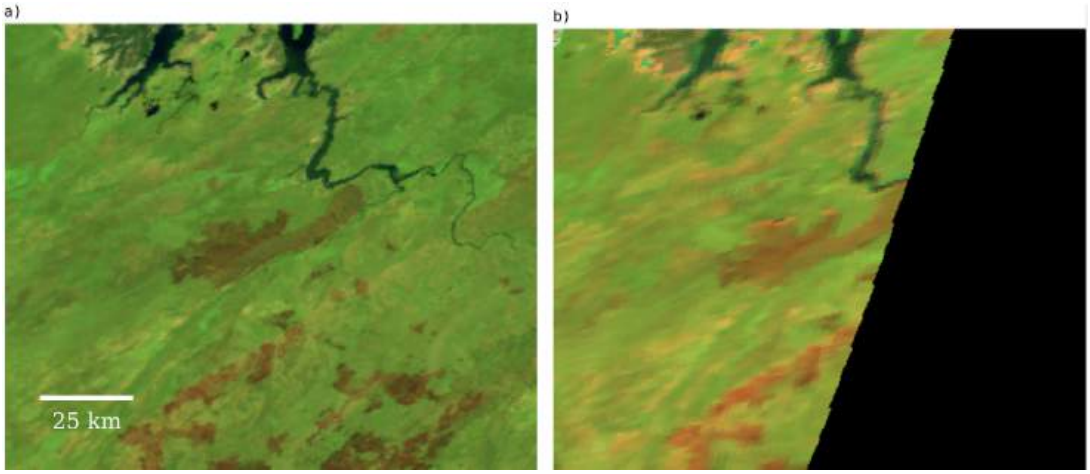


Figure 4.3. Impact of field of view (FOV) on spatial resolution of imagery. MODIS false colour composites (2130nm, 858nm, 555nm) over a burn scar in northern Australia, 24th April 2008. a) Collected by MODIS Terra b) collected by MODIS Aqua. The large FOV of MODIS alters the spatial area sampled towards the edge of the swath (black area in b).

observed signal between adjacent pixels. Huang et al. (2002) indicated that the blurring due to the sensor PSF reduced the accuracy of land cover classifications by around 5%. The combined sampling of the sensor detectors define a field of view (FOV) of the land surface that the sensor observes at a given moment. The coarse-resolution instruments used for global burned area mapping have large FOVs (and correspondingly large swath widths). These instruments, however, show an increasing spatial "footprint" of the pixel (a larger PSF) towards the edge of the swath (see figure 4.3).

Instruments obtain measurements of an area of the land surface intermittently in time and from different acquisition geometries of the Sun and sensor. This variation in the angular geometry of subsequent observations also provides information on the scattering anisotropy of the surface. This anisotropy is described by the bidirectional reflectance distribution function (BRDF). As a result, wide-swath instruments have provided invaluable global observations of land surface albedo (Schaaf et al., 2002). However, the directionality of the reflectance measurements also complicates the detection of surface changes in the land surface related to fire (Roy et al., 2005b). Zhang et al. (2003) found that changes in the viewing geometries between pre- and post-fire reflectance resulted in enhanced difficulty of identifying burned areas in boreal forests. Similarly, Stroppiana et al. (2002) considered the implications of BRDF on burned area identification and a derived simple correction scheme. To reduce the directionality of the reflectance measurements, vegetation indices (VIs) have been used across many burned area mapping algorithms. However, it should be appreciated that VIs do not fully remove reflectance directionality (Meyer et al., 1995).

The temporal sampling of the land surface is another key feature in the ability to resolve burned areas. Most significant for burned area mapping is the relationship between observation opportunity and the persistence of the burn signal on the land surface. This persistence is determined by the characteristics of the post-fire recovery of vegetation, as well as the dissipation of ash and char from the burn site. In boreal forests, an observable burn signal may last many years, while savannas typically register a persistent signal for a few weeks, and the subsequent ploughing of croplands may remove evidence for burning within a week (Sukhinin et al., 2004; Trigg and Flasse, 2000; Hall et al., 2016). The timely observation of the land surface pre- and post-fire then serves as a key determinant on the successful detection of burned areas. Melchiorre and Boschetti (2018) indicated that the median global persistence of an observable burn signal to be 29 days, and that within 48 days 87% of global burned area is undetectable. This indicates that as the temporal sampling provided by the instrument reduces, it can be expected that the burn signal may be missed. This results from a smaller spectral distinction between burned and unburned pixels in both time and space (see figure 4.4).

4.4.3 Algorithm uncertainties

As chapter 3 indicated, the decision structures of burned area mapping algorithms play a considerable role in the error properties of mapped products. Mapping algorithms necessarily require auxiliary information that is used to produce the retrieval. Most typically, these involve thresholds which are applied to assist the discrimination between burned and unburned surfaces. Other parameters are also used to set restrictions on the compositing of observations, constraints on fire spatial processes, and the utility of active fire observations for the identification of definitely burned pixels (Roy et al., 2005b; Alonso-Canas and Chuvieco, 2015; Giglio et al., 2008). The values of these parameters are often determined *ad hoc* or empirically during the development of the algorithm. In such cases, algorithm developers prototype algorithms over selected sites with known burned areas from additional reference information — for example higher quality burned area maps from high-resolution sensors. Parameters are then adjusted to improve the classification accuracy until the parameters provide the best possible match to the known burned area. The value of these parameters are then set globally or regionally based on assumptions of similar fire characteristics to these calibration datasets. The suitability of such parameter values for other regions and fire regimes relies therefore on the representativeness of the calibration dataset. Nevertheless, there is evidence to suggest that such an approach does not lead to globally optimal estimates, with present products showing regional disparities in accuracy (Padilla et al., 2015). Alternatively, parameters are locally calibrated to the fire characteristics of the region. This is typically achieved by learning the local fire characteristics from active fire observations, in

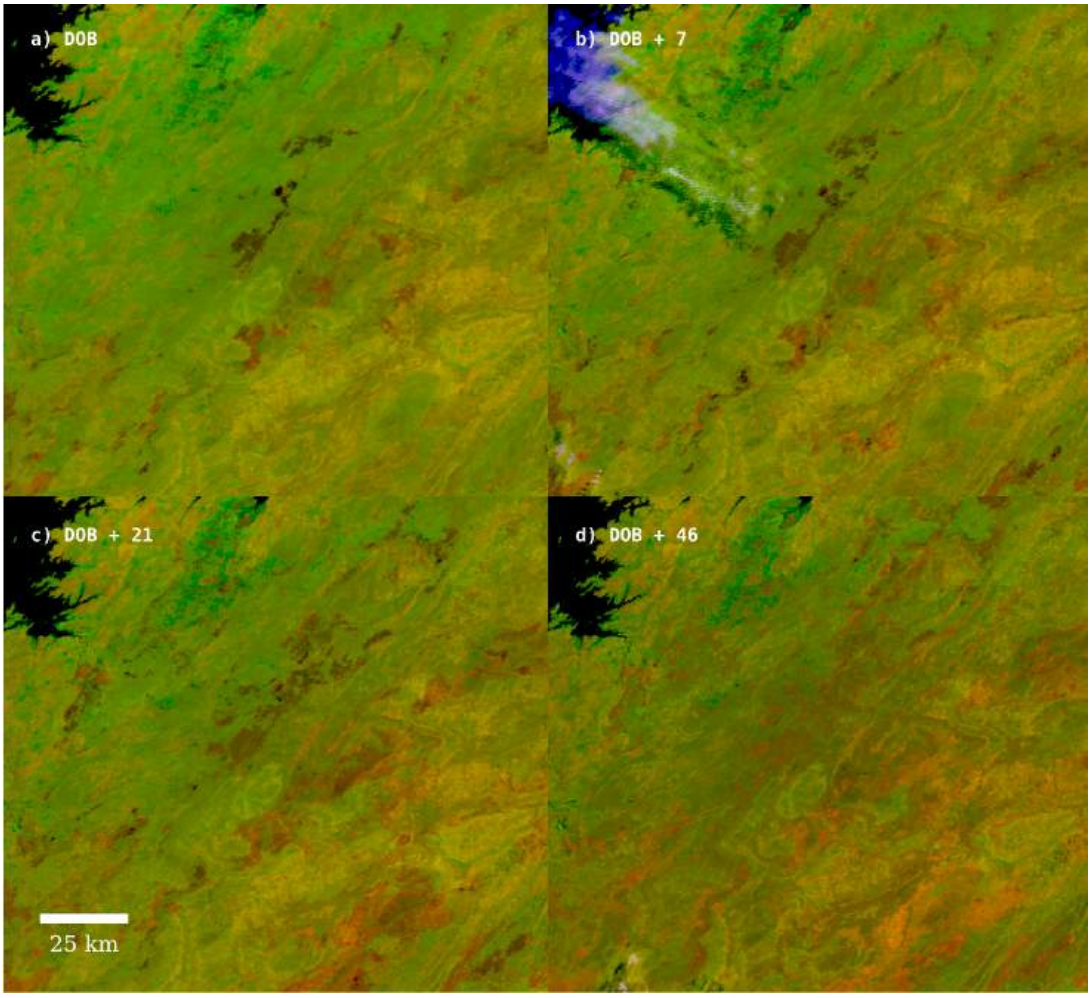


Figure 4.4. Temporal persistence of the burn scar. MODIS false colour composites (2130nm, 858nm, 555nm) over a burn scar in northern Australia, for four dates. The burn scar is located in the centre of the image. The ability to discern the burn scar reduces in the weeks after the fire.

so-called “hybrid” algorithms. Most typically, active fire products are used to learn the distributions of burned and unburned surfaces and therefore provide a locally calibrated classifier (Alonso-Canas and Chuvieco, 2015; Giglio et al., 2008; Fraser et al., 2000). These hybrid algorithms have an advantage therefore in that the local calibration of the algorithm should improve the accuracy of retrievals. However, the quality of this localisation is determined by the sampling provided by active fire products, which for products derived from polar-orbiting instruments varies substantially with latitude (Giglio et al., 2006).

Given these sources of uncertainties, a renewed focus is required on generating uncertainty quantified burned area estimates. Recent examples of best practice algorithm uncertainty quantification can be found in the terrestrial Earth observation community. Kaminski et al. (2012) developed a complete uncertainty traceable algorithm for the retrieval of Leaf Area Index (LAI) and Fraction of Absorbed Photosynthetically Active Radiation (FAPAR). This required the formulation of the retrieval algorithm within a Bayesian framework in which LAI and FAPAR are retrieved according to uncertainties in the observations and the model parameters. The mathematical frameworks necessary to produce such retrievals are now well developed across several remote sensing fields (Rayner et al., 2016; Kaminski and Pierre-Philippe, 2017). But two challenges unique to burned area mapping are the discrete nature of the retrieved parameter, and the lack of a rigorous mathematical formulation of the “burned area retrieval problem” up to present. These challenges require a unique solution for generating uncertainty characterised burned area datasets.

4.5 Basic principles of uncertainty characterisation

The purpose of uncertainty characterisation is to characterise and quantify how the properties described above determine the uncertainties in the retrieval. The metrological community have established a set of best practice for uncertainty characterisations as described in the *Guide to expression of uncertainty in measurement* (GUM, ISO/BPIM, 2008). The definitions of uncertainty specified in the GUM have been recognised within remote sensing and adopted by the European Space Agency’s (ESA) Climate Change Initiative (Hollmann et al., 2013). The GUM has also been adopted within the CEOS LPV group as the basis for the validation of burned area products (CEOS-LPV, 2019). Table 4.3 provides a summary of the uncertainty definitions provided by the GUM relevant to remote sensing retrievals. The GUM categorises uncertainty characterisation methods into two primary types:

Type A methods of uncertainty evaluation by the statistical analysis of a series of repeated observations.

Type B methods of uncertainty evaluation other than the statistical analysis of a series of observations (e.g. prior knowledge, scientific judgement, propagation of uncertainties).

The interpretation of this classification of uncertainty methods is generally two approaches. Type A evaluations involve estimating the uncertainty from a set of repeated measurements y_i ($i = 1 \ldots n$) of the same measurand Y. In this case, the standard uncertainty $u(y)$ is the experimental standard deviation:

$$u(y) = \sqrt{\frac{1}{n-1}\sum_{i}^{n}(y_i - \bar{y})^2} \tag{4.3}$$

where $\bar{y}$ is the arithmetic mean of the set of n measurements. In contrast with Type A evaluations, Type B evaluations involve characterising uncertainty when there are no repeated measurements. Instead, $u(y)$ is characterised from scientific judgement and all available sources of information in a best practice manner.

Given the adoption by the CCI programme and the CEOS LPV group of the GUM, this section proposes methods in accordance with the GUM to produce uncertainty quantified burned area products. This section therefore provides progress towards producing burned area products which will meet the requirements of CEOS level 3 & 4 validation (CEOS-LPV, 2019). Chapter 5 presents an empirical method to produce uncertainties for burned area estimates from three global products. The method is a 'top-down' approach in that uncertainties are estimated from the collocation of three burned area products using a method that follows the Type A approach (e.g. equation4.3). The method provides globally unique uncertainties for the three products but only at coarse spatial scales. Chapter 6 then develops a best practice framework for producing uncertainty characterised burned area algorithms from the initial satellite observations. This involves defining methods for propagating uncertainties through burned area algorithms and defining suitable uncertainty definitions for binary satellite data products. This framework provides a suitable basis for the generation of future burned area products which meet requirements for CDRs. The approach also has the advantage that uncertainties are verifiable in the sense of being traceable through a mapping algorithm to the pixel scale, but also provides additional challenges over empirical methods.

Table 4.3. Language of GUM relevant to the definition of uncertainties in remote sensing (adapted from Datla et al., 2010).

Quantity	**Symbols and Definitions**
Measurand	Y, quantity to be measured
Measurement	y, best estimate of the expected value of Y
Standard uncertainty	$u(y)$, uncertainty of the measurement y, expressed as a standard deviation.
Expanded uncertainty	$U = k \cdot u(y)$, defines the confidence interval of the measurement $[y \pm k \cdot u(y)]$. k is a coverage factor which determines the fraction of the distribution represented, e.g. for $k = 1$ the coverage is 65% and for $k = 2$ is 95% (assuming the normal distribution).
Type of uncertainty evaluation	Type A (statistical) and Type B (non-statistical) of uncertainty characterisation.

Chapter Five

Theoretical uncertainties for global satellite-derived burned area estimates

This chapter addresses the requirement for uncertainties on global satellite-derived burned area by estimating the uncertainties of three widely-used burned area products. This chapter forms the basis of the research published in Brennan et al. (2019). Section 5.1 describes the theoretical uncertainty estimation procedure used here. Section 5.2 then presents the results of the uncertainty model and compares the uncertainty estimates against two other available estimates of burned area uncertainties. The assumptions of the error model are considered in section 5.3 and section 5.4 discusses potential mechanisms for the reported uncertainties.

5.1 Uncertainty estimation procedure

5.1.1 Burned area datasets

The present chapter estimates theoretical uncertainties for three global burned area products. The Moderate Resolution Imaging Spectroradiometer (MODIS) Collection 6 burned area product (MCD64C6) provides a global record of burned area for the MODIS period (i.e. 2000–present). The algorithm uses active fire observations to refine a classifier based on the application of a temporal change spectral index derived from MODIS short-wave infrared channels 5 (1230–1250nm) and 7 (2105–2155nm) (Giglio et al., 2018).

The MODIS Collection 5.1 burned area product (MCD45C5.1) was produced with a different algorithm and provides a global record of burned area for a reduced period covering 2000–2016. The now deprecated MCD45C5.1 product uses a multi-temporal modelling algorithm which flags for

changes in the land surface based on differences between predicted and observed reflectance. The algorithm then filters changes to those that match the expected reflectance characteristics of burned surfaces in the near-infrared (841–876nm) and short-wave infrared (1230–1250nm). The algorithm does not utilise active fire observations (Roy et al., 2005a).

The ESA Climate Change Initiative Fire product (FireCCI50) provides global burned area for 2001-2016. The algorithm uses changes in MODIS near infrared (841–876nm) surface reflectance with a classifier that like MCD64C6, is locally trained with active fire observations from the MODIS sensors (Chuvieco et al., 2018). The product is novel in that it provides burned area at a spatial resolution of 250m compared to the 500m spatial resolution of the other two products. This limits the algorithm to using the MODIS red and near-infrared spectral channels.

5.1.2 Computation of uncertainties

Stoffelen (1998) first proposed triple collocation (TC) as a method to estimate uncertainties in three collocated data products. The method has now been used across a considerable range of remote sensing derived geophysical variables, including soil moisture, precipitation, leaf area index and fraction of photosynthetically absorbed radiation (Gruber et al., 2016; Roebeling et al., 2012; Fang et al., 2012; D'Odorico et al., 2014). Consider three observational records X_1, X_2, X_3 of a variable with an unknown but true value T. The TC error model specifies that each observational record may be related to the truth via a linear measurement equation:

$$X_1 = \alpha_1 + \beta_1 T + \epsilon_1 \tag{5.1}$$

$$X_2 = \alpha_2 + \beta_2 T + \epsilon_2 \tag{5.2}$$

$$X_3 = \alpha_3 + \beta_3 T + \epsilon_3 \tag{5.3}$$

where α and β represent additive and multiplicative biases respectively. ϵ denotes the residual (random) errors of the relation.

As posited, the three measurement equations provide a linear system that is under-determined. However by making three assumptions, the system can be solved to provide estimates of the random errors of each product. First, each product is assumed to have zero mean residual errors ($E[\epsilon] = 0$). Second, the errors of each product are assumed to be uncorrelated (but not necessarily independent) with each other. Finally, the random errors ϵ are assumed to be uncorrelated with the true value T, as systematic errors are incorporated into β. The last assumption is not met for geophysical variables which show random errors that are functionally related to the magnitude of the signal (Tian et al., 2013). Figure 5.1 shows mean annual burned area of the three products (resampled to a shared 1°

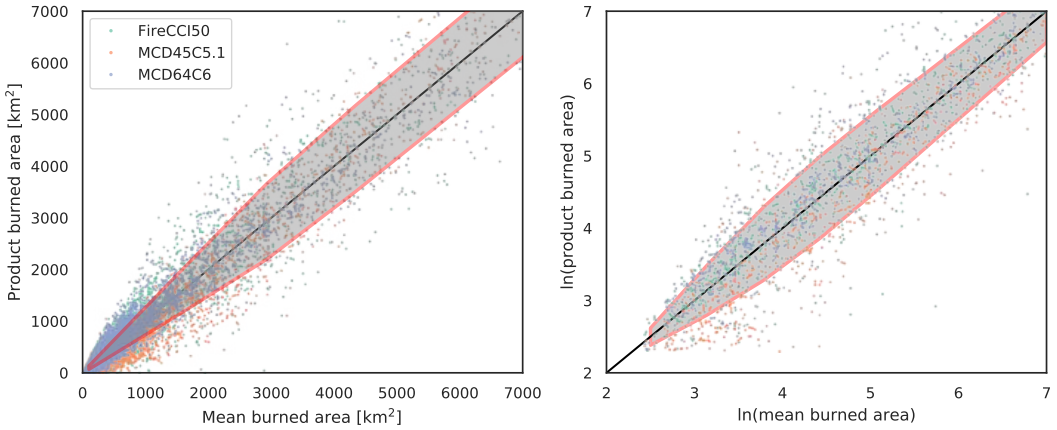


Figure 5.1. Differences between the burned area reported by the three products and the mean of the three products. Also shown is the standard deviation of the products (grey) binned by the mean burned area of the three products. Each point corresponds to monthly burned area reported by a product in a 1° grid cell for 2001–2014. Increasing standard deviation with the magnitude of burned area implies heteroscedastic errors , while log-transformed burned area has errors which are more homoscedastic.

grid) against individual product estimates for 2001–2014. The shaded area represents the standard deviation between the products binned by the mean of the three products. It can be observed that the divergences between the products grow with the magnitude of burned area reported. This indicates that the constraint imposed on the burned area becomes more uncertain with the magnitude of burned area detected. This occurs because the random errors in burned area are *heteroscedastic* (Giglio et al., 2006). The TC model in equation 2-4 assumes however that the random errors ϵ are homoscedastic — in that the error variance model $\epsilon = \mathcal{N}(0, \sigma^2)$ is not a function of the true (unobserved) burned area T. This feature of the errors is common to several other geophysical variables (e.g. precipitation, above-ground biomass) (Tian et al., 2013; Alemohammad et al., 2015; Gonzalez de Tanago et al., 2018).

In log-space however the differences between products do not increase with logarithmic burned area and are closer to being homoscedastic. Alemohammad et al. (2015) proposed that for heteroscedastic datasets, an alternative TC error model is suitable in which the random error is a multiplicative signal on the truth T. Instead, the error model for X can be defined as:

$$X = \alpha T^{\beta} e^{\epsilon} \tag{5.4}$$

where α is a multiplicative error, β is the deformation error and e^{ϵ} is the residual (random error). Taking the natural logarithm of equation 5.4 leads to an additive measurement model:

$$ln(X) = \alpha + \beta ln(T) + \epsilon \tag{5.5}$$

with the assumption that in the log-space, the random errors are normally distributed $\epsilon = \mathcal{N}(0, \sigma^2)$. Representing $x = ln(X)$ and $t = ln(T)$, equation 5.5 is equivalent to:

$$x = \alpha + \beta t + \epsilon \tag{5.6}$$

which provides a linear system equivalent to equation 5.1. Given the same assumptions of the classical TC method, the residual error estimates of each product (in log-space) can be derived from the following manipulations of the sample covariance matrix of the three log-transformed products $\mathbf{C}$ (McColl et al., 2014):

$$\sigma_1^2 = \mathbf{C}_{11} - \frac{\mathbf{C}_{12}\mathbf{C}_{13}}{\mathbf{C}_{23}} \tag{5.7}$$

$$\sigma_2^2 = \mathbf{C}_{22} - \frac{\mathbf{C}_{12}\mathbf{C}_{23}}{\mathbf{C}_{13}} \tag{5.8}$$

$$\sigma_3^2 = \mathbf{C}_{33} - \frac{\mathbf{C}_{13}\mathbf{C}_{23}}{\mathbf{C}_{12}} \tag{5.9}$$

The TC method requires that the three datasets explicitly cover the same temporal and spatial domain and are of the same variable (Yilmaz and Crow, 2014). To achieve this, the three burned area datasets were aggregated to a shared temporal and spatial grid. The three products were aggregated from the original pixel resolution products to a common lower 1° resolution sinusoidal grid g. For each 16-day period between January 2001—December 2014, the burned area reported by each product within the cell $g(t, x, y)$ was aggregated to form a full temporal record for each cell through time of length N_t. The temporal span of the datasets provided potentially $N_t = 286$ observations. A feature of solving the multiplicative error model in log-space is that any product that reports no burned area will prevent the estimation of the covariance matrix $\mathbf{C}$. As a result, any 16-day period where at least one product reported no reported burned area was excluded. This meant that approximately 40% of cells globally had no agreed burned area between the products, and therefore do not have uncertainties. Nevertheless, the major fire regions are well sampled across the record (see figure 5.2). The TC method is able to sample the majority of the reported fire activity by the products. The 40% of cells without uncertainties contain less than 0.5% of the total burned area (2001–2014) reported by each product.

5.1.2.1 Annualised uncertainties

Annualised uncertainties on the total burned area are also of particular interest to the users of burned area products. First it is necessary to transform the random errors in log-space ϵ to the real-space to produce 16-day uncertainties in burned area. The random errors back-transformed into burned area are defined by a log-normal distribution specified by Log-normal($\mu = 0, \sigma^2$). Therefore the

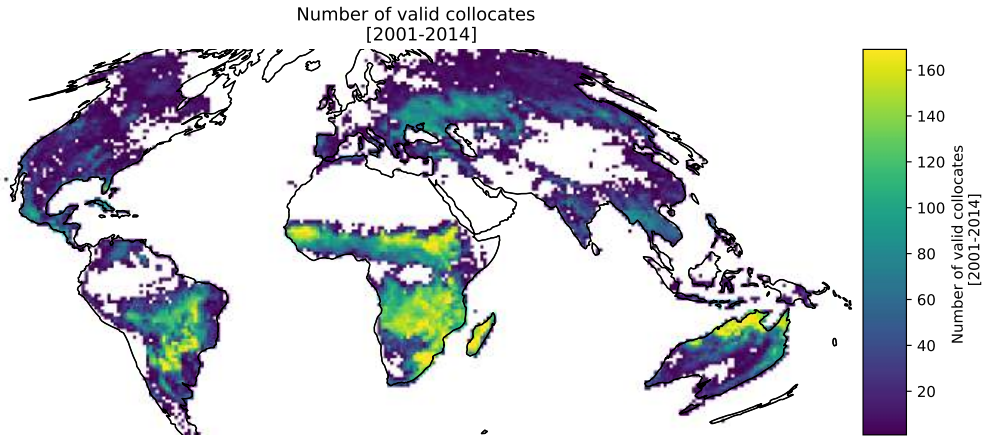


Figure 5.2. Number of valid collocates for 2001–2014.

distribution of 16-day burned area $P(X)$ can be defined in reference to equation 5.4 as:

$$P(X) = X_o e^{\mu+\sigma Z}, \tag{5.10}$$

where X_o is the observed burned area for the product and Z the standard normal distribution. To produce an annualised uncertainties, each 16-day burned area distribution $P(X)$ was sampled from and integrated over the year to provide a distribution of annual burned area for each grid cell. The independence assumption of individual observation errors in this scheme is also a necessary requirement of the TC method (Gruber et al., 2016). To summarise the annual distribution, it was then approximated as a normal distribution based on matching the moments of the samples. Figure 5.3 shows an example of the procedure for producing 16-day and annualised uncertainties for an area covering Northern Australia. Large absolute uncertainties are associated with the peak in the burning season here.

Given the regional variability in absolute burned area, the relative magnitude of the annual uncertainties to the reported burned area of each product was also considered. The relative uncertainty in mean annual burned area is defined by:

$$\text{relative uncertainty } [\%] = 100 \times \frac{\sigma_{\text{year}}}{\text{BA}_{\text{year}}} \tag{5.11}$$

where BA_{year} is the total burned area reported by the product for the grid cell.

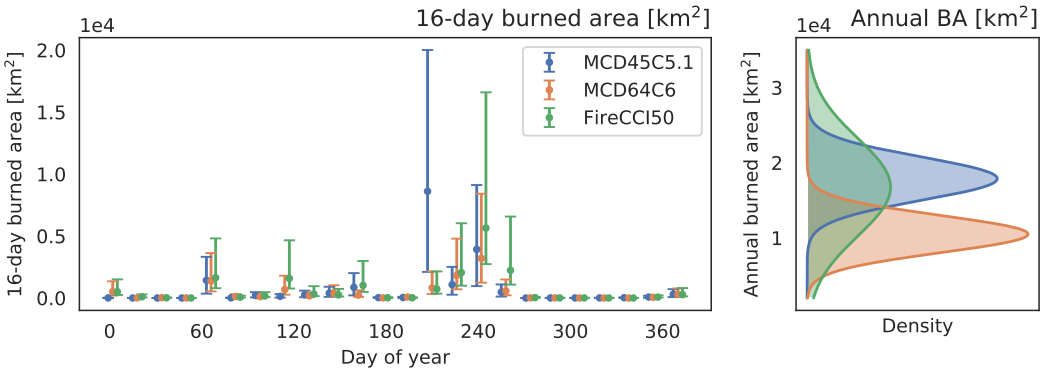


Figure 5.3. 16-day and annual uncertainties for a grid cell covering Northern Australia. Left) The multiplicative error model provides unique uncertainties on each 16-day observation for each product (95% confidence intervals shown). Right) To produce an annual uncertainty on the reported burned area, these are aggregated to produce an annual distribution which is then approximated as a normal distribution.

5.1.2.2 Regional and global uncertainties

Given that it may be expected that the performance of each product will vary with the local fire-vegetation interaction, the uncertainties were considered with regard to land cover provided in the MODIS Collection 6 land cover product (MCD12Q1.006) (Friedl et al., 2010). The University of Maryland (UMD) land cover classification was simplified into five more primary categories of 1) forest including all forest types, 2) croplands, 3) shrublands including both open and closed shrublands, 3) savannas (including woody savannas), and 5) grasslands. The simplified land cover product was then aggregated to the sinusoidal 1° resolution grid by considering the dominant land cover type in each cell. Product errors were also quantified for the 14 fire regions specified by GFED which have been previously used for regional comparisons of burned area products (Giglio et al., 2013b)

A complicating feature of the aggregation to regional scales is that the spatial correlation of the uncertainties at the grid cell level is unknown. It would generally be expected that the uncertainties in adjacent grid cells may be similar, due to correlations in the error characteristics controlled by e.g. land cover, cloud statistics and algorithmic limitations. The integration of grid cell level uncertainties via an independent quadrature summation would imply a strong constraint on there being no spatial correlation in the uncertainties (Bellprat et al., 2017). Instead, to produce the regional estimates, 16-day burned area for each product is aggregated for the whole region or land cover stratification and the TC error model is then solved. This allows for the effective spatial error correlation in the products to

be present in the regional uncertainties while requiring no additional assumptions about the error structure.

5.2 Results

Figure 5.4 displays global maps of the residual errors ϵ (in log-space) for each product. Spatial patterns are different from the spatial distribution of burning, indicating that systematic errors are not leaking into the random errors. The largest random errors for each product are located in Eastern China corresponding to regions of agricultural fires. Here errors are greater than 1 for all products, which indicates a random error of greater than 100% in the detected burned area. This would indicate that the level of agreement between the products is lower than the precision of the products.

Local patterns of the errors then diverge for each product. MCD45 has larger random errors in central and eastern Europe, in regions with predominantly agricultural fires. The lowest uncertainties are found in savanna ecosystems of southern hemisphere Africa and northern Australia. MCD64 shows the largest uncertainties in agricultural and tundra regions of eastern Eurasia. It also has the largest uncertainties in Western Africa, in areas where deforestation fires are common. MCD64 has larger uncertainties in savannas relative to MCD45 and lower random errors in areas with agricultural burning. FireCCI has smaller errors in agricultural regions of eastern Eurasia compared to the other two products. FireCCI also has smaller random errors in regions of agricultural burning and deforestation areas around the Amazon compared to MCD45 and MCD64.

Figure 5.5 displays global maps of mean annual burned area and associated uncertainties for the three products. Between the products, similar spatial distributions in burned area and TC-estimated uncertainties can be observed. The heteroscedastic nature of burned area uncertainties is apparent with standard uncertainties scaling with the magnitude of burned area. Absolute uncertainties for each product are largest in sub-Saharan Africa and northern Australia which corresponds to regions with the greatest burned area. Greater disagreement in the magnitude of burning occurs in regions with less frequent burning or typically compounding factors on detection. In equatorial Asia, MCD64 and FireCCI50 detect respectively 1310% and 940% more burned area than MCD45. Greater detection by MCD64 here has been associated with the use of active fires (Humber et al., 2018). These higher estimates are also better constrained with relative uncertainties of 35% and 36% respectively for FireCCI50 and MCD64; compared to a greater relative uncertainty for MCD45 (70%). FireCCI50 detects 66% more burned area in the agricultural burning regions of central and eastern Europe than MCD64 and 48% more than MCD45. However, the large uncertainties on these estimates indicate them to be

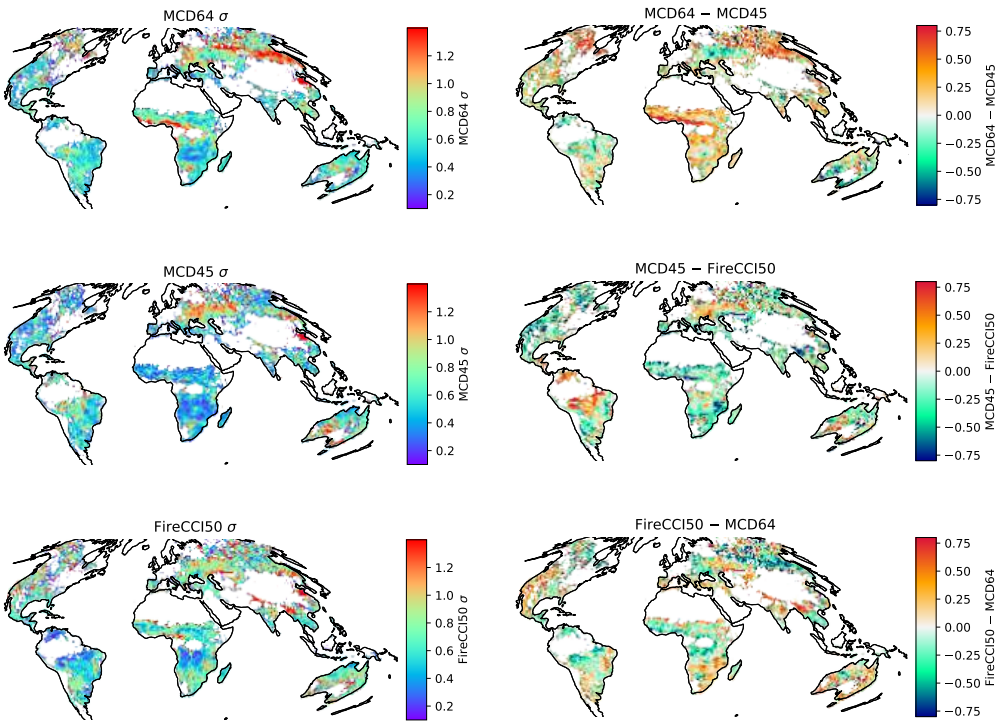


Figure 5.4. Left) TC random errors ϵ for the three products and right) differences between product random errors.

consistent within the uncertainties: with relative uncertainties exceeding 100% for FireCCI50 (141%), MCD45 (168%) and MCD64 (95%). Regions where MCD45 reports no burning prevents the estimation of TC-uncertainties due to the requirement of the multiplicative error model used here. This is most noticeable in equatorial Asia and South America.

Globally, MCD64 reports the greatest mean annual burned area $3.76 \pm 0.15 \times 10^6$ km^2. This is followed by FireCCI50 which reports $3.70 \pm 0.17 \times 10^6$ km^2 and MCD45 $3.31 \pm 0.18 \times 10^6$ km^2. In terms of relative uncertainties, MCD64 has the smallest relative uncertainty of 3.9%, FireCCI50 4.5% and MCD45 has the largest (5.5%). MCD64 and MCD45 provide consistent estimates of mean annual burned area for 76% of grid-cells with TC-estimated uncertainties. In these locations the confidence intervals specified by one standard error (one-sigma) provided by the TC method overlap. MCD64 and FireCCI50 agree across a slightly broader spatial extent, with 80% of available cells agreeing within the uncertainties of each product. MCD45 and FireCCI50 have the lowest agreement of the three products, with consistent estimates (overlapping one-sigma confidence intervals) across 72% of TC-cells. Figure 5.6, shows locations where all three products within one-sigma and two-sigma confidence intervals on

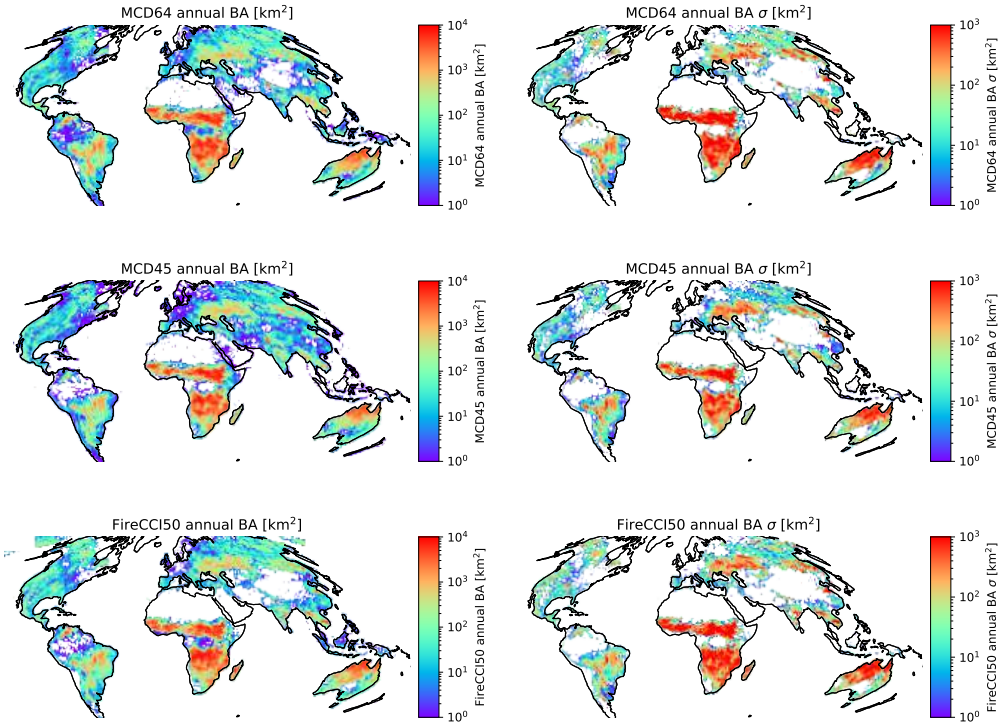


Figure 5.5. Left) Mean annual burned area km^2 and right) associated standard errors of mean annual burned area km^2 .

reported mean annual burned area. For a one-sigma confidence interval, all three products agree for 60% of the TC cells. For the broader two-sigma confidence interval, the three products agree across 85% of the valid cells. Regions where the products do not agree within a two-sigma confidence interval are concentrated in equatorial Asia, the northern Amazon region, the south-eastern United States, and parts of the Indian subcontinent.

Figure 5.7 shows a regional breakdown of mean annual burned area and uncertainties stratified by land cover. Globally, burned area estimates are most uncertain for cropland and shrublands for all products. All three products perform comparatively better in savannas and grasslands. For nearly all land covers, MCD45 has the largest relative uncertainties of the three products. It has the largest uncertainties in shrublands, with a relative uncertainty of 25%, followed by FireCCI50 (13%) and then MCD64 (8%). The uncertainty for the MCD45 product in shrublands is contributed to in large part by a poor constraint on burning in Australian (AUST) shrublands where the relative uncertainty exceeds 40% ($1.29 \pm 0.56 \times 10^5$ km^2), compared to 15% and 8% for the FireCCI50 and MCD64 products respectively. FireCCI50 uncertainties in shrublands are driven by large uncertainties on comparatively

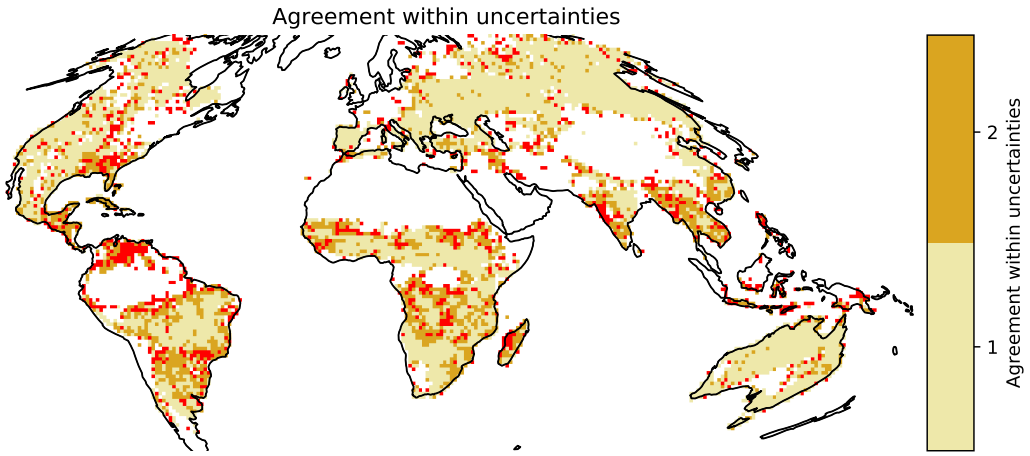


Figure 5.6. Consistency of mean annual burned area for the three products. Light brown regions correspond to regions where all three products have overlapping one-sigma confidence intervals as provided by the TC method. Brown regions correspond to overlapping two-sigma confidence intervals. Red regions report areas where the three products do not agree within their two-sigma confidence intervals.

small reported shrubland burned area in Central America (CEAM) 765 ± 1846 km^2 and Temperate North America (TENA) 1175 ± 2115 km^2. This contrasts with much smaller uncertainties on a similar reported burned area from MCD64 in Temperate North America (TENA) 1172 ± 449 km^2.

All products have a poor constraint on global cropland burning with relative uncertainties of 8–10%. MCD45 generally has the largest relative uncertainties on cropland burning across all fire regions, with 95% (two-sigma) confidence intervals larger than the magnitude of reported burned area for Europe (EURO), boreal Eurasia (BOAS), and equatorial Asia (EQAS). Exceptions are found in temperate North America (TENA) and southeast Asia (SEAS) where MCD45 reports the most cropland burning and also has the lowest relative uncertainties.

An interesting feature occurs in boreal forest ecosystems, where MCD45 and FireCCI have smaller uncertainties in boreal Eurasian (BOAS) forests compared to boreal North American (BONA) forests. Uncertainties for MCD45 are around two times larger in BONA forests, and 40% larger for FireCCI50 in BOAS as compared to BONA forests. Alternatively, MCD64 has lower relative uncertainties in BONA compared to BOAS, with uncertainties 70% larger in boreal Eurasia.

In the key burning regions of Northern Hemisphere (NHAF) and southern hemisphere Africa (SHAF), MCD45 typically has the most constrained estimates of burned area. The three products provide consistent estimates in grasslands and savannas in both regions, with reported area burned

being within the shared uncertainty envelope. The uncertainties are still considerable, however, with relative uncertainties for all three products exceeding 13% in both savannas and grasslands for NHAF and 8% in SHAF. This leads to broad standard errors on each product in NHAF, with reported mean annual burned area of $1.03 \pm 0.19 \times 10^6$ km^2 for MCD64, $1.07 \pm 0.13 \times 10^6$ km^2 for MCD45, and $0.99 \pm 0.27 \times 10^6$ km^2 for FireCCI50. Table 5.1 summarises the mean annual burned area and uncertainties by fire region.

5.2.1 Comparison against other uncertainty estimates

5.2.1.1 GFED4 uncertainties

The uncertainties from the TC method were then compared with two other available uncertainty estimates. First in relation to the MCD64 product, the uncertainties provided with the GFED4 burned area product were considered. The GFED4 burned area and uncertainty are derived exclusively from the MCD64 product for the period considered here (using the method previously discussed in chapter 4). GFED4 however utilised the older MCD64 Collection 5.1 product, which detects significantly less global burned area than the present Collection 6 product (Giglio et al., 2018). Nevertheless, in the absence of other uncertainty estimates, it is sensible to consider the relative uncertainties for the GFED4 product against the TC method. To align the uncertainties with those provided by the TC method, total annual burned area uncertainties were considered. To produce annual uncertainties for GFED4, the monthly variances provided by the GFED4 product were added in quadrature.

Figure 5.8 shows differences between mean annual relative uncertainties in GFED4 vs TC derived uncertainties. TC uncertainties generally exceed GFED uncertainties in most regions. The global median for TC uncertainties is 38% vs GFED 34%; however mean global GFED uncertainties exceed those provided by the TC method. Mean global GFED uncertainty is 65% compared to 45% provided by the TC method, though this figure is skewed by a greater range in the GFED uncertainties (GFED interquartile range (IQR): 15%–80% vs TC IQR: 26%–57%). Areas of higher TC uncertainties are found in the agricultural burning regions of northern China and eastern Russia, where TC uncertainties exceed GFED by 70–100%. TC uncertainties also exceed GFED uncertainties in western Africa (90%) and areas of North America, especially in boreal forest regions of eastern Canada. Conversely, GFED uncertainties exceed TC uncertainties in several regions. For example, GFED uncertainties are larger in boreal Eurasia (40–60%), eastern India (30–70%) and parts of South America (35–65%).

Two probable causes for differences between the two uncertainty estimates are conceived. Primarily, GFED4 is based on an older collection of the MCD64 product which detected globally around 26%

	Burned area [$\times 10^3$ km^2]			**Standard uncertainty** [$\times 10^3$ km^2]			**Relative uncertainty** [%]		
	FireCCI50	**MCD45**	**MCD64**	**FireCCI50**	**MCD45**	**MCD64**	**FireCCI50**	**MCD45**	**MCD64**
AUST	514.39	394.02	476.27	37.19	73.84	33.43	7.23	18.74	7.02
BOAS	94.47	64.70	86.26	53.96	20.87	66.34	57.12	32.26	76.90
BONA	22.37	14.63	20.61	11.97	9.85	13.57	53.53	67.33	65.87
CEAM	22.75	13.75	25.68	9.43	8.25	5.95	41.47	60.02	23.16
CEAS	209.51	190.49	194.89	53.58	42.11	30.49	25.58	22.10	15.64
EQAS	9.18	0.88	12.47	3.22	0.61	4.52	35.08	69.25	36.29
EURO	13.92	11.63	10.52	11.87	13.23	4.94	85.31	113.81	46.94
MIDE	10.01	16.22	12.55	5.70	9.57	0.95	57.00	59.03	7.56
NHAF	987.23	1077.43	1032.15	266.58	126.32	188.68	27.00	11.72	18.28
NHSA	51.68	10.50	45.14	7.27	12.66	10.83	14.07	120.65	23.99
SEAS	119.73	91.29	117.40	59.26	38.28	67.35	49.49	41.93	57.37
SHAF	1397.68	1227.17	1413.56	103.21	145.44	167.46	7.38	11.85	11.85
SHSA	215.15	169.19	279.61	5.44	48.83	42.77	2.53	28.86	15.29
TENA	31.68	25.77	24.42	4.05	4.45	3.39	12.78	17.26	13.90
WORLD	3701.72	3309.44	3755.80	165.55	183.38	146.05	4.47	5.54	3.89

Table 5.1. Mean annual burned area [$\times 10^3$ km^2], standard uncertainty [$\times 10^3$ km^2] and relative uncertainty [%] for the products by fire region.

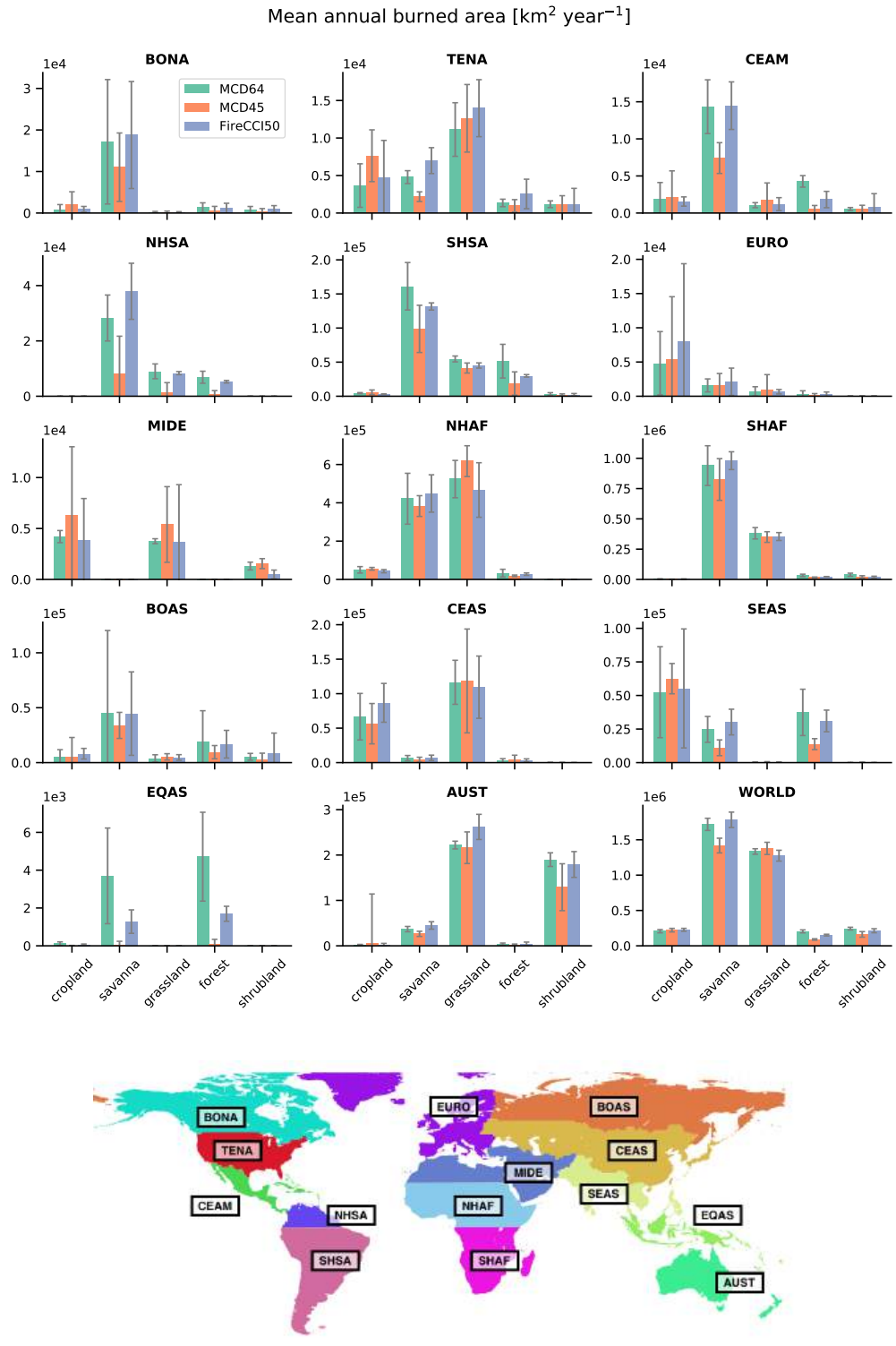


Figure 5.7. Mean annual burned area and uncertainties $\mathrm{km}^2/\mathrm{year}$ stratified by land cover for each of the fire regions. BONA) Boreal North America, TENA) Temperate North America, CEAM) Central America, NHSA) Northern Hemisphere South America, SHSA) Southern Hemisphere South America, EURO) Europe, MIDE) Middle East, NHAF) Northern Hemisphere Africa, SHAF) Southern Hemisphere Africa, BOAS) Boreal Asia, CEAS) Central Asia, SEAS) Southeast Asia, EQAS) Equatorial Asia, AUST) Australia & New Zealand.

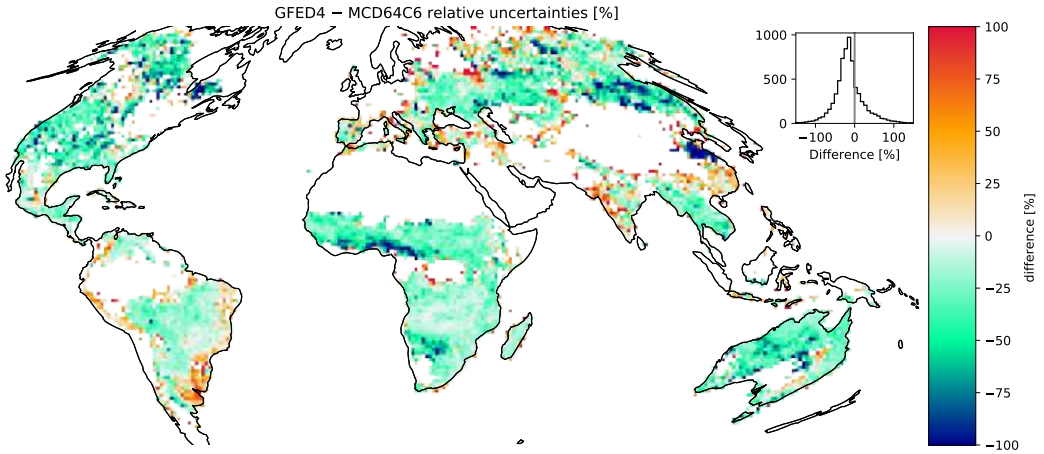


Figure 5.8. Differences in relative uncertainties between GFED4 and TC-estimated relative uncertainties. Blue corresponds to larger TC relative uncertainties and red to larger GFED4 uncertainties.

less burned area than the present Collection 6 product (Giglio et al., 2018). An equally important consideration is that the uncertainty assumptions of the two methods are different. For the GFED uncertainties, Giglio et al. (2010) indicated that these are likely to be conservative due to the potential cancelling of omission and commission errors in the total reported burned area, with the effect being that GFED uncertainties are also likely over-estimated for the MCD64 Collection 5.1 product. Whereas the TC method accounts for any potential cancelling of errors by focusing on the observed burned area irrespective of the error source.

5.2.1.2 FireCCI50 product uncertainties

The FireCCI50 Climate Model Grid (CMG) product also provides standard errors per grid cell at the coarse spatial resolutions considered here. These are produced from an aggregation of individual uncertainties in the 250m pixel product to produce fortnightly standard errors in burned area. In the same manner as with the GFED4 uncertainties, annual uncertainties from the FireCCI50 product were produced by adding the uncertainties in quadrature for each fortnightly product.

The uncertainties provided with the FireCCI50 product represent the first attempt to provide a full uncertainty traceability chain for burned area datasets (Chuvieco et al., 2018; Brennan et al., 2018). The reported uncertainties are considerably smaller than those provided by the TC error model as well as the uncertainty estimates provided by GFED4. Figure 5.9 shows a comparison of relative

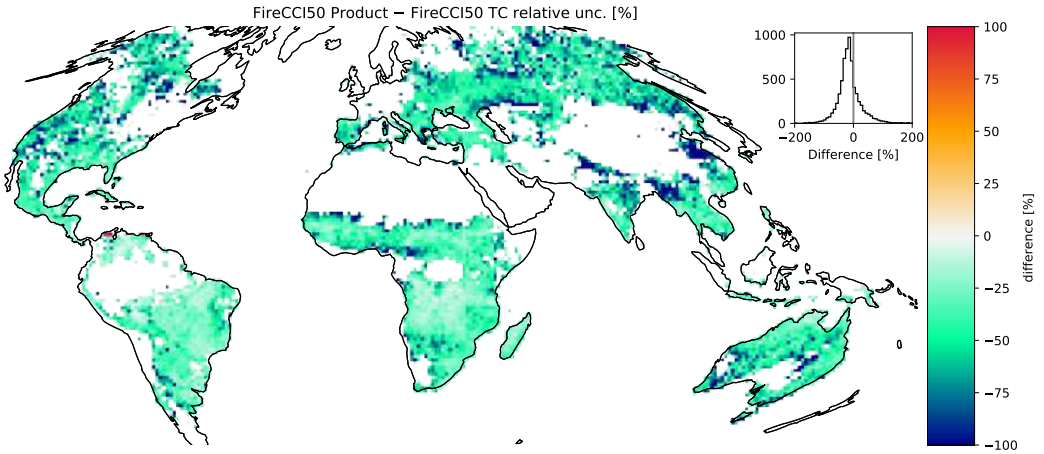


Figure 5.9. Differences in relative uncertainties between product uncertainties for FireCCI50 and TC-estimated relative uncertainties. Blue corresponds to larger TC relative uncertainties and red to larger FireCCI50 product uncertainties.

uncertainties for TC-derived uncertainties and the uncertainties provided with the FireCCI50 product. TC uncertainties exceed product uncertainties in 98% of the valid grid cells. Globally, the median relative uncertainty implied by the product is 2% compared to 41% from the TC uncertainties. The product uncertainties have a much smaller global range (IQR: 1–5%) compared to the TC estimate (IQR: 27%–58%). The difference between TC uncertainties and product uncertainties are largest in cropland areas of northern China (150–200%), eastern Russia (50–100%) and eastern India (60-120%). TC uncertainties are also around (70–100%) larger in regions of the western United States.

Figure 5.10 shows an example of the pixel-level uncertainties provided with the FireCCI50 product. Reference burned area is overlaid from the analysis of two Landsat acquisitions. It can be seen that the product correctly detects the larger burn scars in the image extent. For these larger burn scars, the provided confidence is 70–100%. However, smaller burn scars which are not classified as burned by the algorithm show burn probabilities which are similar to the unburned background (20–40%). These values do not correspond well with the likely fire signal at these locations, with the apparent pattern in unburned confidence values arising from the interpretation of the composited observations used within the algorithm.

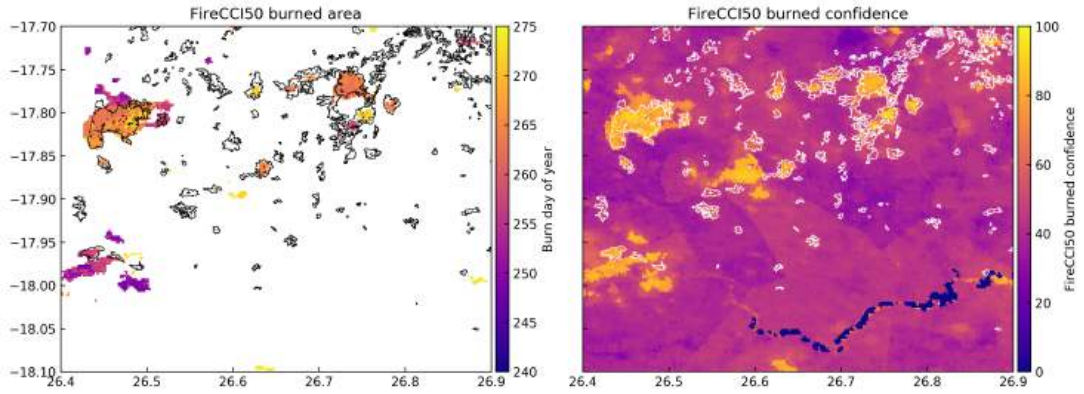


Figure 5.10. Example of the pixel level uncertainties (burned confidence) provided with the FireCCI50 product. The area covers northern Zimbabwe for the period September 2008. Landsat derived burned area is overlaid.

5.3 Considerations of the TC error model

As previously indicated in section 5.1.2, the TC error model has several key assumptions which must be considered. An initial requirement of the TC method is that the three products correspond to three temporally and spatially collocated data products. Here, this was achieved by considering the products at coarse spatial and temporal scales. The aggregation of daily pixel products to 16-day windows should help to reduce the influence of differences in reporting dates of fires between products. Similarly, the aggregation to a 1° spatial resolution grid reduces the chance of highly local differences in reported burned area, and therefore should provide more robust estimates for each product. Nevertheless, due to the requirements of the TC method, around 40% of global land cells do not have uncertainties — although this figure includes desert regions. Zwieback et al. (2012) indicated that the relative error in uncertainty estimates from the TC method can be approximated by $\sqrt{\frac{5}{n}}$, where n is the valid number of collocated observations used to compute the product covariance matrix. Users should be aware that the accuracy of uncertainties in regions with less frequent burning will therefore be lower than in those regions with longer fire seasons. Given the available temporal span of the products, the mean global relative uncertainty in TC error estimates is expected to be around 33%.

The most significant assumption of the TC method here is that the products do not have error cross-correlations (ECC) (Zwieback et al., 2012; Gruber et al., 2016). ECC structures between burned area products may occur due to 1) the use of the same satellite instruments, 2) shared observation opportunity at the 1° spatial scale and 3) similarities in the retrieval algorithms. Each is now considered. A key concern is that the three products all utilise observations from the MODIS instruments. All three

products utilise MODIS surface reflectance measurements; with FireCCI50 and MCD64 additionally using MODIS active fire detections. In terms of the second ECC source, grid cell uncertainty estimates may also be affected by the general observational opportunity available within the TC cell. The MODIS active fire products have a better sampling at higher latitudes relative to the equator (Giglio et al., 2006), and persistent cloudiness may introduce additional error correlations between the products. Finally, similarities within the mapping algorithms may introduce additional ECC sources. For example, similar thresholds on fire-related changes in reflectance may cause error correlations between the products. In regards to the magnitude of the effect of each potential ECCs, we judge that product uncertainties are more likely driven by differences in the mapping algorithms for each product. This is because the three algorithms use considerably different decision structures for mapping the pixel level burned areas. For example, while MCD64 and FireCCI41 both use active fire observations, the two algorithms utilise distinct expectations of fire properties in different spectral regions. Similarly, several intercomparison activities of these three products have indicated considerable differences between estimates at both the pixel level product and regional burned area estimates despite the use of the same input data(Humber et al., 2018; Padilla et al., 2015).

It should also be stressed that the uncertainties estimated with the TC method likely represent a lower bound on the true uncertainties of these products. The TC measurement model can explicitly estimate random errors but not systematic errors (i.e. bias) present in the data products from fires which are undetectable. The under-estimation bias observed for these coarse-resolution products in validation studies indicates that the products likely do have considerable systematic errors. Chuvieco et al. (2018) have estimated that the FireCCI50 product has global omission errors of 70% and MCD64C6 62%, which are partially balanced by commission errors of 50% and 35% respectively. Roteta et al. (2019) also indicated that a higher spatial resolution 20m burned area product provided 80% more burned area than the MCD64C6 product for sub-Saharan Africa, which while not providing a true validation indicates considerable biases in coarse-resolution products. Users should be aware therefore that the likely systematic biases in coarse resolution products mean that the TC uncertainties provide a lower bound on the true uncertainty.

5.4 Discussion

This study has estimated theoretical uncertainties for three global satellite-derived burned area datasets. This study provides an update on ongoing efforts to provide quantitative uncertainties for remotely sensed global burned area estimates initiated with GFED4 (Giglio et al., 2006) and continued within

the FireCCI products (Chuvieco et al., 2018; Brennan et al., 2018). Within the four-stage validation scheme developed for land remote sensing products developed by CEOS Land Product Validation (Table 4.1 in the previous chapter), the majority of current burned area products have only achieved stage three validation (Boschetti et al., 2009; Morisette et al., 2006; Chuvieco et al., 2018; Boschetti et al., 2016; Padilla et al., 2017). Meeting the stage four requirement for statistically robust and validated uncertainties remains an open challenge for the burned area community. While new large-scale validation datasets of burned area have been recently developed (Chuvieco et al., 2018; Padilla et al., 2017), these only provide regional-to-global commission/omission error statistics which need to be interpolated with a statistical model of the measurement process to provide explicit spatiotemporally dense uncertainties (such as is done in GFED4). Specifying and then parameterising spatially and temporally such models is a considerable challenge. Instead, the presented triple collocation (TC) error model provides a data-driven method to independently and automatically estimate uncertainties in three global burned area products post hoc, and which is suitable for inclusion as part of stage four validation campaigns.

A feature of the TC analysis shown here is the large relative uncertainties for the products in croplands and shrublands. The large relative uncertainties in shrubland burning have not been previously highlighted for global satellite burned area products. A potential mechanism for this is a detection threshold associated with the limited and discontinuous fuel bed in shrublands. The limited vegetation density in shrublands will limit the magnitude of the radiometric burn signal pre-to-post fire — limiting the change signal the algorithms use to classify burning. Combining the limited vegetation signal with the general sparseness of vegetation ground cover in shrublands will lead to this 'patchiness' of the burn signal which when observed at 500m will fall around the detection thresholds of the mapping algorithms considered here (Roy and Landmann, 2005). The large relative uncertainty for MCD45 recorded in Australian shrublands is potentially a feature of the limited performance of the algorithm over surfaces with bright soils (de Klerk et al., 2012; Roy et al., 2005a). This represents a promising area for future research. Cropland burning has been a persistent problem for coarse-resolution burned area products. Particular features which obscure detection in croplands are the transient nature of the burn signal before ploughing, and the highly fragmented nature of burning on the land surface. Given these circumstances, the ability to detect cropland burn scars from MODIS resolution data has been previously questioned (Hall et al., 2016). Zhu et al. (2017) indicated omission errors for the MCD64 product greater than 60% for small cropland fires. Similarly, MCD45 has been reported to considerably under-report cropland burning globally (Roy et al., 2008b). However, discrepancies between the products are likely to still be driving the TC uncertainties, for example,

observed commission errors by MCD64 for harvesting in Eurasia and MCD45 in Australia (Humber et al., 2018; Giglio et al., 2009). It remains an open question whether the higher spatial resolution available in the FireCCI50 products improves performance over croplands, with some evidence that it might (Chuvieco et al., 2016). The FireCCI50 product detects the greatest magnitude of cropland burning globally and has the smallest relative uncertainties. Future studies may be better able to indicate whether the increase in spatial resolution has produced this.

Previous validation activities have indicated that satellite-derived burned area products typically perform best in regions where fire activity is more prevalent (Padilla et al., 2015). We also find that the smallest relative uncertainties are typically found in the frequently burning savannas and grasslands of Africa, Australia and South America. Nevertheless, relative uncertainties in burned area estimates for these regions were found to be in excess of 8–10%. Given the predominance of fire activity in these areas, they contribute considerably to the uncertainty on reported global burned area. In areas with more infrequent burning or more barriers to detection, relative uncertainties were found to be higher. In such circumstances, the particular limitations of each detection algorithm are most likely to drive the differences observed. For example, differing observational requirements of the products drives large uncertainties in equatorial Asia (EQAS) where persistent cloud reduces the area actually mapped (either burned or unburned) by of all the algorithms. The MCD45 algorithm has been found to suffer uniquely in cloudier regions due to the greater sampling requirement of the algorithm as well as over-restrictive cloud masking conditions (Roy et al., 2002; Humber et al., 2018; Giglio et al., 2010). Changes made to the MCD64 Collection 6 product, including relaxations on cloud masking have increased the mapped area in these cloudier regions (Giglio et al., 2018).

As noted previously, MCD64 reports the greatest burned area globally $3.76 \pm 0.15 \times 10^6$ km^2; followed by FireCCI50 $3.70 \pm 0.17 \times 10^6$ km^2 and then MCD45 $3.31 \pm 0.18 \times 10^6$ km^2. In terms of the global agreement between products, figure 5.11 shows the distribution of mean annual burned area for the three products. A higher level of agreement between the FireCCI50 and MCD64 products can be observed with the two products agreeing well within one standard deviation. The MCD45 product disagrees most with the MCD64 product and slightly less with the FireCCI50 product. The three products overlap within two standard deviations. Even so, the degree of discrepancy on global burned area estimates would indicate that the previously used confidence bounds (i.e. from the range of product (Rabin et al., 2017)) provide an under-estimate in the global burned area uncertainty.

Estimates of the mean annual burned area from the three products agree within their respective uncertainties in around 60% of valid TC-estimates. Nevertheless, while estimates are consistent, regional estimates remain poorly constrained by the products considered. Uncertainties above 10%

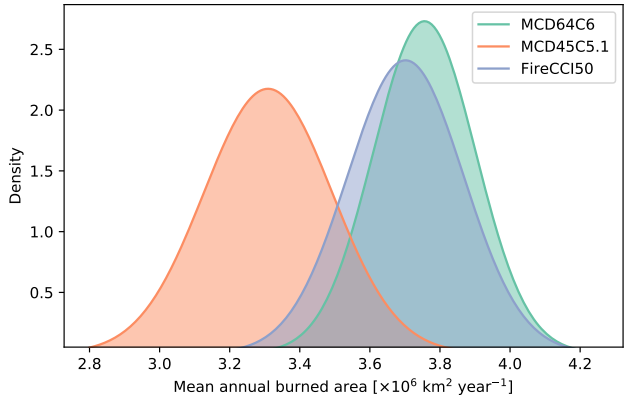


Figure 5.11. Constraints on global mean annual burned area $\mathrm{km}^2/\mathrm{year}$ provided by the three products.

are found for all products in at least one land cover, including uncertainties >24% for MCD45 in shrublands, 11% for MCD64 in croplands and 13% in shrublands for FireCCI50. Regional uncertainties are often larger than these figures, with relative uncertainties of over 100% for MCD45 in croplands and grasslands in central America and boreal Asia; and for forests in Europe and boreal North America. Relative uncertainties larger than 100% for MCD64 are also found in forests and croplands in boreal and central Asia. FireCCI50 also has relative uncertainties >100% for croplands and forests in Australia, boreal North America and Europe. As these products are often also used at national-to-regional scales, it is important to consider the precision of the current products at these scales (Liu et al., 2019; Zhu et al., 2017; Roy and Boschetti, 2009). The uncertainty estimates here are therefore useful for these users to discern any limitations of products at the appropriate scale. While the TC-estimated uncertainties can not directly provide information on uncertainties at the pixel level, users should also be encouraged to consider the quality assurance (QA) information provided in these products.

The presented TC uncertainties have many applications. The uncertainties could, for example, be used to drive development and refinement of parameters in dynamic vegetation models (DVGMs) related to fire processes or to improve optimisation routines for parameter selection (Poulter et al., 2015; Forkel et al., 2019). They could also be used to better constrain uncertainties on emission estimates derived from 'bottom-up' inventory approaches (Randerson et al., 2012; French et al., 2004; Knorr et al., 2012; van Der Werf et al., 2017). Explicit uncertainties per observation additionally allow for the development of more advanced assimilation of the satellite observations into models through mathematical frameworks in data assimilation. Similarly, they open up the ability to calibrate model parameters x against observations of burned area. For example, assume a DGVM has a fire model that

predicts burned area at a time t ($\mathrm{BA}_{\mathrm{model}}(t)$) as a function of e.g. meteorological drivers, vegetation parameters and some fire-related parameters I (e.g. Thonicke et al. (2010); Mangeon et al. (2016)):

$$H(x, I, t) = \mathrm{BA}_{\mathrm{model}}(t). \tag{5.12}$$

Under the assumption that the burned area estimates are normal, one could derive the (log)likelihood function $L(\mathrm{BA}_{\mathrm{obs}} \mid x)$ as:

$$L(\mathrm{BA}_{\mathrm{obs}} \mid x, t) \propto \frac{[H(x, I, t) - \mathrm{BA}_{\mathrm{obs}}(t)]^2}{2\sigma_{\mathrm{TC}}(t)^2}. \tag{5.13}$$

Minimisation of this function would result in the parameters that provide the closest fit to the observations, weighted by how much one could trust these observations.

5.5 Conclusion

The wide application and interpretation of remote sensing products of burned area require explicit estimates of the uncertainties of these products. This chapter has presented theoretical uncertainties for three global satellite-derived burned area products. This chapter therefore partially addresses the aim (II) stated in chapter 1 section 1.2 to "develop uncertainty quantification methods for burned area products". A triple collocation error model was applied to produce unique, near-global, uncertainties for the MCD64 Collection 6, MCD45 Collection 5.1, and FireCCI50 burned area products. While products were found to provide consistent estimates in a majority of the sampled global fire extent, the constraint on burned area in many regions was found to be poor with uncertainties in each product exceeding 8–10% in the regions with the greatest burned area. Uncertainties on burned area in regions with less burned area were also found to be considerable. Individual products were shown to have uncertainties exceeding 100% in specific regions and land covers. The present study would suggest that previous estimates of uncertainty in global burned area from satellite products appear to be under-estimates. Users of these products should therefore be aware of the uncertainties both in these products, and the regional and land cover specific differences in product confidence as provided by these uncertainties.

Chapter Six

Uncertainty traceability considerations for burned area: towards a best practice framework

The previous chapter addressed key aims by considered an empirical method to estimate uncertainties in global burned area products. This was achieved through a 'top-down' estimation of the uncertainties using the statistical assumptions of the triple collocation method. The inherent limitations of such an approach are of course that uncertainty has to be estimated after the production of three products and only at coarse spatial scales. This chapter builds on the previous work by proposing a framework to develop traceable estimates of BA uncertainty using metrological principles. This addresses the second part of the aim specified in chapter 1 to "develop uncertainty methods for burned area algorithms following metrological best practice".

As was described in chapter 4, burned area products lie at the top of a measurement chain. The set of transformations of the initial sensor measurement eventually lead to the burned area classification, with errors potentially entering the process at all levels (Merchant et al., 2017). An important feature of any uncertainty characterisation within remote sensing Climate Data Records (CDRs) is the traceability of the measurement. Traceability refers to a requirement that the errors contributed at each step of the measurement chain should be understood and included in the computation of the measurement uncertainty (Merchant et al., 2017). The definition and quantification of these uncertainties is referred to as developing an uncertainty budget.

This chapter proposes a best practice framework for achieving uncertainty quantified burned area estimates with traceability across spatial scales. As burned area mapping involves a classification

task the required approach for uncertainty characterisation differs somewhat from methods specified in the GUM. This arises because the GUM is primarily focused on the measurement of continuous measurands, and specifies a standard uncertainty $u(y)$ determined as the standard deviation of values reasonably attributable to the measurement. Instead, proposed here is a probability statement of the uncertainty in the burned area inference $\mathrm{P_b}$, which aligns with the uncertainty recommendations for discrete CDRs as described in Merchant et al. (2017).

6.1 Uncertainty in burned area for climate modeller grids

Firstly, it is necessary to define the different types of burned area products. Burned area products are primarily provided as level-3 and 4 products. Level-3 products are pixel-level data sets provided typically at the spatial resolution of the level-2 gridded surface reflectance product used. These products are typically at spatial resolutions of 100m–1km with a daily temporal resolution. They provide a binary burned/unburned mask for all pixels considered burnable (vegetated/non-water pixels) and an estimate of the day of burn (DOB) for burned pixels. These pixel-level products are next aggregated to lower spatial and temporal resolutions to provide level-4 "climate model grid" (CMG) products. CMG products are typically provided at spatial resolutions of 0.05-0.5° and at daily-to-monthly temporal intervals. These products provide for each grid cell the burned area km^2 over the mapping period. The aggregation is typically calculated from the summation of the burned pixels in the pixel-product within the spatial and temporal extent of each CMG grid cell.

Consider a climate model grid (CMG) grid cell g. In terms of the original level-3 product, this grid cell defines the set of higher-resolution pixels p_i ($i = 1 \ldots K$) that fall within it $\{p_i \ldots p_K\}$, over the spatial s and temporal t coverage of the grid cell. The measurement of burned area B km^2 for g then involves the aggregation of pixels in $\{p_i \ldots p_K\}$ which have been mapped as burned. This aggregation involves computing the sum of burned pixels n_b in $\{p_i \ldots p_K\}$. The nature of this aggregation process indicates some features of B. First, that B is computed as the sum of burned pixels demonstrates it to be a discrete random variable. This means that B can only take integer values implied by the summation of the n_b burned pixels. This also implies that a pixel p_i cannot be considered fractionally burned — but instead only be considered fire-affected or not fire affected. This binary choice is common to all present global burned area products. That B is a discrete random variable implies that it must be represented by a probability mass function (PMF), with all possible values of n_b represented by a probability mass defined by the PMF $\Pr(n_b)$.

The uncertainty in burned area B arises from the inability to correctly infer the value of n_b. This results from the algorithm making commission and omission errors on some of the pixels in $\{p_i \dots p_K\}$. An immediate PMF scheme for $\Pr(n_b)$ arises from the case in which the probabilities of omission and commission errors can be quantified as $\mathrm{P_o}$ & $\mathrm{P_c}$ respectively. In such an approach, the total misclassification error is $\mathrm{P_e} = \mathrm{P_o} + \mathrm{P_c}$, as commission and omissions represent disjoint events for the pixel $\{p_i \dots p_K\}$. Conversely, the probability that a pixel is correctly identified as burned $\mathrm{P_b}$ is then $\mathrm{P_b} = 1 - \mathrm{P_e}$. Then for the K pixels in the set $\{p_i \dots p_K\}$, a PMF for the number of burned pixels n_b in $\{p_i \dots p_K\}$ would be described by the binomial distribution:

$$\binom{n_b}{K} \mathrm{P_b}^K \, (1 - \mathrm{P_b})^{n_b - K} \tag{6.1}$$

Such estimates of $\mathrm{P_c}$ and $\mathrm{P_o}$ would naturally have to come from validation of BA estimates against a known reference data set. Such an approach has already been implemented implicitly to provide uncertainty estimates of burned area in (Giglio et al., 2010). In Giglio et al. (2010) an estimate of the variance in grid cell burned area was provided from:

$$\sigma_A^2 = c_B B \tag{6.2}$$

where B is the burned area and c_B a "binned-burned-area uncertainty coefficient". The coefficient c_B km^2 provides a scaling of the expected error per unit of detected burned area B km^2. c_B therefore encodes the net uncertainty for the grid cell in a manner similar to the binomial approach above — if B was represented as the number of burned pixels n_b, c_B would effectively be $(\mathrm{P_c} + \mathrm{P_o})$. This approach allows for an estimate of the total uncertainty derived from both the properties and resolving ability of the algorithm. There are, however, several limitations to such an approach. Primarily, this approach represents only an expectation of the uncertainty in the burned area B. While regional values of $\mathrm{P_b}$ can be derived from regional validations, the true representativeness of such a value may be limited. The generation of suitable and sufficiently voluminous validation data for such an approach is intensive. For example, Giglio et al. (2010) only estimated three unique regional values for c_B. Further, such validation data is naturally biased towards fires which are easier to distinguish: larger fires, as well as those, occurring when there are sufficient acquisitions from the high-resolution instrument (i.e. Landsat) to produce a validation data set. As a result, such validation data sets may miss smaller fires in regions with persistent cloud cover which also correspond to the most poorly constrained regions for BA estimates i.e. tropical regions (Humber et al., 2018).

Instead of global values for $\mathrm{P_c}$ and $\mathrm{P_o}$, consider that for each pixel p_i in $\{p_i \dots p_K\}$, the potential error of commission or omission could be quantified. Such probabilities would be a feature of the

suitability of the signal to the expected properties of burned pixels as defined by the algorithm, as well as the quality of the observations for the pixel. Given that the probability that the pixel is burned $\mathrm{P_b}$ is $\mathrm{P_b} = 1 - (\mathrm{P_e})$, it is equivalent to talk in terms of the belief that the pixel is burned. With a unique probability for each pixel such that $\mathrm{P_b}(x, y, t)$, the PMF of the number of burned pixels $\Pr(n_b)$ is provided by the Poisson binomial distribution:

$$\Pr(n_b = n) = \sum_{A \in F_N} \prod_{i \in A} \mathrm{P_b}^{(i)} \prod_{j \in A^c} (1 - \mathrm{P_b}^{(j)}) \tag{6.3}$$

where F_{n_b} is the set of all subsets of the K pixels comprising the CMG cell that can be selected from $\{1, 2, 3 \ldots N\}$. As an example, if $K = 3$, the set of possible combinations of two burned pixels $n_b = 2$ is $F_2 = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$. A^c is the complement of the set A, i.e. the pixels not considered burned in F_2: $\{\{3\}, \{2\}, \{1\}\}$.

Unfortunately, the computation of the PMF in equation 6.3 is unfeasible for even moderate values of N. This results from the requirement to enumerate all potential subsets in F_N. Fortunately, for larger values of K, a normal approximation to the PMF is suitable given the central limit theorem (Volkova, 1996; Hong, 2013). Further, the first two moments of equation 6.3 are simple to calculate:

$$\bar{n_b} = \sum_{i}^{K} \mathrm{P_b}^{(i)} \qquad\qquad \sigma_b^2 = \sum_{i}^{K} \mathrm{P_b}^{(i)} (1 - \mathrm{P_b}^{(i)}) \tag{6.4}$$

where $\bar{n_b}$ is the expected mean number of burned pixels and σ_b^2 the variance of number of burned pixels in $\{p_i \ldots p_K\}$.

6.2 Uncertainty quantification at the pixel scale

It is now appropriate to turn to the task of uncertainty characterisation at the pixel level through the *per-pixel* probability of burning $\mathrm{P_b}$. Chapter 4 discussed how uncertainty in both the observations and the specification of the burned area mapping algorithm lead to a limited precision in the ability to delineate burned areas. Consider that an algorithm $\mathbf{G}$ provides a mapping from observations d to a likelihood statement that the surface has burned b:

$$b = \mathbf{G}(d, w) \tag{6.5}$$

where w represents the auxiliary information (e.g. algorithm parameters) used within the algorithm to perform the classification. In fact, this mapping is in-exact — errors are present in both the observations and parameters of the algorithm:

$$b = \mathbf{G}(d, w) + \epsilon \tag{6.6}$$

For simplicity consider that both the parameters and observations can be considered as $\theta = [w, d]$. Given that $\mathbf{G}$ provides a mapping between θ and b, a small perturbation to the values of θ, $\theta + \Delta\theta$ will produce a perturbation to the likelihood b:

$$b + \Delta b = \mathbf{G}(\theta + \Delta\theta) \tag{6.7}$$

Clearly certain perturbations to $\theta + \Delta\theta$ are more likely than others and therefore the natural manner to encode the uncertainty in the values of θ is by specifying the probability density function (PDF) $\Pr(\theta)$. Given the expected perturbations $b + \Delta b$, the estimates provided from the algorithm are also represented by the PDF $\Pr(b \mid \theta)$.

The posterior distribution of the algorithm $\Pr(b \mid \theta)$ therefore encodes how much for a given set of values of θ, the algorithm believes that the pixel has experienced a fire. However, to arrive at $\mathrm{P_b}$ the focus is not on the sensitivity of the algorithm to any given set of θ values but rather the amount of variance introduced into the classification $\mathbf{G}(\theta)$ from the variance in $\Pr(\theta)$. The expected value of the posterior $\Pr(b \mid \theta)$ is a suitable way to encode this uncertainty as $\mathrm{P_b}$:

$$\mathrm{P_b} = \int_{\Theta} \mathbf{G}(\theta) \Pr(\theta) \, \mathrm{d}\theta \tag{6.8}$$

where Θ is the domain of θ. Functionally, the outputted likelihood of burning b for a given input θ is weighted/moderated by the probability density of θ, and as a result $\mathrm{P_b}$ responds to the uncertainty in the classification task.

6.3 Basic principles of uncertainty propagation relevant to burned area detection

It is now suitable to consider some of the practicalities of specifying the probability distributions for the data d and algorithm parameters w; as well as uncertainty traceability through a mapping algorithm. It is not possible to provide a full coverage of burned area methods, primarily due to the lack of consistent methodologies between algorithms, but general principles can be drawn. An immediate challenge is the representation of uncertainty in the observations d. As was discussed in section 4.4.1 & 4.4.2, observations of the land surface (i.e. surface reflectance) contain errors and have the property of being temporally and spatially incomplete. A widely applied distribution for observational uncertainties in remote sensing is the multivariate normal distribution:

$$\Pr(d) = \frac{1}{\sqrt{(2\pi)^N \, |\mathbf{C}_{\mathrm{obs}}|}} \exp\left(-\frac{1}{2}(d - d_\mu)^T \mathbf{C}_{\mathrm{obs}}^{-1}(d - d_\mu)\right) \tag{6.9}$$

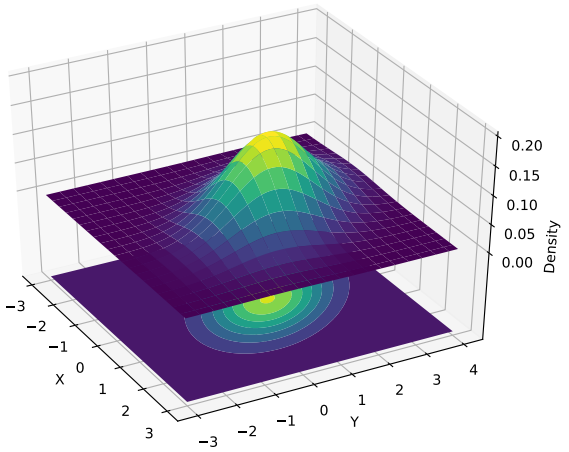


Figure 6.1. A two variable X & Y multivariate normal distribution.

where d_μ is the recorded value in the data set, and $\mathbf{C}_{\text{obs}}$ is the observational covariance matrix (see figure 6.1 which displays the PDF of a multivariate normal distribution). The full specification of $\mathbf{C}_{\text{obs}}$ is, in many cases, difficult. The knowledge of suitable magnitudes for variance terms in $\mathbf{C}_{\text{obs}}$ requires a full uncertainty propagation from the initial signal recorded by the sensor (Merchant et al., 2017). Such information may not be available for the L2 reflectance data typically used within burned area algorithms. Similarly, observational errors in nearby pixels may have complicated correlation structures due to residual atmospheric correction effects or the properties of the land surface itself (Povey and Grainger, 2015). Such spatial and temporal correlations mean that $\mathbf{C}_{\text{obs}}$ may have significant covariance terms (Bellprat et al., 2017; Merchant et al., 2017; Pinnington et al., 2016).

Burned area algorithms very rarely directly utilise 'untreated' observations of surface reflectance as part of the detection. Instead, algorithms conduct considerable pre-processing of the observations to produce a more amenable signal on which to classify burned area. As described above in section 4.4, many mapping algorithms make use of some compositing scheme in which observations sampled through time are merged to produce a spatially complete composite of reflectance. Giglio et al. (2008) for example collect observations over eight days to ensure a sufficient sampling from the two MODIS sensors, and Roy et al. (2005a) composite observations over a period of up to 32 days to get a sufficient sampling of the pre-fire surface reflectance. Such compositing can reduce the uncertainty in the signal, as the averaging effect reduces sensitivity to random errors — but may not remove error covariances.

As was discussed in section 4.4.2, mapping algorithms often make use of spectral indices (also termed vegetation indices; VIs) to help reduce systematic error sources such as residual atmospheric

noise or reflectance directionality due to the angular sampling geometry of the observations. The computation of a spectral index also involves a transformation of the measurements — and therefore requires careful consideration of the propagation of uncertainties. Consider that the production of the index VI involves the transformation of reflectance observations $\vec{\rho}$ as $\text{VI} = f(\vec{\rho})$. The GUM suggests that for transformations that are linear or nearly linear uncertainty propagation can be achieved by analytic means (ISO/BPIM, 2008). In such cases the uncertainty in VI can be characterised by a covariance matrix:

$$\mathbf{C}_{\text{VI}} = \mathbf{G}\ \mathbf{C}_{\text{obs}}\ \mathbf{G}^T \tag{6.10}$$

where $\mathbf{C}_{\text{obs}}$ is the error covariance matrix of the reflectance $\vec{\rho}$. $\mathbf{G}$ is a matrix of the first-order partial derivatives of f:

$$\mathbf{G} = \begin{bmatrix} \frac{\partial f_1}{\partial \rho_1} & \cdots & \frac{\partial f_1}{\partial \rho_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial \rho_1} & \cdots & \frac{\partial f_m}{\partial \rho_n} \end{bmatrix} \tag{6.11}$$

This approach is typically referred to as the *moment propagation of errors* because the method is based on propagating the second moment of the distribution $\Pr(\vec{\rho})$ through f (Smith, 2013). With the first order approximation of f provided by G, the first two moments are therefore propagated as $\mathbb{E}[\text{VI}] = f(\mathbb{E}[\vec{\rho}])$ and $\text{var}[\text{VI}] = \mathbf{G}\ \mathbf{C}_{\text{obs}}\ \mathbf{G}^T$. For non-linear transformations, $\mathbf{G}$ also provides the best linear approximation to the transformation at the point $\vec{\rho}$.

However, typical spectral indices are not linear transformations of input reflectance and therefore the quality of such approximations should be tested. The simplest method to do this is to consider Monte Carlo propagation methods in which samples are drawn from the input distribution and propagated through the index to sample the output distribution. Figure 6.2 illustrates the general principle of this process. The left panel shows the representation of the uncertainty in the reflectance as both a multivariate distribution and a set of samples from this distribution. The isolines of the index are also shown, which record the values of the index for variations in the input reflectance. At the red dot, a linearisation of the index $\mathbf{G}$ is taken and used to propagate the uncertainty into the index. The right panel then shows the error distribution propagated through to the values of the index. It can be seen that the linear propagation under-estimates the true variance of the uncertainty in the spectral index. This occurs because the linearisation $\mathbf{G}$ cannot capture the non-linear features of the index around the point.

In addition to uncertainty quantification for just observations, the uncertainty in the parameters w used within the algorithm should be characterised. As was discussed back in chapter 4, often with

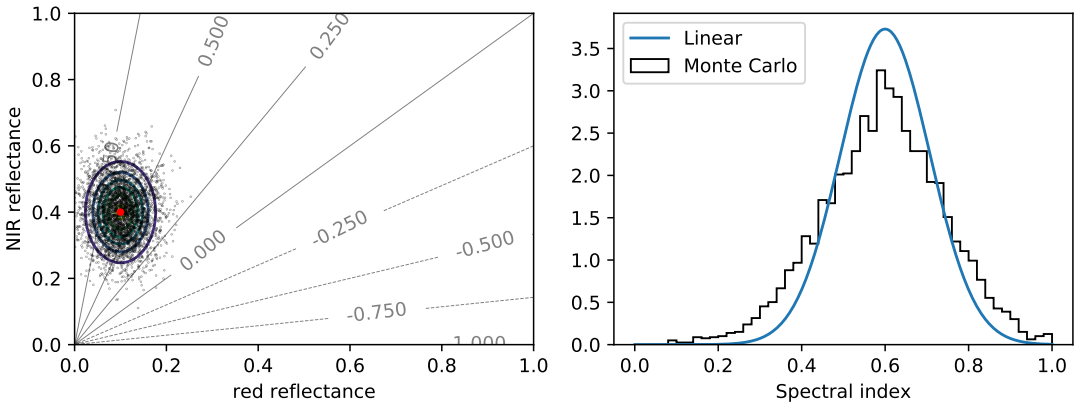


Figure 6.2. Uncertainty propagation methods through a spectral index. Left) representation of the uncertainty in red and NIR reflectance as a set of samples from a multivariate normal distribution. The isolines of the spectral index are also shown. Right) error distribution propagated to the index with Monte Carlo samples and via the linearisation of the index.

remote sensing retrievals the value of auxiliary parameters are poorly defined. This is also sometimes the case in burned area mapping algorithms. The importance of characterising the uncertainty in these retrieval parameters depends on how sensitive the retrieval is to their value. For example, small changes in the value of a parameter $w_1 + \Delta w_1$ within a mapping algorithm may lead to a large change in the belief that a pixel is burned. The nature of these sensitivities was explored back in chapter 3 with the sensitivity analysis of the six global burned area algorithms.

To demonstrate the complexity of algorithm parameterisations, table 6.1 provides the parameters and key algorithm conditions used within the MODIS collection 6 burned area product (MCD64). Giglio et al. (2018) indicate that these parameters were chosen empirically and that point out that this is common for most global burned area mapping algorithms. What can be remarked is the general complexity of specifying distributions for these parameters and algorithm conditions. Considering the nature of some of these parameters provides insight into the difficulty of uncertainty characterisation of burned area algorithms.

An initial challenge is that several of the algorithm conditions are based on threshold tests. As an illustration, the algorithm uses a threshold condition on the maximum change reported in a temporal change index S^* to attribute unburned pixels based on $S^* > 2$. Non-linear transformations — such as threshold tests — present challenges to uncertainty propagation because even small errors may cause the step to be switched (Merchant et al., 2017). A suitable method to deal with such transformations is to replace them with smooth approximations. For example the test $S^* > T$ can be replaced with a

Table 6.1. Key processing and classification parameters or variables in the MCD64 Collection 6 algorithm (Giglio et al., 2018). Variables are reported with a — in the value column.

		Description	Value
Processing parameters			
	W	Compositing window length.	8
	t^*	Date of largest spectral change.	—
	t_f	Date of active fire detection.	—
	S	Temporal change index.	—
	S^*	Maximum change in temporal index.	—
	ΔVI	Change in vegetation index.	—
	ΔVI*	Largest change in vegetation index.	—
	D_b	Distance to nearest burned pixel.	—
	σ_p	Distance decay constant for D_b.	5km.
	σ_k	Standard deviation of kernel density estimator for burned/unburned classes.	0.02
	N_b	Number of training pixels for burned class.	—
Classification parameters			
	$s^* < 2$	Unburned spectral change. Threshold separating burned and unburned.	2
	σ_t^* >8	Unburned temporal metric. Threshold separating burned and unburned.	8
	$D_b > R_d$	Distance decay for unburned pixels. Threshold separating burned and unburned.	5km
	$N_b > 100$	Required number of burned training pixels to estimate distribution of burned class.	100
	$Q_l^{(50)}$	median of burned/unburned VI classes. Median estimated from many samples.	50%
	$\Delta Q_l^{(50)} \geq -0.05$	Separability of burned/unburned classes test. Threshold separating burned and unburned.	-0.05
	P_{min}	Prior probability minimum. Fixed prior probability.	0.01
	P_{max}	Prior probability minimum. Fixed prior probability.	0.5
	$\sigma_t^* < Q_l^{(98)}(\sigma^* \mid B)$	Classify burned if σ_t^* exceeds 98% percentile of burned class distribution. Relative/adaptive threshold.	98%
	$\text{VI}_{\text{post}}^* < Q_l^{(98)}(\text{VI}_{\text{post}}^* \mid B)$	Classify burned if $\text{VI}_{\text{post}}^*$ exceeds 98% percentile of burned class distribution. Relative/adaptive threshold.	98%

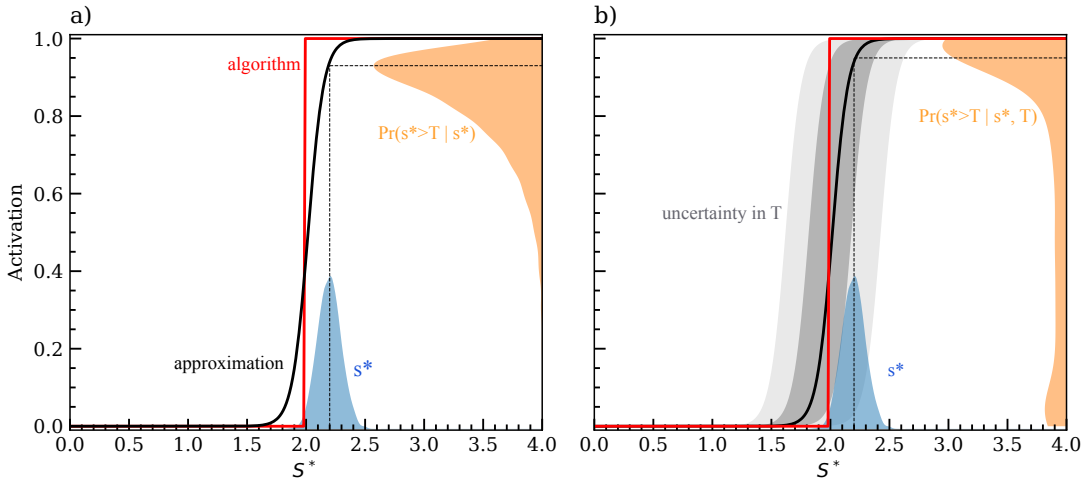


Figure 6.3. Example of propagating uncertainty in a parameter through a threshold function. The distribution for S^* is shown in blue and the activation of the function is shown by the orange distribution. a) The threshold function (red line) can be approximated with a smooth differentiable function (black line). b) The uncertainty in the value of the threshold T can also be incorporated by considering the smooth transformation as a represented by the conditional distribution $\Pr(S^* > T \mid S^*, T)$ shown as the grey shaded region.

smooth approximation:

$$\frac{1}{1 + e^{-\alpha(S^* - T)}} \tag{6.12}$$

where α provides a "tuneable" parameter which determines the sharpness of the transition. Figure 6.3a) depicts how this function (black line) approximates the non-linear threshold function (red line) and allows for propagation of the uncertainty in S^* (blue distribution) to provide $\Pr(S^* > T \mid S^*)$ (orange distribution). In addition to just propagating uncertainty through the transformation, it is also important to encapsulate the uncertainty in the belief of the value of the threshold T with a distribution $\Pr(T)$. Figure 6.3b) shows how introducing uncertainty into the value of T produces a broader mapping of the activation $\Pr(S^* > T \mid S^*, T)$. This reveals additional features of the problem: with $\Pr(S^* > T \mid S^*, T)$ now being bi-modal; with the mode of the distribution not equal to the propagated $\mathbb{E}(S^*)$ value through the linearisation.

Other non-linear transformations also present issues. The decay constant σ_p specified in table 6.1 determines the rate at which a pixel is classified as burned based on the whether nearby pixels are burned. This is processed through the Mahalanobis distance:

$$f(\text{dist}(x, y)^2, \sigma_p) = \exp\left(-\text{dist}(x, y)^2 / 2\sigma_p^2\right) \tag{6.13}$$

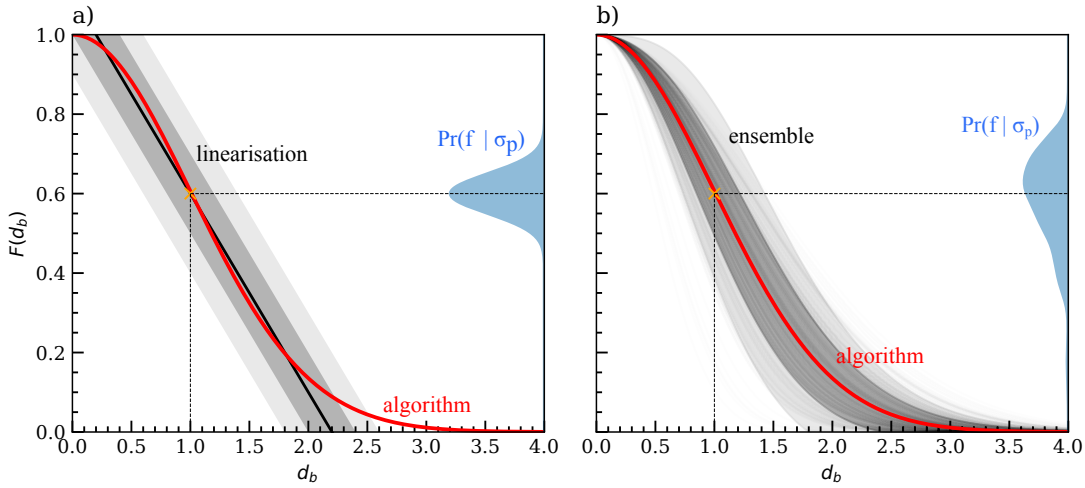


Figure 6.4. Comparing the quality of a) linearisation and b) ensemble-based for uncertainty propagation for a non-linear transformation. The red curve represents the exponential decay relationship specified in the algorithm. Here the shaded areas represent the 1σ and 2σ confidence intervals of the relationship determined by $\Pr(f \mid \sigma_p)$. In a) the function is linearised at $d_b = 1$ and the uncertainty in f due to σ_p quantified with a normal approximation (shaded region). At $d_b = 1$, the distribution of $\Pr(f \mid \sigma_p, d_b = 1)$ is displayed (blue curve). Alternatively in b) samples from $\Pr(\sigma_p)$ are propagated through f at $d_b = 1$ and a kernel density estimated (blue curve).

where $\text{dist}(x, y)^2$ is the squared Euclidean distance to the nearest burned pixel or active fire observation. The value of σ_p therefore determines the rate at which f decays – higher values of σ_p correspond to larger values of f and a greater spatial correlation between pixels. With the assumption that the distance to the pixel (referring to $\text{dist}(x, y)$ as d_b) is perfectly known, there are several approaches to provide UQ for σ_p. An immediate option is to produce an linearisation of f at $f(d_b)$. This effectively linearises the exponential decay to a linear decay at the linearisation point. Figure 6.4a) demonstrates this approach. The linearisation of f at $d_b = 1$ is made (the black line) and the uncertainty due to σ_p propagated based on an normal distribution approximation for $\Pr(\sigma_p)$. This then provides an normal approximation to the conditional distribution $\Pr(f \mid \sigma_p, d_b = 1)$. As before, a more flexible alternative to the moments propagation is to consider propagating a Monte Carlo ensemble through f. Figure 6.4b) shows the propagation of uncertainty based on an ensemble from $\Pr(\sigma_p)$ and estimating the conditional distribution $\Pr(f \mid \sigma_p, d_b = 1)$ with a kernel density estimate. The ensemble method has a larger variance than the moment propagation, but also a mode of 0.65 as compared to 0.6 for the linearisation. As with the VI example above, the difference occurs because the linearisation cannot capture the non-linearity of the exponential function.

Other processing parameters directly determine the quality of the retrieval. For example, the number of burned training pixels N_b (which come predominantly from active fires) determines the quality of the kernel density estimate for the burned class used in the classification stage of the algorithm. Giglio et al. (2018) indicate that if $N_b < 100$ there are insufficient samples to derive this distribution and the algorithm can not process the data. Suitable methods can be developed for reducing the influence of these parameters. For example, a prior distribution for the burned class negates the need for the $N_b < 100$ test as this distribution can be used as a 'back-up' and updated from whatever limited samples are available. In the most difficult cases, sensitivity analyses can be carried out to consider the influence of parameters. Giglio et al. (2018) indicate that the compositing window length W was chosen after such a sensitivity analysis.

6.4 Demonstration

It is now demonstrated how the proposed approach can be used to generate uncertainty quantified products. This is done by proposing a simple experimental burned area retrieval algorithm. To provide a sufficiently general while informative approach here the mapping algorithm is formulated as a logistic regression problem — although any other probabilistic classifier could be used instead. The formulation of the algorithm is now briefly described. A multi-temporal change based algorithm is examined here, as these are most common across the literature. In these type of algorithms, rapid changes in the land surface between two dates t_0 and t_1 are attributed to burning against other potential sources of change.

Logistic regression provides a technique for fitting a discriminative model for the classification $\Pr(\text{burned} \mid d)$. The general form of logistic regression for burned pixels is provided by:

$$\Pr(\text{burned} \mid d, w) = \text{Ber}(\text{burned} \mid g(w^T d)) \tag{6.14}$$

where Ber is the Bernoulli distribution and $g(\cdot)$ is the sigmoid function:

$$g(w^T d) = \frac{1}{1 + e^{-(w^T d)}} \tag{6.15}$$

where d are the observations and w is a vector of coefficients typically with two elements — an offset which determines where $g(\cdot) > 0.5$ and a rate parameter which determines how quickly the logistic is activated from 0 to 1. Logistic regression, then, involves the application of a linear model to the data $w^T d$ which is mapped to a probabilistic output $\text{burned} \in [0, 1]$ by the sigmoid function.

As was discussed in section 4, mapping algorithms typically make use of VI transformations of reflectance to aid the detection problem. A normalised difference VI which responds to the differential

scattering of burned areas between two SWIR channels (1200nm and 2100 nm respectively) is used:

$$\mathrm{VI} = \frac{\rho_{1200} - \rho_{2100}}{\rho_{1200} + \rho_{2100}} \tag{6.16}$$

This VI is also used within the MODIS Collection 6 burned area product (Giglio et al., 2009) and has been shown to have good burned-unburned discrimination Roy et al. (2005). Following a fire, the VI drops significantly due to a reduction in ρ_{1200}. Therefore the change in the VI pre- to post-fire should provide a suitable change-based signal to detect burning:

$$\Delta\mathrm{VI} = \mathrm{VI}_{\mathrm{pre}} - \mathrm{VI}_{\mathrm{post}} \tag{6.17}$$

Returning to the classification, ΔVI provides a feature space $d \in \mathbb{R}$ on which to discriminate burned areas. The model parameters w are used to encode our beliefs about the burned conditional distribution of $\Pr(\Delta\mathrm{VI} \mid \mathrm{burned})$. Fundamentally, for burned surfaces it is expected that $\Delta\mathrm{VI} < 0$. This constraint can be imposed into the classification by augmenting d with a constant offset term such that $d = [1, \Delta\mathrm{VI}],\ d \in \mathbb{R}^2$. The model activations $w \in \mathbb{R}^2$ encode the regions of the feature space d which correspond to high probabilities of the pixel being burned.

While rather simple, this algorithm captures some common characteristics of most burned area algorithms. Primarily, a logistic classifier represents a suitably generic template for a probabilistic algorithm. Primarily most present algorithms are solved as an explicit two-class classification problem in which information about the class conditional distributions of burned and unburned pixels are described and used to classify new observations. Similarly, many present algorithms involve discriminative classifiers in which the decision boundary between the burned and unburned classes is determined either by *a priori* expectation or via a training methodology with active fire observations. Logistic classifiers have also been used in many burned area detection studies (Koutsias and Karteris, 2000; Fraser et al., 2003; Bastarrika et al., 2011).

6.4.1 Uncertainty characterisation

The algorithm proposed highlights some of the challenges for uncertainty quantification in burned area algorithms. A principal challenge is the propagation of uncertainty in the reflectance measurements to ΔVI. After this, the transformation through the sigmoid can produce complex posterior distributions for $\Pr(\mathrm{burned} \mid w, d)$. The challenges of uncertainty propagation for such an algorithm and the computation of the uncertainty quantity $\mathrm{P_b}$ is now examined. As previously discussed, a suitable PDF for the reflectance observations R is provided by the multivariate normal distribution:

$$R_{\mathrm{obs}}(\Omega, \Omega', \lambda) = R_{\mathrm{True}}(\Omega, \Omega', \lambda) + \mathcal{N}(\mathbf{0}, \mathbf{C}_{\mathrm{obs}}(\lambda)) \tag{6.18}$$

where R_{True} is the true but unknown reflectance.

To consider the importance of the pre-processing stages of the algorithm (i.e. temporal sampling and BRDF-effects) a BRDF correction scheme is also implemented and UQ for this pre-processing now described. Semi-empirical models for the BRDF of coarse-resolution satellite reflectance measurements are now well developed and have been previously used within BA algorithms (Roujean et al., 1992; Roy et al., 2005a). These models account for the angular dependence of the reflectance measurements as a linear combination of three scattering kernels: an angular invariant isotropic kernel and two angular dependent kernels which represent idealised scattering processes in volumetric media; and the shadowing environment within a simple crown canopy. To provide sufficient angular sampling to perform the BRDF inversion, observations are required from multiple acquisitions. For polar-orbiting sensors such as MODIS, this means using observations collected typically over a period of 16 days (Schaaf et al., 2002). Given this requirement, two separate pre- and post-fire BRDF inversions are performed from the observations collected in the pre-fire and post-fire compositing windows, t_- and t_+ respectively. As the kernels represent a linear model, uncertainty propagation is straightforward, and the posterior covariance matrix C_f is provided from:

$$\mathbf{C}_f = \left(\mathbf{K}^T {\mathbf{C}_{\text{obs}}}^{-1} \mathbf{K}\right)^{-1} \tag{6.19}$$

where $\mathbf{K}$ is a matrix containing the BRDF kernels of dimension $[3, N]$. $\mathbf{C}_f$ will respond to the quality of the sampling provided by the available observations. For example, the variances in $\mathbf{C}_f$ will increase as the sampling (both in terms of the number of observations and angular sampling) degrades (Lucht and Lewis, 2000). $\mathbf{C}_f$, therefore, provides the crucial link between the observations of directional reflectance and the quality of the pre-fire and post-fire estimates of the land surface — and therefore encodes the uncertainties for the pre-processing stage within the algorithm.

The BRDF correction provides a nadir BRDF adjusted reflectance (NBAR) $\rho_{\text{iso}}(\lambda)$ for each compositing window, as well as an uncertainty in the NBAR reflectance $\sigma^2_{\text{iso}}(\lambda)$. To make the reflectance signal more amenable to discriminating burned areas, the NBAR composites are then mapped to the VI. Consider that the VI is a non-linear function of $\rho_{\text{iso}}(\lambda)$:

$$\text{VI} = f[\rho_{\text{iso}}(\lambda)] \tag{6.20}$$

Given that the mapping through $f[\cdot]$ does not represent a linear transformation of the reflectance, the distribution of $\Pr(\text{VI})$ will not be normally distributed. However, if it can be accepted that a normal approximation is sufficient, the uncertainty can be approximated through a first order Taylor

expansion of f. A Taylor expansion can then be used to approximate σ^2_{VI} as:

$$\sigma^2_{\text{VI}} \approx \sigma^2_a \left(\frac{\delta f[a,b]}{\delta a} \right)^2 + \sigma^2_b \left(\frac{\delta f[a,b]}{\delta b} \right)^2 \tag{6.21}$$

where $a = \rho_{1200} - \rho_{2100}$ and $b = \rho_{1200} + \rho_{2100}$. σ^2_a and σ^2_b are similarly calculated from the propagation rules as $\sigma^2_a = \sigma^2_b = \sigma^2_{\text{iso}}(\lambda = 1200) + \sigma^2_{\text{iso}}(\lambda = 2100)$ with the assumption that the uncertainties in each reflectance channel are uncorrelated.

The algorithm uses the pre-to-post change signal $\Delta\text{VI} = \text{VI}_{\text{pre}} - \text{VI}_{\text{post}}$ to perform the classification. As this is a linear mapping, and the pre- and post-fire measurements can be considered independent, the propagation is straightforward:

$$\sigma^2_{\Delta\text{VI}} = {\sigma_{\text{VI-pre}}}^2 + {\sigma_{\text{VI-post}}}^2 \tag{6.22}$$

The representation of uncertainty in the classification algorithm is now considered. As specified in equation 6.15, the classifiers maps ΔVI to a likelihood statement that the pixel is burned $\text{burned} \in [0, 1]$. The offset in ΔVI, w_0 specifies the location at which $\text{burned} > 0.5$, and therefore indicates that the pixel is considered more likely to be burned than unburned. w_1 specifies the rate at which values of ΔVI are mapped to greater probabilities, with larger values of w_1 providing a sharper distinction in the mapping.

Typically within BA algorithms some parameters are considered known *a priori* — for example the global thresholds present in many algorithms (Giglio et al., 2008; Alonso-Canas and Chuvieco, 2015). Other parameters are instead learnt locally based on a sampling of burned pixels provided by active fire observations. In either case, the uncertainty in the classification algorithm can be encoded by considering w to be presented by a PDF $\Pr(w)$. The distribution $\Pr(w)$ is specified as an multivariate normal distribution $\mathcal{N}(w_\mu, \mathbf{C}_w)$. The covariance matrix $\mathbf{C}_w$ therefore encodes the degree of confidence placed in any two values of w, with the most likely classification specified by w_μ. The mapping through the classification sigmoid $g[\cdot]$ then involves the inner product of the two model parameters with the observations as $d = [1, \Delta\text{VI}]$. Consider that $a = w^T d$ produces a scalar PDF $\Pr(a)$. As the inner product of two multivariate normal distributions, $\Pr(a)$ is not itself necessarily normally distributed. However, following the uncertainty propagation laws, an approximation to the uncertainty in the inner product can be produced as:

$$\sigma^2_a \approx (w_0 \cdot \Delta\text{VI})^2 \cdot \left[\left(\frac{\sigma^2_{\Delta\text{VI}}}{\Delta\text{VI}} \right)^2 + \left(\frac{\sigma^2_{w_0}}{w_0} \right)^2 \right] + (w_1 \cdot \Delta\text{VI})^2 \cdot \left[\left(\frac{\sigma^2_{\Delta\text{VI}}}{\Delta\text{VI}} \right)^2 + \left(\frac{\sigma^2_{w_1}}{w_1} \right)^2 \right] \tag{6.23}$$

Given that distributions are defined for both the model parameters and the observations, it is possible to define the posterior probability density function of $\Pr(\text{burned} \mid w, \Delta\text{VI})$. As discussed

above, as $g[\cdot]$ is a function of the inner product $a = w^T d$, the posterior can be expressed in terms of a only as $\Pr(\text{burned} \mid a)$. Given that $\Pr(a)$ is approximated as a normal distribution, and $\Pr(a \mid \text{burned})$ is simply the sigmoid function $g[a]$ the analytic form of the posterior is:

$$\Pr(\text{burned} \mid a, \mu_a, \sigma_a^2) = \frac{1}{1 + \exp(-a)} \cdot \frac{1}{\sqrt{2\pi\sigma_a^2}} \exp\left(-\frac{1}{2}\frac{[a - \mu_a]^2}{\sigma_a^2}\right) \tag{6.24}$$

As outlined in section 6.2, the posterior distribution $\Pr(\text{burned} \mid a)$ can be integrated to a probability of fire occurrence $\mathrm{P_b}$ by marginalising the uncertainty in the observations and algorithm parameters against the algorithm likelihood function:

$$\mathrm{P_b} = \int_{a \in A} \Pr(\text{burned} \mid a) \Pr(a) \, \mathrm{d}a \tag{6.25}$$

Computing $\mathrm{P_b}$ therefore requires evaluating an one-dimensional integral over a. Linking the components of the algorithm via the uncertainty characterisation at each step provides a traceability chain — which provides a diagrammatic description of the uncertainty characterisation through the algorithm (Scanlon et al., 2015; Nightingale et al., 2018). Such diagrams help to ensure the consistency of the uncertainty methods and are also of use for users to better understand the algorithm (Nightingale et al., 2019). A traceability chain of each step and representation of uncertainty is presented in figure 6.5.

6.4.2 Experimental setup

To demonstrate the importance of uncertainty quantification to estimates of burned area, three scenarios are advanced. As a frame of reference, we aim to estimate burned area for a region of northern Australia for a temporal interval spanning 23/04/2008—28/04/2008. Observations of surface reflectance from the MODIS Aqua and Terra satellites (MOD09GA/MYD09GA) are used. To perform the BRDF correction, observations for a period covering 16 days before and after the temporal interval were collected. Observations which are corrupted by cloud or cloud shadow are removed with the quality assurance (QA) data sets provided, as well as those covering non-land surfaces. Occasionally observations affected by residual cloud contamination pass through the QA and may lead to a poor solution of the BRDF correction. To prevent this, the quality of each inversion was evaluated. The residual between the predicted surface reflectance and the observed reflectance $e_\lambda = \rho_{\text{pred}}(\lambda, \Omega, \Omega') - \rho_{\text{obs}}(\lambda, \Omega, \Omega')$ provides information on the quality of the prediction. When e_λ^2 is large relative to the uncertainty in the observations, as provided by $\mathbf{C}_{\text{obs}}$, it is likely that the observation contains residual cloud or cloud shadow effects. Observations where e_λ^2 was larger than $5\sigma_\lambda^2$ were excluded and the BRDF correction re-computed (the per-channel uncertainties σ_λ for MOD09 are provided in table 6.2).

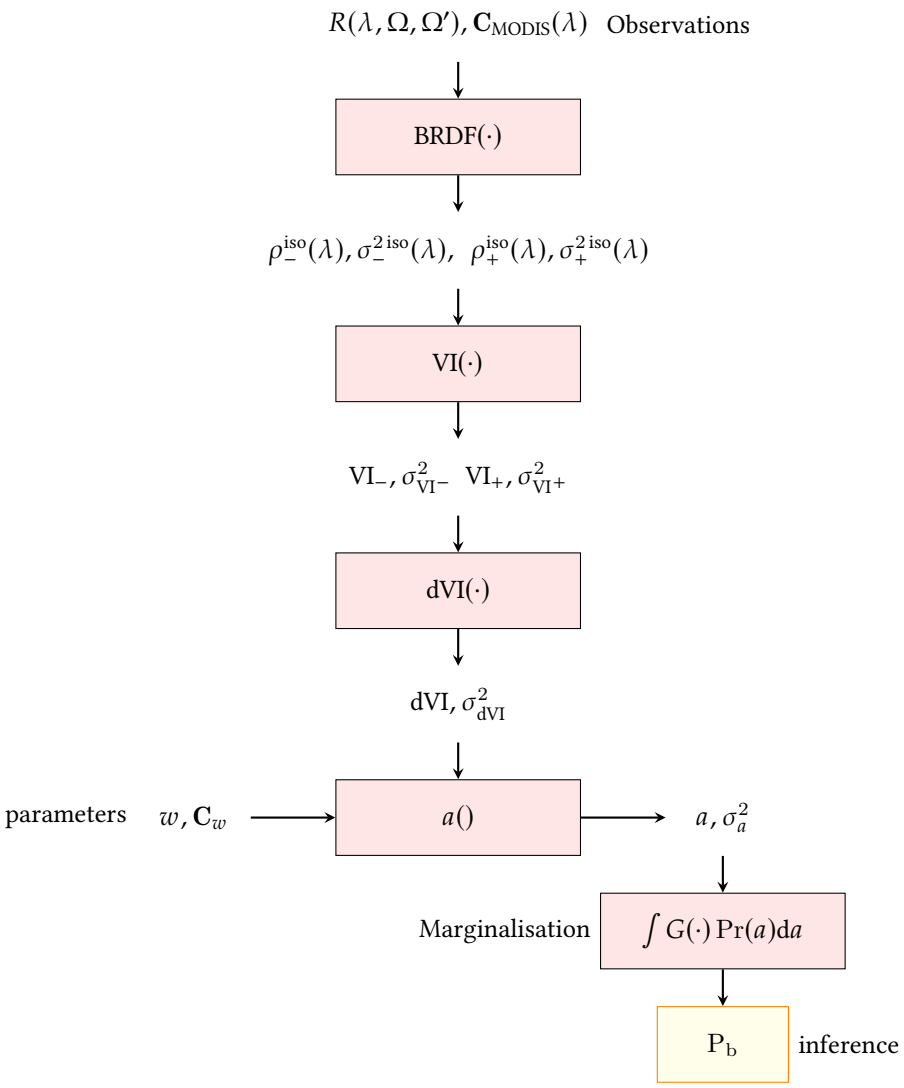


Figure 6.5. Uncertainty traceability chain for the demonstration algorithm. Each transformation within the algorithm is recorded and output quantities with associated uncertainties are declared. Such traceability chains provide a tool to help ensure consistent estimation of uncertainties.

Table 6.2. Approximate per-pixel uncertainty for MODIS channels. Per-pixel uncertainties are not available for the MODIS Collection 6 surface reflectance products so estimated uncertainties are used (see Roy et al. (2005a)).

MODIS channel	5	7
central wavelength λ	1240nm	2130nm
σ^2_λ	0.013	0.006

Table 6.3. Experiments considered.

	Name	Description
1	Standard experiment	Reflectance observations from both Aqua and Terra. Reflectance channel uncertainties as specified in table 6.2.
2	Larger observational uncertainties	Reflectance observations from both Aqua and Terra. Reflectance channel uncertainties are increased by a factor of 3 to those specified in table 6.2.
3	Only Aqua	Reflectance observations from only the Aqua instrument. Reflectance channel uncertainties as specified in table 6.2.

Given that the sampling of observations is crucial for the estimate of the pre- and post-fire NBAR reflectance, the impact of the temporal sampling of the observations on uncertainties was examined. The uncertainty in the BRDF correction is partially determined by the number of observations available N, reducing proportional to $1/\sqrt{N}$ (Lucht and Lewis, 2000). To consider how $\mathrm{P_b}$ changes with observational sampling, the algorithm was also implemented with observations only collected by the Aqua instrument. This reduces the effective number of samples in half and also alters the angular sampling of the land surface which determines the uncertainty of the NBAR product through the BRDF kernel models (Lucht and Lewis, 2000).

Separately to the sampling, consider that the observations provided by the EO sensor are now less precise. There may be many different sources of errors in the surface reflectance observations. Older satellite instruments may have greater noise due to higher electrical noise or a less granular quantization of photon counts. Larger systematic errors may also be present in the observations as a result of poorer atmospheric correction (e.g. from more limited spectral sampling), larger gridding artefacts or less precise geo-registration of observations. To consider the impact of these uncertainties on $\mathrm{P_b}$, the algorithm was also implemented with an inflation of the theoretical uncertainties of the MODIS channels by a factor of 3. Table 6.3 summarises the three experiments.

6.4.3 Aggregation of per-pixel uncertainties

Following the estimation of pixel-level uncertainties, synthetic lower resolution CMG products were generated. To propagate the uncertainty from the pixel to grid-scale, the uncertainty aggregation framework outlined in section 6.1 was used. Pixels which are unmapped by the algorithm — which occur when there are insufficient samples to perform both BRDF inversions — are excluded from the aggregation. Aggregated CMG products were created for the three scenarios outlined above.

The independence assumption will lead to an overestimate of the mean and underestimate of the variance in areas of relatively uniform $\mathrm{P_b}$ values. This is because uncertainties at the typical resolutions provided by EO sensors will be spatially and temporally correlated (Bellprat et al., 2017). For example, for unburned pixels, adjacent $\mathrm{P_b}$ values will be highly correlated with each other. Only within burn scars and at the perimeters will a higher level of independence in adjacent $\mathrm{P_b}$ values be expected. In unburned areas, the effective number of samples n_e in the grid cell g is considerably less than the number of EO pixels. As a result, the inclusion of many highly correlated values of $\mathrm{P_b}$ will reduce to the variance of the aggregated burned area in the grid cell due to the assumption of independence.

The specification of the spatial correlation of the $\mathrm{P_b}$ will be dependent upon both the clustered nature of fire in the land surface and the spatial correlation of systematic errors in the observations and algorithm. This makes it difficult to parameterise the spatial auto-correlation of the uncertainties — for example via a covariance matrix. Instead, an attempt can be made to infer the effective number of samples n_e within the grid cell to improve the estimates of $\bar{n_b}$ and σ_b^2. For example, in areas of high spatial auto-correlation in $\mathrm{P_b}$ values, it is expected that $n_e < N$. Practically, reducing N increases the variance σ_b^2 in the grid cell.

A method to estimate n_e is to consider reducing the number of pixels within an area as a function of the variance in $\mathrm{P_b}$ values. A variance preserving scheme based on a Quadtree compression of the $\mathrm{P_b}$ uncertainty values is now proposed. If $\mathrm{P_b}$ values within a spatial extent S are similar as defined by a threshold in the spatial variance of $\mathrm{P_b}$, the extent is considered effectively one larger segment with one probability of burn $\mathrm{P_b}$. The Quadtree scheme therefore prunes areas of low spatial variance at 500m into a reduced number of larger segments and therefore approximates the spatial auto-correlation of uncertainties. This reduces the number of samples and increases the variance σ_b^2. The aggregation estimates from both the independent 500m pixel and effective segmentation scheme were then compared. An example of the two schemes is shown in figure 6.6.

6.4.4 Results

Figure 6.7 shows the influence of the three scenarios on the retrieved uncertainties $\mathrm{P_b}$. Panel a) shows $\mathrm{P_b}$ under the ideal scenario. $\mathrm{P_b}$ matches at least visually well to the official MODIS burned area product (shown in panel d). Large values of $\mathrm{P_b}$ (i.e. approaching unity) are found within the burn scars. Unburned areas in the bottom of the extent show $\mathrm{P_b}$ less than 0.05. At the perimeters of the burn scars identified by MCD64 $\mathrm{P_b}$ are closer to 0.5 — indicating that the algorithm has little confidence if these pixels are burned or unburned. Panel b) shows the influence of the experiment with a tripling of

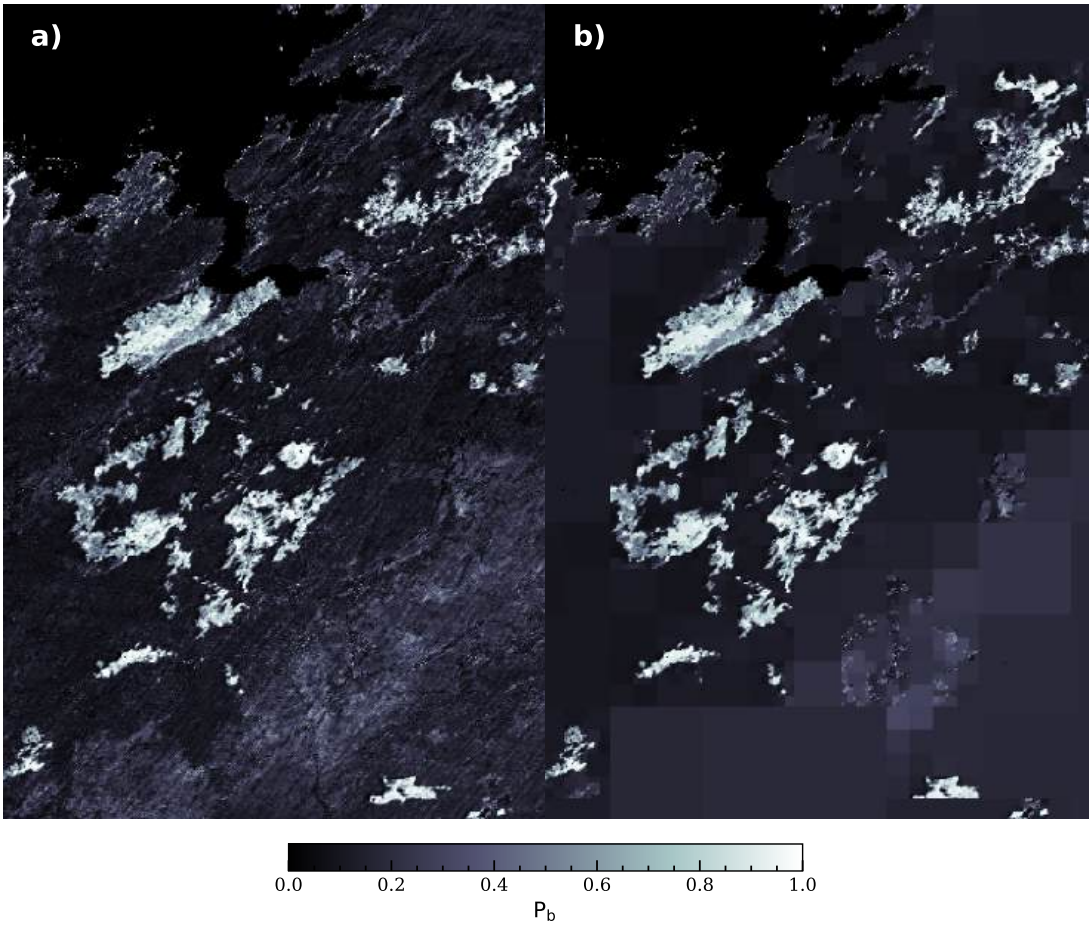


Figure 6.6. Two pixel uncertainty gridding schemes. a) independent pixel assumption assumes no auto-correlation in uncertainties for 500m pixels. b) effective pixel scheme (i.e. segmentation scheme) aggregates areas of low variance in P_b values into segments at a lower resolution than 500m.

the observation uncertainties. Pixels that were either high or low P_b in panel a) have been moderated towards 0.5. For example, the burned pixels have P_b values closer to 0.7–0.8 indicating less confidence in the detection. Finally panel c) shows the influence of running the retrieval with observations only from the Aqua instrument. Large areas are also now unmapped (shown in grey) where there are less than the required number of observations to perform the BRDF inversion. There is also some visible striping in P_b values as a response to changes in sampling.

Figure 6.8 shows how the distribution of P_b is affected for the three scenarios. P_b associated with unobserved areas (for example in the only Aqua scenario) are not included in the distributions. Most apparent is the relative similarity in the distributions for the ideal case and the 'Only Aqua' scenarios. The mode of P_b for both scenarios is comparable (0.1 for the idea case and 0.12 for the

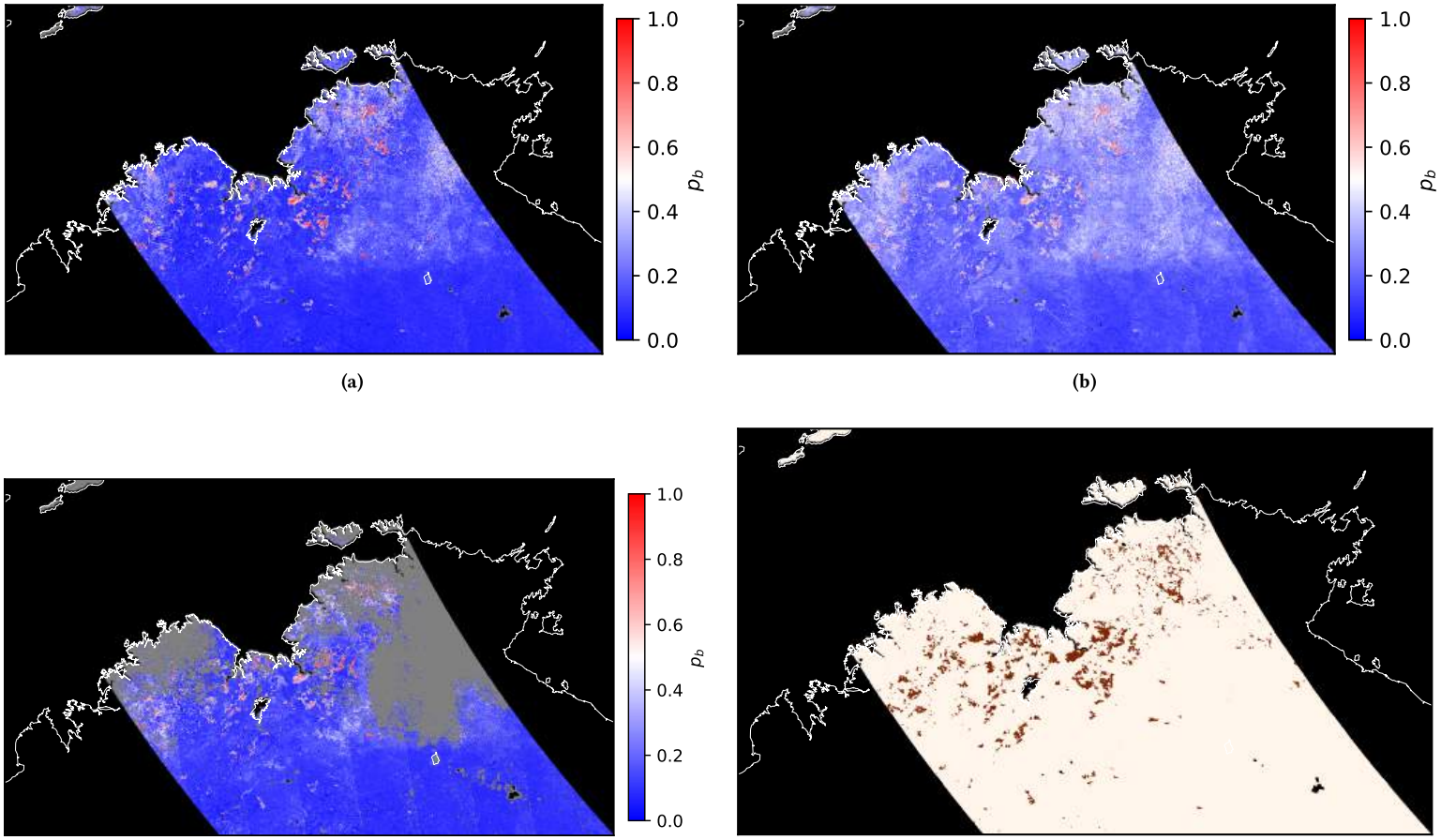


Figure 6.7. Pixel level uncertainties P_b for the three scenarios. a) ideal case b) scenario with a tripling of the reflectance uncertainties c) scenario with reduced sampling with observations only from the Aqua instrument. d) MCD64 burned area for the mapped period.

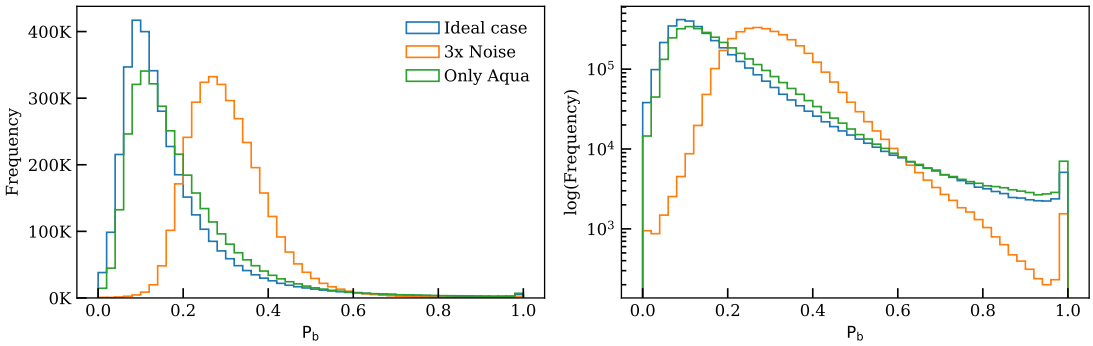


Figure 6.8. Frequency distribution of P_b for the three scenarios. Left) frequency distribution of P_b for the three scenarios. Right) logarithmic frequency distribution shows change in $P_b > 0.6$ for the three scenarios.

only Aqua scenario). The right panel provides a logarithmic plot of the frequency distributions so that the distributions for P_b values above 0.6 are more easily compared. The distribution for only Aqua shows a small shift compared to the ideal case. There are more mid-range values of P_b (0.2-0.5) for only Aqua and slightly fewer higher $P_b > 0.8$ values. This corresponds to the concept above of the P_b values being drawn away from the extremes of the distribution (zero and unity) towards 0.5 with increased uncertainty. The frequency distribution for the tripling of observation uncertainties (the '3x Noise' scenario) is considerably different from the other two distributions. First, the mode of the distribution is shifted to 0.28. In the right panel, the effect on high $P_b > 0.8$ frequencies can be observed most clearly. The increase in observational uncertainties reduces the number of $P_b > 0.8$ pixels away from the ideal case far more than the only Aqua scenario. The number of near-certain fires (where $P_b > 0.95$) is reduced by 75% from the ideal case. This again demonstrates that P_b provides a versatile signal which adapts to changes in uncertainty in the mapping problem.

Figure 6.9 shows some of the additional information that P_b can bring to the problem. The figure shows a subset within the tile with a mixture of large and small fires. It can be perceived that the MODIS product (MCD64) is unable to detect some of the smaller burns, even in the cases of active fire observations (shown in panel c). P_b, on the other hand, shows some sensitivity to the burning, with values expressing high uncertainty $P_b = 0.5$ or marginal evidence for burning (P_b probabilities around 0.6–0.7).

The two-pixel aggregation schemes are now considered. Figure 6.10 shows the 0.25° aggregated statistics for the independent pixels scheme for the three scenarios. In the ideal case, areas of definite fire activity — where the algorithm can resolve burn scars — results in higher mean estimates (figure 6.10a) and lower variance estimates (figure 6.10b). Figure 6.10c and d show the mean and variance

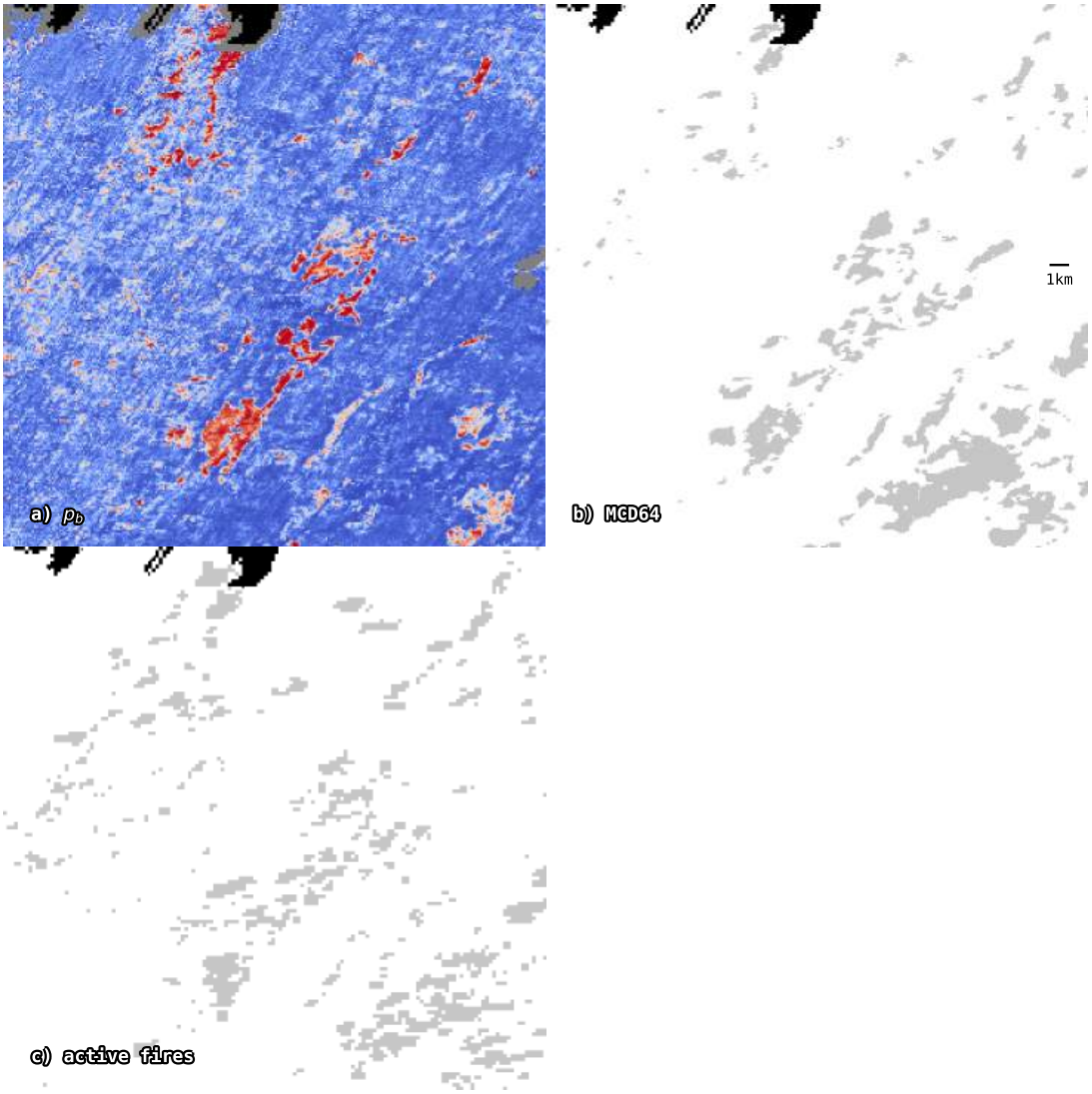


Figure 6.9. Information in P_b. Subset of an area from N. Australia with a mixture of large and small fires. a) P_b estimate, red pixels correspond to high values of P_b, blue to low values and white to values around 0.5. b) MODIS Collection 6 burned area product (MCD64) for the same temporal and spatial extent. c) Active fire detections from the MODIS Aqua and Terra satellites (MYD14/MOD14) collected over the mapping period.

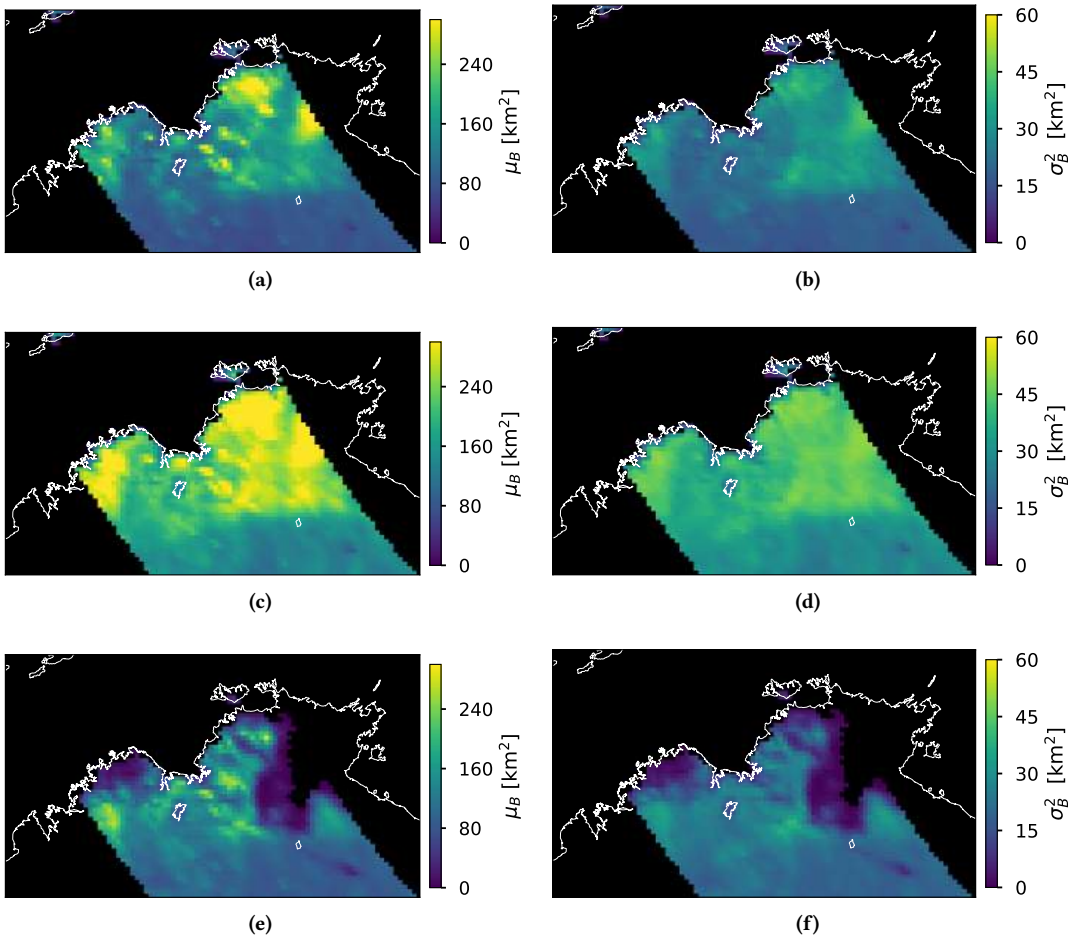


Figure 6.10. Independent pixel scheme aggregation 0.25° mean burned area μ_B [km^2] and variance μ_B^2 [km^2] for the three scenarios. a-b) ideal case c-d) scenario with a tripling of the reflectance uncertainties e-f) scenario with reduced sampling with observations only from the Aqua instrument.

in burned area km^2 for the scenario of inflated observational uncertainties. The aggregation process leads to higher estimates of the mean and variance for areas with higher uncertainties at the pixel level. Finally, figure 6.10 (e & f) show the aggregated estimates for the scenario of running the algorithm with just the Aqua instrument. It can be seen that larger areas now have both low mean BA km^2 and correspondingly low variances in BA km^2. These correspond with poorly observed grid cells. Such information about the fraction of observed pixels would necessarily have to be provided to users to distinguish these poorly observed grid cells.

Figure 6.11 shows the 0.25° aggregated statistics for the effective segmentation-based aggregation scheme for the three scenarios. These show a larger dynamic range relative to the independent pixel

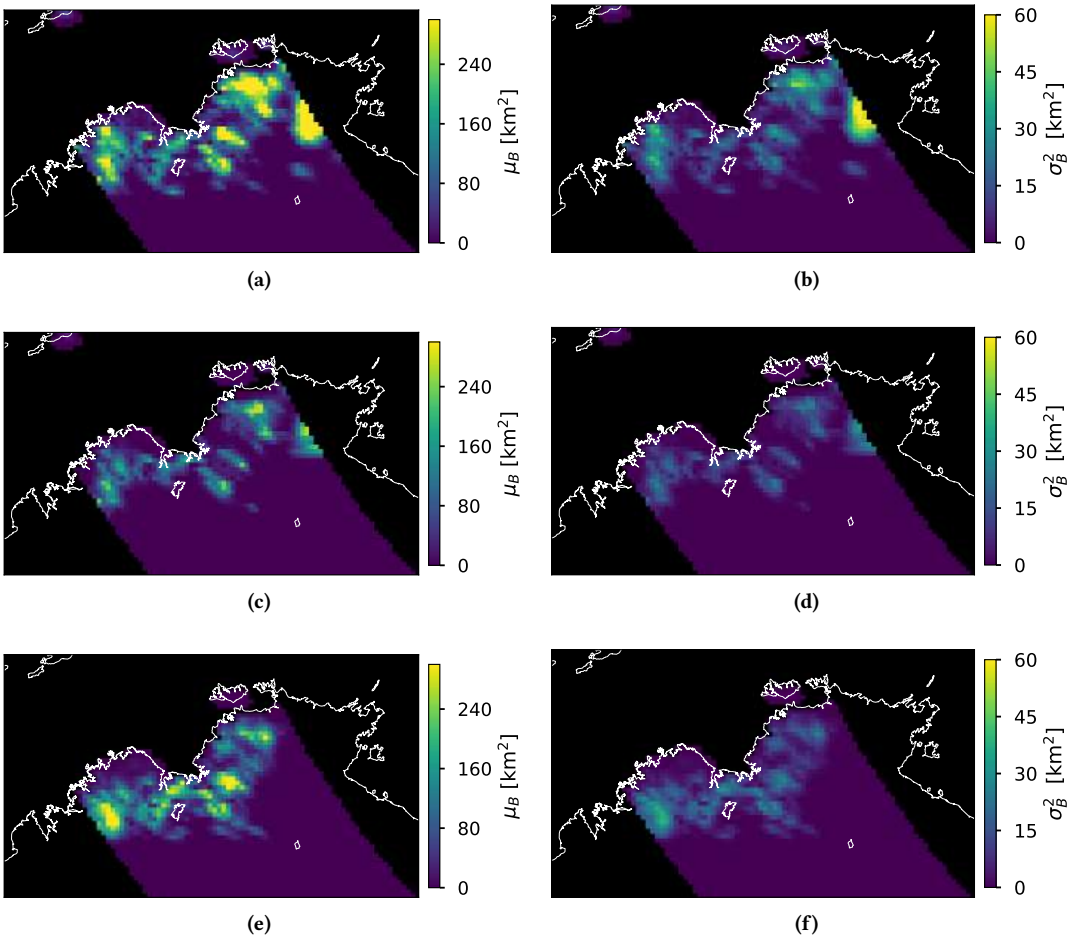


Figure 6.11. Effective pixel scheme aggregation 0.25° mean burned area μ_B km^2 (left column) and variance μ_B^2 km^2 (right column) for the three scenarios. a-b) ideal case c-d) scenario with a tripling of the reflectance uncertainties e-f) scenario with reduced sampling with observations only from the Aqua instrument.

scheme. Areas of low pixel P_b values now result in more realistic mean and variance statistics at the grid-scale compared to the independent aggregation scheme. Visually it can be seen that the grid cells with the highest μ_B are now much more concentrated to areas that actually have burning at the pixel scale (i.e. as provided by MCD64). Areas with little evidence at the pixel scale now have μ_B values much closer to 0 as should be expected when the grid cell is primarily composed of pixels with P_b values close to 0. The grid cell variances μ_B^2 also align much better with areas of actual fire activity or those where the pixel uncertainty is largest (i.e the eastern extent where P_b is around 0.5).

6.5 Validation methodology for uncertainties

An important consideration is the validation of any uncertainty scheme. Any uncertainty characterisation scheme will often be approximate for several reasons. First, typical linearisation schemes as described for the algorithm above provide only approximate uncertainty characterisations (as was demonstrated in section 6.3). The non-linear nature of many algorithms will mean that approximate methods will need to be developed for the next generation of burned area products. A validation scheme for uncertainties at both the pixel and grid-scale is now designed and evaluated.

6.5.1 Validation of pixel uncertainties

To validate pixel-level uncertainties, the expected value of the integral in equation 6.8 can be approximated with Monte Carlo methods. To do this, K realisations of the algorithm are run, each with a unique sampling of both the initial reflectance observations and classifier parameters. The estimate of the true uncertainty $\hat{\mathrm{P}_\mathrm{b}}$ is then computed by the Monte Carlo expectation of the integral in equation 6.8:

$$\hat{\mathrm{P}_\mathrm{b}} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{G}(w_k,\ R_k(\lambda)) \tag{6.26}$$

The Monte Carlo sampling estimate of $\mathrm{P_b}$ will capture the non-linear features not included with the linearisation and moment propagation methods described above. The quality of the UQ can then be evaluated by considering the difference between the two estimates (for example with a Brier score i.e. Brier (1950)).

6.5.2 Validation of aggregation schemes

For the climate grid cell g, each realisation k of the product $b^{(k)}$ can be converted into a binary burned/unburned map $m^{(k)}$. Typically this will involve choosing a threshold $T \in [0, 1]$ which discriminates burned areas from unburned areas (i.e. $\mathrm{burned}^{(k)} > T$), but this value can also be marginalised. Each mapped product $m^{(k)}$ then provides a different record of the number of burned pixels $n_b^{(k)}$. Each realisation of $n_b^{(k)}$ hence provides samples from the distribution of the cell's burned area km^2. The first and second moments of this distribution can be computed from the estimators:

$$\hat{\mu_B} = \frac{1}{K}\left(\sum_{k=1}^{K} n_b^{(k)}\right) \qquad\qquad \hat{\sigma_B^2} = \frac{1}{K-1}\left(\sum_{k=1}^{K}\left[\ n_b^{(k)} - \hat{\mu_B}\ \right]^2\right) \tag{6.27}$$

To demonstrate the validation methodology, the uncertainty estimates for the ideal scenario described above were validated. Samples of the surface reflectance observations were generated according to

Table 6.4. Validation statistics for the mean μ_B km^2 and variance σ_B^2 km^2 of the two aggregation schemes.

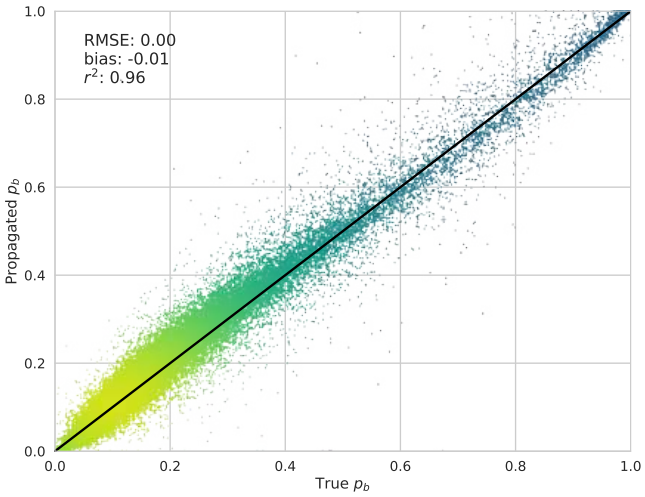
	R^2		RMSE		Bias	
	μ_B	σ_B^2	μ_B	σ_B^2	μ_B	σ_B^2
Independent pixels	0.75	0.51	4.59	0.95	90.39	18.94
Effective segmentation-based	0.86	0.82	3.88	0.54	51.70	7.18

the multivariate normal distribution specified in equation 6.18. The parameters of the classification algorithm are likewise generated from the distribution $\mathcal{N}(w_\mu, \mathbf{C}_w)$. The algorithm is then run on each realisation k of the observations and classifier and the burned/unburned map m produced according to the classification threshold of $\mathrm{burned} > 0.5$, which represents the decision boundary of the logistic classifier. The pixel-level estimate of $\hat{\mathrm{P}_\mathrm{b}}$ is then produced from the sample mean of the k realisations — which is the equivalent marginalisation of the uncertainties for an ensemble. The agreement between P_b and $\hat{\mathrm{P}_\mathrm{b}}$, therefore, indicates the quality of the uncertainty characterisation for correctly propagating the uncertainty through the algorithm. To produce aggregation statistics, each burned/unburned map m is aggregated to the same grid as the aggregation scheme via a simple summation. This provides for each grid cell k samples from the distribution of burned area km^2. The sample mean and variance of the grid cell are then calculated from these samples. To consider the quality of both the pixel P_b and aggregation statistics, the relative mean bias, root-mean-square error (RMSE) and coefficient of determination R^2 between the estimates and the validation were computed.

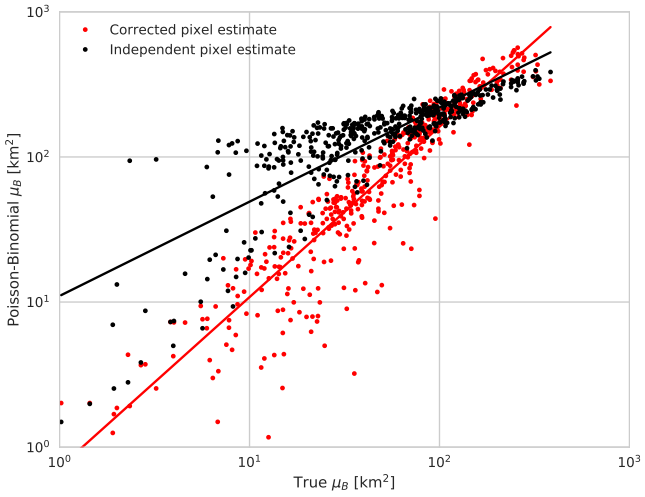
Figure 6.12 shows that the agreement between the uncertainty characterisation and the Monte-Carlo based uncertainty propagation for P_b and the aggregated statistics. The true and propagated values of P_b are in agreement ($R^2 = 0.96$) with a bias (-0.01) and overall RMSE (10^{-5}). Figure 6.12 shows the validation of the aggregation mean and variance for the two aggregation schemes. Principally, it can be seen that the effective segmentation-based scheme reduces the bias in the mean and variance for grid cells with low burned areas relative to the independent pixel scheme (see Table 6.4). The bias of the mean aggregation is reduced from 90.39 km^2 to 51.70 km^2 with the effective segmentation scheme. The bias in the variance is approximately halved from 18.94 to 7.18 km^2. The RMSE is also reduced for both the aggregated mean and variance for the effective segmentation scheme relative to the independent pixel scheme. The coefficient of determination R^2 is higher for the effective scheme also.

6.6 Summary

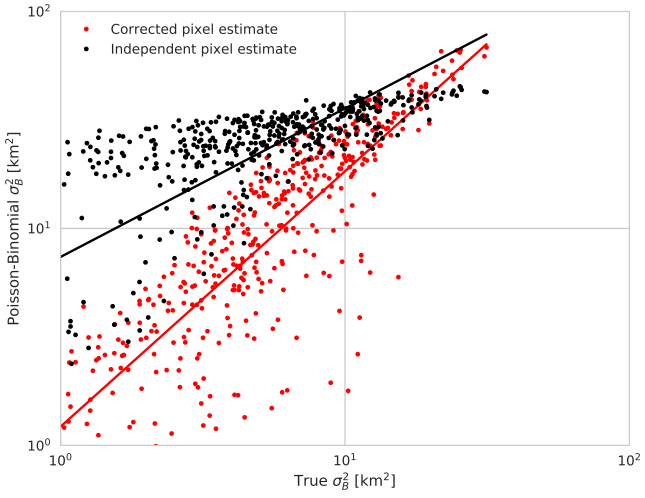
Uncertainty quantification (UQ) is a necessary and fundamental part of scientific measurement. A growing focus on the production of CDRs from satellite remote sensing has highlighted the need for better UQ of burned area products. This is mirrored in the GCOS requirement for burned area products. Until recently burned area products have not provided quantitative estimates of uncertainty at either the pixel or CMG scales. This section has explored the requirements for UQ of burned area products and presented two approaches which meet GUM guidelines for UQ. The section has worked to address the aim stated in section 1.2 to "develop uncertainty quantification (UQ) methods for burned area products and algorithms following metrological best practice".



(a)



(b)



(c)

Figure 6.12. Validation of a) pixel uncertainties P_b, b) aggregated burned area mean km^2, c) aggregated burned area variance km^2.

Part III

Prototyping an algorithm for consistent burned area mapping

Chapter Seven

Prototyping an algorithm for monitoring fire-affected areas from multiple sensors

7.1 Introduction

The previous section developed approaches for uncertainty quantification (UQ) in burned area climate data records (CDRs). Uncertainty quantification was detailed so thoroughly because it allows users to better ascertain the value of the products created with remote sensing data. But the UQ methods developed previously also open up exciting avenues for deriving consistent long-term data records of burned area from multiple sensors. This chapter details the prototyping of a new algorithm for this purpose. Chapter 8 then demonstrates the algorithm on a true multi-sensor system using observations from Suomi-NPP VIIRS, Sentinel-3 OLCI and MODIS.

7.2 The sensor record

Figure 7.1 shows the availability of optical instruments suitable for producing a global long-term data record of burned area. Proceeding back to the 1980s are the moderate resolution Thematic Mapper (TM) and Enhanced Thematic Mapper Plus (ETM+) instruments aboard the Landsat platforms (4-7). Since the launch of Landsat-1 in 1972, the seven successful Landsat missions have provided in excess of 6 million images (Markham et al., 2018). With Landsat-8, launched in 2013, the optical and thermal capabilities of the Landsat sensors have been further upgraded. The Operational Land Imager (OLI) provides a heritage channel configuration to match previous TM/ETM+ sensors but with improved signal-to-noise characteristics (Markham et al., 2018; Irons et al., 2012).

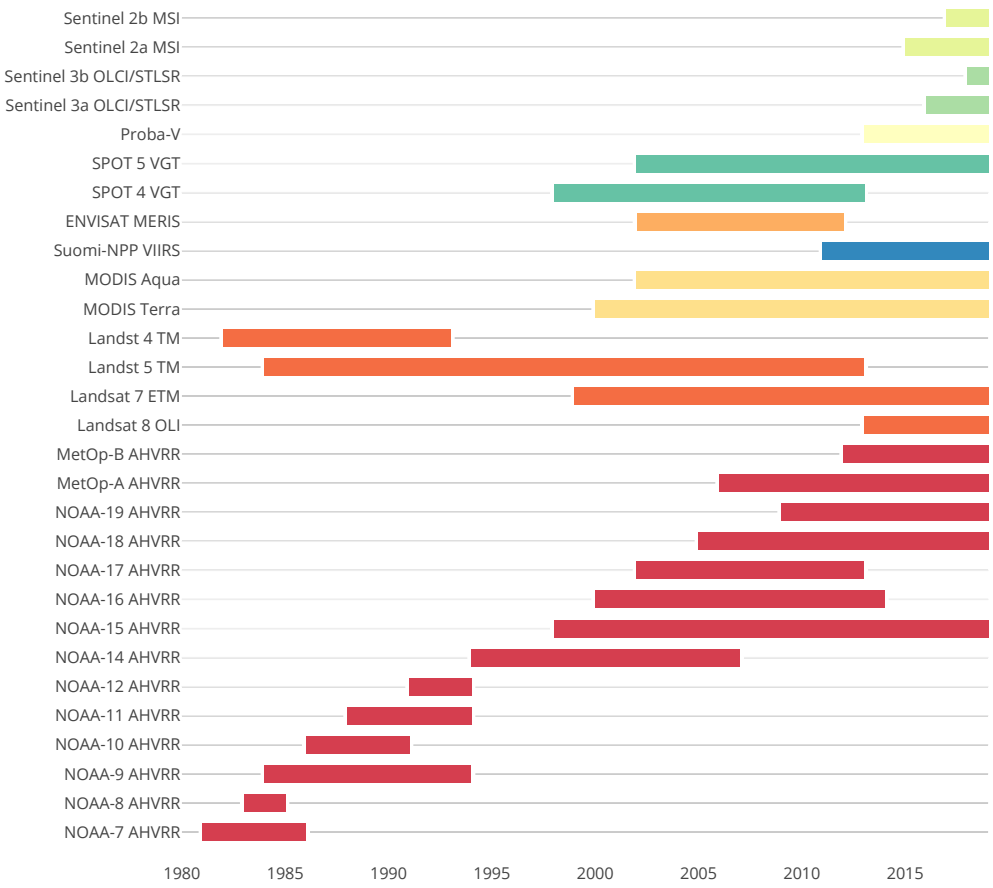


Figure 7.1. Satellite instruments suitable for the production of a long-term data record (1981–present) of burned area.

The Advanced Very High Resolution Radiometer (AVHRR) instruments aboard multiple satellites proceed back to NOAA-7 AVHRR (1981-1986) and up to the MetOp-A/B satellites carrying the latest generation of these instruments. The AVHRR instruments have a nominal pixel resolution of 1.1km and limited spectral sampling (see figure 7.2). Due to limitations of onboard storage and downlink capacity, the full global AVHRR record is only available presently at 4km resolution. For the newer METOP AVHRR sensors, 1.1km resolution observations are available globally (Xiong et al., 2018). An effort has been put into producing a CDR of surface reflectance from the AVHRR sensors from NOAA-7 to the present MetOp AVHRR sensors, providing a global plus 30-year record of surface reflectance (Franch et al., 2017). These two instrumental records (AVHRR and Landsat TM/ETM+/OLI) provide approaching 40-years of coarse and moderate-resolution measurements.

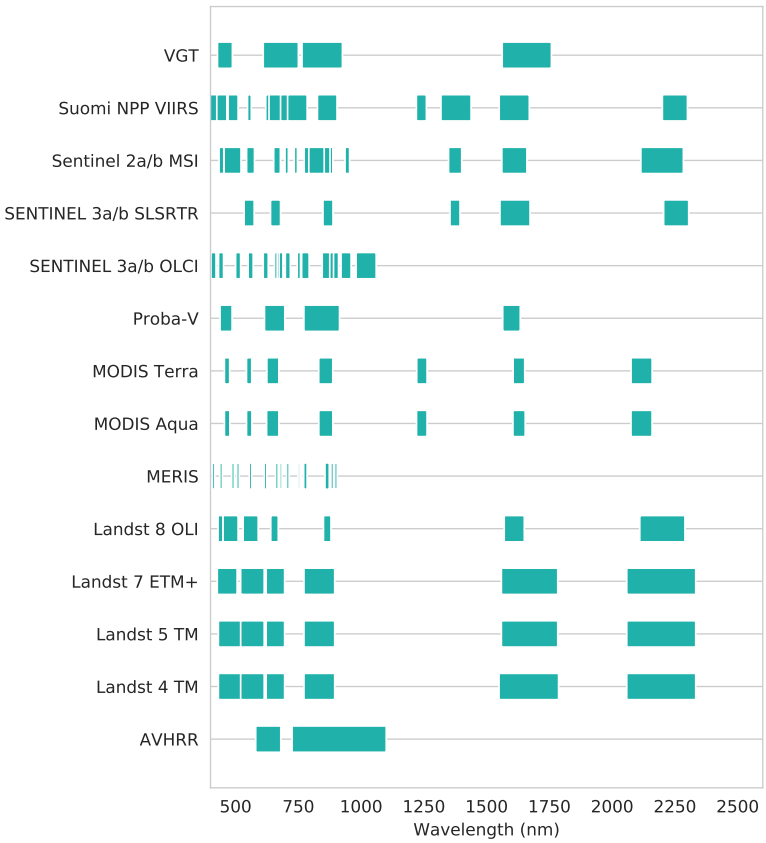


Figure 7.2. Spectral channels in the optical region (400–2600nm) available on the sensors suitable for the production of a long-term data record (1981–present) of burned area.

The MODIS instruments aboard the NASA Aqua and Terra satellites have provided almost 20-years of daily observations at 250m-1km. The twin instruments collect measurements across 36 spectral channels ranging in wavelengths from $0.4\,\mu m$ to $14.4\,\mu m$. Given the moderate resolution and daily acquisition, MODIS has been foundational to the development of most present burned area CDRs (Roy et al., 2005a; Giglio et al., 2018; Chuvieco et al., 2018). MODIS has also been used extensively for the production of active fire data products (Justice et al., 2002; Giglio et al., 2003, 2016).

In an attempt to continue the heritage of the MODIS programme, the Visible Infrared Imaging Radiometer Suite (VIIRS) instrument was launched in 2011. VIIRS provides continuity to the MODIS record through a similar channel configuration in the optical domain (Murphy et al., 2001). The sensor has several improvements beyond the MODIS sensors. A wider swath (3000km) than MODIS (2330km) provides more frequent acquisitions from only one sensor, although fewer acquisitions overall than

the two MODIS instruments. VIIRS also has a considerably lower cross-track expansion of the pixel point spread function (Hillger et al., 2013; Lee et al., 2006).

The SPOT-VGT sensor was launched onboard the SPOT-4 platform in 1998 and provides observations in four optical channels at approximately 1km spatial resolution (Maisongrande et al., 2004). SPOT-VGT was used to produce the first global burned area estimates for 2000 (Tansey et al., 2004). The launch of PROBA-V (Project for On-Board Autonomy – Vegetation) in 2013 provides continuity with VGT. PROBA-V has four spectral channels configured to provide continuation with SPOT-VGT, but provides observations at a higher spatial resolution of 300m (Sterckx et al., 2014; Dierckx et al., 2014).

The Medium Resolution Imaging Spectrometer (MERIS) flew on the ESA Envisat mission from 2002–2012. The MERIS instrument was configured with fifteen spectral channels, providing a dense spectral sampling exclusively over the visible ($0.39\,\mu m$) to the near-infrared region ($1.40\,\mu m$). A swath width of 1150km provided global coverage at a spatial resolution of 300m every three days (Rast et al., 1999).

To provide continuity with the Envisat record, the ESA Sentinel-3 mission serves to provide both optical and thermal measurements globally at daily intervals (Donlon et al., 2012). The Ocean Land Colour Imager (OLCI) serves to provide continuity with the Envisat MERIS instrument. Twenty-one spectral channels provide narrow sampling across the visible to the near-infrared spectral region at 300m. In addition to OLCI, Sentinel-3 carries the Sea and Land Surface Temperature Radiometer (SLSTR). SLSTR carries six channels for the visible to short-wave infrared spectral regions and five thermal channels. SLSTR also has two dedicated channels for monitoring high-temperature events without saturation which makes them highly suitable for detecting small actively burning and smouldering fires (Wooster et al., 2012).

Sentinel-2A (launched 2015) and Sentinel-2B (launched 2017) now provide medium resolution imaging at previously unmatched spatial, temporal, and spectral detail (Drusch et al., 2012). The two satellites also have local overpass times which are similar to the Landsat satellites, helping to ensure consistency with the Landsat record (Jutz and Milagro-Pérez, 2018). Carrying only one instrument, the Multi Spectral Instrument (MSI) provides 13 optical channels. Three spectrally narrow channels in the so-called 'red-edge' spectral region provide novel information on vegetation state at a spatial resolution of 20m (Clevers and Gitelson, 2013; Fernández-Manso et al., 2016). The spatial resolution of the channels varies between 10m and 60m (Drusch et al., 2012). The 290km swath of the instrument coupled with the two Sentinel-2 satellites provides a global revisit frequency of five days.

7.3 Processing models of multi-sensor CDRs

Given the abundance of satellite observations, what are the options for producing multi-sensor CDRs of burned area? It is possible to broadly consider two processing models of the CDR generation process. These two models can be referred to as the *pre-merging* and *post-merging* models. For pre-merging methods, observation streams from different sensors are first merged to produce a merged surface reflectance and active fire product. One burned area algorithm is then run on these CDRs to produce a consistent multi-sensor CDR of burned area. Alternatively, the *post-merging* model consist of running burned area algorithms on several distinct observational records and then retrospectively merging these. The two approaches are now considered in more detail. As these two models can be more generally applied to many level-3 satellite retrievals, other ECVs are used to contextualise the challenges of the two approaches.

7.3.1 Pre-merging processing model

The first model arguably represents the ideal approach. The computational model is schematically shown in figure 7.3. First observations are collected from each sensor and then merged to produce a consistent surface reflectance or active fire CDR. These initial 'sensor-agnostic' CDRs are then used to produce the burned area CDR. This represents the ideal processing model, primarily because, the maximum amount of information about the land surface is extracted from the observations. The most obvious example of this relevant to burned area detection is the potential improvements to pre- and post-fire sampling as variations in the acquisition time of different sensors will provide a greater chance of cloud-free acquisitions of the surface (Li and Roy, 2017). This improves the chances of observing the surface immediately before and following the fire as observations from different sensors are combined.

There are a few examples where this processing model has been used to produce global products. European sensors have been used in tandem to produce a global albedo and reflectance CDR for 1998-2011 within the Globalbedo project (Muller et al., 2012). The project used observations from SPOT-VGT and MERIS in combination with auxiliary information from the MODIS sensors to produce a multi-sensor reflectance CDR. This involved correcting level-1 data from each sensor to directional broadband surface reflectance and gridding these to the shared MODIS sinusoidal grid. Production of multi-year BRDF adjusted reflectance and albedo products involved the application of additional prior information from the MODIS sensors to constrain the solution using optimal estimation methods (Lewis et al., 2012). Using these optimal estimation methods allowed for uncertainty characterisation

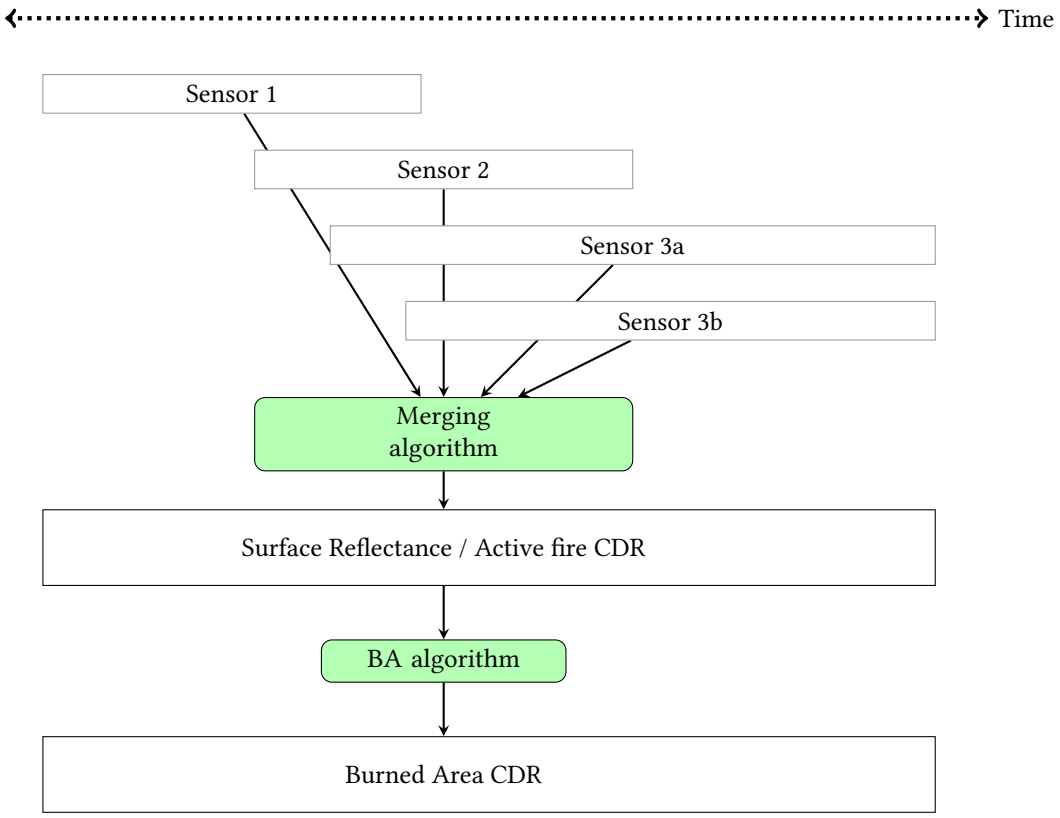


Figure 7.3. Pre-merging model of burned area CDR generation. Observation streams from different sensors are first merged to produce a merged surface reflectance and active fire CDR. One burned area algorithm is then run on these CDRs to produce a consistent multi-sensor CDR of burned area.

of the reflectance CDR following metrological principles. Recently, the development of Sentinel-3 Synergy products provides an example of the possibility of generating such CDRs. The two Sentinel-3 satellites both carry instances of the OLCI and SLSTR instruments. These instruments provide comprehensive coverage of the optical—thermal spectrum, with OLCI providing 21 channels in the visible to near-infrared and SLSTR providing channels in the short-wave infrared. The Sentinel-3 Synergy products, therefore, aim to provide visible to SWIR coverage in one gridded daily reflectance product at 1km (Donlon et al., 2012).

Progress has also been made for moderate-resolution instruments. The combination of Landsat and Sentinel-2A/B sensors provides a considerable opportunity for moderate-resolution terrestrial monitoring at near-daily intervals. Li and Roy (2017) considered the potential revisit intervals of combinations of Sentinel 2 and Landsat 8 in “virtual constellations”. They indicated a global median average revisit interval of 2.9 days — comparable to previous coarse-resolution instruments such

as MERIS (Li and Roy, 2017). The opportunity of a virtual constellation of moderate-resolution instruments has led to the development of the Harmonised Landsat 8 and Sentinel-2 (HLS) project by NASA which aims to produce a surface reflectance record from both sensors (Claverie et al., 2018). Central to the processing is the joint atmospheric correction scheme used on both Landsat 8 and Sentinel-2 which ensures greater consistency in the surface reflectance products. At present, the HLS is in early development and covers around 7.3% of the global terrestrial surface.

Despite this progress, the production of multi-sensor CDRs of surface reflectance or active fire has been challenging. A primary challenge is the differences in spectral channels provided by different sensors. Even superficially equivalent spectral channels (i.e. with equivalent central wavelengths) will have variations in the spectral response function (SRF) of the channels. Trishchenko et al. (2002) simulated the impact of differences in SRF in similar moderate resolution sensors. They found that the variations in SRF led to considerable discrepancies in NIR reflectance as well as vegetation indices between the sensors. Beyond such simulation studies, empirical 'spectral band adjustment factors' have been developed. These typically involve deriving calibration curves for similar spectral channels between different sensors over training sites (Zhang et al., 2018; Lacherade et al., 2013; Steven et al., 2003) and/or via comparison with hyperspectral sensor systems (Chander et al., 2010; Claverie et al., 2018). Errors in the positioning information of the instruments combined with differences in the geolocation routines may also lead to spatial misregistration of level-1 swath data between different sensors. Storey et al. (2016) noted a 38m sensor-to-sensor misregistration between Landsat 8 and Sentinel-2 due to errors in the collected ground control points for Landsat. For the most ideal of cases, the production of each CDR would take sensor top of atmosphere radiance and then process each measurement stream to produce a consistent CDR. This would additionally involve using the same atmospheric correction routines between sensors. Even with such consistent processing routines, differences may remain due to variations in the information content of the sensors. Zhang et al. (2018) highlighted the challenge of producing consistent retrievals for Landsat 8 and Sentinel-2. In their study, level-1 observations from both sensors were projected to a shared 30m grid, and identical atmospheric and BRDF correction algorithms applied. Despite these steps, reflectance from Sentinel-2 MSI was systematically higher than from Landsat 8 OLI.

Because of this complexity, present efforts for producing CDRs of surface reflectance are generally piecemeal — focusing on the combination of similar sensors (e.g. Sentinel-2 and Landsat 8, or OLCI and STSLR); or in early development (e.g. the low geographic coverage of the HLS product). They are not suitable, therefore, for the production of a global long-term data record of burned area.

7.3.2 Post-merging processing model

Given the considerable challenges of producing a multi-sensor surface reflectance CDR — which is ultimately only an intermediate product for burned area — an alternative approach is the production of a burned area CDR from multiple sensors independently. The general principle of this approach is diagrammatically shown in figure 7.4. With this processing model, burned area products are produced independently from each sensor record, and then retrospectively merged to produce a long-term CDR of burned area.

This model has been used within the FireCCI project to produce several burned area CDRs. Within the FireCCI product, coarse-resolution burned area products have been produced from MERIS (MERIS FireCCI41 and FireCCI31), MODIS (MODIS FireCCI51), and also AVHRR (FireCCILT10) (Chuvieco et al., 2018; Alonso-Canas and Chuvieco, 2015; Otón et al., 2019). An effort was also put into producing burned area from VGT/PROBA-V in anticipation of producing a burned area product from Sentinel-3 (Campagnolo et al., 2019). These products have overlapping temporal extents, for example, 2001-2016 for MODIS FireCCI51 and 1982-2017 for AVHRR FireCCILT10. While these products have been provided to users separately, no single merged CDR has been produced.

VIIRS was partially designed to be a successor to the MODIS sensors, with similar spectral channels, and spatial and temporal resolutions. With the MODIS sensors likely to be decommissioned in the 2020s, a focus has been on producing MODIS-successor geophysical products in a consistent manner. For example, the NASA science team have been developing operational surface reflectance products from VIIRS gridded into the MODIS sinusoidal grid (JPSS, 2013). The VNP09GA products provide surface reflectance at 500m and 1km on the MODIS sinusoidal grid along with auxiliary information similar to the MODIS M0D09GA products. This link has enabled the generation of a NASA VIIRS Burned Area product (SNP64A1) built on the heritage of the MODIS burned area products (Giglio et al., 2013). The SNP64A1 algorithm adapts the MODIS MCD64 algorithm to be suitable for the VIIRS sensor. Issues exist around differences in the actual spectral channels of the two sensors and therefore the extent to which the products will provide a consistent long-term burned area CDR still needs to be examined. In a similar manner, active fire products from VIIRS are being generated following the MODIS active fire product heritage. The 750m and 375m VIIRS active fire products are processed from algorithms based upon the multi-spectral contextual algorithm developed for MODIS active fire products but with additional changes for the VIIRS sensor (Schroeder and Giglio, 2018).

There are considerable challenges to producing a merged burned area CDR from such individual products. The first issue is the merging of burned area products from different sensors (the ‘merging

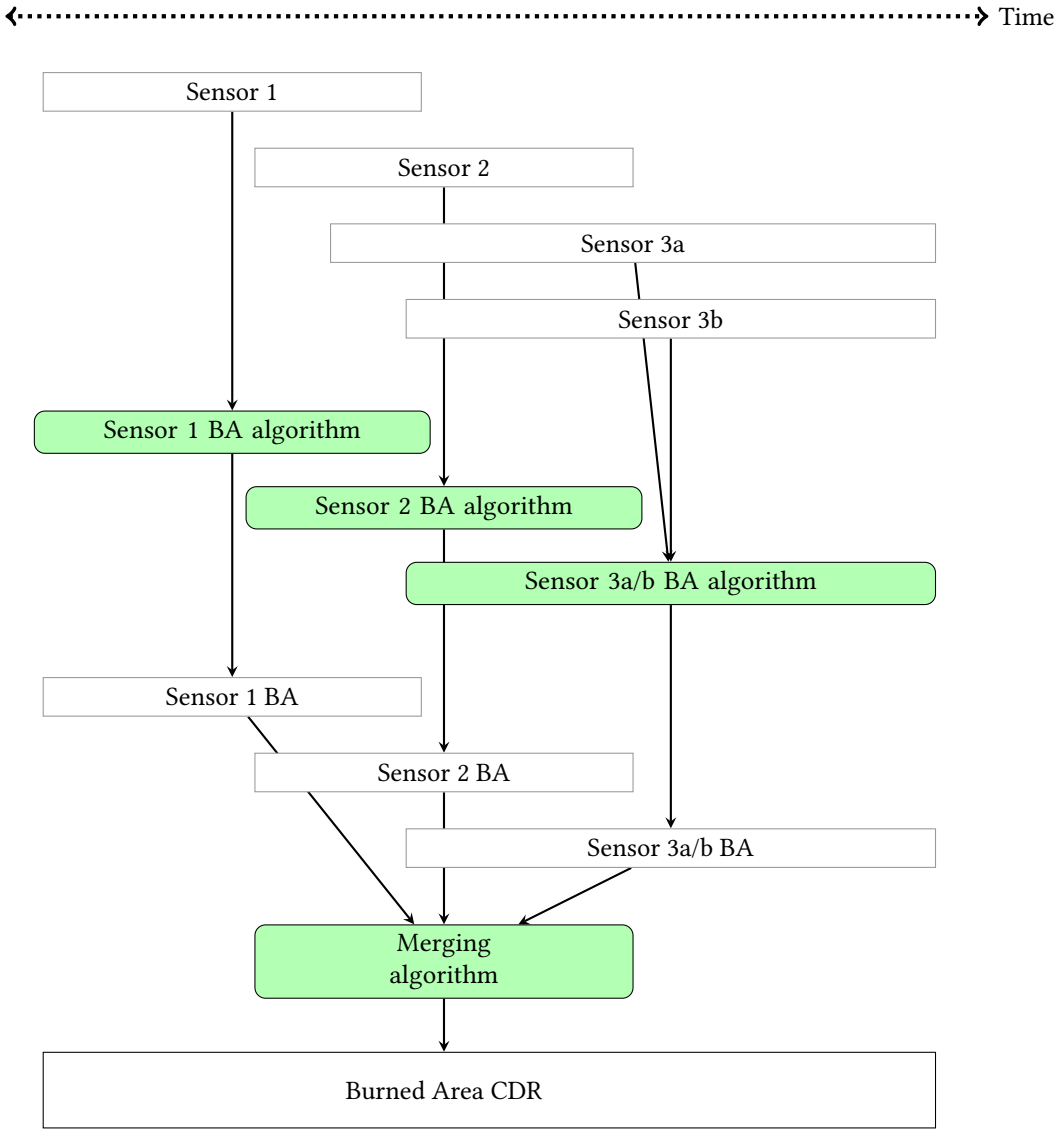


Figure 7.4. Post-merging model of burned area CDR generation. This consists of running burned area algorithms on several distinct observational records and then retrospectively merging the outputted burned area into one burned area CDR.

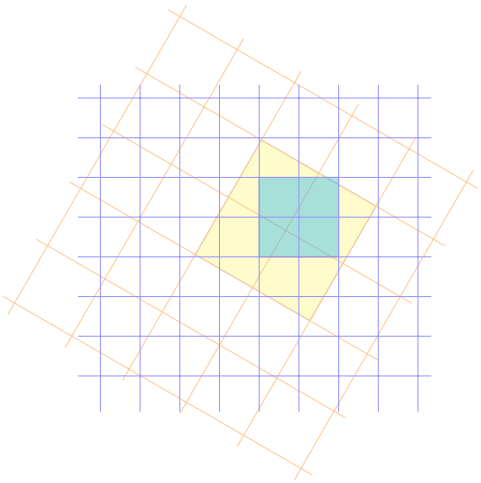


Figure 7.5. Challenges of merging products. Merging overlapping satellite burned area products is presently challenging due to 1) different spatial resolutions and gridding systems, 2) lack of uncertainty information to combine burned area.

algorithm' in figure 7.4). Level-2 surface reflectance products from different sensors may be in different spatial grids and at different spatial resolutions (see figure 7.5). The individual burned area products then need to be mapped into a shared grid to produce one burned area CDR that is consistent and uncertainty quantified. This merging process requires quantitative information on the quality of each product at the pixel level and at daily intervals. Even for periods where burned area is only retrieved from one sensor (e.g. AVHRR pre-1998), such quantitative uncertainty information is needed to provide users with information on the precision of burned area in these more poorly constrained periods of the satellite record (Nightingale et al., 2019). The natural formulation for this is a per-pixel uncertainty. Indeed this was the focus of chapter 6. The framework presented in chapter 6 was developed to be suitable to the problem of merging burned area products of differing precision and provided at different spatial/temporal resolutions.

Figure 7.6 highlights these issues. It shows burned area for an area of Northern Queensland, Australia for November 2011 from two FireCCI products. There are differences between the products, most notably, the larger burned area detected by the FireCCI51 product. More subtle distinctions also exist. For example, differences in the timing information of burned pixels between the products, and variations in the unobserved areas and inland water bodies in the MODIS FireCCI51 product. The two products are also provided at different spatial resolutions — the FireCCI51 product at nominally 250m and the FireCCI41 product at 300m.

A more fundamental question surrounds the scientific quality of such a merged product derived from considerably different algorithms. While the merging process described above treats random errors in the products it cannot at present deal with systematic uncertainties or biases between the products that arise out of different assumptions within each algorithm. For example, while the various

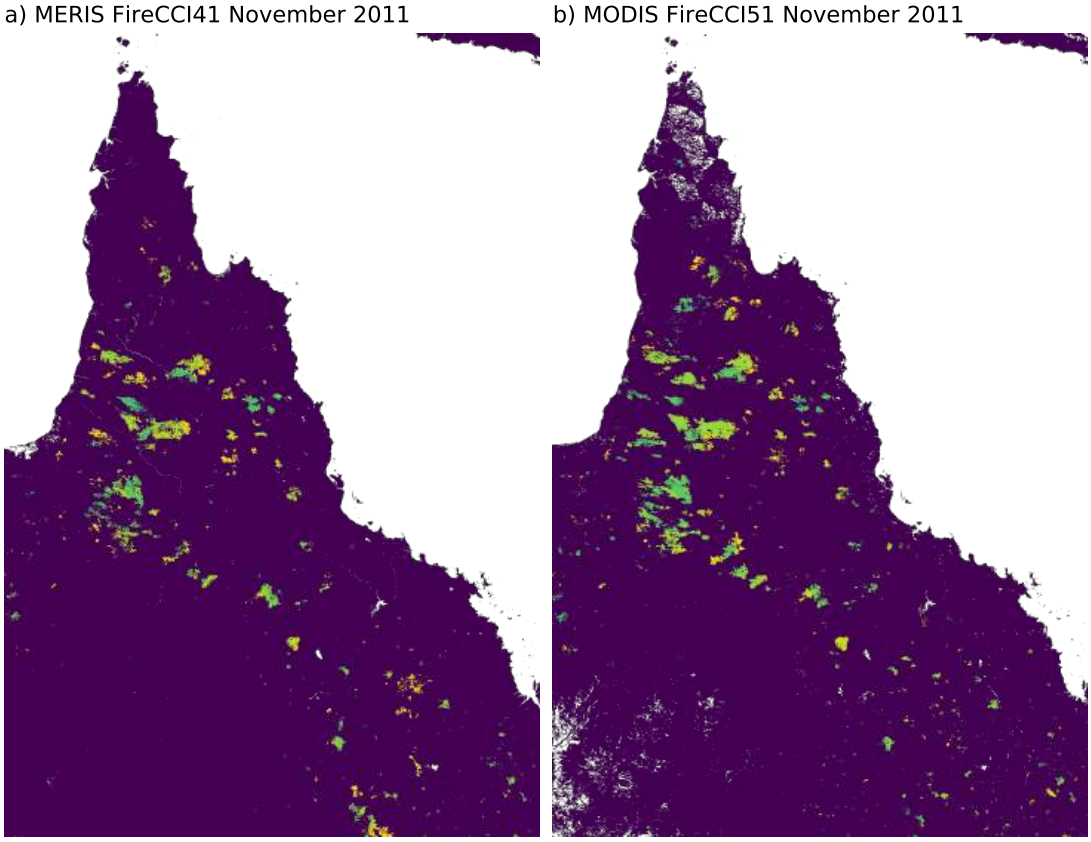


Figure 7.6. Challenges of merging products. Burned area in a region of northern Queensland, Australia for November 2011 from two FireCCI products. a) MERIS FireCCI41 b) MODIS FireCCI51. There are differences in the reported burned area from the two products, as well as differences in day of detected burning. The products are also provided at different spatial resolutions.

FireCCI products could be merged into one burned area CDR, differences in the algorithms used for each sensor mean that the scientific assumptions vary through the merged CDR. Similarly, while the MODIS MCD64A1 and VIIRS SNP64A1 could be merged as the algorithm is at least superficially the same, the lack of per-pixel uncertainty information in both products makes it difficult to optimally merge the two products or provides users with information on the uncertainty of the merged result.

To address this point, it is possible to envisage a different post-merging generation model. In this model (see figure 7.7), the burned area algorithm is applied consistently across sensors. The merged burned area product is then constructed from combining probabilistically each sensor inference on burned area. This approach relies on two components. First, it requires an algorithm which can consistently apply the same assumptions across sensors which have varying spectral, temporal and spatial characteristics. Secondly, the merging process requires a probabilistic treatment in which the

burned area inference is treated as a probability. Sections 7.4 & 7.5 now go on to develop such an algorithm.

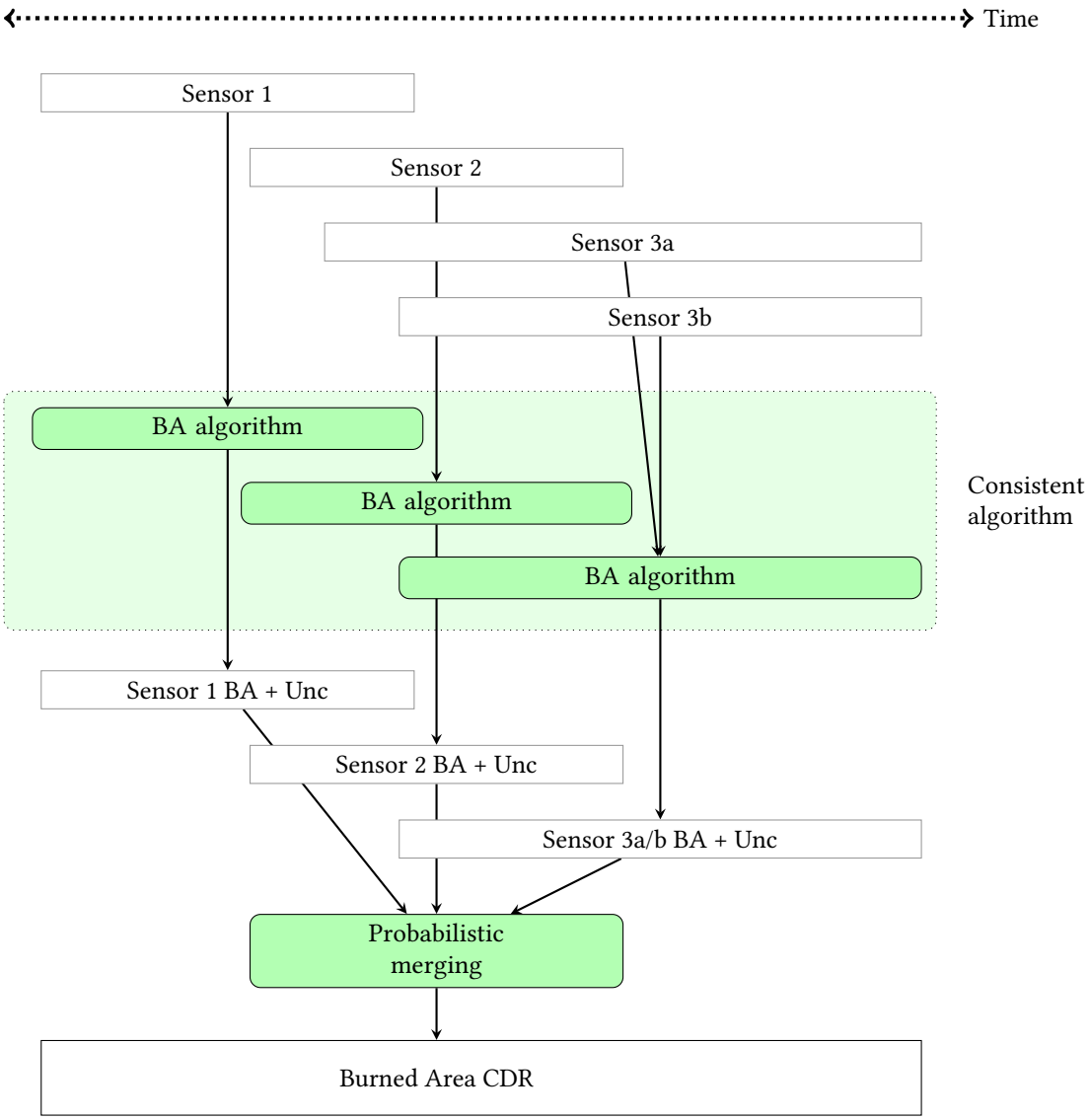


Figure 7.7. A better post-merging model of burned area CDR generation. As with figure 7.4, this consists of running burned area algorithms on several distinct observational records and then retrospectively merging the outputted burned area into one composited burned area CDR. However, the burned area algorithm is *consistent* across different sensor records which then provides the same assumptions into each product. This then ensures better consistency when the products are merged into a burned area CDR.

7.4 Ensuring spectral consistency with a multi-spectral model of fire impact

The fcc spectral model has been introduced previously in chapter 3. For presentation, it is restated here as:

$$\Delta\rho(\lambda) = fcc \cdot \rho_{\text{burn}}(\lambda) + [1 - fcc] \cdot \rho_{-}(\lambda) \tag{7.1}$$

where $\Delta\rho(\lambda)$ is the change between the pre-fire $\rho_{-}(\lambda)$ and post-fire $\rho_{+}(\lambda)$ reflectance (Roy and Landmann, 2005). $\rho_{\text{burn}}(\lambda)$ is the spectral end-member of the pixel when the vegetation is considered fully radiometrically burned. The spectral end-member $\rho_{\text{burn}}(\lambda)$ is composed of two primary surface elements: 1) a dark char/bright ash spectrum and 2) an exposed soil spectrum. $\rho_{\text{burn}}(\lambda)$ is therefore modelled as:

$$\rho_{\text{burn}}(\lambda) = a_0 + a_1 S(\lambda) \tag{7.2}$$

where $S(\lambda)$ is a quadratic approximation to a soil-spectrum (Gomez-Dans et al., 2013).

Formulating the three parameters of the fcc model as $\theta = [a_0, a_1, fcc]$, and referring to the fcc model $\mathbf{K}(\theta)$, solving equation 7.1 can be achieved via minimising the functional $J_{\text{obs}}(\theta)$:

$$J_{\text{obs}}(\theta) = \left[\Delta\rho - \mathbf{K}(\theta)\right]^{\mathrm{T}} \mathbf{C}_{\text{obs}}^{-1} \left[\Delta\rho - \mathbf{K}(\theta)\right] \tag{7.3}$$

$J_{\text{obs}}(\theta)$ is a quadratic cost function that is minimised when the prediction from the model $\mathbf{K}(\theta)$ matches best to the observations $\Delta\rho$ moderated by the observational uncertainties provided by the observation covariance matrix $\mathbf{C}_{\text{obs}}$. The uncertainty in the retrieval θ can be approximated from the curvature of $J(\theta)$ at the solution to provide a posterior covariance matrix $\mathbf{C}_{\theta}$.

The fcc model presented above provides a model of the spectral burning signal that can be applied to any optical sensor. Because $\rho_{\text{burn}}(\lambda)$ is defined for all wavelengths, the parameters of the fcc model are effectively 'spectrally invariant'. This means that any combination of spectral channels can be used to retrieve fcc, a_0 and a_1. The fcc model hence provides the opportunity to strongly enforce the consistency of spectral assumptions about burning across sensors with different spectral sampling.

While the spectral invariance principle determines that θ is retrievable from any optical sensor, the quality of the inversion of equation 7.3 will be determined by the available spectral sampling of a sensor. Figure 7.8 displays a synthesised vegetation canopy spectrum experiencing a fire as modelled by the fcc model. The spectral channels available on different sensors are also shown, as well as the two component spectra of $\rho_{\text{burned}}(\lambda)$: the a_0 and a_1 spectral basis functions. VIS—NIR instruments such as OLCI have no sensitivity to SWIR changes due to fire. This effectively limits their sensitivity

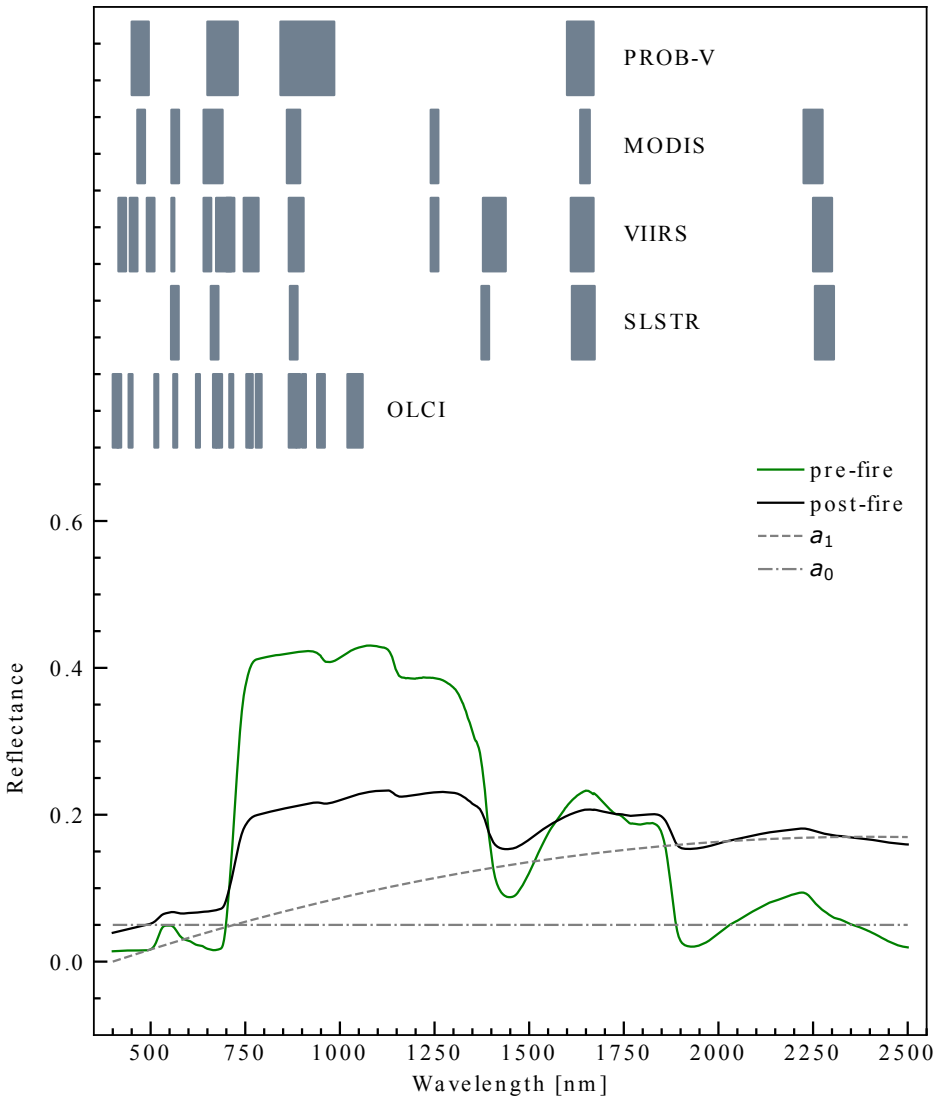


Figure 7.8. Spectral characteristics of the fcc model and spectral sampling of different coarse-resolution optical instruments. Depicted is the modelled pre-fire and post-fire reflectance from the fcc model, and the spectral end-members of $\rho_{\text{burn}}(\lambda)$: a_0 and a_1.

to detect fire related changes in the land surface and accurately infer the values of the θ from the observations alone.

A feature of the spectral shape parameter is that they display spatially consistent patterns which are related to the local properties of the vegetation, soil and burning characteristics which align well to properties of regional fire regimes (Archibald et al., 2013). To demonstrate this, the fcc model was

solved for each burned pixel reported by the MODIS Collection 5 (MCD64C5) burned area product for 2000–2014. For each pixel recorded burned in MCD64C5, the fcc model was inverted using MODIS surface reflectance from Aqua (MYD09GA) and Terra (MOD09GA). This 14-year record of fcc, a_0 and a_1 retrievals was then aggregated to produce a climatology of the spectral parameters on a coarse 0.1° resolution grid. This climatology was characterised with a multivariate normal distribution $\mathcal{N}(\theta_{\text{Clim}}, \mathbf{C}_{\text{Clim}})$. The parameters of the distribution were estimated for each grid cell from the sample mean and sample covariance of individual pixels:

$$\hat{\theta_{\text{Clim}}} = \frac{1}{N}\sum_{i=1}^{N}\theta_i \tag{7.4}$$

$$\hat{\mathbf{C}_{\text{Clim}}} = \frac{1}{N-1}\sum_{i=1}^{N}(\theta_i - \hat{\theta_{\text{Clim}}})(\theta_i - \hat{\theta_{\text{Clim}}})^{\text{T}} \tag{7.5}$$

To ensure the quality of the parameter estimates, robust statistical methods were used (Rousseeuw and Driessen, 1999).

Figure 7.9 depicts statistics for parameters of the fcc model globally. Regional patterns in the spectral parameters can be observed. For example, low values of μ_{a_0} and μ_{a_1} correspond to spectrally very dark post-fire reflectance associated with the deposition of dark char from burning in the boreal forests of North America and northern Eurasia. Larger values of μ_{a_1} in the Sahel and southern Africa correspond to burning regimes with bright soils and low vegetation cover leading to a post-fire signal constituted primarily of exposed soil. These variations in μ_{a_0} and μ_{a_1}provide localisation of the spectral fire signal adjusted for factors such as land cover and local pyrome. The variance of the spectral parameters also provides information on the local variability of the spectral fire signal. Large values of σ_{a_0} and σ_{a_1} feature in areas of inhomogeneous land cover, including the boreal-tundra interface in northern Eurasia and the forest-grassland interface in eastern Canada.

The spectral invariance properties and regional patterns in θ shown in figure 7.9 provides the route to ensuring consistent inferences of fcc from any optical sensor. This involves enforcing additional information about the coefficients for optical sensors with more limited spectral sampling. If a prior distribution for θ can be established, the observation functional $J_{\text{obs}}(\theta)$ can be complemented by an additional functional to form a constrained cost function:

$$J(\theta) = \underbrace{\left[\theta - \theta_{\text{prior}}\right]^{\text{T}} \mathbf{C}_{\text{prior}}^{-1} \left[\theta - \theta_{\text{prior}}\right]}_{J_{\text{prior}}} + \overbrace{\left[\Delta\rho - \mathbf{K}(\theta)\right]^{\text{T}} \mathbf{C}_{\text{obs}}^{-1} \left[\Delta\rho - \mathbf{K}(\theta)\right]}^{J_{\text{obs}}} \tag{7.6}$$

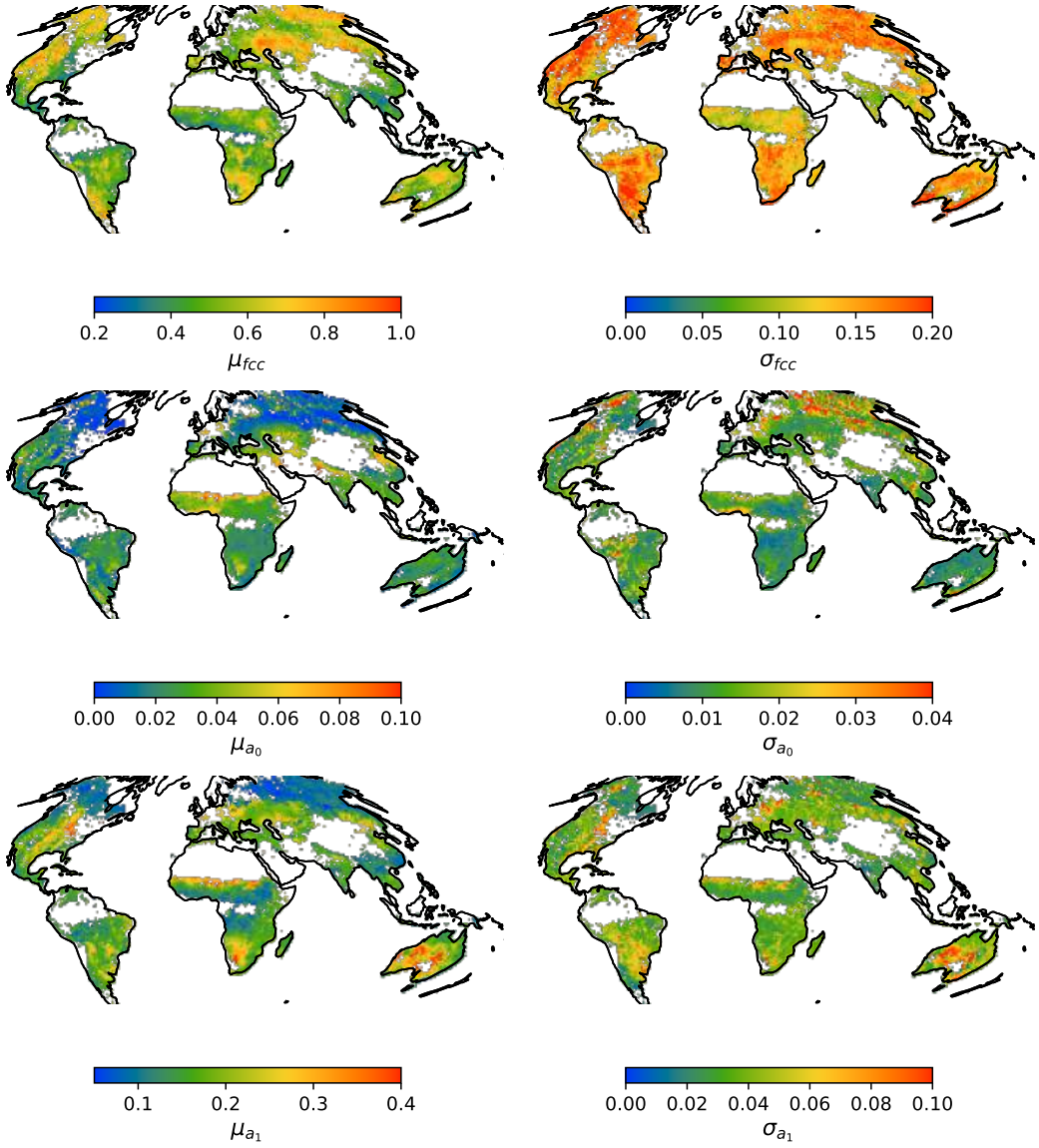


Figure 7.9. Regional patterns in the parameters of the fcc model. Sample mean (left column) and standard deviation (right column) of fcc and spectral shape parameters a_0 and a_1 derived from MODIS observations (2000—2014).

such that when the estimated parameters θ are in broad agreement with the prior θ_{prior}, $J_{\text{prior}}(\theta)$ is minimised. The covariance matrix $\mathbf{C}_{\text{prior}}$ determines the precision of the prior distribution and therefore how far the retrieval can move away from the prior θ_{prior}.

The inclusion of additional prior information into $J(\theta)$ presents a natural method to encapsulate information about the parameters of the fcc model which can aid the retrieval from spectrally deficient instruments. To distinguish these, a classification is introduced of more informative spectral sensors (MISS) and less informative spectral sensors (LISS). MISS sensors are those which have sufficient spectral sampling to more accurately invert the fcc model. These are characterised as those that have $m > 3$ channels (i.e. to retrieve the three model parameters) and those that have sampling that covers the VIS, NIR and SWIR spectral regions. Such sensors include MODIS and VIIRS. Conversely, LISS sensors do not meet these conditions, because they do not have spectral channels in the SWIR for example. Examples of LISS sensors are the OLCI, MERIS and AVHRR instruments. This concept of MISS and LISS instruments is central to ensuring the consistent retrieval of burned area between different sensors within the algorithm. Fortunately for the modern coarse-resolution record, MISS instruments are always available and will continue to be due to the operation designation of VIIRS (Hillger et al., 2013).

7.4.1 Demonstration with a modelled canopy

To demonstrate the method of the spectrally constrained retrieval described above consider a synthetic example of a canopy experiencing a fire. A pre-fire canopy reflectance $\rho_{-}(\lambda)$ was generated with the PROSAIL canopy radiative transfer model (Jacquemoud et al., 2006). The post-fire reflectance $\rho_{+}(\lambda)$ was then modelled explicitly with the fcc model in equation 7.1, with parameters $fcc = 0.6$, $a_0 = 0.05$, $a_1 = 0.15$. The pre- and post-fire reflectance, and the spectral burn signal ρ_{burn} are shown in figure 7.10. The fire leads to a reduction in NIR reflectance, an increase in the SWIR and a small increase in the visible red and blue. The spectral burn signal corresponds to a monotonically increasing spectrum that appears similar to a soil spectrum. From the synthetic pre and post-fire reflectance, the fcc model was then solved for two sensors. An MISS sensor (i.e. MODIS) with seven spectral channels covering the VIS-NIR-SWIR, and a LISS sensor with only two spectral channels — a visible red and a NIR channel (i.e. AVHRR).

Figure 7.11 shows the predicted post-fire reflectance from inverting the fcc model for both sensors and the prediction of the reflectance change signal. For the MODIS-like instrument, the fcc retrieval predicts the reflectance change signal with an RMSE of 8e-06. The retrieved parameters are almost identical to the truth ($fcc = 0.60, a_0 = 0.049, a_1 = 0.15$). This is generally expected — the information content of the seven MODIS channels is sufficient to solve the fcc model correctly, especially as the synthetic experiment represents an ideal case (i.e. no model error or observational errors).

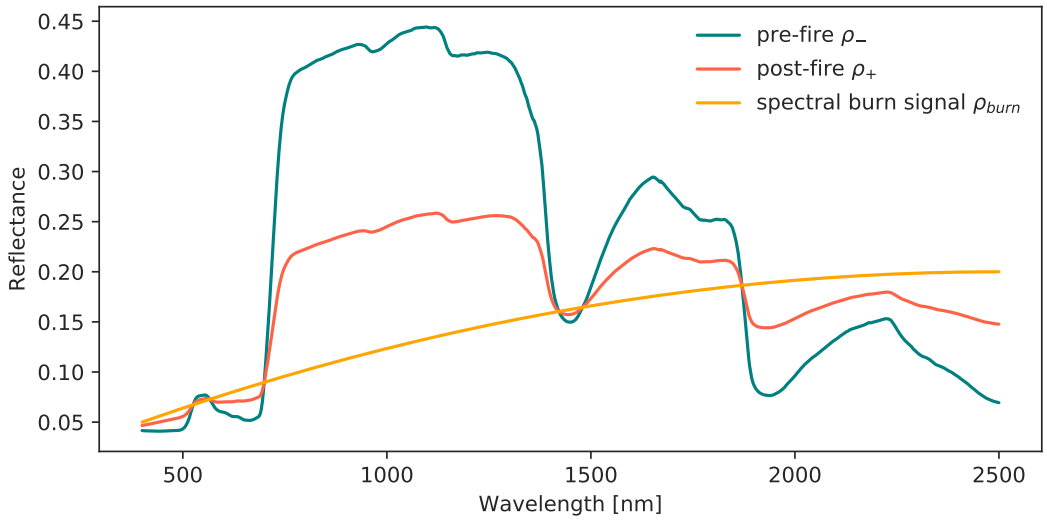


Figure 7.10. Modelled canopy reflectance before and after a fire event. The pre-fire reflectance was modelled with PROSAIL and the post-reflectance modelled from the fcc model with parameters $fcc = 0.6$, $a_0 = 0.05$, $a_1 = 0.15$. Also shown is the spectral burn signal ρ_{burn}.

For the AVHRR-like sensor, the two methods for fcc (equations 7.3 and 7.6 respectively) were used. First the conventional unconstrained least squares method was used to estimate $[fcc, a_0, a_1]$ with the AVHRR-like spectral channels (also shown in figure 7.11). The solution has a small RMSE (6e-17) as can be evidenced by the good fit to the reflectance change in the two channels. However, the retrieved parameters bear little resemblance to the true values. An fcc of 0.46 is 30% lower relative to the true fcc of 0.6. The spectral shape parameters ($a_0 = 0.14, a_1 = -0.33$) are also dissimilar from their true values with a_1 being non-physical.

The effect of the poor retrieval of a_0 and a_1 from the LISS instrument can be seen if these parameters are used with the fcc model to predict the post-fire reflectance in the unobserved MODIS channels. This can be done because the a_0 and a_1 parameters are spectrally invariant and when combined with ρ_{burn} provide a model of the post-fire reflectance for all optical wavelengths. For the MODIS channels, the prediction of the post-fire reflectance is poor and non-physical in the SWIR. The consequence of the large difference with the MODIS retrieval indicates that the parameters are not retrieved in any consistent manner, and attempting to infer burned area from the two retrievals would likely provide different interpretations.

The retrieval with the spectral constraint for the AVHRR-like instrument is also shown in figure 7.11. The spectral prior θ_{prior} was defined using the MODIS retrieved parameters and an inflation

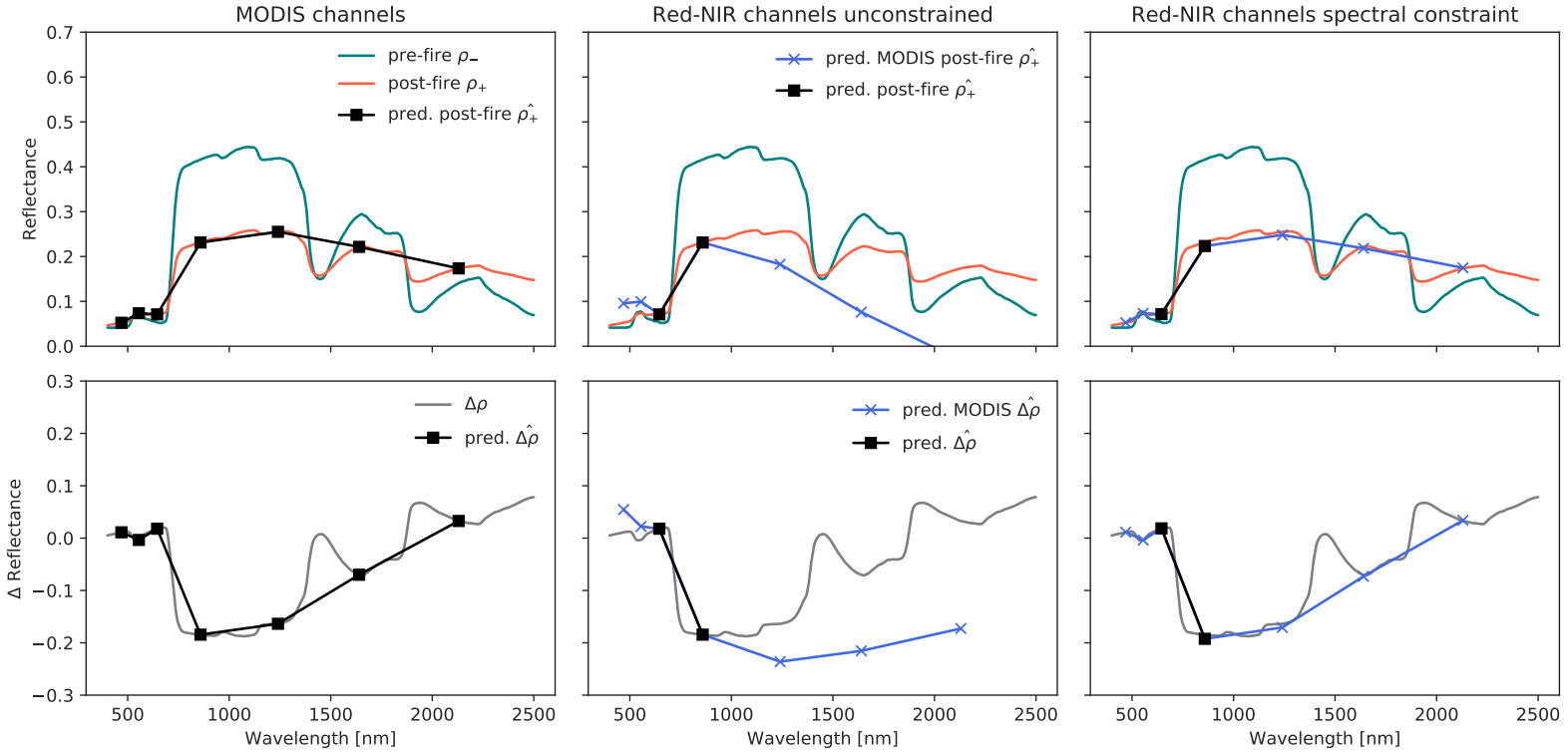


Figure 7.11. Demonstrating the spectrally constrained fcc retrieval on a synthetic fire from two sensors. Top row) synthetic pre-fire ρ_- and post-fire ρ_+ reflectance and predicted post-fire reflectance $\hat{\rho}_+$ from a MODIS-like and AVHHR-like sensor. Bottom row) true change in reflectance between pre- and post-fire $\Delta\rho$ and predicted reflectance change $\hat{\Delta\rho}$. Left column) retrieval from the MODIS-like instrument matches well to the true pre- and post-fire reflectance and the reflectance change signal. Central column) The retrieval from the AVHHR-like instrument predicts the red-NIR reflectance well but due to the poor estimates of a_0 and a_1 provides an un-physical post-fire reflectance for other wavelengths. Right column) Using information from the MODIS-like retrieval into the AVHRR-like retrieval makes the prediction align more closely with the true post-fire reflectance and change signal across all spectral channels.

of the posterior covariance of the MODIS retrieval by a factor of 100. This inflation prevents the solution from being overly restricted by the MODIS retrieval. As with the unconstrained solution, the predicted reflectance change in the two channels matches well to the observed signal. However, the spectral constraint has pushed the retrieved parameters towards values which are consistent with the MODIS retrieval. fcc is retrieved as slightly higher (0.63) than the MODIS and the true fcc. The spectral parameters are also much closer to the true values ($a_0 = 0.049, a_1 = 0.149$). The effect of this improved estimate of the spectral parameters can be seen in the much better prediction of the MODIS channels.

To contemplate the role of the spectral prior more closely, figure 7.12 shows the cost function of the constrained method (equation 7.6). Shown in the figure is J_{obs} — the cost function associated with the observation misfit, J_{prior} the cost associated with disagreement to the prior, and $J_{\text{obs}} + J_{\text{prior}}$ — the posterior cost of the observation likelihood combined with the prior.

The broad shallow functions of J_{obs} for a_0 and a_1 indicates that the two channels provide a poor constraint on the parameters — i.e. a large range of values have similar costs and therefore a range of values do not improve the fit to the observations. For a_0 and a_1, most of the information is instead contained within the spectral prior and so the minimum of the posterior corresponds strongly with the minimum of J_{prior}. The limited information about the parameters in the observations is still included — as evidenced by the slight shift of the posterior minimum from the prior minimum.

Conversely, for fcc, J_{prior} is very broad — indicating that the prior does not overly constrain the value of fcc. Unlike the spectral parameters, the observations do provide a better constraint on more likely values of fcc as evidenced by the narrow minimum for J_{obs}. The important feature here is that by providing prior information on $[a_0, a_1]$ the retrieval of the fcc model from LISS sensors (i.e. AVHRR-like) is much more consistent with the MISS sensors and indeed the true values. Importantly, fcc does not need to be heavily constrained — instead the change in reflectance $\Delta\rho(\lambda)$ from the LISS sensor is still informative on fcc when provided with the prior on $[a_0, a_1]$.

The aforementioned optimal estimation framework provides a path to a consistent application of the fcc model across MISS and LISS instruments. The system is reliant on having an appropriately informative prior on $[a_0, a_1]$ to constrain the fcc model for LISS instruments. While the retrieval from LISS instruments will likely be of a lower quality due to the more limited spectral sampling the proposed method will be consistent with the MISS estimates.

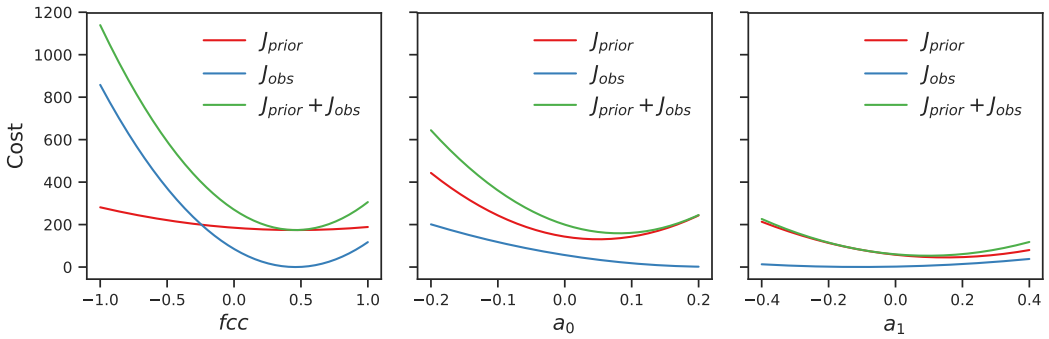


Figure 7.12. The cost function (equation 7.6) of the synthetic LISS retrieval for the three parameters of the fcc model.

7.5 Implementation of the algorithm

The spectral invariance properties of the fcc model, combined with the constrained spectral retrieval system developed above provide a suitable retrieval scheme for inferring burned area from multiple sensors. The algorithm is implemented to perform burned area detection on a month-by-month basis while utilising three months of observations (see figure 7.13). The use of an extended processing window helps to ensure that sufficient observations are available from each sensor to run the algorithm. To provide accompanying illustrations of the algorithm, a study region of approximately 40000 km^2 located on the coastline of north-eastern Mozambique is used for the mapping period of September 2012. Figure 7.14a) locates the study region and b) shows a real colour composite from MODIS for September where burn scars are moderately visible, active fire detections from MODIS are also superimposed in red. The study region provides a challenging case because the burned areas are typically small (<15km^2) as shown in figure 7.14b). Further, the area is home to low-density shrubland vegetation which has senescenced in the dry season (which occurs from April to December) prior to the burning event. The combination of small burn scars and vegetation senescence means that the post-fire change in reflectance is more limited, providing a small signal-to-noise ratio for detection.

7.5.1 Pre-processing of optical observations

As was discussed in sections 6.4 & 2.5, observations from coarse-resolution sensors exhibit changes in reflectance between acquisitions ('BRDF effects') which are a result of the sampling geometry of the acquisition. The fcc model requires observations of surface reflectance which are corrected for these

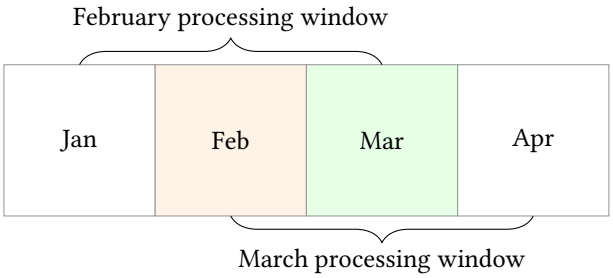


Figure 7.13. Processing strategy of the algorithm. The algorithm operates on a month-by-month basis but utilises three months of observations from the sensor. This means that for the month of February observations from 1st January til 31st March are used.

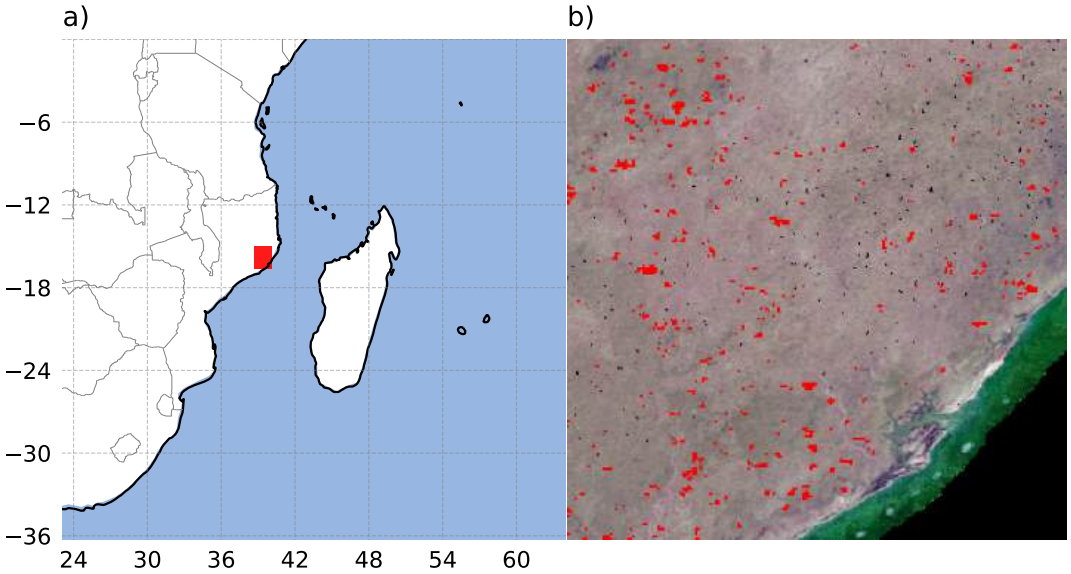


Figure 7.14. Study region considered. a) locates the study region and b) shows a true colour composite from MODIS for September where burn scars are visible as dark patches, active fire detections collected by MODIS for September are superimposed in red.

angular effects. Linear BRDF (i.e. 'kernel' models) models discussed in section 6.4.1 are well suited for this purpose (Wang et al., 2018; Schaaf et al., 2002; Lucht and Roujean, 2000).

An issue identified in section 2.5 about these models is requirement for the collection of several observations (e.g. typically more than seven) within a period of a few weeks. This requirement meant that previous burned area mapping algorithms which used these models performed poorly in persistently cloudy regions (Roy et al., 2005a). Quaife and Lewis (2010) developed a BRDF correction scheme suited to situations with sparse observation sampling. Based upon data assimilation principles, the method weights between the fit of the kernel model $\mathbf{K}$ to the observations and the principle that the trajectory of the parameters should be smooth in time:

$$J(f) = ||\mathbf{K}f - \rho(\Omega, \Omega')||^2_{\mathbf{C}_{\text{obs}}} + \lambda^2 \sum_t \varphi(f_{t-1} - f_t) \tag{7.7}$$

The second term in equation 7.7 is a summation of functions φ which operate on first order finite differences of the parameter trajectories $f_{t-1} - f_t$. $\varphi(f_{t-1} - f_t)$ serves to keep the differences between subsequent times t small and therefore enforce temporal smoothness. The parameter λ^2 determines the weighting between the fit to the observations and the smoothness constraint.

Gomez-Dans et al. (2013) built on the method of Quaife and Lewis (2010) by proposing a relaxation of the global smoothness constraint. The algorithm involves an iterative refinement of an edge process w which relaxes the smoothness constraint λ^2:

$$J(f) = ||\mathbf{K}f - \rho(\Omega, \Omega')||^2_{\mathbf{C}_{\text{obs}}} + \sum_t \left[\frac{\lambda^2}{w_t}\right] \cdot \varphi(f_{t-1} - f_t) \tag{7.8}$$

w is a vector of [0,1] weights which therefore reduce the influence of $\varphi(f_{t-1} - f_t)$ in periods of rapid change in the land surface. w is iteratively estimated as:

$$w = \frac{1}{\sqrt{1 + \left[f_{t-1} - f_t\right]^2}} \tag{7.9}$$

and equation 7.8 iteratively solved with the updated weights w. The algorithm presented by Gomez-Dans et al. (2013) was adapted for use here. As formulated in Gomez-Dans et al. (2013), the algorithm operates on all three kernel weight trajectories. Given the short period of interest (i.e. one month), an alternative scheme is implemented which operates only on observations that are BRDF corrected with a residual BRDF correction formulated as:

$$f_{\text{iso}} = \rho(\Omega, \Omega') - \left[f_{\text{vol}} K_{\text{vol}}(\Omega, \Omega') + f_{\text{geo}} K_{\text{geo}}(\Omega, \Omega')\right] \tag{7.10}$$

where $f_{\text{vol}}, f_{\text{geo}}$ are solved via equation ordinary least squares (Lucht and Roujean, 2000; Schaaf et al., 2002). This residual formulation means that the geometric and volumetric weights are considered

constant over the temporal window. This scheme also presents a major efficiency advantage in that the size of the solution space to optimise is now only N time steps as opposed to $3N$ in the original scheme presented by Gomez-Dans et al. (2013). The value of λ^2 is important because it defines the degree of global smoothness for the early iterations of the algorithm. For the process of detecting edges, it is preferable to initially over-smooth the solution so that the edge process in equation 7.9 responds only for cases with disturbance.

Figure 7.15 shows results of the retrieved daily Nadir BRDF adjusted reflectance (NBAR) and uncertainties for the MODIS channels for a pixel experiencing a fire in the study region. The NBAR trajectories show a reduction in the angular effects apparent in the MODIS reflectance measurements. The sudden adjustment in reflectance associated with the fire is also preserved in the NBAR retrieval. The BRDF scheme also shows some outlier rejection with the solution unaffected by the largest outliers in the reflectance measurements.

7.5.2 Consistent multi-sensor fcc retrieval

The central aim of the algorithm is the consistent estimation of fcc from multiple sensors. As was shown in section 7.4, constraining the spectral-shape parameters $[a_0, a_1]$ of the fcc model provides a method for producing consistent fcc estimates from sensors with different spectral channels. The retrieved daily NBAR from each sensor can be used to sequentially update the multi-sensor estimate of fcc from each sensor.

For each pixel p and time-step t, the potential pre-fire ρ_- , post-fire ρ_+ and change signal $\Delta\rho$ NBAR is retrieved from the NBAR time-series:

$$\rho_- = \rho_t \tag{7.11}$$

$$\rho_+ = \rho_{t+1} \tag{7.12}$$

$$\Delta\rho = \rho_+ - \rho_- \tag{7.13}$$

The uncertainty in the NBAR change is derived by considering each time-step to have independent uncertainties in the NBAR:

$$\sigma^2_{\Delta\rho} = \sigma(\text{NBAR})^2_t + \sigma(\text{NBAR})^2_{t+1} \tag{7.14}$$

For MISS instruments such as MODIS, θ is estimated for each pixel by minimising the cost function (equation 7.3) which provides no prior constraint on fcc or the spectral-shape coefficients.

Figure 7.16 shows the daily retrieval of the fcc model on the time series of MODIS surface reflectance detailed in figure 7.15. A notable feature is the general sparseness of the retrieved fcc

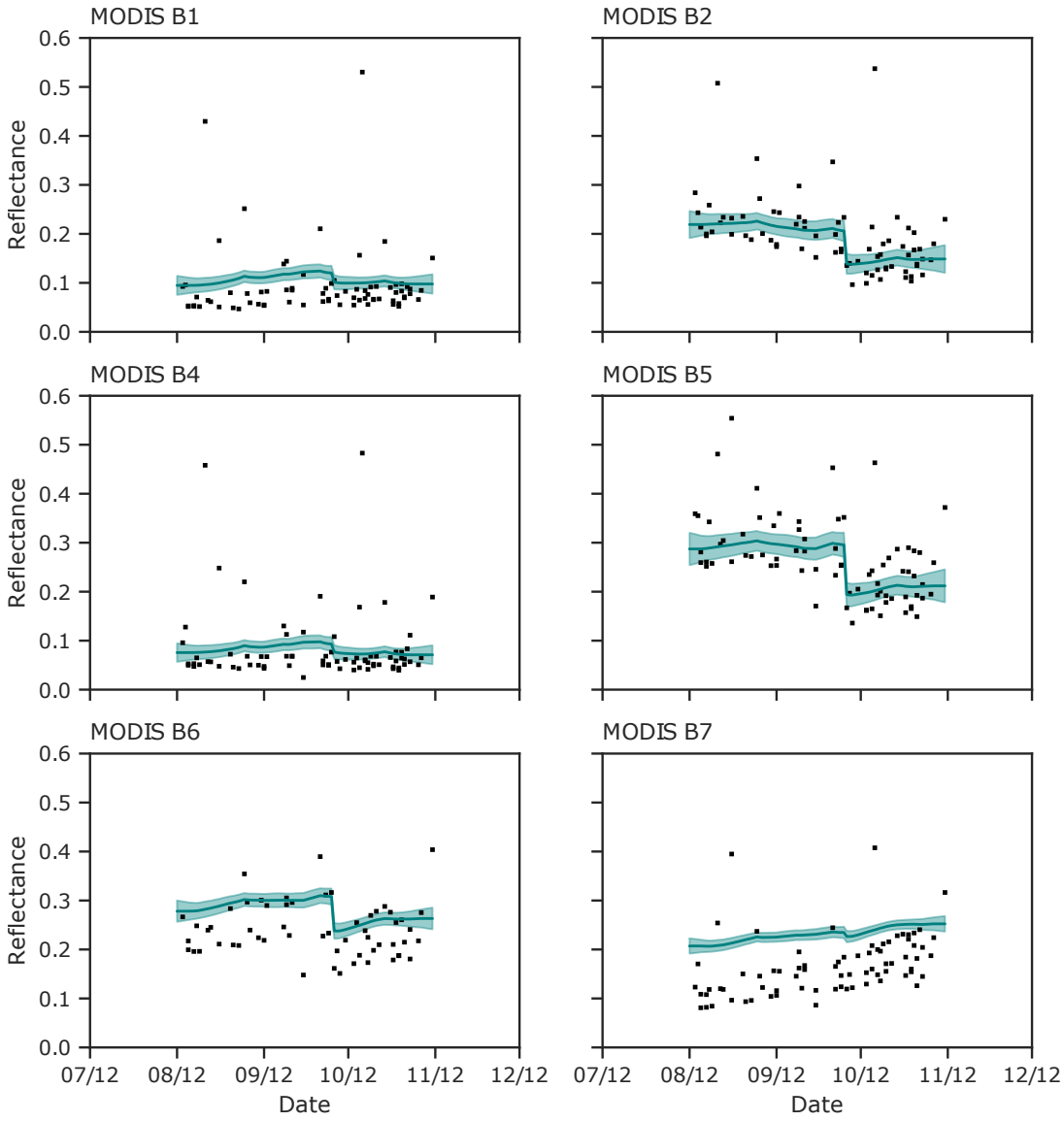


Figure 7.15. Time series of surface reflectance (black) collected by MODIS Aqua and Terra for August–November 2012 for a fire-affected pixel in the study region. Retrieval of daily isotropic surface reflectance (blue) and uncertainties (light blue).

time series. Only on the day of the fire is a high magnitude fcc reported. On the day of the fire, fcc is estimated to be 0.86±0.08, a_0 is relatively small 0.05±0.001 and a_1 larger 0.2±0.003. On other dates, the uncertainties on the three parameters are much larger. For example before and following the fire date, uncertainties on a_0 and a_1 also exceed the likely range of these parameters — indicating the lack of evidence for a fire on these dates. An interesting feature is an observable increase in the uncertainty

in the retrieved fcc following the fire. This occurs when both the pre- and post-fire reflectance match well to the shape of the spectral burn signal ρ_{burn}.

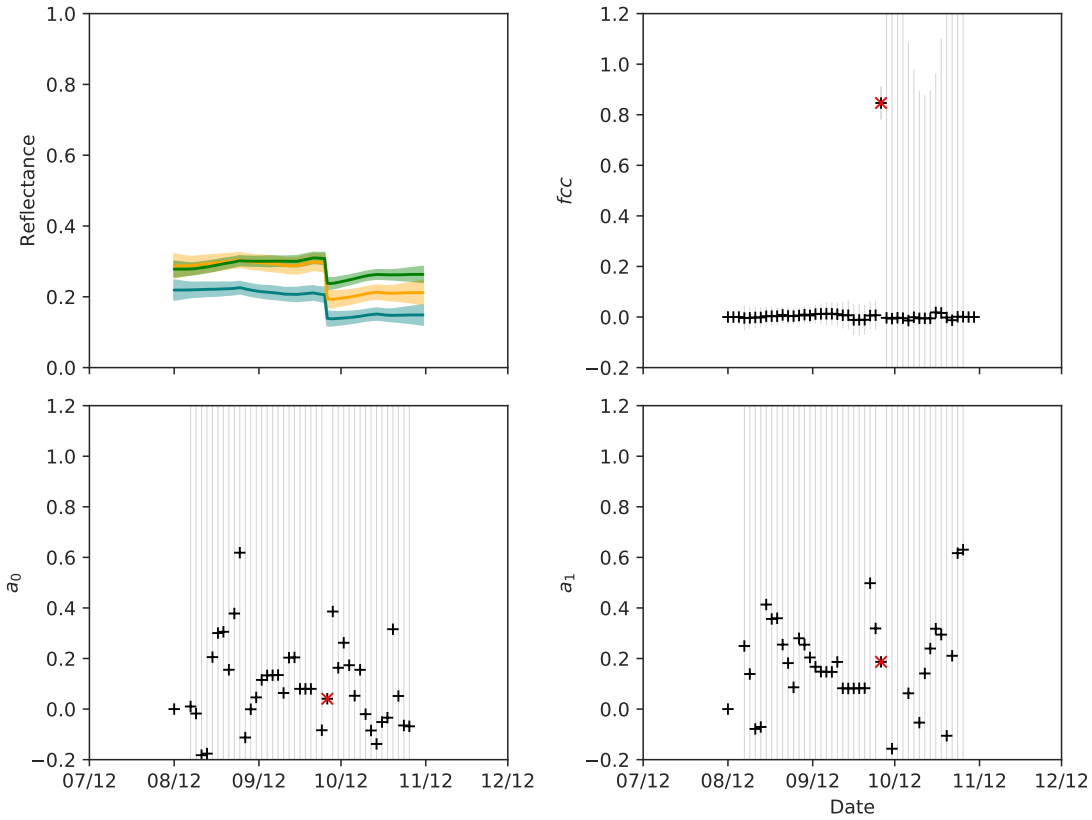


Figure 7.16. Retrieval of the fcc model for a fire-affected pixel previously shown in figure 7.15. 1σ uncertainties are shown in grey. Note only NBAR for MODIS channels 2, 4 and 6 are shown in panel a.

Figure 7.17 shows a composite of the maximum reported fcc for mapping period. fcc values within burn scars range from around 0.15 to 1, while non-fire-affected areas report maximum fcc values typically less than 0.1. The contrast discernible in fcc implies that fcc alone provides an initial constraint on burned area. The spectral parameters a_0, a_1 associated with the maximum fcc are also shown in figure 7.17. Burned areas are identifiable by the spatial consistency of the parameters. Within the burns, a_0 ranges between 0–0.1 and a_1 between 0.15–0.4. The speckle patterns in unburned areas are indicative of poorer quality retrievals of a_0 and a_1. This occurs as the spectral end-members of the fcc model are designed to be a poor fit to green vegetation and non-fire affected surfaces (Gomez-Dans et al., 2013).

Turning to the retrieval of the fcc model from LISS instruments, θ is retrieved from the minimising the cost function in equation 7.6. The solution to equation 7.6 is constrained by the prior distribution

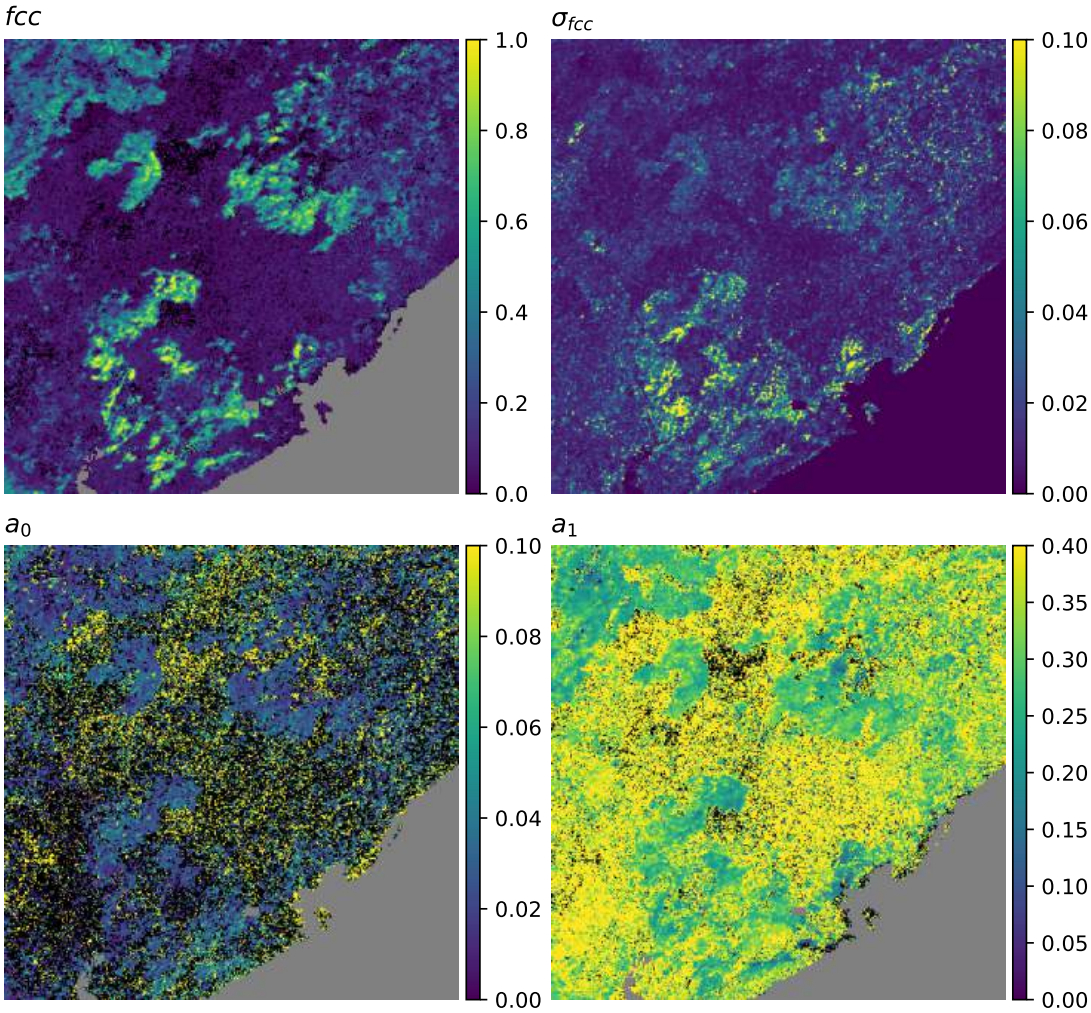


Figure 7.17. Composite of maximum fcc and associated uncertainties σ_{fcc} for September 2012 over the study region. Also shown are the values of a_0 and a_1 associated with this fcc value.

on θ. Section 7.4 showed that providing a prior for the shape parameters $[a_0, a_1]$ is sufficient to resolve consistent fcc values. Further, a_0 and a_1 as semi-physical mixtures of ash and char should be consistent between different sensors and were shown to be consistent in space (see figure 7.9).

The prior distribution for θ is considered multivariate normal $\mathcal{N}(\theta_{\text{prior}}, \mathbf{C}_{\text{prior}})$ as implied in the cost function equation 7.6. There are primarily two sources of information for this prior. First, the retrievals of θ from the MISS instrument provide the best available information on θ for the mapping period. Pixels where the is probable evidence of fire (i.e. where fcc provides an indication of a change in the NBAR) are used for this. Pixels where $fcc > 0.3$ are selected to provide sampling from which a sample mean θ_{MISS} and covariance matrix $\mathbf{C}_{\text{MISS}}$ are generated.

There may be situations in which there are no suitable pixel retrievals of θ from the MISS instrument. These for example occur when the MISS instrument either does not observe the surface (i.e. due to persistent cloudiness) or no high fcc retrievals are estimated by the MISS instrument. In this situation, a suitable solution is to use the historical prior developed for MODIS in section 7.4.

Merging these two sources of information on θ can be formulated as Bayesian filtering operation: the climatology probability density is updated with the new information provided from the MISS retrieval. Given that the historical prior and MISS retrieval distributions are specified as multivariate normal distributions, the standard Kalman update provides an analytic method to perform this update:

$$\mathbf{K} = \mathbf{C}_{\text{Clim}} \left[\mathbf{C}_{\text{Clim}} + \mathbf{C}_{\text{MISS}}\right]^{-1} \tag{7.15}$$

$$\theta_{\text{prior}} = \theta_{\text{Clim}} + \mathbf{K} \left(\theta_{\text{MISS}} - \theta_{\text{Clim}}\right) \tag{7.16}$$

$$\mathbf{C}_{\text{prior}} = \left(\mathbf{I} - \mathbf{K}\right) \mathbf{C}_{\text{Clim}} \tag{7.17}$$

The matrix $\mathbf{K}$ typically referred to as the Kalman gain provides the magnitude of the update necessary to θ constructed on comparing the covariance matrices of the historical prior $\mathbf{C}_{\text{Clim}}$ and the MISS retrieval $\mathbf{C}_{\text{MISS}}$. This means that when the MISS retrieval is better than the climatology, θ_{prior} is shifted considerably towards the MISS retrieval. Conversely when the MISS retrieval is uncertain — for example when the MISS instrument detects no probable fires — θ_{prior} remains much closer to the historical prior.

One modification is then made to this prior distribution. While the shape coefficients $[a_0, a_1]$ are spatially uniform, fcc varies at the pixel level in response to fire impact. fcc is therefore specified with a broad uninformative prior ($fcc_{\text{prior}} = 0$ and $fcc(\sigma)_{\text{prior}} = 10$). This also negates the bias introduced by selecting pixels where $fcc > 0.3$. With θ_{prior} and $\mathbf{C}_{\text{prior}}$ now specified, the LISS retrieval is conducted using equation 7.6.

Following the retrieval of θ from each sensor, an optimal merged estimate is then made to combine the information arriving from each sensor. Variations in the 'clear-sky' sampling from each sensor may lead to differences in the detected edge process which then leads to differences in the reported timing of fire events. To avoid this, the merging is conducted on the most probable fire detection for each sensor for the mapping period, which was considered to be the date with the greatest retrieved value of fcc.

The selected composite of θ from each sensor are now suitable for producing a merged optimal estimate of θ_{merged}. Recognising the estimate from each sensor as a multivariate normal probability distribution function (PDF) means that each observational record can be combined using conditional

probability. For example for two sensors, the conditional PDF is formed from Bayes rule as:

$$\Pr(\theta_{\text{sensorA}} \mid \theta_{\text{sensorB}}) = \Pr(\theta_{\text{sensorA}}) \Pr(\theta_{\text{sensorB}}) \tag{7.18}$$

Given that θ_{sensorA} and θ_{sensorB} are defined as multivariate normal distributions the merged posterior θ_{merged} can be analytically calculated from iteratively applying the following to each successive sensor record i:

$$\mathbf{K}_i = \mathbf{C}_{i-1} \left(\mathbf{C}_{i-1} + \mathbf{C}_i\right)^{-1} \tag{7.19}$$

$$\theta_{i|i-1} = \theta_{i-1} + \mathbf{K}_i(\theta_{i-1} - \theta_i) \tag{7.20}$$

$$\mathbf{C}_{i|i-1} = (\mathbf{I} - \mathbf{K}_i)\mathbf{C}_{i-1} \tag{7.21}$$

The merged solution θ_{merged} and posterior uncertainties $\mathbf{C}_{\theta\ \text{merged}}$ provide now a sensor-agnostic signal of evaluating evidence of fire.

7.5.3 Burned area detection

Optical remote sensing is only one such signal with which to identify burned area. Active fire products detect flaming areas at the time of the satellite acquisition from anomalously hot thermal signatures (Giglio et al., 2016; Schroeder et al., 2014). By requiring the fire to be actively burning or smouldering, active fire products provide only an incomplete sampling of burned area both in space and time (for an example, see figure 2.3 in chapter 2). Nevertheless, the information active fires can bring to mapping burned area from optical reflectance changes is considerable and has meant that active fires are routinely used in many burned area mapping algorithms (Giglio et al., 2018; Chuvieco et al., 2018; Alonso-Canas and Chuvieco, 2015; Fraser et al., 2000).

A spatiotemporal statistical model is specified for interpreting active fire detections and spreading these across the full extent of the processing tile. The statistical model works on the principle that pixels close in space are more likely to have burned, and burned within a few days of the active fire detection. As the nature of this relationship will be inexact, the dynamic model is specified with uncertainty. The model is specified as a Beta distribution Beta(α, β), such that the active fire inference on burned area is specified:

$$L_{\text{af}}(\text{burned} \mid \alpha, \beta) \sim \text{Beta}(\alpha, \beta) \tag{7.22}$$

A Beta distribution provides a suitable distribution to model probabilities due to it being bounded between 0 and 1. The two parameters of the distribution specify how the probability mass is weighted between 0 and 1. The active fire inference model works on modifying the value of α depending on

both the spatial and temporal distance to the active fire detection. Locations near an active fire receive a large value of α, and pixels further away receive $\alpha \rightarrow 1$. The β parameter is always specified to be unity so that at pixels where α is approximately unity, L_{af} is a uniform distribution.

The details of the process model for α are now detailed. The spatial distance between the pixel and the active fire determines the influence it should have on α. This is built on the assumption that fire events are clustered in the landscape and therefore the likelihood of fire event far away from an active fire is diminished. In a similar manner, the date of the active fire t_{af} is informative about the likelihood of a fire on a different day. Both the spatial distance and temporal date can be incorporated into a distance model:

$$\alpha(x, y, t) = 1 + C \cdot \alpha_{\text{max}} \exp\left(-\frac{d_t}{\sigma_t}\right) \exp\left(-\frac{d_{xy}}{\sigma_{xy}}\right) \tag{7.23}$$

where d_t is the absolute difference in days between the active fire date t_{af} and the mapping date t. d_{xy} is the Euclidean distance between the pixel and the active fire detection. The parameters σ_{xy} and σ_t specify the rate at which the influence of the spatial and temporal distances respectively reduce. For example, larger values of σ_{xy} indicate the assumption that fires are more uniformly spread out in space.

The parameter C is the associated confidence for the active fire detection. This parameter for example comes from the MODIS active fire product and is determined by possible sources of commission error such as sun-glint (Giglio et al., 2016). This confidence level C provides a suitable way to propagate the uncertainty in the active fire observations into the algorithm. α_{max} specifies the maximum value of α which allows the distribution to skew more heavily towards 1. α_{max} was chosen to be 10 so that at the location and on the day of an active fire detection, the mode of $L_{\text{af}}(\text{burned} \mid \alpha, \beta)$ is effectively C. Choosing σ_t, σ_{xy} is more challenging because these determine the level to which the individual detections are spread in space and time. It unlikely that global values for these are appropriate due to observed variations in fire spread rate and duration for different regions (Andela et al., 2019). In a similar model to the one proposed here Giglio et al. (2018) provide different decay constants to account for larger fire sizes in Africa.

Andela et al. (2019) recently presented the The Global Fire Atlas (GFA), which provides useful information for choosing σ_t and σ_{xy}. The dataset provides estimates of fire size, duration, daily expansion and direction of spread for 2003-2016. The atlas lists 13.3 million unique fires which have also been aggregated to provide coarse-resolution 0.25° gridded summary products of fire characteristics. The fire duration estimates from the GFA are used to estimate σ_t. The mean fire duration D over 2003-2016 for each 0.25° grid cell was estimated from the GFA product. It was then

assumed that $\sigma_t = \frac{1}{2}D$. This means that if a value of 4 days has been set for σ_t, the contribution of individual detections decreases 50% every 2 days. In a similar manner, the fire size estimates from the GFA were used to choose σ_{xy} globally. As before, the 13-year mean fire size S km^2 was used to specify $\sigma_{xy} = \frac{1}{2}\sqrt{S}$ km. This means for a fire size of 100 km^2 , the contribution decreases by 50% every 5km, and beyond 10km the detection contributes less than 10% to α.

Figure 7.18 depicts the global climatology of σ_{xy} and σ_t derived from the Global Fire Atlas datasets. Larger fire sizes in Australia and southern Africa lead to high values of σ_{xy}. The highest values of σ_t are found in boreal forests in North America and Eurasia. Deriving this information from the Global Fire Atlas therefore provides an improvement over more coarse continental averages as used in Giglio et al. (2018).

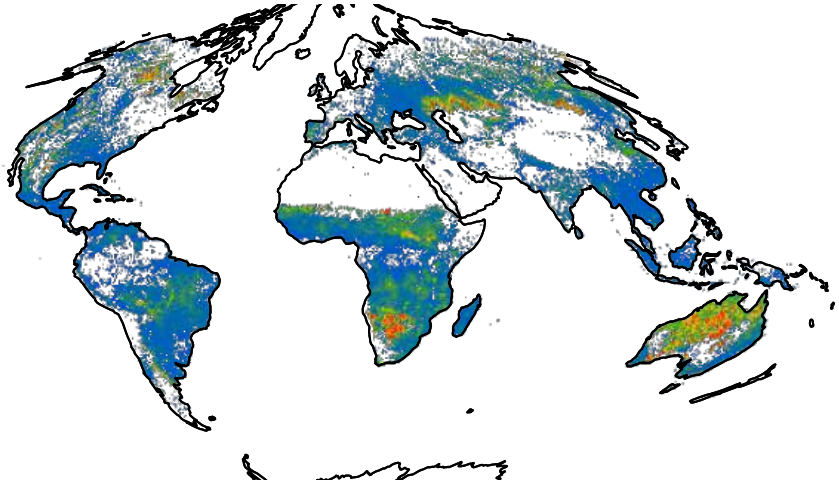
Figure 7.19 shows the mode and standard deviation of $L_{\text{af}}(\text{burned} \mid \alpha, \beta)$ over the study region for September 2012. Shown in the panels are the mode (left column) and standard deviation (middle column) of $L_{\text{af}}(\text{burned} \mid \alpha, \beta)$ for three dates. Shown in the right column are the active fire detections by MODIS Aqua and Terra for each of the three dates. Here, the Global Fire Atlas indicates fires which are temporally short in duration (i.e σ_t is around 3-4 days), and while large areas are burned, the individual fire size is relatively small (σ_{xy} of 2-4 km). This is also evidenced by the patchy and small clusters of active fires detected on each date. Most pronounced is the clustered nature of high probabilities towards areas with many active fire detections. At greater distances from these, the mode of the distribution returns to 0.5 and the standard deviation so that $L_{\text{af}}(\text{burned} \mid \alpha, \beta)$ returns to a uniform distribution — implying that the active fire detections provide no evidence of burning for these pixels. The temporal evolution of the distribution is also apparent, with the areas of the highest mode varying between the three dates. These areas also correspond to the smallest variances.

The inference of fire affected pixels from θ_{merged} is now considered. There are primarily two statistical tests considered in the formulation of the burned area inference:

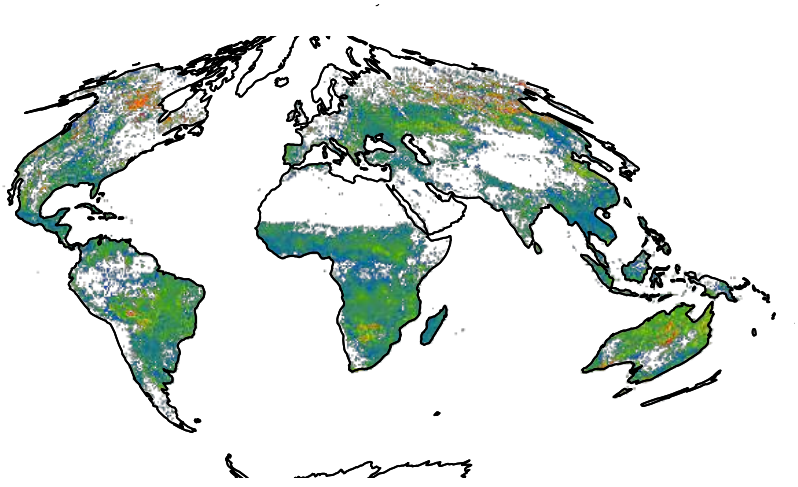
1. The spectrally integrated change signal fcc is sufficient and robust as evidenced by its uncertainty σ_{fcc}.

2. The spectral nature of the change signal $[a_0, a_1]$ matches well to the expected spectral characteristics of fire at the pixel.

The estimate of fcc provides an integrated metric of the fire radiometric severity. As per definition, fcc should range between 0 and 1, with values of fcc closer to 1 indicating near-complete combustion of all vegetation in the pixel (Roy and Landmann, 2005). Small changes in the land surface between subsequent days may result in small positive estimates of fcc not attributable to burning. In addition



(a) σ_{xy}



(b) σ_t

Figure 7.18. Global maps of a) σ_{xy} and b) σ_t as derived from the Global Fire Atlas (Andela et al., 2019).

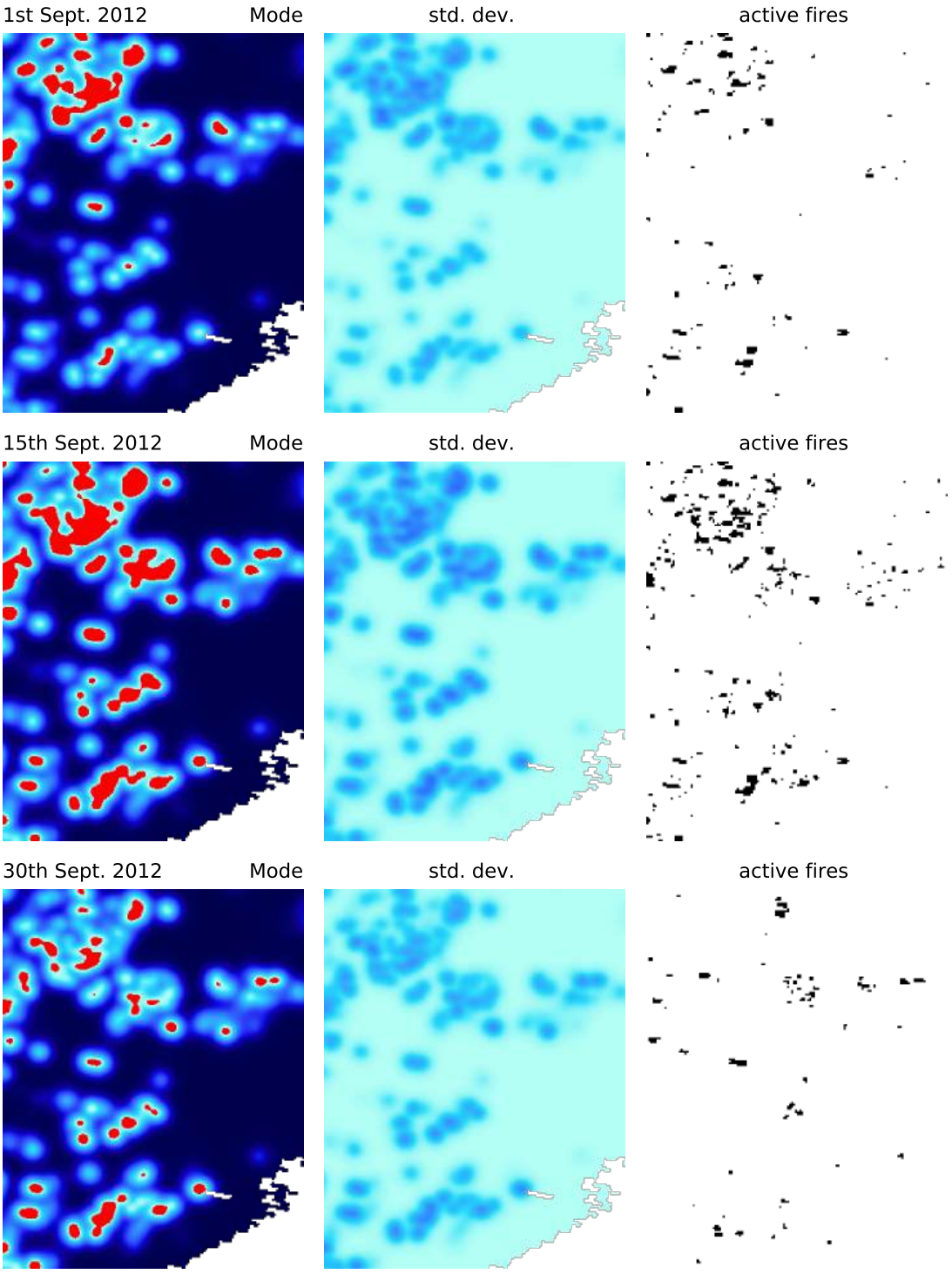


Figure 7.19. Moments of the active fire inference distribution $\Pr(\text{burned} \mid \alpha, \beta)$ over the study region for three dates. Left column) Mode of $\Pr(\text{burned} \mid \alpha, \beta)$ for three dates. Middle column) Standard deviation of $\Pr(\text{burned} \mid \alpha, \beta)$ for the three dates. Right column) active fires (black) detected by MODIS Aqua and Terra for the three dates. The mode images scale between 0.5 (black) to 0.85 (cyan), above 0.85 is highlighted in red. The standard deviation is scaled between 0 (black) and 0.25 (cyan).

to this, a lower bound on the observable fire signal is determined by the spatial resolution of the coarse-resolution data and uncertainty in the reflectance signal $\Delta\rho$. This suggests an effective signal-to-noise ratio (SNR) of the fcc retrieval below which values of fcc are not to be considered reliable evidence of fire occurrence (Roy et al., 2019). To discriminate between real burning and spurious retrievals, a threshold condition is enforced, with the inference that if $fcc > T$ the pixel has likely burned. With a change of variables, $z = fcc - T$, $\sigma_z^2 = \sigma_{fcc}^2 + \sigma_T^2$, the likelihood that fcc exceeds T is:

$$L(\text{burned} \mid fcc) \propto \exp\left(-\frac{1}{2}\frac{z^2}{\sigma_z^2}\right) \tag{7.24}$$

Phrasing the SNR test this way ensures that uncertainty in fcc *and* the SNR threshold T is accounted for. For the purposes of prototyping the algorithm T is specified with $T_\mu = 0.2$ and $\sigma_T = 0.03$, implying a relative uncertainty in the value of T of 15%.

While fcc is determined by the magnitude of the change signal it does not provide conclusive evidence of burning. The $[a_0, a_1]$ coefficients provide information on the spectral shape of the change signal and have been shown to be spatially consistent for burned areas (i.e. figure 7.9). Figure 7.20 shows the estimated spectral parameters classified by 'high' $fcc > 0.3$ and low $fcc < 0.3$ values. Also displayed are the 1–3σ contours of the spectral prior distribution from the MODIS archive (see section 7.5.2). The estimate of a_0 and a_1 for high fcc pixels are clustered mostly within 1 standard deviation of the prior mean. Conversely, a_0 and a_1 values associated low fcc values show more limited correspondence with the prior. In panel b), a random selection of $[a_0, a_1]$ pairs are chosen and the 1σ uncertainties plotted. For high fcc retrievals, the uncertainties are typically much smaller than those from low fcc pixels and also much smaller than the prior variance.

Large departures from the expected values of $[a_0, a_1]$ are therefore indicative of spectral changes inconsistent with fire. The likelihood in equation 7.24 can then be complemented by an additional term:

$$L(\text{burned} \mid \theta) \propto \exp\left(-\frac{1}{2}\left[(a - a_{\text{prior}})^{\mathrm{T}}\mathbf{C}_{\text{a prior}}^{-1}(a - a_{\text{prior}}) + \frac{z^2}{\sigma_z^2}\right]\right) \tag{7.25}$$

where a_{prior} and $\mathbf{C}_{\text{a prior}}^{-1}$ are provided from the spectral shape database. As a departs further from the spectral expectation, the likelihood exponentially decreases.

$L(\text{burned} \mid \theta)$ provides a mapping from θ to the evidence of fire occurrence conditional on θ and the uncertainties. In a similar manner, $L_{\text{af}}(\text{burned} \mid \alpha, \beta)$ provides a mapping from the active fire evidence to a burned area inference. Because the two inferences originate from different sources (i.e. optical and thermal observations) they can be considered independent so that the combined inference is simply:

$$L(\text{burned} \mid \theta, \alpha, \beta) = L(\text{burned} \mid \theta)\, L_{\text{af}}(\text{burned} \mid \alpha, \beta) \tag{7.26}$$

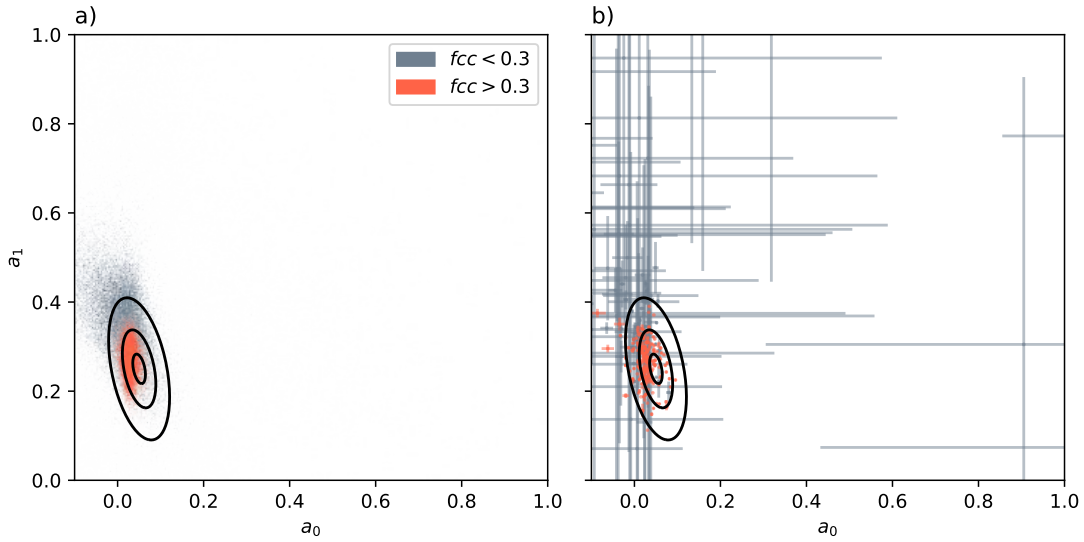


Figure 7.20. Agreement between a_0 and a_1 with the spectral parameter prior. a) spectral parameters for likely-burned ($fcc > 0.3$) and unlikely-burned ($fcc < 0.3$) pixels for the study region. Likely-burned pixels fall within the prior from the MODIS record. b) random selection of the pixels from panel a) depicted with 1σ uncertainties. Pixels that are likely to be burned show have much smaller uncertainties which fall within the prior distribution.

As described in section 6.2 in chapter 6, the expected value of the likelihood (which is specified between [0,1]) marginalises the uncertainties in the state θ and algorithm parameters α, β, T provides the probability of fire occurrence $\mathrm{P_b}$:

$$\mathrm{P_b} = \int L(\text{burned} \mid \theta, \alpha, \beta)\, \mathrm{d}\theta\, \mathrm{d}\alpha\, \mathrm{d}\beta \tag{7.27}$$

Figure 7.21 displays the maximum $\mathrm{P_b}$ from the daily $\mathrm{P_b}$ estimates for the study region. $\mathrm{P_b}$ shows good discrimination of burned areas, with areas that the algorithm has detected as burned ascribed with high values of $\mathrm{P_b}$, while unburned areas provided low probabilities ($\mathrm{P_b} < 0.01$). Producing explicit burned area maps as is done currently then involves a simple delineation of areas where $\mathrm{P_b}$ is considered to provide sufficient evidence of burning. Figure 7.21 also depicts conventional day of burning (DOB) map for pixels where the maximum $\mathrm{P_b} > 0.5$.

7.6 Synthetic demonstration with MODIS

To consider how the consistent fcc retrieval scheme operates, the algorithm was processed over a synthetic multi-sensor configuration. The two MODIS instruments onboard Aqua and Terra provide

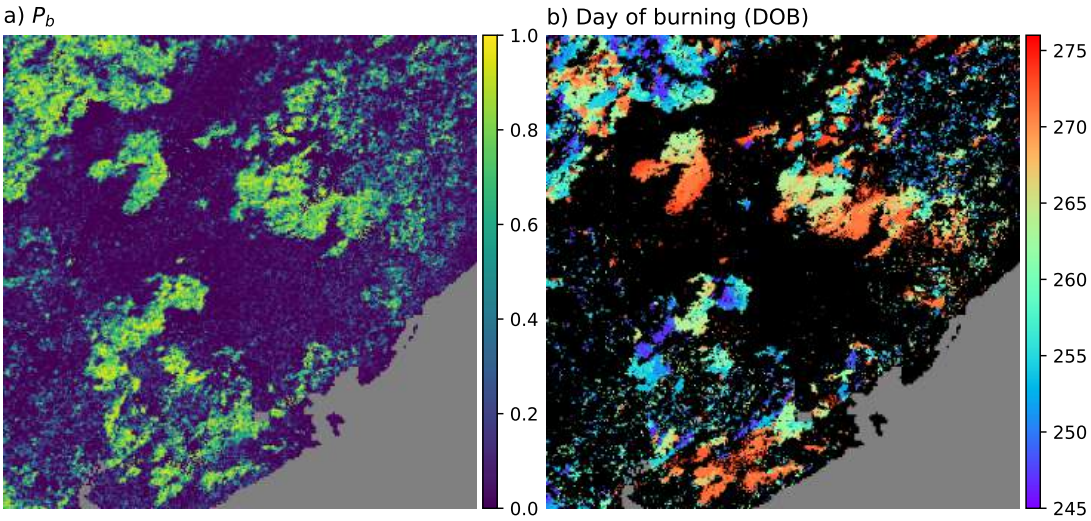


Figure 7.21. a) Maximum $\mathrm{P_b}$ from the daily $\mathrm{P_b}$ estimates for the study region. b) An explicit day of burning (DOB) product produced from $\mathrm{P_b} > 0.5$

two independent measurement systems on which to prototype a synthetic multi-sensor processing chain for the algorithm.

To consider a MISS and LISS sensor processing chain, observations of surface reflectance from MODIS Aqua (MYD09GA.C6) were degraded to provide a four-channel sensor with three visible channels and a NIR channel. This four-channel instrument provides an approximation to the Sentinel3 OLCI instrument. While OLCI has 21 spectral channels, these are spectrally narrow and occupy only the visible-to-NIR region (400nm–1020nm). Given the generally high level of correlation which can be expected between these channels, a four-channel representation provides a broad approximation to the information contained in the OLCI channels for fire monitoring (especially given the spectrally smooth end-members for ρ_{burn} within the fcc model — shown in figure 7.8). The two OLCI sensors provide a revisit of approximately 1.1 days at the equator, which is comparable to the revisit of the MODIS Aqua instrument. This LISS sensor is named synOLCI.

Observations from MODIS Terra (MOD09GA.C6) can be considered broadly similar to a VIIRS-like observational record. The MISS instrument is referred to as synVIIRS. Active fire detections are provided by the MODIS Collection 6 active fire product from both MODIS Terra (MOD14.006) and Aqua (MYD14.006). The purpose of this synthetic experiment is to answer two questions:

1. How does the consistent fcc retrieval scheme perform for the LISS sensor?

2. How much additional information does the LISS sensor provide?

To address the first question, the properties of the constrained fcc retrieval for synOLCI were examined. Figure 7.22 shows a comparison of the retrieval of fcc, a_0, a_1 from synOLCI with the unconstrained fcc retrieval method suitable for MISS sensors (equation 7.3) and the spectrally constrained fcc retrieval designed for LISS instruments (equation 7.6). The unconstrained fcc retrieval for synOLCI (panel a) shows some discernible burned areas. There is however considerable speckle to the retrieval which degrades the quality of the burned area inference possible from it. Utilising the spectral constraint improves on this (panel b), with an better spatial consistency of fcc for burned and unburned pixels and less spurious fcc retrievals (i.e. speckle). The spectrally constrained retrieval improves the spatial coherence of the fcc signal even though retrieval operates independently in each pixel. The spectral parameters (panels c-f) also show much better spatial consistency over burned areas (as shown earlier in figure 7.17) for the constrained retrieval.

Evaluating the second question is relatively straightforward due to the statistical framework of the algorithm. The reduction in uncertainty in the fcc retrieval from the combination of the MISS and LISS sensors compared to just either sensor reveals how much additional information has been introduced from the multi-sensor framework. Similarly, inferring the burned area from 1) only the synVIIRS retrieval and 2) the synMERGED retrieval indicates how the information rendered from the LISS sensor improves the inference. This improvement is encapsulated in the change in the probability between $\mathrm{P_b}$(synVIIRS) and $\mathrm{P_b}$(synMERGED).

Figure 7.23 illustrates how the inclusion of the LISS sensor synOLCI improves the estimate of fcc. Panel a & b show fcc estimated from synVIIRS only, and the merged synVIIRS and synOLCI inference. The overall effect of the inclusion of synOLCI is to reduce the estimated fcc in some pixels as depicted in panel c. This is not uniform across all burn scars, with several of the burned pixels along the coast showing an increase in fcc in the merged fcc. Panel d) shows the impact of the inclusion of synOLCI on the fcc posterior uncertainty. The ratio of the posterior uncertainties σ_{fcc}(synMERGED)/σ_{fcc}(synVIIRS) indicates a reduction in uncertainty when less than unity. The greatest shrinkage in the posterior uncertainty can be observed along in burned pixels detected by synVIIRS along the coast. Within areas of high fcc detected from synVIIRS, the additional information from synOLCI leads to a ratio of around 0.4–0.7, indicating that the inclusion of synOLCI improves the quality of the merged fcc.

Figure 7.24 depicts the posterior probabilities of fire occurrence $\mathrm{P_b}$. Areas of high probabilities (defined as $\mathrm{P_b} > 0.75$) are depicted in red. Most notable is the improvement in the confidence with greater areas of high probability for $\mathrm{P_b}$(synMERGED) than $\mathrm{P_b}$(synVIIRS). Figure 7.24d) shows how this additional information changes the mapped burned area. Shown in teal are areas where

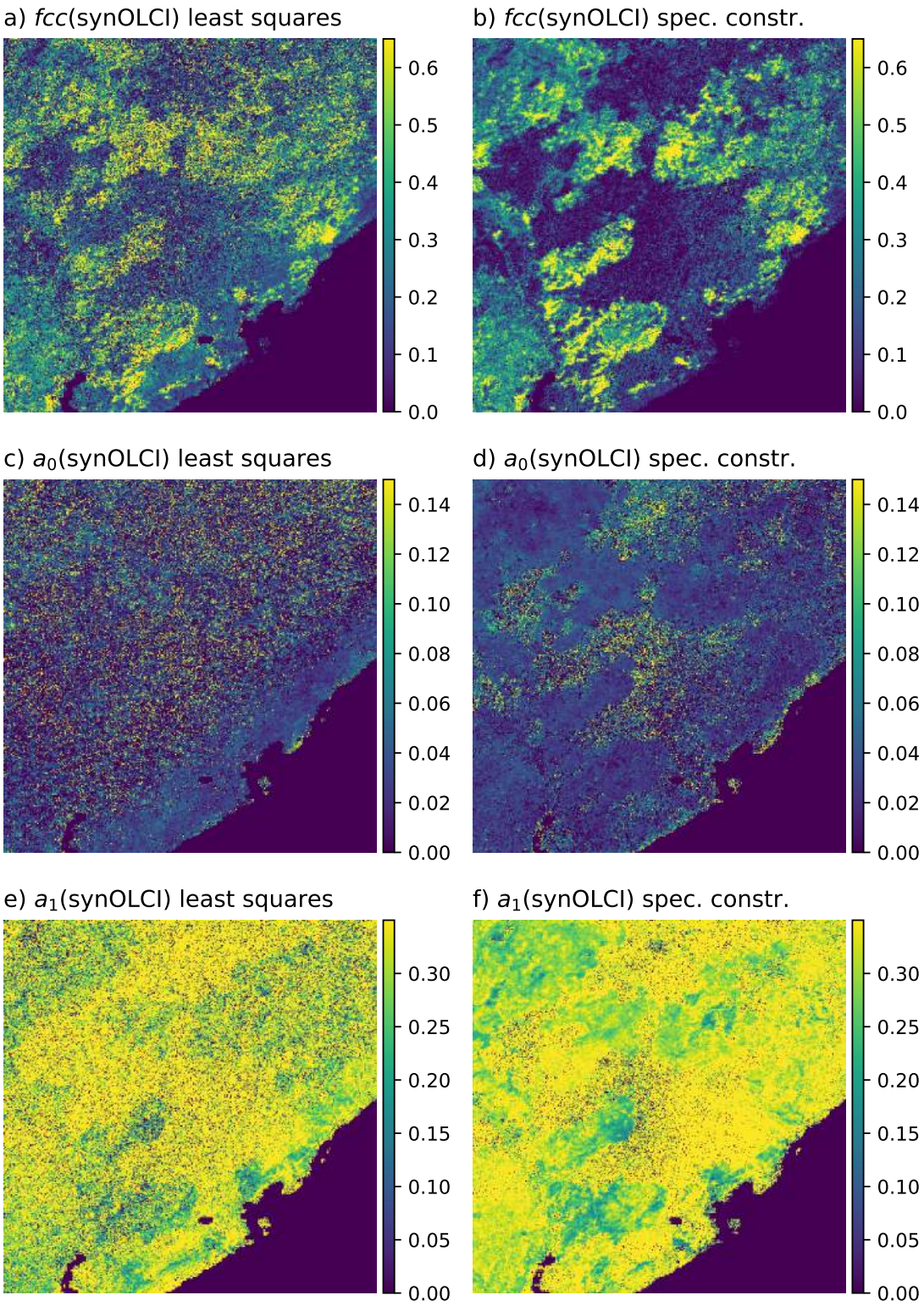


Figure 7.22. Comparison of the two fcc retrieval methods for the synOLCI observations. Left column) depicts fcc, a_0, a_1 for an unconstrained least squares fcc retrieval suitable for MISS sensors as posited in section 7.4. There is considerable noise in the retrieval of all parameters compared to the retrieval conducted by the two MODIS sensors in figure 7.17. Right column) depicts fcc, a_0, a_1 for the fcc retrieval scheme with a prior derived from the MISS retrieval which is suitable for LISS instruments.

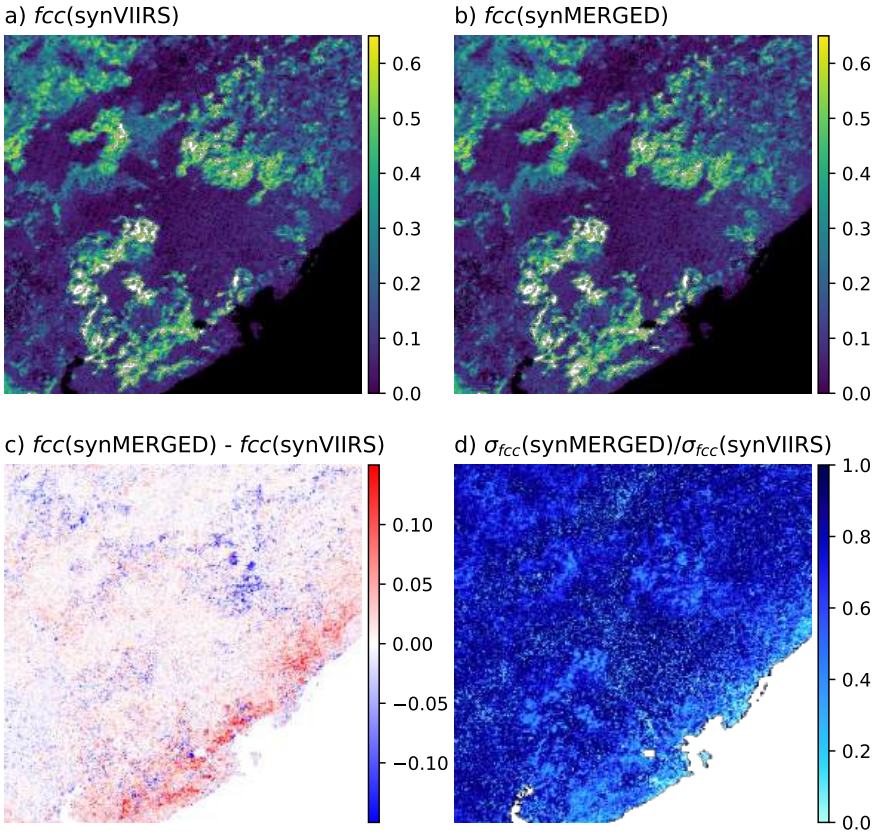


Figure 7.23. Improvements to fcc retrieval with inclusion of a LIS sensor synOLCI. a) fcc from synVIIRS only. b) Merged fcc from both synVIIRS and synOLCI. c) Difference in fcc between synVIIRS and synMERGED. d) Reduction in the posterior uncertainty of fcc due to the merging. Values less than 1 indicate a reduction in the posterior uncertainty in fcc.

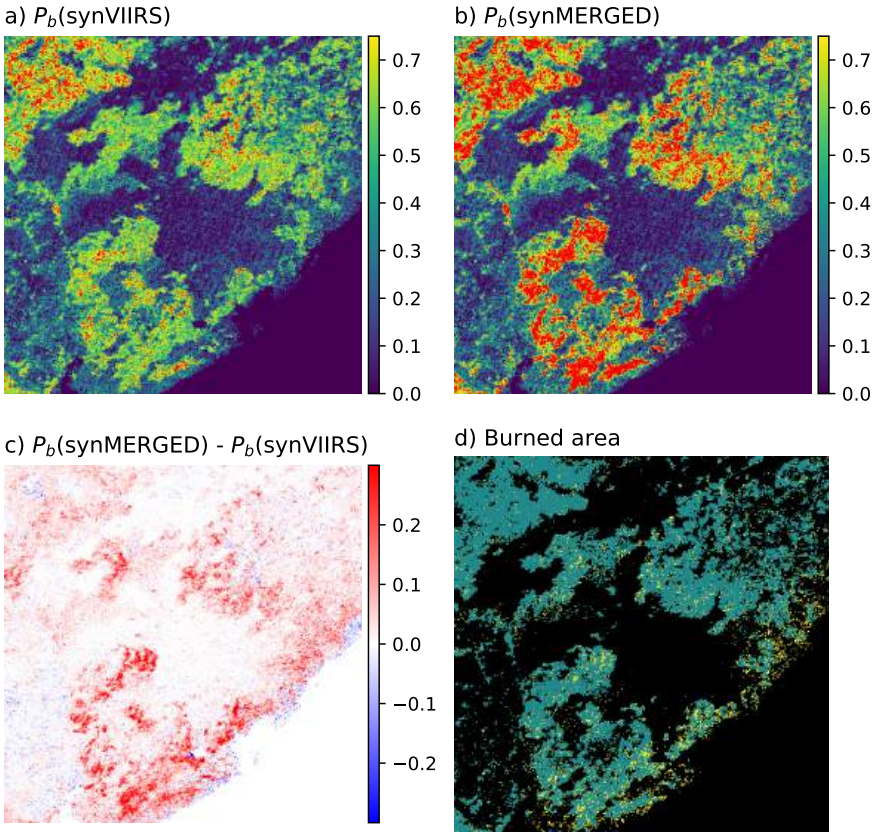


Figure 7.24. Additional burning inference from inclusion of LIS sensor synOLCI.

P_b(synVIIRS) already exceeds 0.5 and are therefore considered burned. Yellow pixels are where the merged product P_b(synMERGED) exceeds 0.5 but not P_b(synVIIRS) indicating that these are mapped burned in the merged product but not from VIIRS exclusively. The effect on the mapped product is fairly limited — the merged product classifies some additional pixels generally clustered within burn scars that had an insufficient fcc retrieval from the synVIIRS-only retrieval. Instead, the larger effect is observed in how the detection confidence is improved with the addition of the LISS sensor. The additional information that the LISS sensor brings to the algorithm leads to an increase in P_b, most particularly over burn scars (shown in 7.24c). Within the burn scars, P_b increases by around 0.1–0.3, while unburned areas mostly show no change or a small reduction in P_b.

This chapter has prototyped an algorithm for monitoring burned area from multiple sensors. The options and challenges for producing multi-sensor algorithms were considered in section 7.3. It was argued that central to this task is the consistent interpretation of the available information from each optical sensor to the task of burned area mapping. This is posited on the use of a multi-spectral model of the optical signal which can compress the information available from different sensors into a single change signal metric fcc and two spectral parameters $[a_0, a_1]$. The tracking of uncertainties through the system ensures that each sensor record can be optimally combined to produce a consistent uncertainty quantified (UQ) BA detection system. The modular framework also allows for the analysis of the contribution of each sensor to the overall inference quality.

This chapter, therefore, contributes to addressing the third aim of this thesis specified in section 1.2 to "prototype an algorithm for the consistent inference of burned area from multiple sensors suitable for generating a long-term CDR of burned area". The next chapter demonstrates the algorithm applied to a real multi-sensor system.

Chapter Eight

Demonstration of the algorithm with observations from several sensors

8.1 Introduction

This chapter completes work towards the third aim of the thesis specified in chapter 1 to "prototype an algorithm for consistent burned area monitoring from multiple satellite instruments". In this chapter, the algorithm is applied to a real multi-sensor combination of observations from Suomi-NPP Visible Infrared Imaging Radiometer Suite (VIIRS), Sentinel-3 Ocean and Land Colour Instrument (OLCI) and MODIS.

8.2 Implementation

For the purposes of prototyping the system, the algorithm was evaluated over three MODIS/VIIRS tiles which cover three distinct regions (shown in figure 8.1). The selection of these sites was limited by the availability of the Sentinel-3 surface reflectance product (January 2019—present), however, they do cover three different fire types. The h20v03 tile covers an area of western Russia which includes a mixture of croplands to the south of the tile and mixed forests to the north of the tile (according to the MODIS MCD12C1 land cover product, Friedl et al., 2010). The h20v08 tile covers a large extent of Democratic Republic of Congo and South Sudan. The tile contains a mixture of tropical forest in the lower extent and savannas and grasslands to the north. The h09v07 tile covers Mesoamerica and contains a mixture of tropical forest, croplands and shrublands.

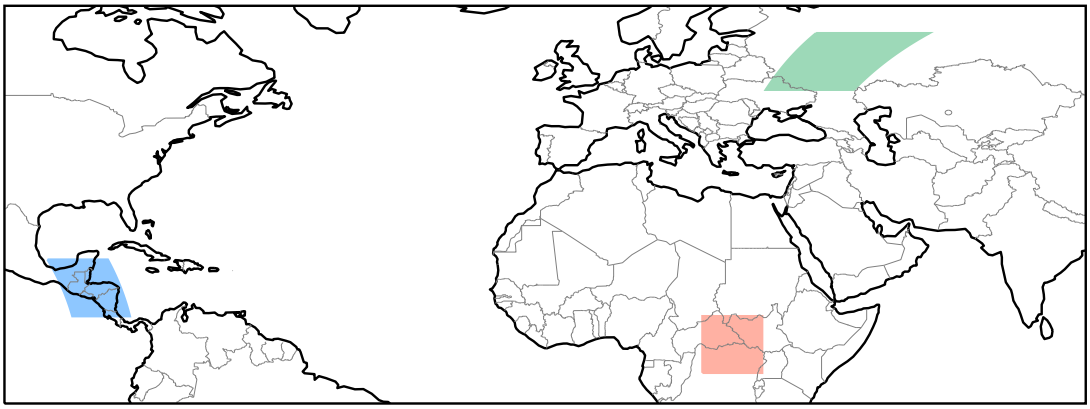


Figure 8.1. The three MODIS/VIIRS tiles used to prototype the algorithm: h09v07 (blue), h20v08 (red), and h20v03 (green). The three tiles were chosen because they cover three different fire types (see main text) and the limited availability of the Sentinel-3 surface reflectance product (January 2019–present).

Surface reflectance observations from the MODIS Aqua (MYD09GA) and Terra (MOD09GA) instruments were used (see section 7.6). These products are provided at a pixel resolution of 500m on the MODIS land sinusoidal grid (Vermote et al., 2011). The products do not provide per-pixel uncertainties in reflectance. Instead an approximate uncertainty is provided as:

$$\sigma_\rho(\lambda) = \sqrt{0.005 + 0.05 \times \rho(\lambda)} \tag{8.1}$$

which is based on the accuracy assessment of the atmospheric correction algorithm used for the product (Vermote et al., 2011). This error model has been shown to be appropriate for the MODIS reflectance product and is derived from validation against AERONET ground data (Vermote and Kotchenova, 2008; Roger et al., 2016). Wang et al. (2011) noted a root-mean-square error (rmse) in MODIS surface reflectance of 0.01âĂŞ0.025 with larger rmse values for spectral channels with higher reflectance. These rmse values correspond to a 4-7% relative uncertainty which is in agreement with the $\sqrt{0.005}$ providing a static 7% relative error in reflectance. The MODIS collection 6 active fire products from Terra (MOD14A1) and Aqua (MYD14A1) were also used within the algorithm. The products provide active fire detections daily at 1km and an associated confidence level based on potential sources of commission error (Giglio et al., 2016).

Surface reflectance and active fire products from Suomi-NPP VIIRS have been developed which provide continuity with the MODIS record and are suitable for use within the algorithm. The VNP09GA product provides daily surface reflectance at 500m-1km (JPSS, 2013). The product is provided in the same sinusoidal projection as MODIS and arranged into tiles which are aligned with the MODIS sinusoidal tile grid (see figure 8.2). The VNP09GA product therefore serves to provide continuity

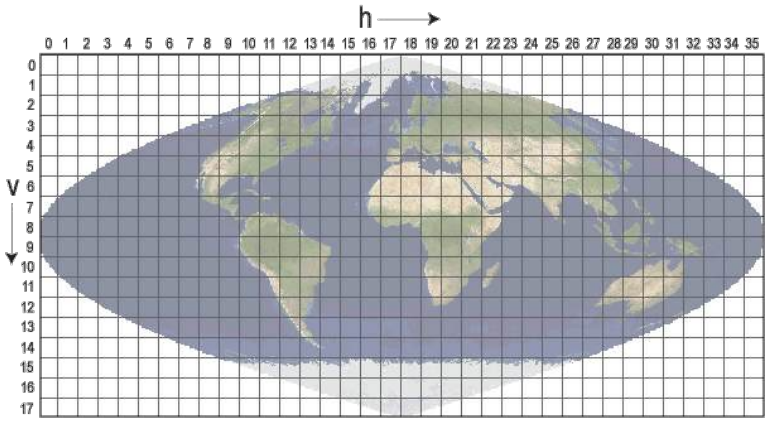


Figure 8.2. The MODIS/VIIRS sinusoidal land tile grid. From MODIS Land Team (source: Vermote et al. (2011)).

with the MODIS daily surface reflectance products MOD09GA/MYD09GA (Vermote et al., 2014). As with MOD09GA, the VIIRS surface reflectance product does not have per-pixel uncertainty estimates. Instead, the uncertainty for VNP09GA is recommended to be the same as the MOD09GA product, i.e. equation 8.1 (JPSS, 2013). Validation of the VIIRS reflectance products indicates that this error model is appropriate (Roger et al., 2016). Cao et al. (2013) observed a 5% bias between the VIIRS M1 spectral channel and the Aqua MODIS band 8 channel which is due to MODIS calibration drift. Vermote et al. (2014) repeated the work of Vermote and Kotchenova (2008) which proposed the error model in equation 8.1 for the VIIRS sensor. Vermote et al. (2014) found through a preliminary validation that the error model provided a good match accuracy of the VIIRS reflectance.

Active fire products have also been developed from the VIIRS sensor which provides a continuation of the MODIS active fire products. The daily active fire products from VIIRS are provided in the MODIS sinusoidal grid and tiling scheme shown in figure 8.2. The 1km VNP14A1 product is derived from an algorithm based upon the multi-spectral contextual algorithm developed for MODIS active fire products but with additional changes to optimise the algorithm for the VIIRS sensor (Schroeder and Giglio, 2018). Conversely, the VIIRS 375m I-Band product is based on an algorithm which diverges more from the MODIS active fire products (Schroeder et al., 2014). The VNP14A1 was chosen for use here due to its heritage with the MODIS active fire products. The attempt to provide continuation of MODIS LTDR products within the VIIRS record provides a suitable baseline to consider sustaining the MODIS fire legacy with the algorithm. Similarly, the use of the MODIS grid provides a suitable format for the production of a consistent burned area product from MODIS and VIIRS in which the level-1 and gridding procedures share the same heritage.

The Sentinel-3 level-2 Synergy product (SY_2_SYN) was used as the input surface reflectance product for OLCI. The product provides atmospherically corrected surface directional reflectance with

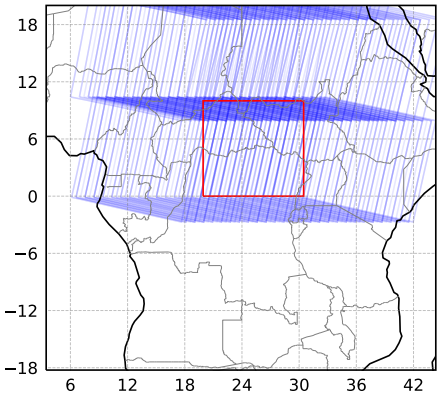


Figure 8.3. Footprints of each SY_2_SYN swath stripe (blue) collected over three months (Jan–Mar 2019) overlapping the MODIS/VIIRS tile h20v08 (red).

associated uncertainties (North, 2010). The product provides reflectance in 16 OLCI channels at 300m resolution. Per-pixel uncertainties are provided for each OLCI channel $\sigma_\rho(\lambda)$. These are estimated as a combined uncertainty from three components:

$$\sigma_\rho(\lambda) = \sqrt{\sigma^2_{\text{sens}} + \sigma^2_\tau + \sigma^2_{\text{RT}}} \tag{8.2}$$

where σ^2_{RT} are the uncertainties associated with the per-channel quantisation and calibration error of the OLCI instrument, σ^2_τ from uncertainties in the atmospheric correction and σ^2_{RT} approximated uncertainties of the atmospheric radiative transfer model (North, 2010). This is a considerable advantage of the SY_2_SYN product over the VIIRS and MODIS reflectance products which have only the recommended uncertainties described by equation 8.1. At present, there has been no published validation or accuracy assessment of the (SY_2_SYN) product beyond North (2010).

The product is provided on the unprojected OLCI swath image grid, which is not compatible with the MODIS/VIIRS sinusoidal grid. Figure 8.3 depicts the footprints of each SY_2_SYN swath image overlapping the MODIS tile h20v08 for the study period. Each product was therefore reprojected to the MODIS–VIIRS 500m sinusoidal grid with a bilinear interpolation reprojection algorithm (GDAL, 2019). For each day several SY_2_SYN products cover the extent of the MODIS tile. The reprojected products were composited to produce a daily surface reflectance product and limited to the MODIS/VIIRS tile extent. Figure 8.4 details this process. This produced a derived reflectance product compatible with the MODIS MOD09GA and VIIRS VNP09A1 products which is referred to here as SYN09.

While the SY_2_SYN product provides OLCI reflectance across 16 OLCI channels, only four channels are retained in the SYN09 product. This was decided upon because the information content of many of these additional OLCI channels for fire monitoring (and more specifically inverting the fcc model) is limited. The spectral sampling provided by the full 16 channels only covers the visible-to-NIR spectral region and therefore the information in these channels is highly correlated for the burn

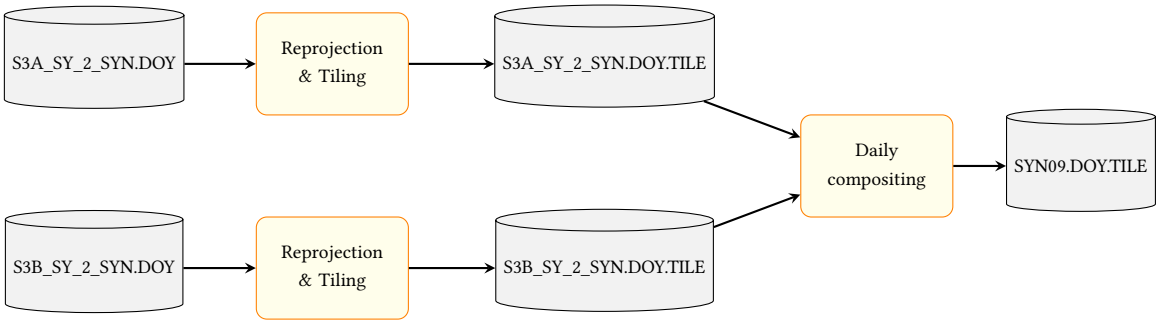


Figure 8.4. Processing chain to produce tiled daily S3-A/B reflectance product SYN09. Synergy reflectance products from Sentinel-3A & B are reprojected and tiled to the MODIS/VIIRS grid. Multiple daily acquisitions are composited to provide a daily surface reflectance product from both sensors SYN09.

signal spectral end-members within the fcc model (see figure 7.10). For the SYN09 product only four channels are retained: Oa03, Oa06 and Oa08 which provides coverage in the visible (Oa03: 442.5nm; Oa06: 560nm; Oa08: 665nm), and the Oa18 channel which covers the NIR (885nm).

Figure 8.5 depicts this SYN09 product for three subsequent dates for the MODIS/VIIRS tile h20v08. Apparent is the variations in the footprint of the OLCI instruments between subsequent dates. As the OLCI 1270km swath is considerably narrower than the VIIRS (3060km) or MODIS (2330km) swath, large areas of the tile are not observed each day by the OLCI instruments. On the final date, acquisitions by both Sentinel-3A and Sentinel-3B are composited to provide full coverage of the tile. Differences in the acquisition geometries (shown is the view zenith angle) of the two satellites leads to a visible BRDF effect in the middle of the image extent. This effect is reduced by the BRDF correction within the burned area mapping algorithm (see section 7.5.1).

Comparing the VNP09 and SYN09 products reveals a degree of spatial coherence between the products. Figure 8.6 shows a visual comparison of the two reflectance products for a small subset of the h20v08 tile which has some visible fires. Good spatial coherence can be observed between the products with matching areas of high and low contrast in and around the burn scars. Changes in the apparent brightness of each image is a result of BRDF effects from changing acquisition geometries of the two sensors.

Following the generation of the SYN09 product, surface reflectance products from each sensor were pre-processed following section 7.5.1. NBAR measurements from each sensor were independently used to retrieve the parameters of the fcc model as specified in section 7.5.2. Retrievals from MODIS and VIIRS NBAR followed the unconstrained retrieval specified in equation 7.3. The more limited spectral sampling provided by the OLCI NBAR requires that the constrained fcc model was used

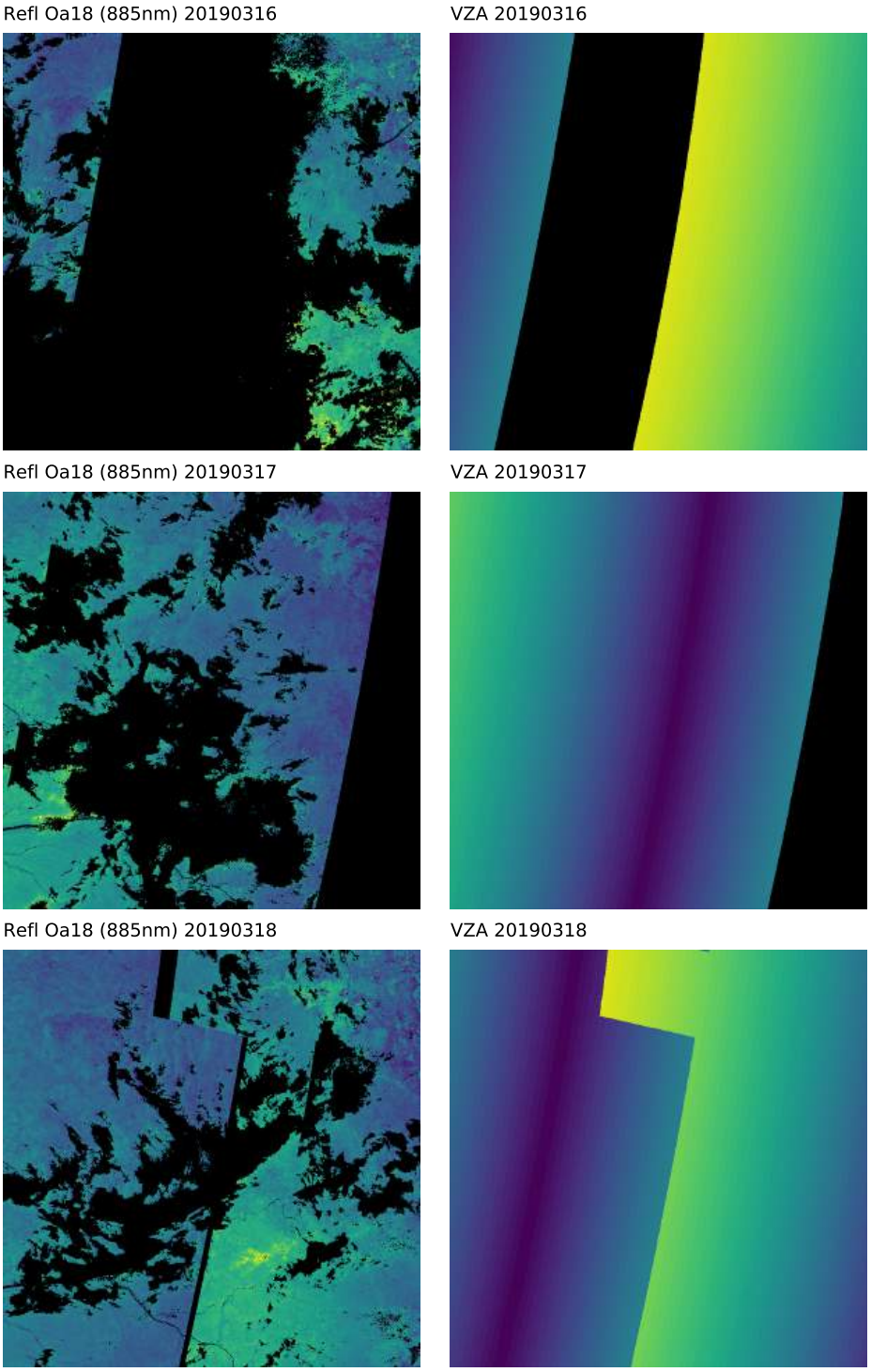


Figure 8.5. The SYN09 reflectance product for three subsequent dates (16/03/2019–18/03/2019) for the tile h20v08. The left column shows surface reflectance for the OLCI Oa18 (885nm) channel and right column the view zenith angle (VZA) between the OLCI sensor and the sun. Two overlapping SY_2_SYN from Sentinel-3A and Sentinel-3B are visible in the last row. The differences in the VZA between the two acquisitions lead to a visible BRDF effect between the two images.

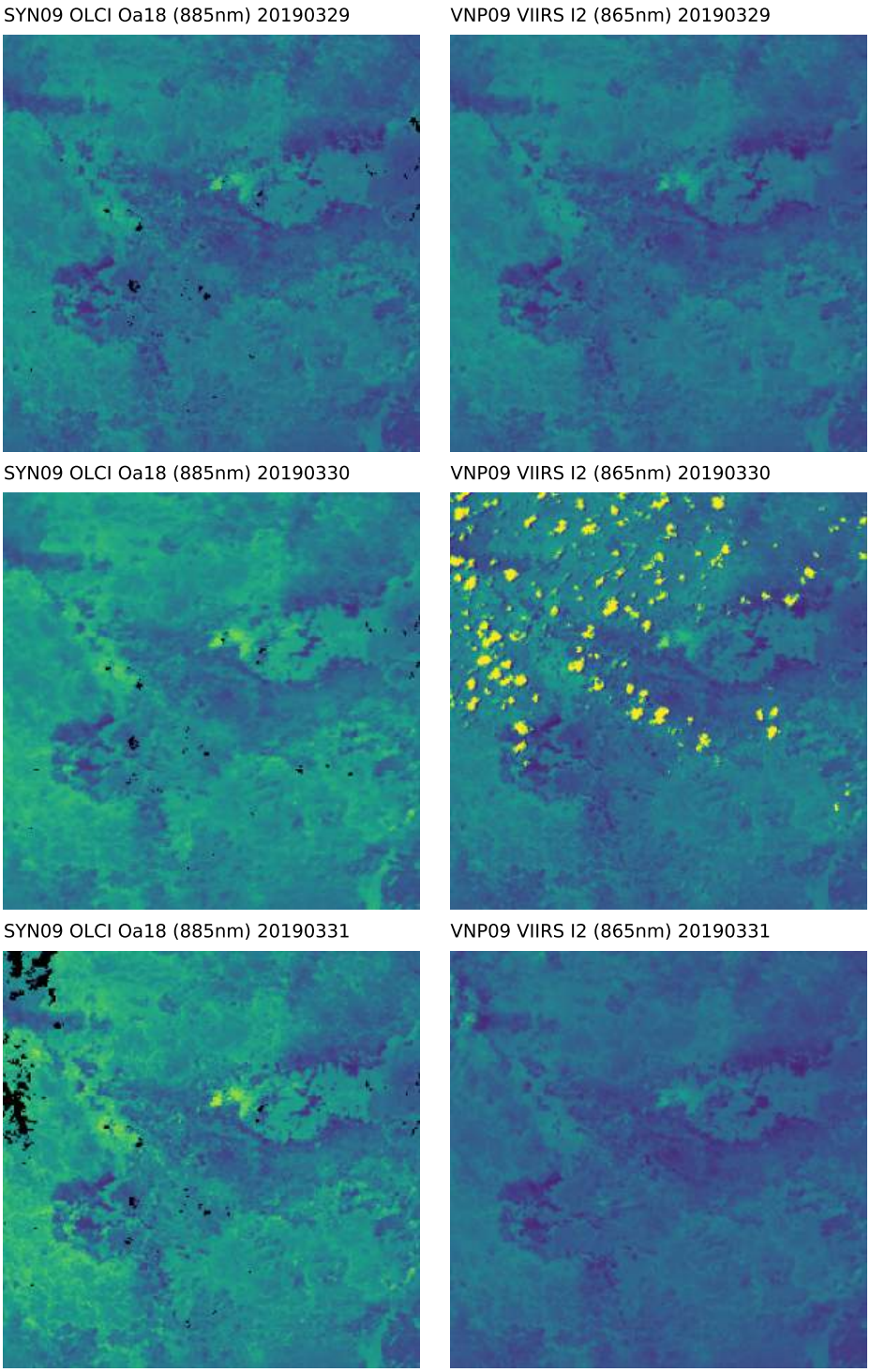


Figure 8.6. Visual comparison of the SYN09 and VNP09 products over a subset of h20v08 with burn scars.

(equation 7.6). Each sensor estimate of fcc, a_0 and a_1 were then optimally merged at the pixel level (section 7.5.2) and the inference on per-pixel burned area conducted (section 7.5.3).

8.3 Results

8.3.1 Multi-sensor surface reflectance tracking and change identification

Figure 8.7 depicts several characteristics of the surface tracking routine for each sensor. Depicted is a region covering 30000 km^2 from the h20v08 tile along the border between South Sudan and the Democratic Republic of the Congo. The grey grid-lines are spaced every 100 MODIS/VIIRS pixels. A–C) depict composites of the maximum reported drop in NIR reflectance ($\Delta\rho$) for OLCI, VIIRS and MODIS respectively. Also plotted in C are active fires detected by the MODIS active fire products (MOD14 and MYD14) in red. Areas where there is no reported drop in NIR reflectance (i.e. $\Delta\rho \geq 0$) are masked as these are unlikely to be pixels experiencing a fire. For each sensor, the largest reductions in NIR reflectance are found in the centre of the large burn scar. Smaller potential burn scars and the perimeter of the large burn scar show more modest reductions in NIR reflectance for each sensor. The reductions in NIR reflectance reported by OLCI are typically 5-10% larger than those detected by MODIS or VIIRS. Also noticeable is evidence of small areas of reflectance reductions detected by OLCI which are not detected by MODIS or VIIRS. Many of these align with isolated detections from the MODIS active fire products. These two features potentially highlight a better sensitivity of the OLCI sensor to smaller burn scars and sub-pixel burning. This could originate in the higher spatial resolution of the OLCI instrument (300m) and corresponding narrower point spread function (PSF). This effect would remain somewhat into the reprojected OLCI 500m sinusoidal product.

D–F) depicts the relative uncertainty in the NIR reflectance change (defined as $\sigma_{\Delta\rho}/\Delta\rho$). Greater relative uncertainties occur at the perimeters of large burn scars where relative uncertainties for MODIS and VIIRS exceed 40%. Within the interiors of potential burn scars, uncertainties from MODIS and VIIRS are lower at around 10-15%. OLCI (D) also shows the same pattern of greater relative uncertainties towards the perimeters of burn scars — though the extent of those that exceed 40% is more limited. The small isolated detections from OLCI are also associated with greater relative uncertainties than pixels within the larger potential burn scars. G–I) shows the day of year (DoY) associated with the change in reflectance t_f for presumed fire-affected pixels. Within the large central burn scar, the sensors detect the change within 1-2 days of each other. An area of the central burn scar is detected inconsistently between the three sensors, with VIIRS detecting a change around day 25. This falls at the edge of the mapping window in which edges are identified.

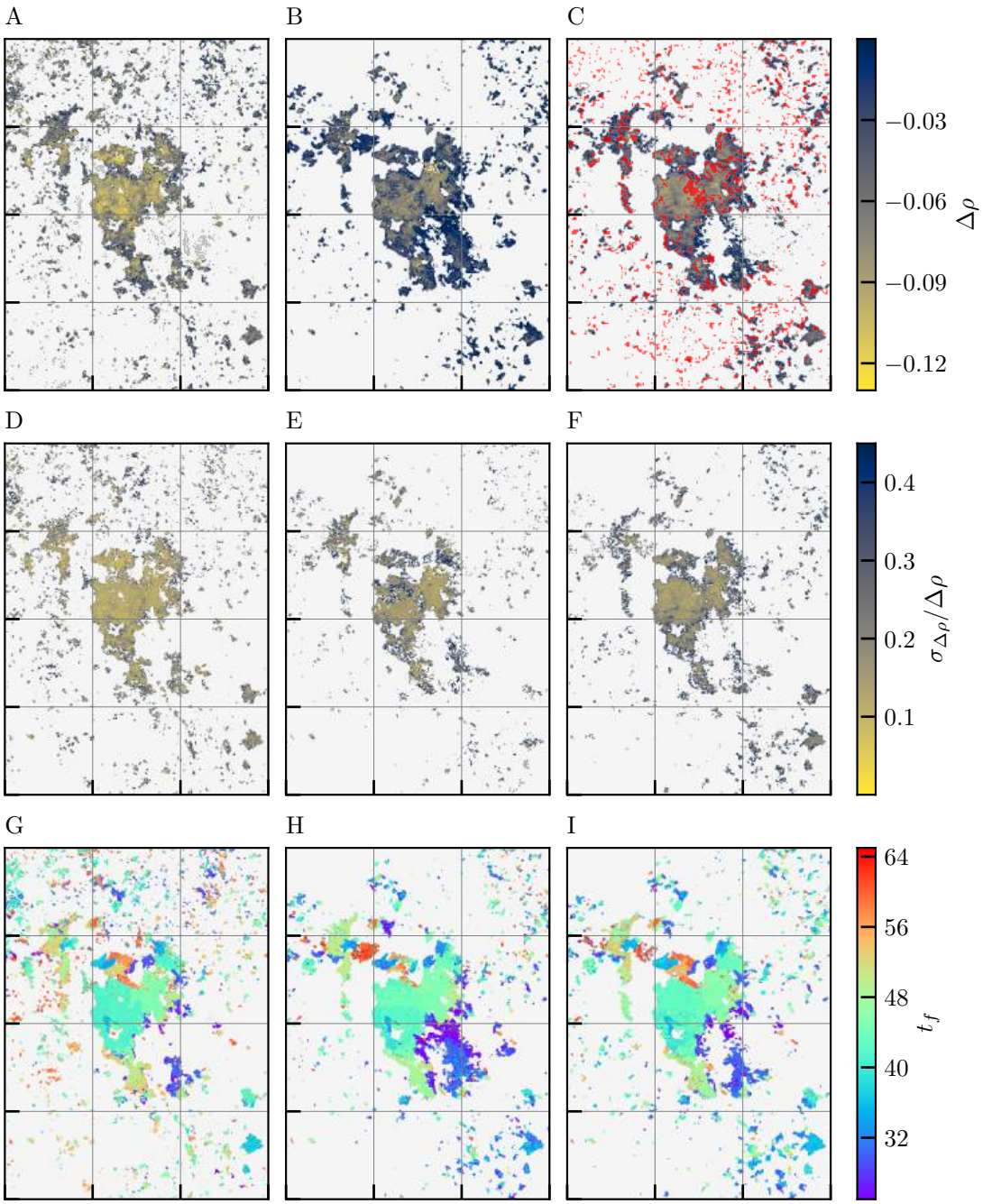


Figure 8.7. Features of the surface reflectance tracking for OLCI (A, D, G), VIIRS (B, E, H) and MODIS (C, F, I). A-C) composite of the maximum drop in NIR NBAR from each sensor. D-F) relative uncertainty of the maximum drop in NIR NBAR. G-I) Day of potential fire event t_f for each sensor. Red dots in panel C are active fire detections from MODIS Aqua and Terra.

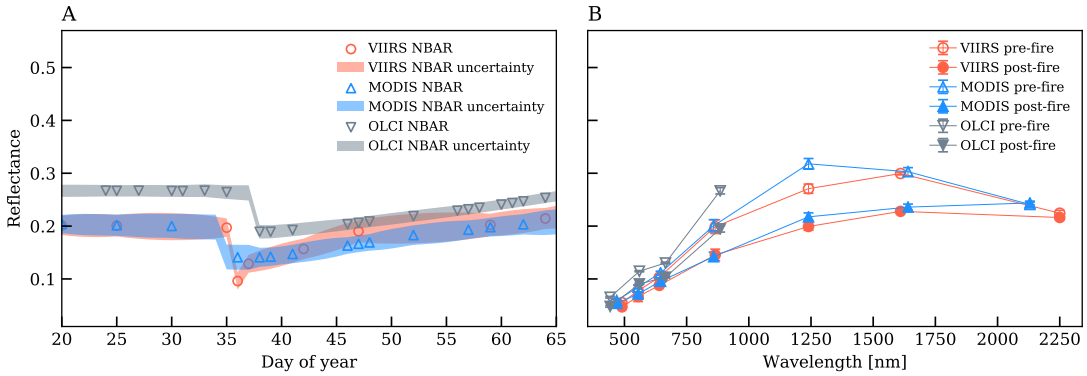


Figure 8.8. A) Time series of NIR Nadir-BRDF adjusted reflectance (NBAR) for OLCI (channel 18; 885nm), MODIS (channel 2; 865nm) and VIIRS (channel I2; 865nm). Also shown is the 95% confidence intervals of the trajectory of NBAR from the temporal model. B) Pre-fire and post-fire spectral NBAR for OLCI, VIIRS and MODIS. Only the four OLCI channels used in the algorithm are shown. 95% NBAR confidence intervals ($\pm 1.96 \cdot \sigma_{\mathrm{NBAR}}$) are also shown (day of year).

Figure 8.8A) a depicts time-series of the estimated BRDF-adjusted (NBAR) observations of NIR reflectance available from each sensor taken from a fire-affected pixel within the area shown in figure 8.7. Also shown is the uncertainty in the temporal trajectory of the surface reflectance (shaded region) from the temporal model specified in section 7.5.1. This corresponds to the 95% confidence interval on the retrieved NBAR and also provides the most probable region of NBAR values on days without acquisitions by the sensor. Because the BRDF correction scheme normalises each sensor to the same view and illumination geometries, the temporal evolution and magnitude of reflectance from each sensor are comparable. The OLCI reflectance has a notable bias from the MODIS/VIIRS NBAR.

The smooth trajectory pf the NBAR time series is interrupted for each sensor by a sudden change in the magnitude associated with the fire. The timing of this change is consistent between MODIS and VIIRS (day 35). The first post-fire acquisition from OLCI occurs on day 38, which leads to a delay in the temporal trajectory (shaded region). Figure 8.8B) depicts the pre- and post-fire BRDF-adjusted spectral reflectance from the three sensors. The change signal for MODIS and VIIRS is consistent, with the most substantial reductions in reflectance in the 865, 1250 and 1630nm channels. The bias in OLCI can also be observed across all spectral channels relative to MODIS/VIIRS, however, the general characteristic of a reduction in reflectance in the NIR (885nm OLCI channel) is still evident.

An important result of the multi-sensor surface tracking and edge detection procedure is the ability to reduce the temporal uncertainty in the reporting date of the change. Figure 8.9 shows retrieved NIR NBAR for days that each sensor attained a clear acquisition and the uncertainty in the temporal evolution of the NBAR from the temporal model described in section 7.5.1. The temporal sampling

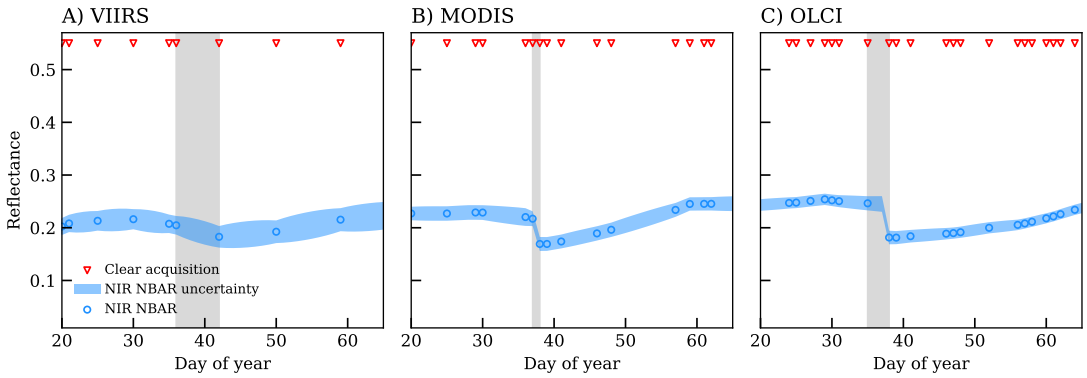


Figure 8.9. Potential improvement in temporal reporting accuracy of the day of the fire t_f. NIR Nadir-BRDF adjusted reflectance (NBAR) and 95% confidence interval for VIIRS (A), MODIS (B) and OLCI (C). The nearest pre-fire t_- and post-fire t_+ sensor acquisition specifies a temporal uncertainty window $[t_-, t_+]$ (grey shaded region).

provided by VIIRS is the most limited of the three sensors, with the pre-fire (DOY 36) and post-fire (DOY 42) observations providing a 6-day temporal uncertainty on the reporting day t_f. The sampling is also too limited to refine the edge process in the NBAR reflectance discernible in the MODIS and OLCI observations. Better temporal sampling by OLCI then refines t_f to range within 3 days and a set of consecutive observations collected by MODIS around the fire event further refines t_f to DOY 37. A larger uncertainty in the temporal evolution of the NBAR from VIIRS can also be observed relative to MODIS and OLCI which occurs because the temporal model within the BRDF-correction algorithm responds to the observational sampling.

Figure 8.10 demonstrates this feature for the region shown in figure 8.7. The day of the nearest available pre-fire t_- and post-fire t_+ acquisition from each sensor determines a temporal window $[t_-, t_+]$ of N_f days. The maximum value of N_f for VIIRS is 21 days, MODIS 16 days and OLCI 14 days. The 75% percentile on N_f for each sensor is greatest for VIIRS (6 days) and 4 days for MODIS and OLCI. As the intersection of the three sensor windows, the merged temporal window $[t_{-\text{MERGED}}, t_{+\text{MERGED}}]$ may be temporally shorter as shown in figure 8.9. Panel D) shows this refinement in the reporting accuracy. Reductions in N_f can be discerned across most of the fire-affected pixels. The extent of the largest values of N_f is reduced (although the maximum N_f remains similar at 20 days). There is also an improvement in 75% percentile of N_f to 3 days.

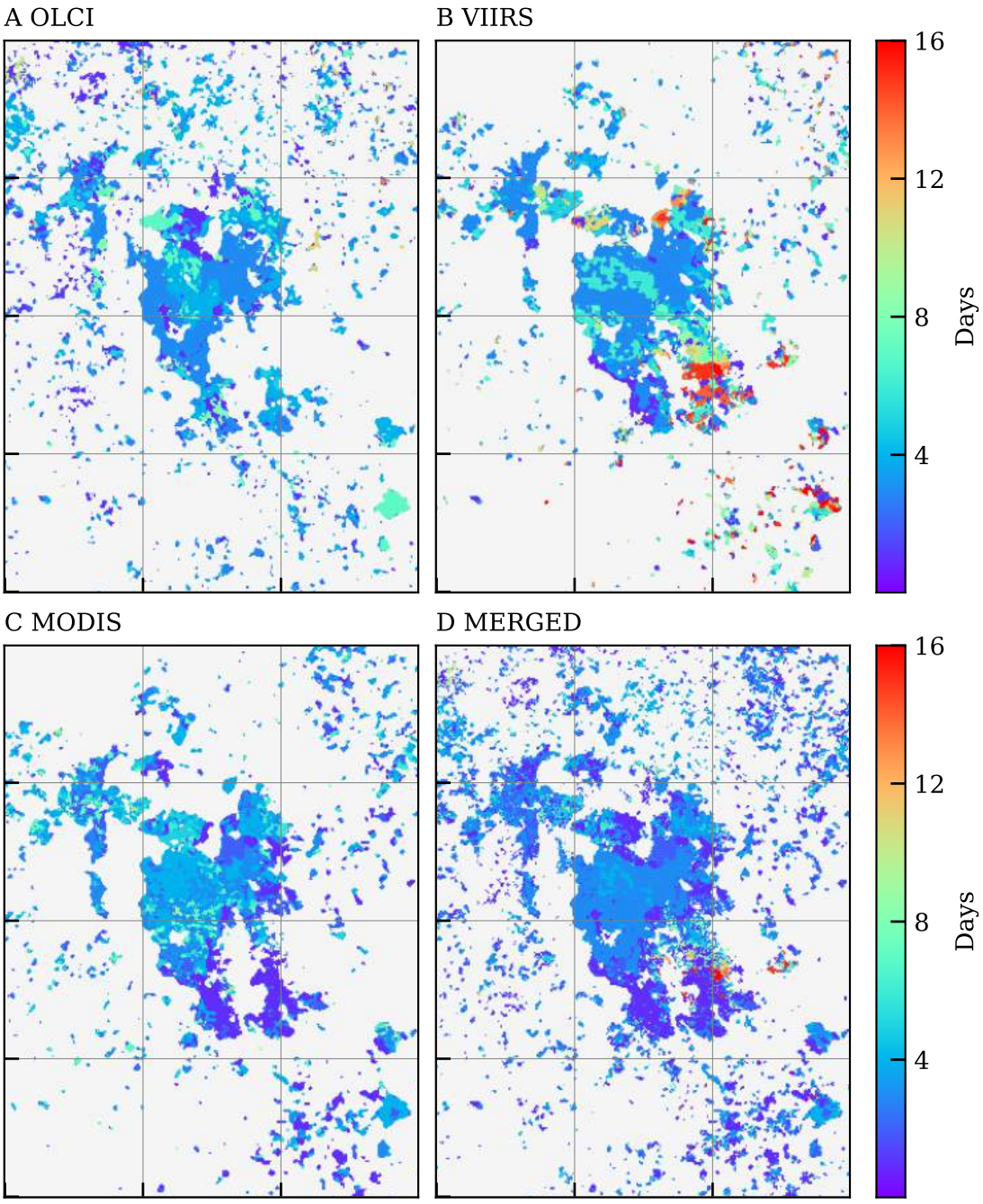


Figure 8.10. Temporal reporting uncertainty N_f [days] for the day of fire occurrence t_f. N_f is determined by the day of the nearest available pre-fire t_- and post-fire t_+ acquisition from the sensor which determine a temporal window $[t_-, t_+]$. Displayed in the panels are N_f for OLCI (A), VIIRS (B), MODIS (C) and the merged N_f (D).

8.3.2 Features of the fcc model retrievals

Figure 8.11 shows composites of the maximum fcc estimated for OLCI (panel A), VIIRS (B) and MODIS (C) through the mapping period. Also presented is the optimally merged estimate fcc_{merged} (D) from the three sensors, over which detected thermal anomalies from the MODIS active fire products are displayed in red. Pixels where the fcc signal is small ($fcc < 0.1$) are not displayed.

fcc_{MERGED} has the greatest extent of potential fire-affected pixels ($fcc > 0.1$), with each sensor contributing to refine the estimate of fcc at these pixels. For fcc_{MERGED} 36% of the pixels in the study region have $fcc > 0.1$, which is greater than fcc_{OLCI} (26%), fcc_{MODIS} (22%) and fcc_{VIIRS} (22%). Results for fcc_{OLCI} indicate the greater sensitivity to fires with smaller NIR reflectance reductions indicated in figure 8.7. In the northern extent of the region, fcc_{OLCI} detects a considerable number of pixels where $fcc < 0.3$ which are not detected by fcc_{MODIS} or fcc_{VIIRS} . Assuming for each 1km MODIS active fire detection each of the four collocated 500m pixels are burned also highlights the improved sensitivity of the sensor merging routine to burned pixels. 46% of active fire detections have $fcc > .1$ for fcc_{VIIRS} . This figure is higher for fcc_{MODIS} (52%), which can be reasonably expected as the active fires also originate from the MODIS instruments. fcc_{OLCI} has the highest correspondence to active fire detections (54%). Optimally combining the fcc from each sensor system raises this proportion to 64% for fcc_{MERGED}. It is worth stating that the active fire detections are not used in the fcc retrieval.

Figure 8.12 shows the relative uncertainty in the maximum fcc composite (σ_{fcc}/fcc) for each sensor and fcc_{MERGED}. As a first-order effect, the relative uncertainty in the fcc retrieval is determined by the magnitude of the fcc reported. The lowest relative uncertainties for each sensor are found in the centre of the large burn scar where the corresponding fcc is highest. In these areas, OLCI has the smallest relative uncertainties (3–10%) of the three sensors. This effect is not exclusive, however — areas of high magnitude fcc retrievals also have larger relative uncertainties as determined by the available sensor sampling (e.g. VIIRS in the large central burn scar). VIIRS and MODIS have larger uncertainties than OLCI and also display a pattern towards larger uncertainties at the perimeter of the large burn scar not evident from OLCI. An identifiable feature is the greater relative uncertainties linked to the relative size of the burn scar, with the smaller clusters of pixels showing greater relative uncertainties in contrast to the larger agglomerations.

The relative uncertainty in fcc_{MERGED} (panel D) as an optimal combination of the information provided by each sensor is the lowest. Within the larger burn scars, relative uncertainties are the smallest (2–5%). There is also an identifiable reduction in the relative uncertainty at the perimeter of

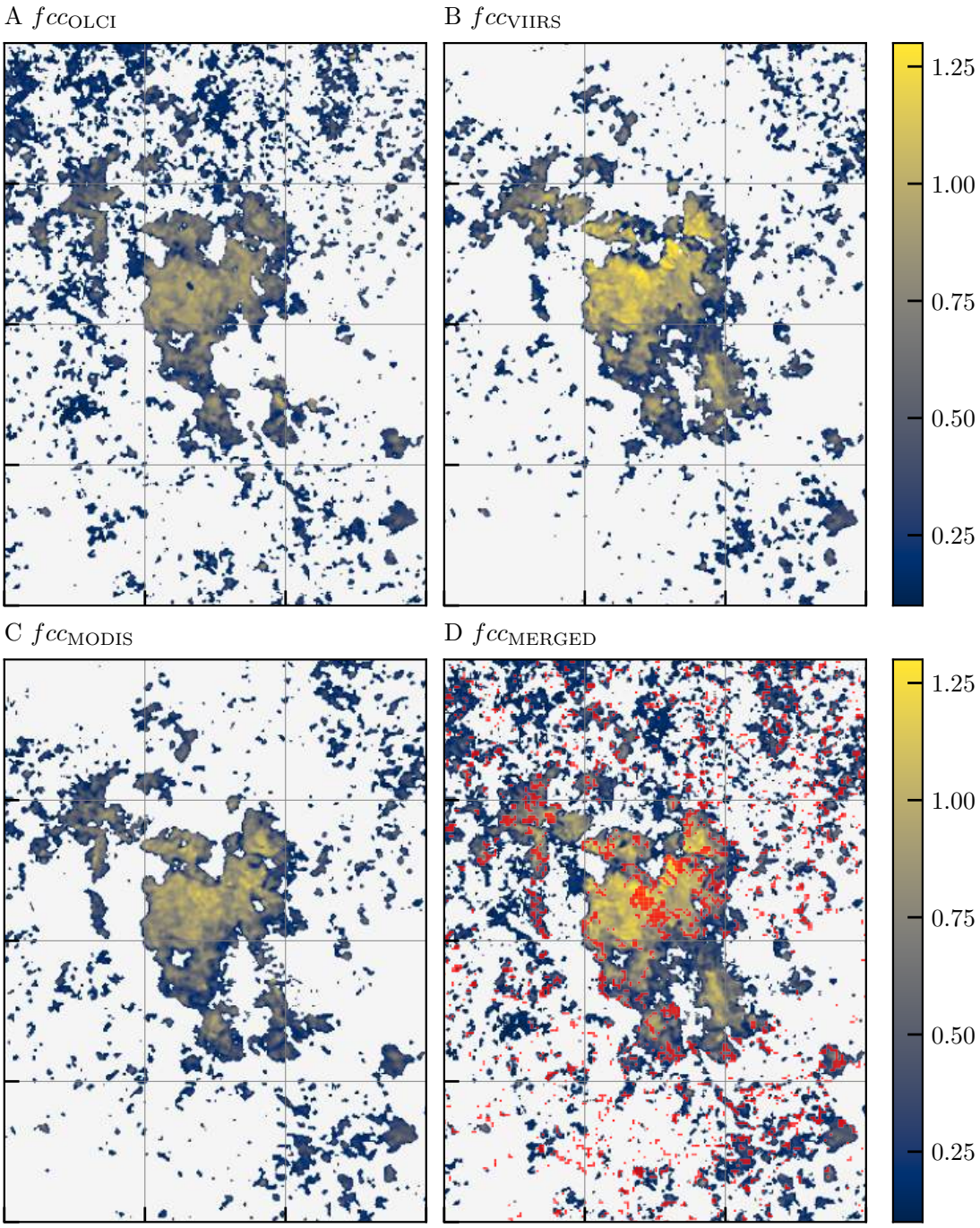


Figure 8.11. Composite of the maximum reported fcc from OLCI (A), VIIRS (B) and MODIS (C). The optimally merged fcc_{MERGED} is also shown in panel D. Red dots in panel D are active fire detections from MODIS Aqua and Terra.

these burn scars where the magnitude of fcc is not particularly changed by the merging. In these areas, the better signal-to-noise ratio of OLCI contributes to reducing absolute uncertainties in fcc. The smaller clusters which are detected by more than one sensor with individually high relative uncertainties show a reduction in the merged result. This occurs because the independent evidence from each sensor contributes to reducing the absolute uncertainty in the merged estimate of fcc.

Figure 8.13 displays the spectral shape coefficients $[a_0, a_1]$ associated with the maximum fcc retrieval over the mapping period for the three sensors. As before, only pixels that can be considered potentially fire-affected ($fcc > 0.1$) are displayed. Each sensor shows the same arrangement of smaller coefficients for the ash/char spectral endmember a_0 and larger coefficients for the exposed soil spectrum endmember a_1. This preference indicates that the spectral nature of the change signal in this geographic region is mostly driven by the exposure of the underlying soil rather than deposition of ash or char onto the surface. This characteristic is sensible as the study area encompasses a region of grassland where burning leads to exposure of underlying bright soils and only limited deposition of char (Disney et al., 2011; Roy and Landmann, 2005). The coefficients are also spatially coherent, with adjacent pixels showing similar values, and areas of higher a_1 values observable in at the edge of the large central burn scar detected by both MODIS and VIIRS. The OLCI coefficients which are constrained by the prior derived from the MODIS archive (see section 7.5.2) show less spatial variation. This indicates the more limited sensitivity of the OLCI spectral sampling for refining the coefficients for each pixel. Nevertheless, the central location of both a_0 and a_1 are consistent with the independently retrieved MODIS and VIIRS values (see section 8.4).

8.3.3 BA attribution

Figure 8.15 shows summary statistics of the prior inference on burned area from active fire detections L_{af} (see section 7.5.3). Active fire detections from MODIS (panel A), VIIRS (B) and both sensors (C) are also displayed. MODIS detects 4770 active fires in the study region (covering approximately 16% of the study area) while VIIRS detects 3031 active fires (10%). The mode of L_{af} (panels D—F) has been discretised into eight probabilities classes to better distinguishes between high probabilities than provided by a uniform colour scale. The spatial model used for the active fire prior also has the impact of joining areas of isolated active fire detections into contiguous spatial areas (i.e. yellow areas where the mode > 0.5). The greater density of active fires collected by MODIS observable within the central burn scar results in a prior with a high mode for these pixels. Merging the active fire records increases confidence in the central burn scar being burned with an increase in the mode (panel F) for these pixels to 0.99 or higher. These areas only align with the areas of greater confidence in the active

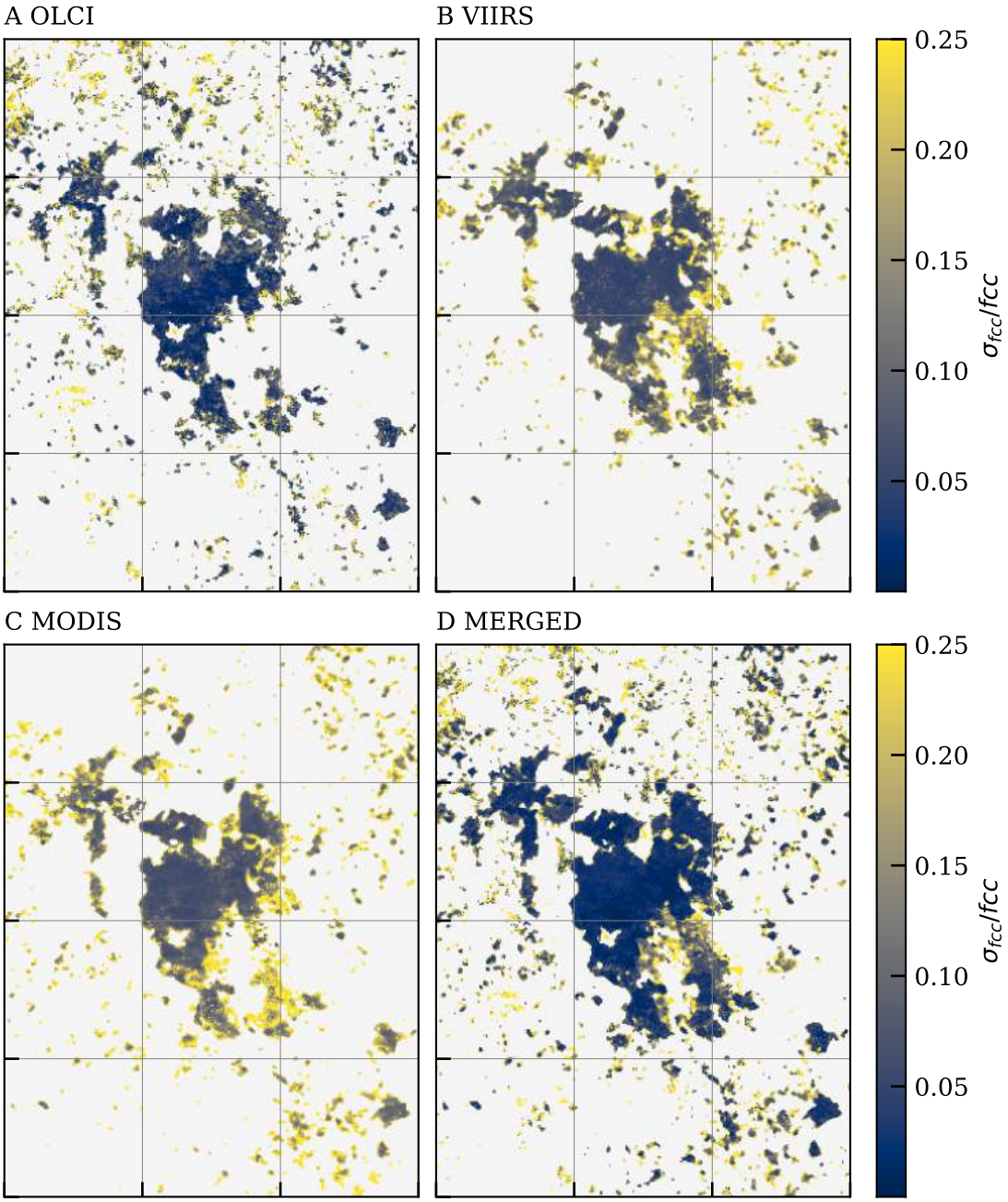


Figure 8.12. Relative uncertainty (σ_{fcc}/fcc) of the composite of the maximum reported fcc from OLCI (A), VIIRS (B) and MODIS (C). The relative uncertainty of fcc_{MERGED} is also shown in panel D.

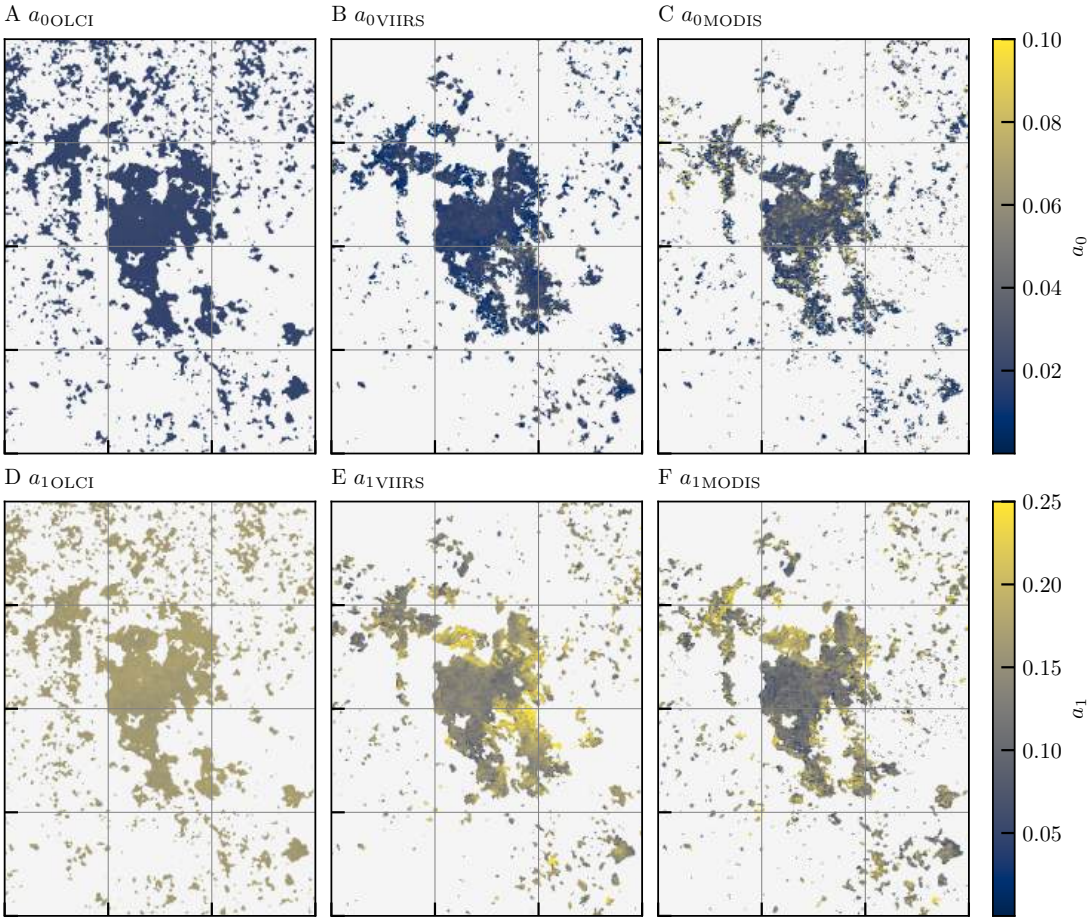


Figure 8.13. Composite of the spectral shape coefficients $[a_0, a_1]$ associated with the maximum reported fcc for OLCI (A & D), VIIRS (B & E) and MODIS (C & F).

fire detections and also those observed by both MODIS and VIIRS. The relative uncertainty u in the prior (defined as $u = \sigma/\mathrm{mode}$) is important because it determines the relative strength of the prior within the algorithm. Areas where u exceeds 0.5, and the beta distribution used replicates a uniform distribution, are masked out. The relative uncertainty u indicates a pattern that would be expected of the active fire statistical decay model proposed. The smallest values of u are located within the central burn scar and in other areas with a high spatial density of active fire detections. Away from these areas, the spatial decay model ensures that the active fire prior becomes less precise and eventually returns to a uniform distribution — indicating no additional information about burning can be inferred from active fire observations for these pixels. The merged relative uncertainty (panel I) is the lowest of the three, with the areas of low uncertainty increased relative to VIIRS or MODIS alone because of the combination of information from both sensors.

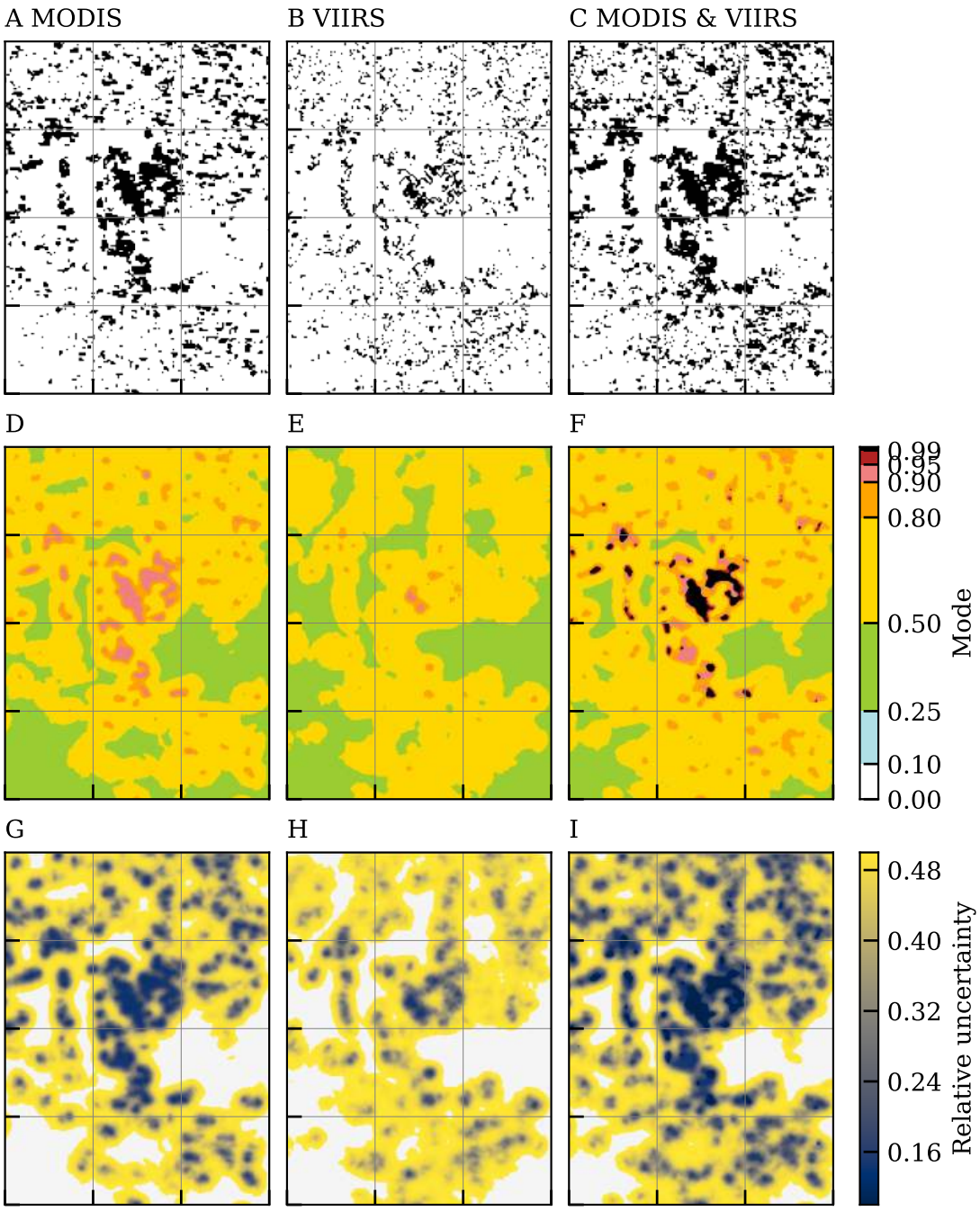


Figure 8.14. Summary statistics of the active fire prior L_{af} for the study region. A) active fire detections from MODIS; B) active fire detections from VIIRS; C) active fire detections from both VIIRS and MODIS. D—F) mode of L_{af} discretised into eight probability classes; G–I) relative uncertainty the in L_{af} (specified as σ/mode).

Figure 8.15 shows results of the probabilistic burned area maps P_b derived from each sensor and also the merged sensor retrieval. It is worth pointing out that individual P_b fields can be calculated from each sensors retrieval of the fcc model parameters $\Pr(\theta_{\text{sensor}})$ because there is nothing sensor-specific about the mapping specified in $L(\theta \mid \text{burned})$ (see section 7.5.3). To better focus on the features of the optical inference, the active fire prior (figure 8.14) has not been included in this figure.

To better discern different levels of confidence in fire occurrence, P_b has been discretised into eight classes. P_b probabilities greater than 0.99 provide near certain evidence of fire occurrence. Several further high probability levels are then defined as 0.95 (i.e. a 95% confidence in burning), 0.9 and 0.8. P_b probabilities between 0.5 and 0.8 provide that there is weak (0.5) to moderate evidence of fire. Values between 0.25–0.5 can be considered as there is some evidence of burning with 0.5 indicating insufficient evidence to classify the pixel as burned or unburned — and maximum uncertainty in the inference. This may occur for example when the pixel fcc is close to the detection threshold T specified in equation 7.24 or the uncertainty in the parameters is sufficiently large that fcc may be above the or below T.

Within the large central burn scar each sensor provides P_b probabilities exceeding 0.95. In these areas the fcc from each sensor is considerably above the detection threshold (figure 8.11) and has relative uncertainties $< 10\%$ (shown in figure 8.12). $P_{b\,\text{OLCI}}$ has small clusters where $P_{b\,\text{OLCI}}$ exceeds 0.99, indicating near certainty of fire occurrence. These particular areas are associated with similar fcc values to MODIS and VIIRS but smaller relative uncertainties in fcc_{OLCI}. The spectral parameters $[a_0, a_1]$ for these high P_b pixels are also consistent with their expected values from the prior. Towards the perimeters of the large central burn scar P_b decreases, most obviously with $P_{b\,\text{MODIS}}$ and $P_{b\,\text{VIIRS}}$. This is associated with a lower radiometric severity at the fire perimeters (as suggested by the lower values of fcc depicted in figure 8.11) where incomplete burning of the vegetation occurs (Roy and Landmann, 2005). The sharper edges in the $P_{b\,\text{OLCI}}$ field are associated with the better spatial resolution of the OLCI instrument compared to MODIS and VIIRS.

The results for $P_{b\,\text{MERGED}}$ clearly show how the available information from each sensor is combined to provide the best possible inference from the three sensors. The spatial characteristics of $P_{b\,\text{MERGED}}$ are also coherent — the large central burn scar is detected by all sensors and associated with the greatest probabilities (> 0.95) and smaller clusters of pixels detected by only one sensor (e.g. OLCI) are assigned probabilities which indicate more uncertainty in the mapped state (0.25–0.5; shown in green). The probabilistic inference, when combined with fcc provides considerably more information than a hard burned/unburned classification.

The number of pixels where $P_{b\,MERGED} > 0.99$ now cover a larger extent of the central burn scar. The smaller burn scars where values of fcc_{MERGED} range between 0.25–0.45 have been confirmed by each sensor are now assigned greater probabilities because of the combination of multiple independent pieces of evidence and the resulting reduction in the relative uncertainty in fcc_{MERGED} . It can also be remarked that the low magnitude fcc_{OLCI} estimates are mapped to values of $P_{b\,OLCI}$ between 0.1–0.5 (shown in green). These probabilities suggest that there is some limited evidence for fire occurrence, with values of 0.5 indicating that the evidence is inconclusive as to whether a fire has occurred or not. When combined with the supplementary knowledge of θ provided by MODIS and VIIRS, these probabilities increase because either the magnitude of the maximum detected fcc increases or the uncertainty in the retrieved fcc model parameters is reduced.

8.4 Consistency of the fcc retrieval system

Figure 8.16 depicts the agreement in fcc for each sensor across the three sites. Shown are scatter plots between pairwise estimates of the maximum fcc reported for each sensor, the ordinary least squares regression between the two estimates (red line) — which provides information on how each sensor combination departs from the ideal case of a one-to-one linear mapping between sensors (dashed line). VIIRS and MODIS have the most similar per-pixel retrievals of fcc, with a coefficient of determination $r^2 = 0.79$ and a regression slope of 1.02. This is in some ways to be expected — these sensors have the most similar spectral sampling. This is followed by VIIRS and OLCI ($r^2 = 0.79$) where there is a bias towards lower values for fcc_{OLCI} relative to fcc_{VIIRS} (regression slope 0.88). MODIS and OLCI have the lowest coefficient of determination (0.69) but a slightly better regression slope (0.94). Table 8.1 provides a summary of comparative statistics of fcc retrievals from each sensor across the three MODIS/VIIRS tiles. The root mean square deviation (RMSD) between each sensor combination are similar, ranging from 0.022 to 0.024, or a percentage difference of 9%. The mean percentage error (MPE) between each sensor combination is greatest for fcc_{MODIS} and fcc_{OLCI} (13%) and smallest in magnitude for fcc_{MODIS} and fcc_{VIIRS} (−4%). Given that fcc is retrieved with no prior information for MODIS and VIIRS, the small bias and good correspondence suggest that fcc can be retrieved consistently from different sensors. Comparing fcc_{OLCI} retrievals to MODIS and VIIRS retrievals confirms that the constrained fcc retrieval system performs acceptably for sensors with limited VIS-NIR spectral sampling. It should be noted again that fcc_{OLCI} is retrieved with only a weak prior on fcc and that the specification of a suitable prior for only the spectral shape parameters $[a_0, a_1]$ is sufficient to achieve this level of agreement (for example see section 7.5.2).

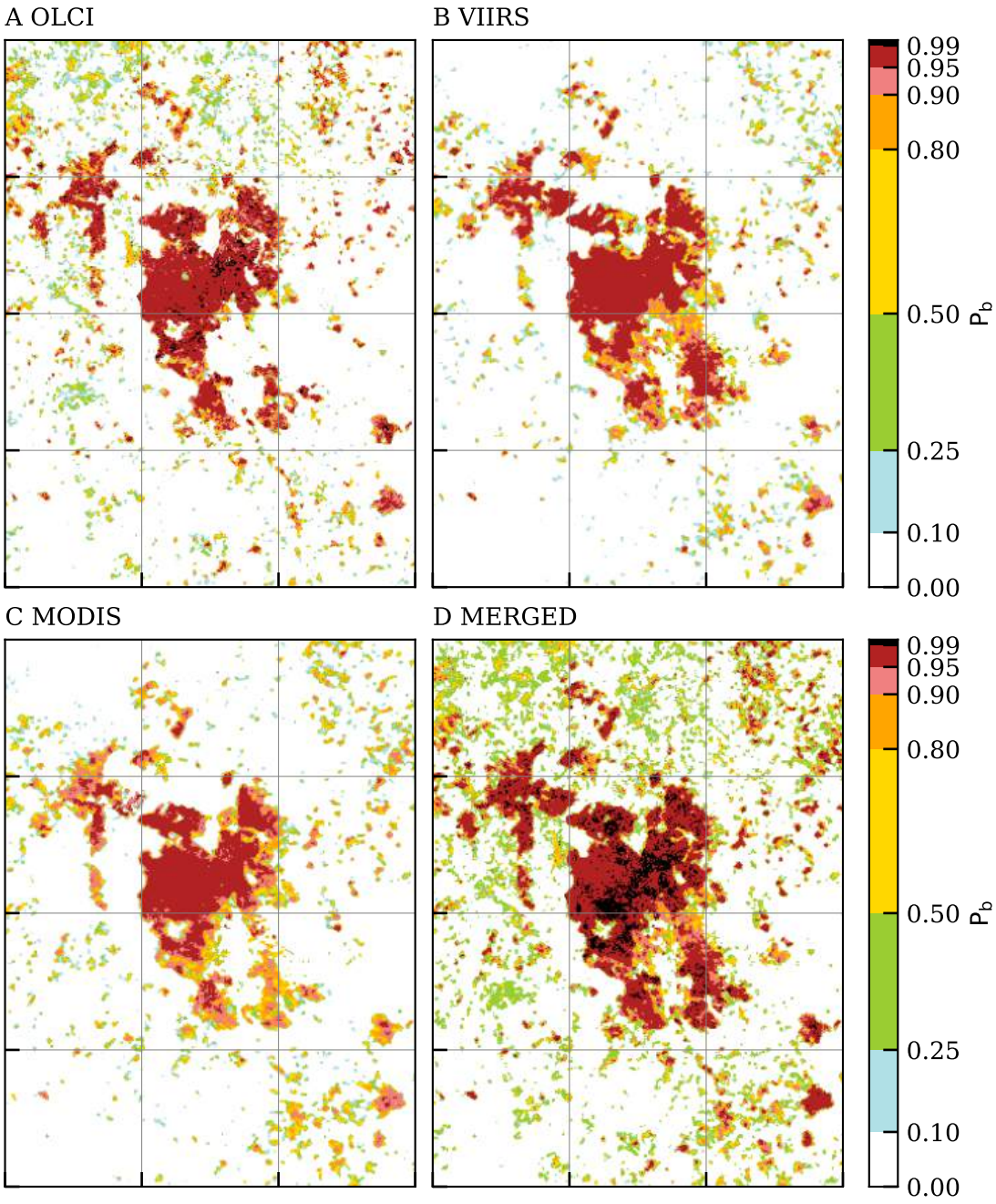


Figure 8.15. Composite of the maximum P_b detected for each sensor and the merged estimate. A) $P_{b\ OLCI}$ B) $P_{b\ VIIRS}$ C) $P_{b\ MODIS}$ and D) $P_{b\ MERGED}$.

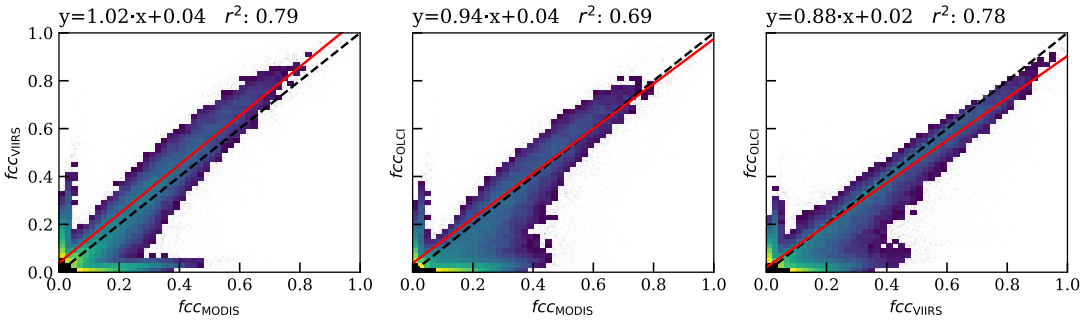


Figure 8.16. Scatter plots of fcc estimated from MODIS, VIIRS and OLCI for every pixel in the three tiles. Left) fcc_{MODIS} vs fcc_{VIIRS}, middle) fcc_{MODIS} vs fcc_{OLCI}, right) fcc_{VIIRS} vs fcc_{OLCI}. Also shown is the regression slope between sensors and coefficient of determination r^2.

Figure 8.17 depicts scatter plots between the spectral shape parameters $[a_0, a_1]$ retrieved from each sensor. Also shown is the ordinary least squares regression between the two estimates (red line) and the 1:1 line (dashed). For each plot, only retrievals where both sensors have an actual change in the reflectance signal (i.e. pixels where fcc>0.1) are shown. This is because when there is no reflectance change signal there is not expected to be any correlation between sensor estimates of $[a_0, a_1]$ as $fcc \cdot \rho_{\mathrm{burn}}$ is undefined when $\Delta\rho = 0$ in the fcc model (see equation 7.1) .

Most apparent in figure 8.17 is the better consistency observable in the sensor estimates of a_1 relative to a_0. Retrievals of a_0 for different sensors show close to no correlation, with coefficients of determination r^2 less than 0.05 for each sensor pairing. Conversely, the a_1 parameter is more consistently retrieved between sensors. The coefficient of determination (r^2) exceeds 0.70 for each sensor pairing and is highest for VIIRS and OLCI (0.72). The regression between VIIRS and OLCI a_1 has the smallest multiplicative bias (regression slope 0.98) but the largest absolute bias towards higher OLCI values (0.04). The poor correspondence in a_0 values between the sensors is unexpected. Given the better correspondence observed for a_1 and fcc itself the most probable explanation is that the a_0 parameter does not sufficiently improve the fit of the fcc model the to the observations. This means that a range of a_0 values provide the same fit to the reflectance change and therefore there is considerably more noise in the a_0 retrievals compared to a_1 and fcc. Alternatively, the poor correspondence may be down to other potential sources of noise such as residual cloud contamination or BRDF-effects which lead to differences in the NBAR time series for each signal which produce greater noise in the a_0 parameter compared to the a_1 parameter. For example, residual cloud shadow will lead to a uniform reduction in reflectance across all wavelengths which matches the flat spectral response of the a_0 end-member.

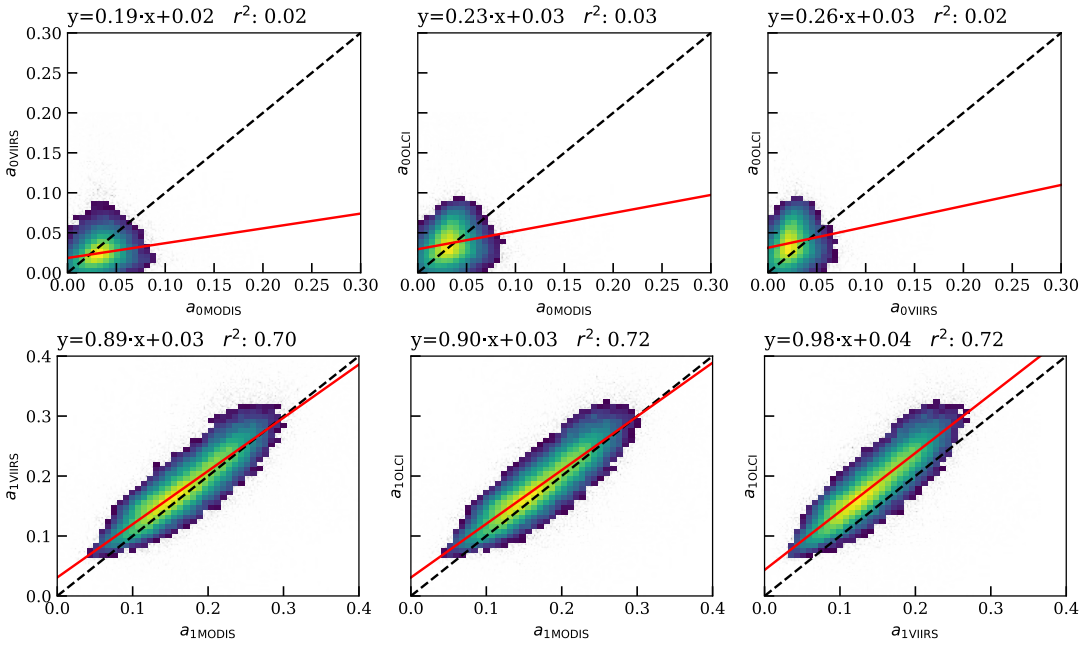


Figure 8.17. Scatter plots of a_0 (top row) and a_1 (bottom row) estimated from MODIS, VIIRS and OLCI for every pixel in the three sites where $fcc > 0.2$ for both sensors. Top left) $a_{0\mathrm{MODIS}}$ vs $a_{0\mathrm{VIIRS}}$, top middle) $a_{0\mathrm{MODIS}}$ vs $a_{0\mathrm{OLCI}}$, top right) $a_{0\mathrm{VIIRS}}$ vs $a_{0\mathrm{OLCI}}$. Bottom left) $a_{1\mathrm{MODIS}}$ vs $a_{1\mathrm{VIIRS}}$, bottom middle) $a_{1\mathrm{MODIS}}$ vs $a_{1\mathrm{OLCI}}$, bottom right) $a_{1\mathrm{VIIRS}}$ vs $a_{1\mathrm{OLCI}}$. Also shown is the regression slope between sensors and coefficient of determination r^2.

Table 8.1 provides additional measures of the agreement between sensor estimates of $[a_0, a_1]$. The mean percentage error (MPE) between sensors is higher for a_0 than a_1. VIIRS and OLCI have the largest relative bias, with OLCI retrievals underestimated relative to VIIRS for a_0 (-66%) and a_1 (-31%).

8.5 Validation & intercomparison with MCD64 Collection 6

For each tile moderate-resolution (20m) images from Sentinel-2 were used for validating the algorithm outputs following protocols developed by Boschetti et al. (2009). The validation protocol dictates that coarse-resolution products are validated from independent data derived from moderate-resolution instruments such as Landsat and Sentinel-2. To follow the protocol, two images were used to ensure that the mapped burned area only occurred within the period of the algorithm processing. The fcc model was then solved over each 20m pixel from the pre- and post-fire Sentinel-2 images. Areas where either the pre- or post-fire Sentinel-2 image were obscured by cloud, or the land surface was covered in snow were excluded. Pixels where $fcc > 0.2$ were labelled as burned and where $fcc < 0.2$ as unburned. Roy and Landmann (2005) indicated a threshold of 0.2 for Landsat 7 below which retrievals

Table 8.1. Consistency of the retrieved fcc model parameters between the three sensors. [1]Root mean square deviation. [2]Mean bias error. [3]Mean percentage error. [4]Slope m and intercept c for the linear model $y = mx + c$ between the two sensors, and coefficient of determination r^2.

		RMSD[1]	**MBE**[2]	**MPE**[3]	**slope**[4] m	**intercept**[4] c	r^2
fcc							
	MODIS vs VIIRS	0.0246	-0.0007	-4.40	1.0245	0.0391	0.7880
	MODIS vs OLCI	0.0252	0.0008	12.51	0.9369	0.0375	0.6938
	VIIRS vs OLCI	0.0224	0.0015	6.57	0.8839	0.0199	0.7772
a_0							
	MODIS vs VIIRS	0.0206	0.0082	23.33	0.1851	0.0184	0.0185
	MODIS vs OLCI	0.0213	-0.0047	-17.95	0.2262	0.0294	0.0254
	VIIRS vs OLCI	0.0210	-0.0134	-66.52	0.2616	0.0312	0.0236
a_1							
	MODIS vs VIIRS	0.0310	-0.0134	-8.55	0.8879	0.0308	0.7037
	MODIS vs OLCI	0.0314	-0.0147	-9.11	0.8954	0.0307	0.7182
	VIIRS vs OLCI	0.0494	-0.0412	-30.81	0.9756	0.0433	0.7183

Table 8.2. Sentinel-2 validation data for the three study tiles.

		S2 tile	**date**	**filename**
h20v03				
	pre-fire	T38UNF	08/04/2019	S2A_MSIL2A_20190408T080611_N0211_R078_T38UNF_20190408T112622
	post-fire	T38UNF	18/04/2019	S2A_MSIL2A_20190418T080611_N0211_R078_T38UNF_20190418T100332
h20v08				
	pre-fire	T35NPJ	10/02/2019	S2A_MSIL2A_20190210T082101_N0211_R121_T35NPJ_20190210T110548
	post-fire	T35NPJ	07/03/2019	S2B_MSIL2A_20190307T081809_N0211_R121_T35NPJ_20190307T130017
h09v07				
	pre-fire	T15QVU	03/04/2019	S2B_MSIL2A_20190403T162839_N0211_R083_T15QVU_20190403T204056
	post-fire	T15QVU	08/05/2019	S2A_MSIL2A_20190508T162901_N0212_R083_T15QVU_20190509T000731

of fcc are unreliable. Roy et al. (2019) also use a threshold 0.2 to map burned areas with a similar fcc model for Sentinel-2. Figure 8.18 displays a subset of the validation granule for the h20v07 tile and associated estimate of fcc. The area covers approximately 640km^2. Areas where either the pre- or post-fire Sentinel-2 image are cloudy are displayed in white. Figure 8.19 shows a small subset from the Sentinel-2 validation images for the h09v07 VIIRS/MODIS tile. The image covers an extent of approximately 36km^2. As before, fcc is mapped between purple and yellow for values [0.2, 1] with areas black areas corresponding to $fcc < 0.2$. Table 8.2 details the three pairs of Sentinel-2 acquisitions used for each tile.

Each Sentinel-2 burned area map was then aggregated to the MODIS/VIIRS sinusoidal grid to provide a 500m burned/unburned validation image covering a small extent of each MODIS/VIIRS tile. While the purpose of the probabilistic retrieval is to provide an uncertainty of the evidence of burning

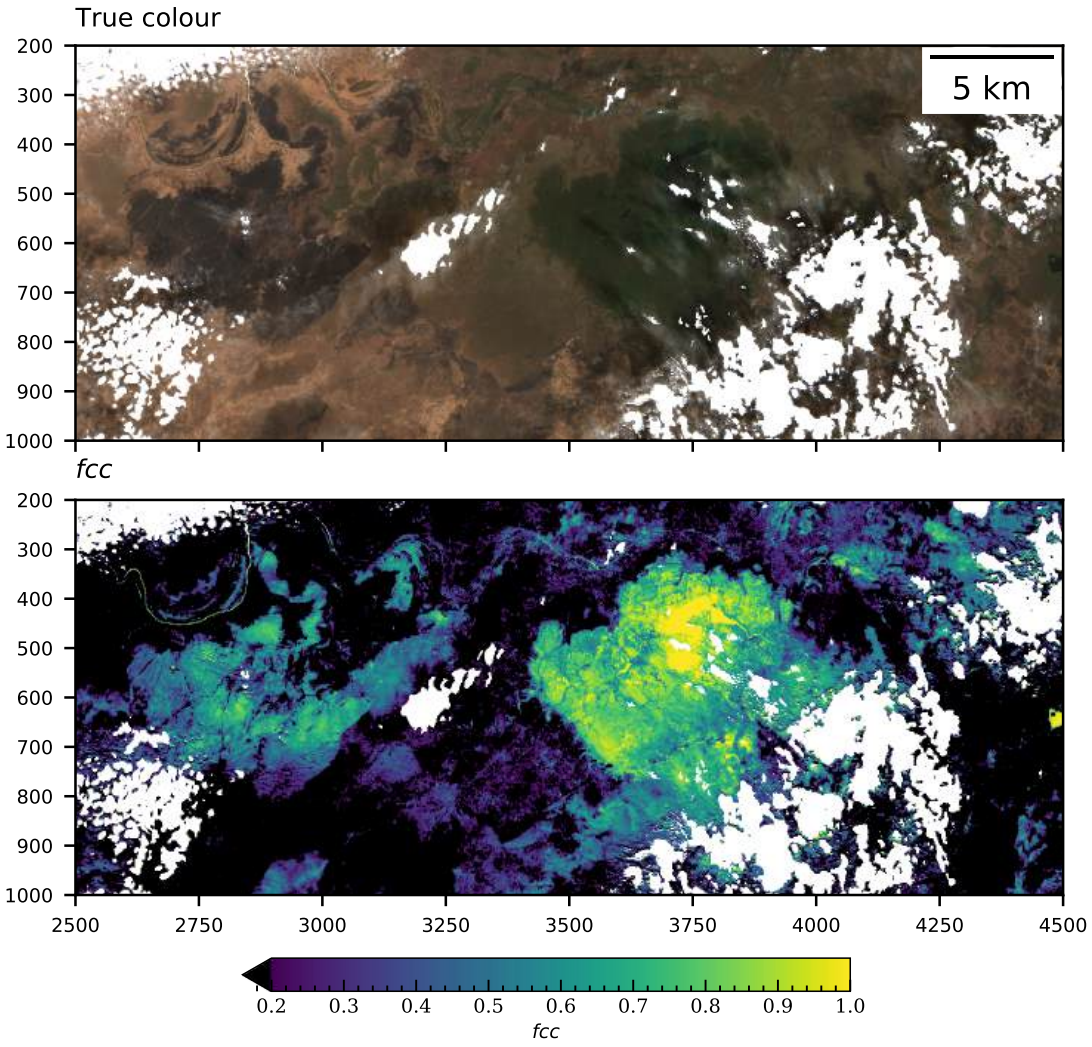


Figure 8.18. True colour composite and fcc for a subset of the Sentinel-2 validation image from the h20v08 tile.

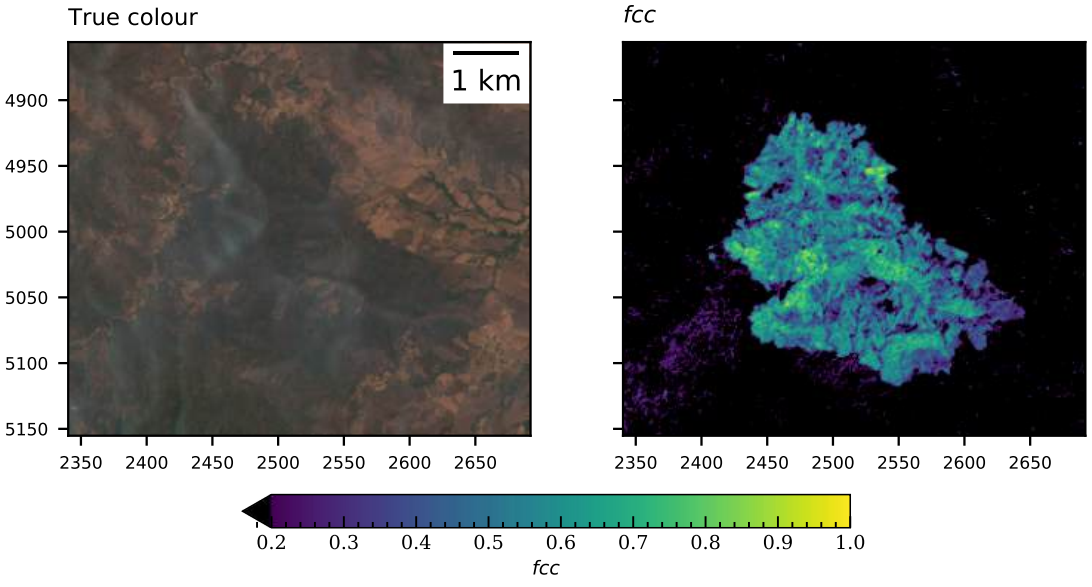


Figure 8.19. True colour composite and fcc for a subset of the Sentinel-2 validation image from the h09v07 tile.

(P_b), discrete burned products can also be generated. To produce a discrete day of burn (DOB) product, the simplest option is to consider pixels where P_b probabilities exceeded 0.5 to be burned. For these pixels a DOB was was provided by the day of maximum reported P_b.

A set of accuracy statistics was generated from the classification error matrix (Congalton and Green, 2002). Each error matrix was constructed from the comparison of the reference Sentinel-2 validation data with one of the mapped DOB products. Omission error (Oe) and commission error (Ce) percentages were calculated from each error matrix. The overall accuracy was not calculated as it is strongly determined by the imbalance in the greater number of unburned to burned pixels (Chuvieco et al., 2018). Instead, an integrated metric of the overall performance of each algorithm is provided by the dice coefficient (DC):

$$\mathrm{DC} = \frac{2E_{11}}{E_{1+} + E_{+1}} \tag{8.3}$$

where E_{11} is the total number of pixels correctly classified according to the reference, and E_{1+} and E_{+1} are the burned area reported by the product and the reference validation data respectively. The dice coefficient therefore expresses the overall performance and has been widely used in most recent burned area product validations (Padilla et al., 2015; Chuvieco et al., 2018; Campagnolo et al., 2019).

Validation statistics were calculated for several scenarios. First, validation statistics were calculated for each sensor (i.e. MODIS, OLCI and VIIRS separately). Second, the inference from VIIRS and OLCI $P_{b\ \mathrm{VIIRS+OLCI}}$ was considered so that a VIIRS+OLCI–only system can be contrasted against MODIS exclusively. Finally, all three sensors were considered from $P_{b\ \mathrm{MERGED}}$. As a point of reference, the validation statistics for the MODIS Collection 6 burned area product (MCD64) were also calculated (Giglio et al., 2018). The FireCCI products have not been processed for 2019 and so could not be included here (Chuvieco et al., 2018).

Table 8.3 reports the accuracy measures for each algorithm implementation and MCD64. The lowest accuracy for all products (including MCD64) is found in the h20v03 tile, expressed by a dice coefficient (DC) of less than 0.1 for all products. Here omission errors (Oe) exceed 90% — though all products have smaller commission errors (<2.1%). From visual inspection of the Sentinel-2 images and VIIRS reflectance time-series, the majority of burning here occurs in croplands shortly following snow-melt. Detection of this type of burning phenomena has been identified as challenging because of a lack of observations of the land surface in the weeks before fire (Hall et al., 2016).

The contribution of each additional sensor within the algorithm is the reduction in omission errors, especially in the h20v08 and h09v07 tiles. In the h20v08 tile, $P_{b\ \mathrm{MERGED}}$ has the lowest omission error (66%), while the individual sensors vary between 67% ($P_{b\ \mathrm{MODIS}}$) to 72% ($P_{b\ \mathrm{OLCI}}$). The best sensor for detection (i.e. in terms of lowest omission error) also varies between tiles. For the h09v07, the lowest

Table 8.3. Validation statistics derived for MCD64 and from $P_{b\ OLCI}$, $P_{b\ VIIRS}$, $P_{b\ MODIS}$, $P_{b\ VIIRS+OLCI}$, $P_{b\ MERGED}$. Reported are per-algorithm omission error rate (Oe), commission error rate (Ce) and the dice coefficient (DC).

	h20v03			**h20v08**			**h09v07**		
	Oe [%]	Ce [%]	DC	Oe [%]	Ce [%]	DC	Oe [%]	Ce [%]	DC
MCD64C6	94.92	2.07	0.03	88.78	0.42	0.08	65.04	0.09	0.22
OLCI	98.01	0.12	0.01	71.53	2.043	0.16	61.55	0.07	0.25
VIIRS	97.51	0.85	0.04	70.25	2.33	0.09	54.29	0.16	0.06
MODIS	98.91	0.14	0.01	67.19	3.17	0.16	62.34	0.32	0.22
VIIRS+OLCI	97.54	0.84	0.02	69.04	2.63	0.16	52.07	0.19	0.26
VIIRS+OLCI+MODIS	97.20	0.88	0.02	66.41	3.57	0.17	49.73	0.40	0.27

sensor omission error belongs to $P_{b\ VIIRS}$ (54%), while $P_{b\ MODIS}$ has the lower omission error (67%) in the h20v08 tile. The merged product $P_{b\ MERGED}$ once again has the lowest omission error (50%) in the h09v07 tile. The merging process does however slightly increase commission errors. For example, $P_{b\ MERGED}$ has commission errors of 4% in h20v08 which is only marginally higher than $P_{b\ MODIS}$ (3%). Algorithm commission errors are greater in h20v08 and h09v08 and lower in h20v03 compared to MCD64. Compared to the MCD64 product, the merged result performs better in the h20v08 and h09v07 tiles in terms of omission errors and dice coefficients. In the h20v08 tile, omission errors for $P_{b\ MERGED}$ (66%) are lower than MCD64 (88%). This is also the case in the h09v07 tile where omission errors for $P_{b\ MERGED}$ (50%) are lower than MCD64 (65%). Dice coefficients for $P_{b\ MERGED}$ (0.17) are better than MCD64 (0.08) in h20v08, and h09v07 (0.27 vs. 0.22).

To provide an intercomparison of the algorithm against burned area from MCD64, each tile was split into 144 smaller cells each 200 x 200km^2 in area. The reported burned area in each cell was then counted, based again on the assumption that pixels where $P_b > 0.5$ are considered burned pixels. The left panel in figure 8.20 shows a scatter plot between the burned area reported by MCD64 and algorithm estimated burned area from $P_{b\ MERGED} > 0.5$ for each tile. Also shown is the ordinary least squares regression (for all tiles) between the two estimates (blue line) and the 1:1 line (dashed). The algorithm detects proportionally 33% more burned area across the three tiles than the MCD64 algorithm (regression slope 1.33). There is an agreement (coefficient of determination r^2=0.6) between burned area from the two products. The greater disagreement in terms of absolute burned are is found the h20v08 tile which has the greatest magnitude of burning of the three tiles. The right panel in figure 8.20 depicts the relation between the estimated burned area from the MODIS-only estimates of burned area ($P_{b\ MODIS} > 0.5$) and the corresponding estimate from only the merging of VIIRS and OLCI ($P_{b\ VIIRS+OLCI} > 0.5$). A strong agreement can be seen between the two estimates

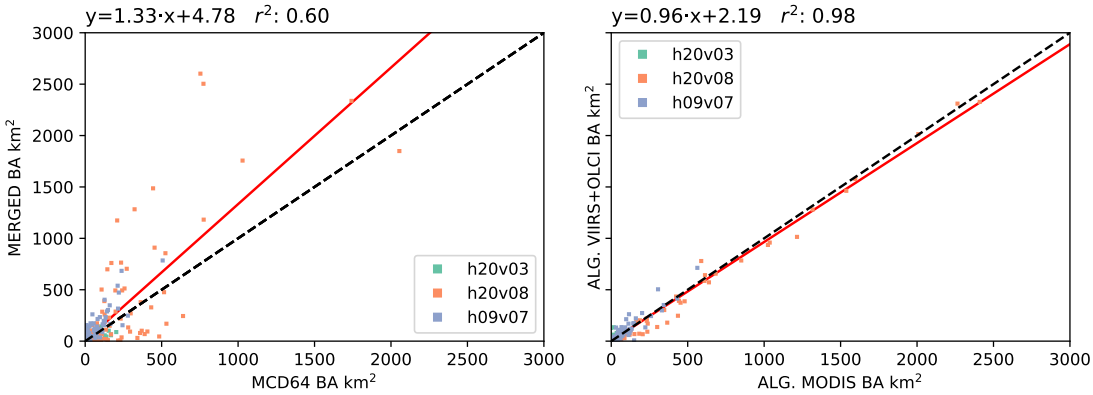


Figure 8.20. Left) Scatter plot between estimate of burned area from $P_{b\ MERGED}$ and MCD64 for each tile. Right) estimate of burned area from $P_{b\ MODIS}$ vs $P_{b\ VIIRS+OLCI}$. Also shown is the ordinary least squares regression between the two estimates (blue line) and the 1:1 line.

(coefficient of determination $r^2 = 0.98$) with a slight overestimate for burned area from MODIS relative to VIIRS+OLCI (regression slope 0.96). The agreement is also comparatively better for the h20v08 tile (when compared to the MERGED estimate). This agreement suggests that a future multi-sensor implementation of the algorithm with only VIIRS and OLCI may provide a consistent estimate of burned area compared with MODIS following the expected degradation of the MODIS sensors in the 2020s.

Relative to the MCD64 product, there is also evidence that the algorithm performs better at detecting fires which are fewer than 1–5 MODIS/VIIRS 500m pixels. Figure 8.21 illustrates examples from each tile where the algorithm detects more small and patchy burn scars than MCD64. In the h09v07 tile, the algorithm detects a greater proportion of the areas of cropland burning which are patchy and are only a few 500m MODIS pixels in size. The MCD64 product reports a much smaller burned area than the algorithm and misses some of the smaller burned areas. The subset from the h20v08 tile shows an example where the MCD64 product shows less sensitivity than the presented algorithm to mapping mixed burned-unburned areas at the perimeters of burn scars. The actively burning fire in the Sentinel-2 subset from the h20v03 shows an example where the MCD64 product detects a greater area than the presented algorithm.

This greater sensitivity to small fires is also suggested by the better detection accuracy by the algorithm to sub-pixel burning. Figure 8.22 shows the proportion of burned pixels correctly identified by MCD64 and the merged algorithm result as a function of the sub-pixel burned fraction f of the MODIS pixel. f was calculated from the aggregation of the 20m Sentinel-2 burned area map to the 500m MODIS sinusoidal grid and therefore provides an approximation to the true area of the 500m pixel

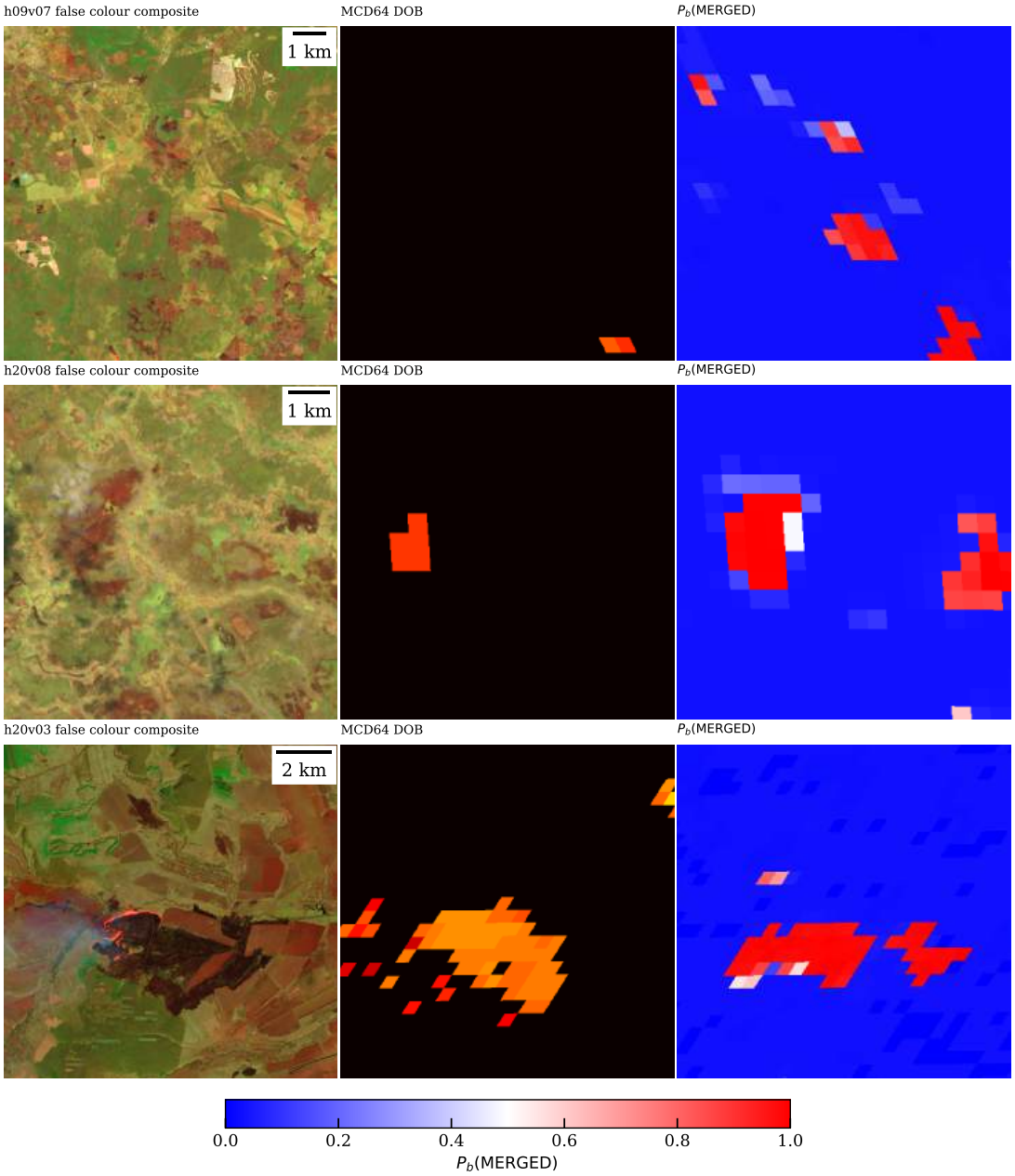


Figure 8.21. Detection of small fires by the algorithm compared to MCD64. Left column) false colour composites of the Sentinel-2 post fire acquisitions (red: B12, green: B8, blue: B3). Middle column) burned area reported by MCD64. Right column) $\mathrm{P_{bMERGED}}$).

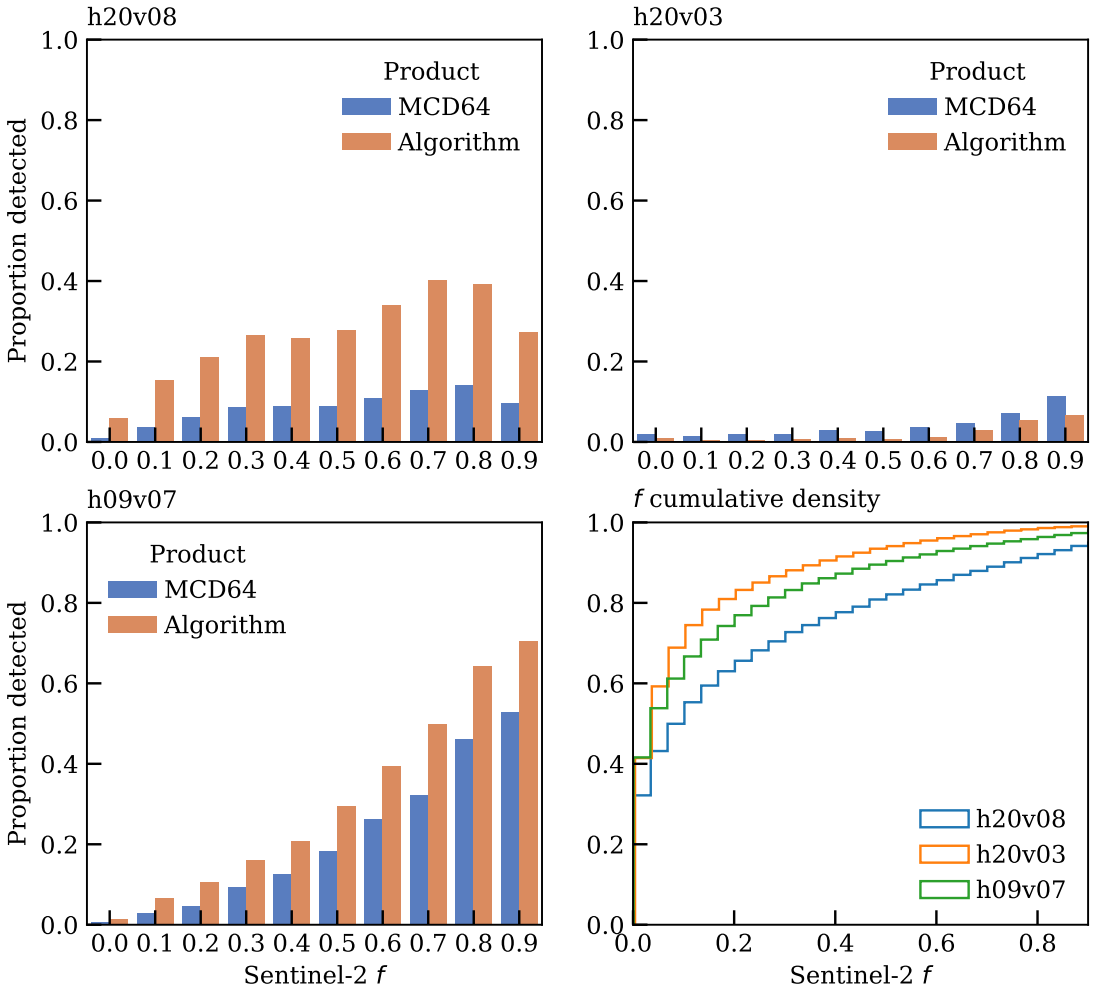


Figure 8.22. Sensitivity of the algorithm and MCD64 to fractional burning of the pixel f. Shown is the proportion of pixels correctly detected by MCD64 and the algorithm binned by the fractional burning of the pixel f. f is calculated from the aggregation of 20m sentinel-2 pixels to provide fractional burning of the 500m MODIS/VIIRS pixels.

affected by fire. For the h09v07 tile, the merged algorithm correctly detects 5–20% more burned area (i.e. where $\mathrm{P}_{\mathrm{bMERGED}} > 0.5$) with fractional burning of the pixel (i.e. $f < 0.9$) than the MCD64 product. For the h20v08 tile, the algorithm correctly detects up to 30% more burned area relative to MCD64 for similarly fractionally burned pixels. This represents a considerable improvement over MCD64 because around 75-90% of burned pixels in the three tiles have fractional burning where $f < 0.5$ (from the cumulative distribution of f in figure 8.22). The exemption is found in the boreal tile h20v03 where MCD64 shows better detection with f, although the relative difference in omission errors with f is smaller ($< 7\%$). Future refinements to the algorithm could focus on improving detection in these areas.

8.6 Discussion

This chapter has demonstrated the algorithm prototyped in chapter 7 on a real multi-sensor system comprising observations from Suomi-NPP VIIRS, NASA MODIS and ESA Sentinel-3 OLCI. This chapter completes the work of addressing the third aim of this thesis specified in section 1.2 to "prototype an algorithm for the consistent inference of burned area from multiple sensors suitable for generating a long-term CDR of burned area".

The validation against Sentinel-2 images shows lower omission errors and higher dice coefficients for two of the three test regions than the MCD64 product. The algorithm detects 33% more burned area than the MCD64 product. There is also evidence that the algorithm is more attuned to detecting sub-pixel burning and small fires (i.e. fewer than five MODIS pixels) than the MCD64 product. Changes made to the Collection 6 MCD64 algorithm have improved sensitivity to small burned patches and isolated burned pixels (Giglio et al., 2018). Nevertheless, the presented algorithm detects up to 30% more fractionally 500m burned pixels as determined by the Sentinel-2 validation images. Previous studies have shown the underestimation of burned area by coarse-resolution products due to small fires is significant. Randerson et al. (2012) reported a 35% increase in global burned area by the inclusion of small fires. A future larger processing of the algorithm could be used to confirm this greater sensitivity to small fires. Beyond the improvement in detection of sub-pixel burning, the multi-sensor characteristic of the algorithm also has the potential to improve the temporal reporting accuracy of burning. The reduction in the reporting uncertainty on the day of burn (DOB) was shown in figure 8.10 to be a result of the overlapping acquisition opportunities of each sensor providing a better refinement on DOB than only one sensor. Improvements in this may help improve the temporal resolution of 'bottom-up' inventory approaches for emission estimates for example (Matthias et al., 2018; Giglio et al., 2013b).

The structuring of the algorithm allows for the direct quantification of the contribution of each sensor to mapping burned area in Bayesian framework through P_b. The contribution of OLCI to the detection of smaller fires with a more limited spectral change signal was identified as one contribution of the better resolution of the OLCI instrument. Similarly, the potential VIIRS+OLCI only system was contrasted with the 'MODIS-only' results. Estimates of burned area from both systems were shown to be in a good agreement ($r^2 = 0.98$), indicating that a future VIIRS+OLCI-only system would provide comparable performance following the expected degradation of the MODIS instruments in the 2020s. Such information is important for assessing whether next generation of high-temporal but coarse spatial resolution optical instruments (i.e. VIIRS and OLCI) are able to continue the 20-year history of

fire monitoring from MODIS (Justice et al., 2002; Chuvieco et al., 2019). The probabilistic formulation of the algorithm also enables more thorough future research into this question via the undertaking of observing system simulation experiments (OSSEs) which simulate the impact of additional sensors on the uncertainties of a remote sensing retrieval (Kaminski and Pierre-Philippe, 2017; Timmermans et al., 2015).

It has been demonstrated the presented framework allows for the retrieval of fcc values from each sensor consistent within a mean percentage error of 13%. The agreement in fcc values between MODIS and VIIRS (mean percentage error -4%) indicates that fcc can be consistently retrieved between different multi-spectral sensors. The constrained retrieval system designed primarily for VIS-NIR instruments was able to retrieve fcc with errors of 13% between MODIS and OLCI, and 7% between VIIRS and OLCI. Using a spectral invariant model of the burn signal was shown to have specific advantages over for example using vegetation indices or changes in individual reflectance channels. First, sensors with different spectral sampling can be used within the algorithm. This is significant because algorithms which rely on, for example, changes in SWIR reflectance, such as MCD64, are unable to use observations from sensors with only visible-NIR spectral sampling. Secondly, the assumptions about the spectral burn signal can be applied consistently to diverse sensor streams as the retrieved parameters explicitly represent the same quantities. The downside of the presented approach is that the retrieval of the corrected surface reflectance is carried out in a sensor-dependent manner. As such it is not possible for the pre- or post-fire reflectance to originate from different sensors which prevents the fullest exploitation of the information available. The merging of observations of surface reflectance from several instruments in this manner will require considerable pre-processing of the measurements, consistent atmospheric correction and thorough uncertainty characterisation (Claverie et al., 2018; Zhang et al., 2018; Merchant et al., 2017; Trishchenko et al., 2002). At present the work necessary to merge these surface reflectance data has not been completed. This can be seen, for example, in the observable biases in the reflectance from OLCI compared to MODIS and VIIRS highlighted in figure 8.8. Merging these reflectance time series may lead to apparent changes in reflectance which are only a product of the sensors used and not true changes in the land surface. This would complicate the detection of true changes in the reflectance signal and lead to potentially greater omission and commission errors. The fcc model is also potentially robust to biases in reflectance so long as the relative change in reflectance (i.e. $\Delta\rho/\rho_{-}$) is comparable between sensors. This is because the model (equation 7.1) rearranges to an effectively a normalised vegetation index $fcc = (\Delta\rho - \rho_{-})/(\rho_{\text{burn}} - \rho_{-})$.

The structuring of the algorithm around two retrieval systems: with high-quality retrievals from spectrally-rich MISS instruments, and lower-quality retrievals assimilated from additional LISS instruments, makes the algorithm adaptable to changes in sensor availability through the satellite record. With the operational designation of the VIIRS instrument (Vermote et al., 2014), coarse-resolution MISS instruments will continue to be available into the future. Additional instruments such as SPOT-VGT, SLSTR, and MERIS can be added into the processing in the future to produce an ECV record of burned area from 1998–present. The concept of the constrained fcc model inversion could also be extended to provide a system for using the 250m MODIS red and NIR channels (Justice et al., 2002). For example, the fcc model could be first solved from the 500m MODIS channels and this solution used as a prior to retrieve fcc and therefore burned area at 250m. This would be preferable to only using the 250m red/NIR channels which limits the spectral information available as done by Chuvieco et al. (2018) or assume that 500m channels can be downscaled to 250m.

The constrained fcc retrieval system also opens up the opportunity to produce the longest possible burned area record through the inclusion of Advanced Very High Resolution Radiometer (AVHRR) data going back to 1982. Otón et al. (2019) produced the first global AVHRR record of burned area spanning 1982—2017. This was achieved via correlating several spectral indices with the MODIS MCD64 product from the overlapping period (2001—2017). In a similar manner, detailed information about fcc and the spectral parameters is available from the MODIS record (as shown in see figure 7.9). The fcc retrieval system could therefore use the MODIS archive and other modern coarse-resolution instruments to provide suitable priors for the AVHRR record before 2000. There are several key advantages of such an approach over that of Otón et al. (2019). First, the greater consistency in the algorithm assumptions (i.e. no requirement for correlating an AVHRR spectral index with the different index used within the MCD64 algorithm). And second an improved consistency between different periods of the satellite record through the uncertainty characterisation features of the algorithm providing a quantitative statement of the quality of the AVHRR inference.

Chapter Nine

Conclusions

This thesis has focused on the development of new techniques for monitoring burned area from remote sensing data and the quantitative evaluation of burned area products. Chapter 1 introduced the central aim of this thesis to provide advancement in the methods for the design and assessment of burned area algorithms, with the ultimate aim of making remote sensing burned area products more useful and scientifically rigorous. This general aim was achieved by considering three objectives:

I Evaluating present practices and developing new methods to assess the qualities of burned area mapping algorithms.

II Developing uncertainty quantification (UQ) methods for burned area products and algorithms following metrological best practice.

III Prototyping an algorithm for the consistent inference of burned area from multiple sensors suitable for generating a long-term CDR of burned area.

9.1 Contributions

Part I New methods for comparing and evaluating burned area mapping algorithms

The first section considered the present science of burned area mapping. Chapter 2 explored the development of these algorithms and their key decision structures. Chapter 3 presented the first 'round-robin' intercomparison exercise of burned area mapping algorithms. The exercise used the principles of statistical emulation to learn the key decision structures of six burned area algorithms. These surrogate

models for the algorithms learned how each algorithm interprets the available information in the observations and revealed the extent to which various characteristics of the observations determined algorithm performance. Practically this involved training random forest (RF) classifiers on mapped omission and commission errors for each algorithm across a globally distributed collection of 93 Landsat ETM+ burned area maps. The RF classifiers were able to replicate the true algorithms, with the accuracies for all algorithm emulators exceeding 84%. Feature importances from the RF classifiers indicated, that for all algorithms, the magnitude of the fire change signal in the reflectance observations was the primary determinant of omission errors. The fractional burning of the pixels also determined algorithm detection. The emulators also revealed information about the decision structures of the algorithms, most notably the presence of implicit sensitivity thresholds to algorithm inputs. By being able to quantify the value of these detection thresholds, they represent useful heuristics to evaluate any improved sensitivity of the next generation of mapping algorithms. The analysis also revealed that algorithms do not achieve the 15% omission/commission error targets except in exceptional cases. The 15% omission error target is only achieved when $fcc > 0.8$, which corresponds to only 3% of the observed burning in the reference data. No current algorithm is within the 15% target at the global level, even for pixels that have complete burning of the pixel (fractional burned area $f = 1$). The methods developed have the potential to routinely implemented with new algorithms compared against these thresholds to provide a framework for monitoring scientific advances in burned area science.

Part II Uncertainty quantification for burned area algorithms and products

Chapter 4 presented a discussion of the principal sources of uncertainty in burned area product generation. Chapters 5 and 6 then presented two new methods for generating uncertainties for burned area products. Chapter 5 presented a method for quantifying uncertainties on burned area from three global products. A multiplicative triple collocation (MTC) model was applied to three global burned area products, providing estimates of the standard errors in each product globally for 2001-2013. The analysis yielded relative uncertainties of 4–5.5% on mean global burned area for the three products, and uncertainties of 8–10% in the most fire prevalent regions of Africa and Australia. Large relative uncertainties revealed in shrublands for the products have not been previously identified. The provided standard error estimates are beneficial to the wider community as they are suitable for use in model-data comparison studies (Poulter et al., 2015; Forkel et al., 2019; Knorr et al., 2012) as well as to calibrate parameters in dynamic global vegetation models (DVGMs) using uncertainty weighted optimisation methods (Thonicke et al., 2010; Mangeon et al., 2016). Chapter 6 then developed

an uncertainty quantification (UQ) framework for burned area products and algorithms. This involved adapting metrology principles as outlined in BIPM et al. (1995) to be suitable for the discrete nature of the burned area measurement. The principles of UQ for satellite retrievals were developed via an example burned area algorithm, and definitions of uncertainty suitable for pixel-level burned area products developed. The scaling of uncertainties from the pixel-level to climate model grid scales was also considered. The developed framework provides the tools on which to develop the next generation of uncertainty quantified burned area algorithms.

Part III Prototyping an algorithm for consistent multi-sensor burned area retrieval

The generation of the longest possible ECV record of burned area will require the merging of observations from multiple sensors. To extract the maximum value from the available observations requires burned area algorithms which can generate consistent products which are transparent about the uncertainties inherent to the process. Chapters 7 and 8 addressed this task.

Chapter 7 first explored the potential options for generating a consistent multi-sensor CDR burned area product and necessary mathematical and statistical frameworks for merging observations from sensors with varying information content. It was argued that inferring evidence of burning from sensors with differing spectral channels requires a spectrally invariant treatment of the fire signal. The fcc model was posited as a method for consistent inference with the addition of Bayesian methods for propagating information between sensors with different spectral sampling. A demonstration of the algorithm on a synthetic multi-sensor system constructed from the two MODIS instruments demonstrated that the algorithm was able to merge information from the two sensors and improve the quality of the retrieved burned area and reduce uncertainties.

Chapter 8 then implemented the algorithm for a real multi-sensor system based on observations from the Suomi-NPP VIIRS, NASA MODIS and Sentinel-3 OLCI. Comparisons between sensors showed that the algorithm could consistently retrieve fcc. Further, the multi-sensor framework was shown to improve the detection of small fires and sub-pixel burning and reduce the temporal reporting uncertainty of burned area. Validation against Sentinel-2 data also indicated that the algorithm had comparable performance to the MCD64 product, with two of the three regions showing lower omission errors.

9.2 Future work

The 'round-robin' intercomparison exercise presented in chapter 3 could be advanced in several ways. Primarily, the exercise could be repeated in the future when new versions of the algorithms are available. This would provide a framework in which to assess algorithm improvement. Similarly, the information that the random forest emulators reveal about algorithm decision structures provides information suitable for comparing algorithm versions. All algorithms displayed a threshold in fractional burning of the pixel f of about 0.5. Reducing this threshold provides a target for each iteration of the round-robin exercise in the future.

Ideally, the repetition of the round-robin exercise would be conducted with input from the community of algorithm producers. The success of the RAdiation transfer Model Intercomparison (RAMI) project provides a model for this. Across the four conducted RAMI phases, developers of radiative transfer models have submitted results for a set of synthetic experiments and automated benchmarking process is undertaken to evaluate and rank model performance (Widlowski et al., 2013). Experiments and evaluation criteria are agreed by the community which therefore help to set a focus towards areas of improvement i.e. in the RAMI case a better prediction of realistic virtual plant canopies (Widlowski et al., 2015).

The uncertainties developed for the three global burned area products with the triple collocation (TC) model in chapter 5 have potentially constructive applications. Most obviously, the uncertainties could be used to drive the development and refinement of fire-related parameters in DVGMs. Initial work would consider a sensitivity analysis of key parameters within the fire process models. The range of acceptable parameter variation could be linked to the range of the predicted global burned area in comparison to the range provided by the observational uncertainties (Gomez-Dans, 2018).

Following this, the TC uncertainties provide the necessary information for the application of more advanced optimisation techniques for the estimation of model parameters. TC uncertainties per observation allows for the assimilation of the satellite observations into fire models through data assimilation. This would be most fruitful with 'empirical' fire models such as the INteractive Fire and Emission algoRithm for Natural envirOnments (INFERNO) and SPITFIRE (Mangeon et al., 2016; Thonicke et al., 2010) which can be run quickly and therefore optimised against observations. There has already been some wider interest in this application. The TC uncertainties were presented at the 5th Fire Model Intercomparison Project (FireMIP) in Frankfurt in 2018. There was interest in the application of the uncertainties for the optimisation and evaluation of fire models within FireMIP (Rabin et al., 2017). Preparatory work has also started on using the TC uncertainties to optimise UK

Earth System Modelling Project (UKESM) models including INFERNO in a manner similar to above. The TC method and uncertainties have been published and made available to the wider community (Brennan et al., 2019).

The UQ framework for burned area mapping algorithms in chapter 6 provides the alternative route to better uncertainty information for burned area products. Chapter 6 provides a framework following metrological best practice which could be used to implement uncertainty quantification steps into present algorithms. Similarly, the definition of the 'probability of burning' P_b as an uncertainty quantified probability provides the opportunity for an agreed definition of uncertainty in these products. The work in chapters 6 and 3 has contributed to uncertainty characterisation within the European Space Agency's Fire Climate Change Initiative (FireCCI). The FireCCI burned area products since FireCCI50 have implemented UQ and provided per-pixel P_b uncertainties. Refinement of the UQ methods (for example the issues detailed in figure 5.10) could be the focus of future work.

The development of a multi-sensor algorithm with quantified uncertainties in section III could also be advanced in the future. Some key results from chapter 8 indicate the algorithm to be a suitable basis for a future global multi-sensor processing. The agreement in burned area from MODIS and VIIRS/OLCI ($r^2 = 0.98$) indicates that the algorithm could provide a consistent data record from 2000—onwards following the expected degradation of the MODIS instruments in the 2020s. At present the development of the NASA MODIS-successor burned area products from VIIRS (VNP64) is still ongoing. Giglio et al. (2013) have highlighted some of the initial challenges of ensuring consistency between VNP64 and MCD64 – primarily differences in the spatial resolution of the SWIR channels onboard VIIRS/MODIS. The inclusion of OLCI and the multi-sensor framework of the algorithm presented in chapter 8 minimises the effect of this because of the higher spatial resolution of OLCI. Once VNP64 is released, a global processing of the algorithm could be used to consider whether the algorithm provides a more consistent record than VNP64/MCD64.

The constrained fcc retrieval system within the algorithm also presents an opportunity to process Advanced Very High Resolution Radiometer (AVHRR) data going back to 1982. The spectral invariance properties of fcc, a_0 and a_1 suggest that with sufficient prior information they can be retrieved from any optical sensor. Detailed information about fcc and the spectral parameters is available from the MODIS record (as shown in see figure 7.9) which may suitable for defining this prior backwards through the AVHRR record. This would provide a long term data record of burned area approaching 40 years. Otón et al. (2019) produced the first global AVHRR record of burned area spanning 1982—2017, based on correlating AVHRR spectral indices with burned area detected by the MODIS MCD64 product for 2001-2017. The proposed method here would be similar – using information from the MODIS

$[fcc, a_0, a_1]$ record to provide suitable priors for the AVHRR record before 2000. The advantage over Otón et al. (2019) would be the greater consistency in the algorithm assumptions – through the use of spectral fire model as opposed to the correlation of AVHRR data against the MCD64 product. Otón et al. (2019) identified that the foremost challenge of this was dealing with the low spatial resolution (4km) of AVHRR data, which inevitably leads to pixels with a mixture of cloud, shadow and burned area. The interpretation of fcc as the combined impact of fractional burning f and combustion completeness cc provides potentially a better model for dealing with this. Initial work should focus on the minimum detectable fcc from 4km AVHRR data for pixels identified as burned from higher-resolution burned area products or validation data.

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